

Enthalpies of Sublimation of Organic and Organometallic Compounds. 1910–2001

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A compendium of sublimation enthalpies, published within the period 1910–2001 (over 1200 references), is reported. A brief review of the temperature adjustments for the sublimation enthalpies from the temperature of measurement to the standard reference temperature, 298.15 K, is included, as are recently suggested values for several reference materials. Sublimation enthalpies are included for organic, organometallic, and a few inorganic compounds. © 2002 American Institute of Physics.

Key words: compendium; enthalpies of condensation; evaporation; organic compounds; organometallic compounds; sublimation; sublimation enthalpy.

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1. Introduction

Sublimation enthalpies are important thermodynamic properties of the condensed phase. Frequently they are used in correcting enthalpies of formation to the gas phase and in evaluating environmental transport properties.^{1,2} Sublimation enthalpy measurements are also useful to studies of polymorphism and predictions of molecular packing. The measurements provide benchmark numbers that can be used to validate the calculations.³ Examination of the data in this compendium will reveal some large discrepancies in reported enthalpies of sublimation. It is likely that some of the discrepancies reported by different laboratories are due to measurements made on different polymorphic modifications.⁴ Sublimation enthalpy measurements also can reveal differences in interactions in chiral solids and their racemic modifications. Very little experimental work has been reported in this respect.^{5,6}

Our interests in sublimation enthalpies goes back nearly 3 decades.⁵ Initially interested in using sublimation enthalpies to correct enthalpies of formation data to a standard state, we have since focused our attention on their measurement,⁷ estimation,⁸ and assessment.⁹ In a parallel study, a compilation of available sublimation enthalpies was initiated in the 1980s.⁵ A reasonably exhaustive version of this database covering the literature up to the mid 1990s is available on line at <http://webbook.nist.gov/chemistry/>. The present version updates this compilation to the year 2001. Although our intent has been to provide an exhaustive coverage of the literature from 1910 to 2001, this listing is probably still far from complete.

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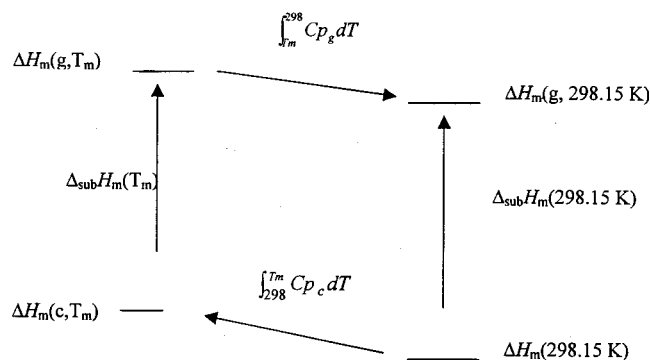


FIG. 1. A thermodynamic cycle for adjusting sublimation enthalpies to 298.15 K.

2. Heat Capacity Adjustments

Sublimation enthalpies are measurements based on mass transport and as such are directly or indirectly dependent upon vapor pressure. The vapor pressure of different solids at the same temperature can vary by many orders of magnitude. In order to obtain a reasonable amount of mass transport, it is frequently necessary to conduct these measurements at temperatures that differ substantially from the standard reference temperature, 298.15 K. The actual temperature of measurement depends on the sensitivity of the instrument or apparatus and the properties of the substance. In addition, these measurements are often conducted as a function of temperature.

The magnitude of the sublimation enthalpy is dependent on temperature. Figure 1 and Eqs. (1) and (2) illustrate the origin of this temperature dependence in terms of a thermodynamic cycle. If the heat capacities of the solid and gas phase are known, C_{p_c} and C_{p_g} , respectively, then the sublimation enthalpy at 298.15 K can be related to the experimental measurements by using Eq. (1). This equation, generally referred to as Kirchhoff's equation, can be used to adjust sublimation enthalpy measurements to any reference temperature. T_m represents either the temperature of measurement for calorimetric measurements or the mean temperature of measurement for experiments conducted over narrow ranges of temperature. Treating the heat capacities of the two phases as independent of temperature and integrating Eq. (1) results in Eq. (2). Since the magnitude of the heat capacity of the gas phase is usually smaller than that of the solid phase (c), sublimation enthalpies increase with decreasing temperature.

$$\Delta_{\text{sub}} H_m(298.15 \text{ K}) = \Delta_{\text{sub}} H_m(T_m) + \int_{298.15}^{T_m} (C_{p_c} - C_{p_g}) dT, \quad (1)$$

$$\Delta_{\text{sub}} H_m(298.15 \text{ K}) \approx \Delta_{\text{sub}} H_m(T_m) + (C_{p_c} - C_{p_g})[T_m - 298.15]. \quad (2)$$

Experimental heat capacities for many solids at 298.15 K are available.¹⁰ Experimental gas phase heat capacities for compound that are solids at 298.15 K are unavailable and generally need to be estimated. Gas phase heat capacities can be calculated from statistical mechanics or estimated by group additivity methods.¹¹ A number of group additivity methods have been developed to estimate gas phase heat capacities.^{11–13} However, group values for some functional groups are not available. This has encouraged the development of other estimation methods. Table 1 briefly summarizes the various equations that have been used in place of the second term in Eq. (2).

Equation (3) can easily be derived by assuming that the gas is ideal and that the Dulong–Petit value of 3RN holds for the solid, where the term R represents the gas constant and N is the number of atoms/molecule.⁵ A similar relationship but characterized by a temperature coefficient of 6R [Eq. (4)] has been suggested by Pedley.¹⁴ Temperature coefficients of 40 J mol^{−1} have been used by Melia and Merrifield,¹⁵ and a value of 60 J mol^{−1} has been used by de Kruif *et al.*¹⁶ for a series of amino acids and peptides.

A major limitation of most of the equations listed in Table 1 is that the heat capacity adjustments are treated as universal constants independent of molecular structure. Only Eq. (7) is sensitive to differences in molecular structure. This equation was derived from a correlation using estimated heat capacities of the solid at 298.15 K.¹⁷ This correlation was developed from the observed dependence of the temperature adjustment on both molecular structure and size.¹⁷ Experimental or estimated values of $C_{p_c}(298.15 \text{ K})$ can be used in this equation.

Previous work has demonstrated that Eq. (7) gives results that are generally as good as or better than the use of the other equations in Table 1.^{7,8(b),18} The use of Eq. (7) should be limited to the temperature range 200–500 K. A standard deviation of $\pm 33 \text{ J mol}^{-1}$ has been associated with the term: $[0.75 + 0.15C_{p_c}(298.15 \text{ K})]$. The total uncertainty of the temperature adjustment depends on both the magnitude of C_{p_c} and T_m . In applications, an uncertainty of one-third of

TABLE 1. Equations for the temperature adjustments of sublimation enthalpies

Corrections for the sublimation enthalpies (J mol ^{−1})	Equation	Reference
$(C_{p_c} - C_{p_g})[T_m - 298.15] = 2R[T_m - 298.15]$	(3)	5
$(C_{p_c} - C_{p_g})[T_m - 298.15] = 6R[T_m - 298.15]$	(4)	14
$(C_{p_c} - C_{p_g})[T_m - 298.15] = 40[T_m - 298.15]$	(5)	15
$(C_{p_c} - C_{p_g})[T_m - 298.15] = 60[T_m - 298.15]$	(6)	16
$(C_{p_c} - C_{p_g})[T_m - 298.15] = [0.75 + 0.15C_{p_c}(298.15 \text{ K})][T_m - 298.15]$	(7)	17

the total temperature adjustment has been arbitrarily chosen as the uncertainty ($\pm 2 \sigma$).⁹

Equations (3)–(6) do not require C_{p_c} values; their use can be an advantage if an appropriate group value or experimental heat capacity is unavailable for a particular substance. Temperature adjustments to 298.15 K are often small and frequently of the same order of magnitude as the uncertainty associated with the measurement. This is the rationale some authors give for not adjusting the measurements for temperature. It should be emphasized that the magnitude of the sublimation enthalpy will increase with decreasing temperature and even though the temperature adjustment may be small, failure to adjust for temperature incorporates a systematic error that can easily be minimized by using one of the equations in Table 1.

The sublimation enthalpies reported in this paper, have not been adjusted to 298.15 K unless done so by the reporting authors. Different authors have used different methods. In some cases experimental data have been used for C_{p_c} and only C_{p_g} has been estimated. The reader is encouraged to refer to the original literature for details. In an effort to provide some assistance to the reader in this regard, a brief discussion of one of the few group additivity methods that are available for estimating the heat capacity of solids is included below.^{12,19} This is followed by an illustration of how this value can be used in conjunction with Eq. (7) to provide temperature adjustments.

3. Group Additivity Values for C_{p_c} (298.15 K) Estimations

Table 2 parts (A) and (B) lists a set of group values that can be used in estimations of C_{p_c} (298.15 K). The groups and their corresponding values are identified by the italics. A hypothetical molecule is given in Fig. 2 that identifies each hydrocarbon group. The functional groups are self-explanatory. The R terms in Table 2 represent unidentified groups and are not included in the value. The use of these group values is illustrated with examples of C_{p_c} (298.15 K) estimations in Table 3. Values in brackets should be considered as tentative assignments. Further details are available in the literature.¹⁹

4. Reference Materials for Sublimation Enthalpy Measurements

Calibration is a fundamental requirement for every sublimation enthalpy measurement. Unlike other thermochemical measurements, uncertainties in sublimation enthalpies can be large, often several kJ mol^{-1} or more, particularly for com-

pounds exhibiting low vapor pressures. While some of the observed differences in reported enthalpies may be due to polymorphism, others are probably due to the lack of a sufficient number of reference compounds that vary in their range of volatility. The ability of an experimental technique to measure vapor pressure in one pressure or temperature region does not in itself guarantee the same accuracy in another. Substantial variations in sublimation values are revealed in the tables that follow. This variance clearly establishes the importance of documenting the accuracy of the measurements through the use of appropriate reference materials that approximate the temperature and pressure regimes of the measurements.

A series of compounds have recently been proposed as reference materials.⁹ These have been classified as primary, secondary, or tertiary reference materials, on the basis of various criteria. The materials classified as primary and secondary reference materials are listed in Table 4. The temperature range, the corresponding vapor pressures, and the recommended values are also included in the table.

5. Sublimation Enthalpy Compendium

The sublimation enthalpies, reported during the time period 1910–2001, are included in Tables 6 and 7. Table 5 contains a listing of the acronyms that are used in these two tables. Table 6 contains sublimation enthalpy data for organic compounds and Table 7 contains data for organometallic and a few inorganic compounds. Information in Table 6 is organized as shown below. Compounds are arranged according to molecular formula. The name of the compound, occasionally a synonym, and the CAS registry number are included on the first line. If the information was available, the first entry on the second line contains information regarding the polymorphic form studied. However, in most cases this information was not available. The range of temperatures studied is the next entry in the table. For measurements performed at a constant temperature or when not specified, this entry is left blank. The sublimation enthalpy at the mean temperature of measurement $\Delta_{\text{sub}}H_m(T_m)$ is the next entry followed by the mean temperature (K), an acronym briefly describing the type of measurement, and the reference to the original work. In some cases the type of measurement was not available, or recorded. In these instances this entry was left blank. If the authors of the work have adjusted their results to 298.15 K, then this information along with the reference is entered on the third line. This information is repeated for multiple measurements. The measurements are arranged in reverse chronological order. A similar format is followed in Table 7 with the major exception that each organometallic compound is arranged alphabetically by element and then according to the Hill system.

TABLE 2. Group values for estimating the C_{p_c} (298.15 K)

(A) Group values for estimating the C_{p_c} (298.15 K) of hydrocarbons. ^a					
Hydrocarbon Groups					
Aliphatic groups			Cyclic aliphatic and olefinic groups		
Description of group	Formula	J K ⁻¹ mol ⁻¹	Description of group	Formula	J K ⁻¹ mol ⁻¹
primary sp^3 C	–CH ₃	36.6	cyclic secondary sp^3 C	–C _c H ₂ –	24.6
secondary sp^3 C	–CH ₂ –	26.9	cyclic tertiary sp^3 C	–C _c HR–	11.7
tertiary sp^3 C	–CHR–	9	cyclic quaternary sp^3 C	–C _c R ₂ –	6.1
quaternary sp^3 C	–CR ₃	–5	cyclic tertiary sp^2 C	–C _c HR–	15.9
			cyclic quaternary sp^2 C	–C _c R ₂ –	[4.7]
Olefinic and Acetylenic Groups			Aromatic Groups		
Description of group	Formula	J K ⁻¹ mol ⁻¹	Description of group	Formula	J K ⁻¹ mol ⁻¹
secondary sp^2 C	=CH ₂	46	tertiary aromatic sp^2 C	=C _{ar} H–	17.5
tertiary sp^2 C	=CH–	21.4	quaternary aromatic sp^2 C	=C _{ar} R–	8.5
quaternary sp^2 C	=C–	6.9	internal quaternary aromatic C	=C _{ar} R–	[9.1]
tertiary sp C	≡C–H	37.1			
quaternary sp C	≡C–	15.5			
(B) Group values for estimating the C_{p_c} (298.15 K) contribution of various functional groups.					
Monodentate functional groups			Tridentate functional groups		
Description of group	Formula	J K ⁻¹ mol ⁻¹	Description of group	Formula	J K ⁻¹ mol ⁻¹
Alcohols, phenols	–OH	23.5	tertiary sp^3 N	–NR ₂	[31.5]
Fluorine	–F	[24.8]	tertiary sp^2 N	=N–	
Chlorine	–Cl	28.7	cyclic tertiary sp^2 N	=N _c –	13.9
Bromine	–Br	32.4	cyclic tertiary sp^3 N	–N _c R–	1.2
Iodines	–I	[27.9]	cyclic tertiary amide	–CONR–	52.7
Nitrile	–CN	42.3	cyclic imide	–CONHCO–	74.1
Carboxylic acid	–CO ₂ H	53.1	phosphine oxide	–(PO)R–	28.5
Acid chloride	–COCl	[60.2]			
Aldehyde	–(C=O)H	[84.5]			
Isocyanate	–NCO	[52.7]			
Nitro group	–NO ₂	56.1			
Secondary sp^3 Nitrogen	–NH ₂	21.6			
Primary amides	–CONH ₂	54.4			
Thiols	–SH	[51.9]			
Sulfonamide	–SO ₂ NH ₂	104			
Substituted urea	–NHCONH ₂	82.8			
Bidentate functional groups			Tetradentate function groups		
Description of group	Formula	J K ⁻¹ mol ⁻¹	Description of group	Formula	J K ⁻¹ mol ⁻¹
Ketones	–CO–	28	quaternary silicon	–SiR ₂ –	32.4
Cyclic ketones	–(CO) _c –	34.3	quaternary tin	–SnR ₂ –	77.2
Ester	–CO ₂ R	40.3	quaternary germanium	Ge	18.9
Lactones	–CO ₂ –	45.2			
Cyclic carbonates	–OCO ₂ –	[68.2]			
Cyclic anhydrides	–CO ₂ CO–	80.3			
Ether	–O–	49.8			
Cyclic ether	–Oc–	9.7			
Secondary sp^3 N	–NH–	–0.3			
Cyclic secondary sp^3 N	–N _c H–	23.9			
Tertiary sp^2 N	=NH	10.7			
Secondary amide	–CONH–	44.4			

TABLE 2. Group values for estimating the C_{p_c} (298.15 K)—Continued

Bidentate functional groups			Tetradentate function groups		
Description of group	Formula	$\text{J K}^{-1} \text{mol}^{-1}$	Description of group	Formula	$\text{J K}^{-1} \text{mol}^{-1}$
Cyclic secondary amide	$-\text{CONH}-$	46.4			
Tertiary sp^3 N	$-\text{NR}_2$	31.5			
Cyclic urea	$-\text{NHCONH}-$	63.6			
Carbamates	$-\text{OCONH}-$	76.1			
Sulfides	$-\text{S}-$	116			
Cyclic sulfides	$-\text{S}_c-$	20.3			
Disulfides	$-\text{S}-\text{S}-$	41			
Sulfoxides	$-\text{SO}-$	47.7			
Sulfones	$-\text{SO}_2-$	88.7			

^aSee Ref. 19.

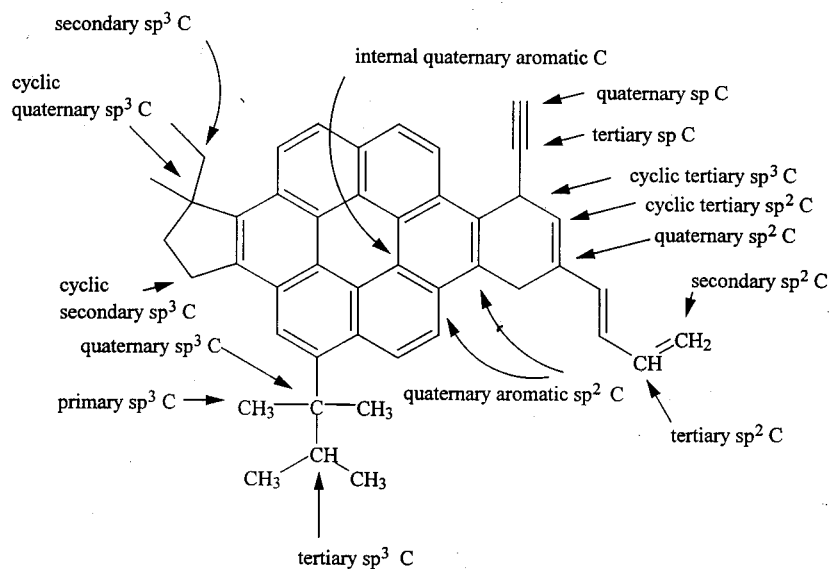
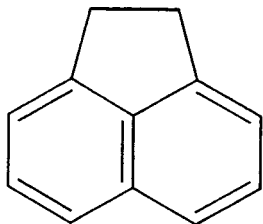


FIG. 2. A hypothetical molecule illustrating the different hydrocarbon groups in estimating C_p .

The authors have made an effort to present the data accurately and without error. However some of the information has been obtained from non-English journals with translations occasionally provided by the author's students. These tables have been compiled over a period of 25 years and

have gone through numerous revisions. Some errors have been corrected; however it is unlikely that all the errors have been detected and corrected. The reader is encouraged to consult the original literature when using this compendium.

TABLE 3. Some estimations of $C_{p,c}$ (298.15 K) using the group values of Tables 2(A) and 2(B)

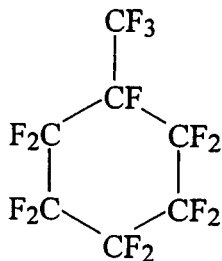
$$C_{p,c}(298.15 \text{ K}) = 6 \cdot 17.5 + 4 \cdot 8.5 + 2 \cdot 24.6$$

$$= 188 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$(190.4 \text{ J mol}^{-1} \text{ K}^{-1})^*$$

$$\Delta_{\text{sub}}H_m(298.15 \text{ K}) = \Delta_{\text{sub}}H_m(T_m) +$$

$$[0.75 + (0.15 \cdot 197)][T_m - 298.15]$$



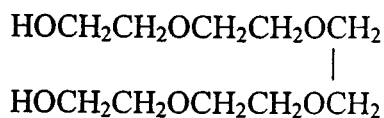
$$C_{p,c}(298.15 \text{ K}) = 6 \cdot 6.1 - 5.0 + 14 \cdot 24.8$$

$$= 378.8 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$(353.1 \text{ J mol}^{-1} \text{ K}^{-1})^*$$

$$\Delta_{\text{sub}}H_m(298.15 \text{ K}) = \Delta_{\text{sub}}H_m(T_m) +$$

$$[0.75 + (0.15 \cdot 379)][T_m - 298.15]$$



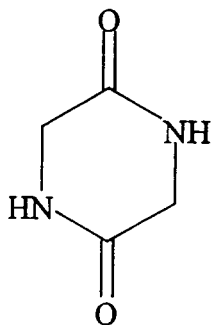
$$C_{p,c}(298.15 \text{ K}) = 10 \cdot 26.9 + 2 \cdot 23.5 + 5 \cdot 49.8$$

$$= 515 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$(515.5 \text{ J mol}^{-1} \text{ K}^{-1})^*$$

$$\Delta_{\text{sub}}H_m(298.15 \text{ K}) = \Delta_{\text{sub}}H_m(T_m) +$$

$$[0.75 + (0.15 \cdot 515)][T_m - 298.15]$$



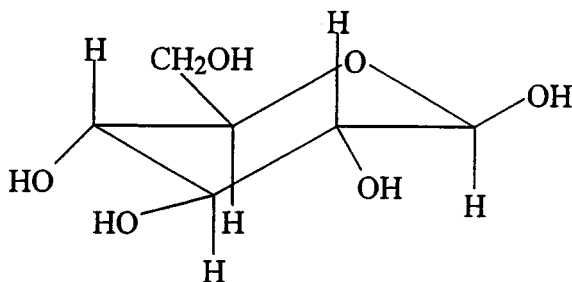
$$C_{p,c}(298.15 \text{ K}) = 2 \cdot 24.6 + 2 \cdot 46.4$$

$$= 142 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$(134 \text{ J mol}^{-1} \text{ K}^{-1})^*$$

$$\Delta_{\text{sub}}H_m(298.15 \text{ K}) = \Delta_{\text{sub}}H_m(T_m) +$$

$$[0.75 + (0.15 \cdot 142)][T_m - 298.15]$$



$$C_{p,c}(298.15 \text{ K}) = 5 \cdot 11.7 + 9.7 + 26.9 + 5 \cdot 23.5$$

$$= 212.6 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$(219.2 \text{ J mol}^{-1} \text{ K}^{-1})^*$$

$$\Delta_{\text{sub}}H_m(298.15 \text{ K}) = \Delta_{\text{sub}}H_m(T_m) +$$

$$[0.75 + (0.15 \cdot 213)][T_m - 298.15]$$



$$C_{p,c}(298.15 \text{ K}) = 15 \cdot 17.5 + 3 \cdot 8.5 + 32.4 + 28.7$$

$$= 349.1 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$(337.6 \text{ J mol}^{-1} \text{ K}^{-1})^*$$

*Domalski and Hearing.¹⁰

TABLE 4. Recommended reference standards for sublimation enthalpy measurements^a

Formula	Substance	Temperature range (K)	Vapor pressure (Pa)	$\Delta_{\text{sub}}H_{\text{m}}(298.15 \text{ K})$ (kJ mol ⁻¹)	Classification
C ₇ H ₆ O ₂	Benzoic acid	298–383	0.1–360	(89 700 ± 1000)	Primary
C ₁₀ H ₈	Naphthalene	250–353	0.1–995	(72 600 ± 600)	Primary
C ₁₀ H ₁₀ Fe	Ferrocene	277–360	0.1–166	(73 420 ± 1080)	Primary
C ₁₄ H ₁₀	Anthracene	338–360	0.1–1.0	(103 360 ± 2670)	Secondary
C ₁₆ H ₁₀	Pyrene	350–420	0.2–50	(100 200 ± 3590)	Secondary
I ₂	Iodine	273–387	4–12 600	(62 440 ± 82)	Secondary

^aSee Ref. 9.

TABLE 5. A list of acronyms used in Tables 6 and 7

A	calculated from the vapor pressure data reported by the method of least squares
B	calculated from the sum of the enthalpy of vaporization at temperature T and the enthalpy of fusion at the melting point
BE	experimental value closest to the results obtained by adding the experimental fusion and vaporization enthalpies
BG	Bourdon gauge
C	calorimetric determination
CATH	cathetometer
GC	gas chromatography
CGC-DSC	combined correlation gas chromatography-differential scanning calorimetry
DBM	dibutyl phthalate manometer
DM	diaphragm manometer
DSC	differential scanning calorimeter
E	estimated
EB	ebulliometer
EM	effusion manometer
EV	evaporation
GS	gas saturation, transpiration
GSM	glass spring manometer
HSA	head space analysis
I	isoteniscope
IPM	inclined piston manometry
KG	Knudsen gauge
LE	Langmuir evaporation
MCV	method of calibrated volume
ME	Mass effusion-Knudsen effusion
MEM	modified entrainment method
MG	McLeod gauge
MM	mercury manometer
MS	mass spectrometry
NA	not available at the time of publication
OM	oil manometer
PG	Penning gauge
QF	quartz fiber
QR	quartz resonator
RG	Rodebush gauge
SG	spoon gauge
SMZG	silica membrane zero gauge
T	tensimeter
TC	thermal conductivity manometer
TB	thermobalance
TE	torsion effusion
TGA	thermal gravimetric analysis
TPTD	temperature programmed thermal desorption particle beam mass spectrometry
TRM	thermoradiometric method
TSGC	temperature scanning gas chromatography
U	unreliable
UV	ultraviolet absorption
V	viscosity gauge
VG	MKS baratron vacuum gauge

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
CBrN	cyanogen bromide (273–308)	45.2 ± 4.2		MM	[506-68-3]
	(256–308)	47		GS	[54/7][70/1]
CBr ₄	carbon tetrabromide				[20/1]
(monoclinic)		54.5 ± 0.7	(298)	C	[558-13-4]
(monoclinic)	(295–319)	54.4 ± 1.3	(307)	BG	[84/3]
(cubic)	(321–329)	49.4 ± 1.3	(325)	BG	[59/5]
(cubic)		48.3	(320)		[59/5]
	(277–363)	51.9	(320)		[55/8]
CClN	cyanogen chloride				[41/4]
	(196–259)	35.7	(228)	A	[506-77-4]
CCl ₄	carbon tetrachloride				[47/2]
		43.3	(226)	B	[56-23-5]
	(209–225)	38.8	(217)		[63/6]
	(227–248)	37.9			[60/1][48/1]
CFN	cyanogen fluoride				[48/1]
	(147–191)	28.9	(176)		[1495-50-7]
	(139–192)	24.4	(166)	A	[87/4][64/17]
	(133–203)	29.3	(168)		[47/2]
CF ₂ N ₂	difluorocyanamide				[31/1]
	(179–198)	20.6 (liq)	(189)		[7127-18-6]
CF ₂ O	carbonyl fluoride				[87/4][66/10]
	(130–159)	23.2	(145)		[353-50-4]
CF ₄	tetrafluoromethane				[87/4][68/3]
[α]	(76–90)	14.7	(83)		[75-73-0]
[β]	(70–76)	16.8	(73)		[87/4][70/25]
	(86–89)	14.7	(88)		[87/4][70/25]
		17.0	(76)		[67/19]
	(80–86)	14.0	(83)	A	[63/6]
CF ₅ N	pentafluoromethyl amine				[33/5]
	(128–141)	18.6	(135)		[335-01-3]
CIN	cyanogen iodide				[87/4][51/18]
	(337–426)	59.9	(352)	GSM	[506-78-5]
	(298–414)	58.6	(356)	A	[87/4][43/2]
	(337–426)	59.8 ± 0.4		GSM	[47/2]
	(278–374)	58.3	(326)		[43/2][70/1]
CN ₄ O ₈	tetranitromethane				[33/2]
	(255–286)	47.4	(271)		[509-14-8]
CO	carbon monoxide				[87/4][41/6]
	(54–61)	7.6	(58)		[630-08-0]
	(51–68)	8.1	(60)	A	[87/4]
	(57–68)	7.9	(62)	A	[47/2]
CO ₂	carbon dioxide				[31/3]
	(198–216)	26.1	(207)		[124-38-9]
	(70–102)	27.2 ± 0.4		LE	[87/4]
	(139–195)	26.3	(167)	A	[74/13]
	(154–196)	26.2	(173)	A	[47/2]
CHF ₃	trifluoromethane				[37/5]
	(89–118)	25.6	(103)		[75-46-7]
CHI ₃	iodoform				[87/4]
	(308–365)	69.9	(323)		[75-47-8]
CHN	hydrogen cyanide				[43/1]
	(244–258)	35.6	(251)	MM	[74-90-8]
	(202–254)	37.6	(228)	A	[87/4][26/4]
CHN ₃ O ₆	trinitromethane				[47/2]
		45.2 ± 2.1	(298)		[517-25-9]
		54.8 ± 4.2			[99/35]
		46.7 ± 0.4			[70/7]
					[67/4][70/1]
					[77/1]
CH ₂ N ₂	cyanamide				[420-04-2]
	(227–289)	75.9	(290)	TE,ME	[83/7]
		75.2	(298)		[83/7]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
CH ₂ N ₄	tetrazole	88.16	(353)	C	[288-94-8]
		87.8 ± 1.4	(369)	ME	[93/8]
	(333–404)	88.0 ± 1.6		ME	[93/8]
	(333–363)	97.5 ± 4.2	(348)	ME	[90/31]
CH ₂ O ₂	formic acid				[51/3][70/1]
	(268–281)	60.5	(275)		[64-18-6]
	(203–218)	62.1 ± 1	(213)	TE,ME	[87/4]
	(265–268)	60.7	(266)		[78/16]
	(253–275)	60.1	(264)	A	[30/1][60/1]
(CH ₂ O ₂) ₂	formic acid dimer				[47/2]
	(203–218)	64.1 ± 1	(213)	TE,ME	[14523-98-9]
CH ₃ Cl	methyl chloride				[78/16]
	(130–172)	31.6 ± 0.1	(151)		[74-87-3]
CH ₃ Cl ₂ OP	methylphosphonic dichloride				[95/23]
		28.0		B	[40/3]
CH ₃ I	methyl iodide				[676-97-1]
	(176–227)	40.2 ± 0.4	(191)	VG	[70/1][55/4]
CH ₃ NO	formamide				[74-88-4]
	(251–273)	72.4	(264)	TE,ME	[82/6]
		71.7	(298)		[43/1][60/1]
		71.7	(276)		[75-12-7]
CH ₃ N ₅	5-aminotetrazole				[83/7]
	(383–443)	112.6 ± 1.2		ME	[83/7]
CH ₄	methane				[79/11]
	(53–91)	9.7	(72)		[4418-61-5]
	(54–90)	9.2	(72)		[90/31]
	(79–89)	10.0	(84)		[74-82-8]
	(48–78)	9.7	(63)	A,MS	[87/4]
	(67–88)	9.62	(77)	A	[63/6][55/2]
CH ₄ N ₂ O	urea				[60/1]
		90.9	(381)		[51/15]
	(338–362)	96.9	(351)	TE,ME	[47/2]
		98.6	(298)		[57-13-6]
		95.4	(361)		[87/5]
	(345–368)	87.9 ± 2.1	(356)		[83/7]
					[83/7]
CH ₄ N ₂ S	thiourea				[78/19]
		88.2	(357)		[56/6][60/1]
		112.0 ± 2	(298)	ME	[70/1][87/4]
		109 ± 2.0	(408)	TE	[53/1]
		111 ± 3.0	(298)		[62-56-5]
	(368–395)	106.6	(384)	TE,ME	[00/23]
		107.6	(298)		[94/20]
		112 ± 1.5	(298)	C	[94/20]
CH ₄ N ₄ O ₂	nitroguanidine				[83/7]
	(402–473)	142.7 ± 2.0	(298)	ME	[83/7]
CH ₅ NO	N-methylhydroxylamine				[78/15]
	(273–308)	56.6	(288)	I	[593-77-1]
CH ₅ N ₃ O	1-methyl-1-nitrosohydrazine				[87/4][57/11]
		79.5 ± 0.4	(298)		[758-19-0]
CH ₅ N ₃ S	thiosemicarbazide				[98/36]
		125.8 ± 1.5	(298)	C	[79-19-6]
CH ₅ O ₃ P	methylphosphonic acid				[82/8]
		48.1 ± 4.2			[993-13-5]
CH ₆ N ₄ S	thiocarbohydrazide				[55/4][70/1]
		152.1 ± 3.0	(298)	C	[2231-57-4]
C ₂ BrCl ₅	bromopentachloroethane				[82/8]
	(383–433)	44.4	(398)		[79504-02-2]
C ₂ Br ₂ Cl ₄	1,2-dibromotetrachloroethane				[87/4][49/11]
					[630-25-1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C_2Br_4	(383–453)	52.5	(398)	A	[87/4][49/11]
	(323–423)	56.7	(373)		[35/4]
	tetrabromoethylene				[79-28-7]
$\text{C}_2\text{Cl}_3\text{F}_3$	(221–310)	44.2	(236)	A	[87/4]
	1,1,2-trichloro-1,2,2-trifluoroethane				[76-13-1]
$\text{C}_2\text{Cl}_4\text{F}_2$	(205–233)	32.9	(219)	A	[47/2]
	1,1,2,2-tetrachloro-1,2-difluoroethane				[76-12-0]
	(235–293)	36.4	(278)		[87/4][47/2]
C_2Cl_6	(237–293)	38.2	(265)	A	[47/2]
	hexachloroethane				[67-72-1]
	(317–345)	58.9	(331)		[87/4]
(melting point 186.6)	(306–459)	48.8	(382)	A	[47/2]
triclinic form	(286–447)	59.1 ± 0.7	(367)		[47/3][60/1]
cubic form	(286–447)	51	(367)		[70/1][41/1]
					[47/3][60/1]
					[41/1]
$\text{C}_2\text{F}_2\text{O}_2$	(335–453)	50.5		GS,A	[35/3]
	(288–333)	59.0	(310)		[30/6]
	oxalyl fluoride				[359-40-0]
C_2F_6	(234–260)	16.7	(247)		[87/4]
	hexafluoroethane				[76-16-4]
C_2N_2		26	(103)		[63/6][48/4]
	cyanogen				[460-19-5]
	(202–239)	33	(224)	A	[87/4]
	(177–230)	33.6	(204)		[47/2]
	(202–245)	34.4	(223)		[39/2]
	(198–240)	32.4	(224)		[25/2][75/10]
$\text{C}_2\text{N}_6\text{O}_{12}$			NA		[16/1]
	hexanitroethane				[918-37-6]
		70.7	(298)	ME	[99/35]
	(293–343)	30.4	(308)		[87/4][63/10]
	(293–313)	70.7 ± 1.7	(303)		[69/10][77/1]
	(293–343)	70.7 ± 1.7			[68/5]
C_2H_2	acetylene				[74-86-2]
	(98–145)	23.5	(130)	H	[87/4]
		21.9	(298)		
	(133–191)	21.8	(162)	H	[60/1]
		20.2	(298)		
	(151–193)	25.2	(193)	A	[56/18]
	(130–189)	22.7	(160)		[47/2]
		21.1	(298)	H	
$\text{C}_2\text{H}_2\text{F}_3\text{NO}$	(89–169)	22.1	(129)	A	[43/4]
	trifluoroacetamide				[354-38-1]
	(288–329)	81	(302)	I	[87/4][78/6]
	(288–329)	77.7 ± 1.4	(298)	I	[78/6]
$\text{C}_2\text{H}_2\text{I}_2$	<i>trans</i> -diiodoethylene				[590-27-2]
	(253–265)	40.7	(258)	ME	[33/1][60/1]
$\text{C}_2\text{H}_2\text{O}_4$	oxalic acid (anhydrous)				[87/4]
	α	93.4	(316)	TE	[144-62-7]
	β	93.3	(318)		[87/4]
	α	98.5			[83/23]
	β	92.5			[83/23]
	α	93.7 ± 1.3	(298)	A	[75/5]
	(orthorhombic)	97.9 ± 2.2	(318)		[53/4][60/1]
	(monoclinic)	93.3	(317)	GS	[53/4][60/1]
		61.8	(306)		[47/6]
		90.6			[26/3]
$\text{C}_2\text{H}_3\text{ClO}_2$	chloroacetic acid				[79-11-8]
		75.3 ± 4.2			[28/1][49/4]
$\text{C}_2\text{H}_3\text{Cl}_3\text{O}_2$					[70/1]
	trichloroacetaldehyde hydrate				[302-17-0]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_2\text{H}_3\text{FN}_2\text{O}_5$	(263–319)	50.9	(291)	A	[47/2]
	2-fluoro-2,2-dinitroethanol				[17003-75-7]
$\text{C}_2\text{H}_3\text{NO}_3$		55.6 ± 2.1			[77/1][68/7]
	oxalic acid, monoamide				[471-47-6]
		108.9 ± 2.1	(298)	ME	[88/18]
$\text{C}_2\text{H}_3\text{NS}$	(355–363)	107.9	(359)	ME	[53/1][60/1]
					[87/4]
$\text{C}_2\text{H}_3\text{NS}$	methyl isothiocyanate				[556-61-6]
$\text{C}_2\text{H}_3\text{N}_3$	(238–293)	31.5	(266)	A	[47/2]
	1,2,4-triazole				[288-88-0]
		80.7 ± 0.5	(298)	C	[99/8]
	(281–296)	84.1	(288)	ME	[89/8]
	(281–296)	84.0 ± 0.7	(298)	ME	[89/8]
		80.6 ± 0.5			[85/6]
$\text{C}_2\text{H}_3\text{N}_3\text{O}_6$		84.1		ME	[61/3]
	1,1,1-trinitroethane				[595-86-8]
		72.0 ± 8.8	(298)		[99/35]
C_2H_4	ethylene				[74-85-1]
	(79–104)	18.4	(91.5)	A,MS	[87/4][51/15]
		15.3	(298)	H	
	(237–289)	44.5 (liq)	(274)		[87/4]
		44.2	(298)	H	
	(77–103)	15.0			[82/19]
	(237–283)	44.3 (liq)	(260)	A	[47/2]
$\text{C}_2\text{H}_4\text{Br}_2$		43.7	(298)	H	
	1,2-dibromoethane				[106-93-4]
	(229–248)	54.8	(239)		[48/1]
$\text{C}_2\text{H}_4\text{I}_2$	(251–281)	49.8	(258)	A	[48/1][47/2]
					[87/4]
	1,2-diiodoethane				[624-73-7]
$\text{C}_2\text{H}_4\text{N}_2\text{O}_2$		65.7 ± 4.1			[54/6][70/1]
	diformylhydrazine				[628-36-4]
	(340–373)	205.1 ± 0.7	(356)	ME	[80/11]
$\text{C}_2\text{H}_4\text{N}_2\text{O}_2$	(370–403)	100.8			[56/6][60/1]
	oxamide				[471-46-5]
		117.3 ± 1.2	(298)	TE,ME	[88/18]
	(370–398)	115.8	(387)	TE,ME	[83/7]
	(353–369)	113.0	(361)	ME	[53/1]
$\text{C}_2\text{H}_4\text{N}_2\text{S}_2$					[60/1][70/1]
	dithiooxamide				[79-40-3]
		103.8	(298)	TE,ME	[88/18]
	(345–372)	105.1	(361)	TE,ME	[83/7]
$\text{C}_2\text{H}_4\text{N}_4$	(360–378)	105.4	(369)	ME	[53/1][60/1]
					[87/4]
	dicyandiamide				[461-58-5]
$\text{C}_2\text{H}_4\text{N}_4$	(420–450)	128.7	(436)	TE,ME	[83/7]
	1-methyltetrazole				[16681-77-9]
$\text{C}_2\text{H}_4\text{N}_4$	(282–311)	86.7 ± 1.9		ME	[90/31]
	5-methyltetrazole				[4076-36-2]
$\text{C}_2\text{H}_4\text{O}_2$	(323–418)	93.8 ± 0.5		ME	[90/31]
	acetic acid				[64-19-7]
	(243–289)	54.5 (liq)	(274)		[87/4]
	(213–230)	67.3 ± 1	(223)	TE,ME	[78/16]
$(\text{C}_2\text{H}_4\text{O}_2)_2$	(213–230)	70 ± 1	(213)	TE,ME	[78/16]
	acetic acid dimer				
	(213–230)	70.2 ± 1	(213)	TE	[78/16]
$\text{C}_2\text{H}_4\text{O}_3$	(213–230)	68.9 ± 1	(213)	ME	[78/16]
	methyl hydrogen carbonate				[7456-87-3]
$\text{C}_2\text{H}_5\text{NO}$	(204–237)	18.2 ± 1.6	(220)		[73/9]
	acetamide				[60-35-5]
		78.5 ± 0.3			[98/38]
	(273–293)	77.8	(284)	TE,ME	[83/7]
		77.2	(298)		[83/7]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
		78.7±0.3			[75/18][77/1]
		80.3±1	(298)		[71/16]
		80.3±1.3	(298)	C	[65/8]
	(298–349)	77.4±0.4	(323)	GS	[59/3][87/4]
	(293–306)	U57.2	(300)		[52/4]
C ₂ H ₅ NO ₂	methyl carbamate				[598-55-0]
	(287–305)	74.5±0.8	(296)	GS	[59/4]
C ₂ H ₅ NO ₂	glycine				[56-40-6]
	(408–431)	136.5±2	(419)	TE,ME	[79/1]
	(325–425)	U 96.2±4	(375)	LE	[77/2]
	(413–450)	138.1±4.6	(298)	C	[77/3]
	(453–471)	136.4±4.0	(462)	ME	[65/1][70/1]
					[64/16]
	(412–417)	130.5±2	(414)	ME	[59/1]
C ₂ H ₅ NS	thioacetamide				[62-55-5]
		83.3±0.3	(298)	C	[82/7][85/5]
		82.8±0.3	(298)	C	[82/17]
C ₂ H ₅ N ₅	1-methyl-5-aminotetrazole				[5422-44-6]
	(379–438)	116.4±1.7		ME	[90/31]
C ₂ H ₅ N ₅	2-methyl-5-aminotetrazole				[6154-04-7]
	(310–373)	90.6±1.1		ME	[90/31]
C ₂ H ₆	ethane				[74-84-0]
	(80–90)	22.6	(85)		[72/26]
		20.5	(90)	B	[63/6]
C ₂ H ₆ N ₂ O	N-methylurea				[598-50-5]
		94.4±0.84	(343)	C	[93/17]
		97.1±0.4	(298)		[93/17]
		94.9±0.6	(337)	C	[90/25]
		93.3±1.2	(355)	TE	[90/5][87/5]
		87.3	(348)		[87/5]
		99.3±0.7			[86/6]
		78.2		E	[82/13]
C ₂ H ₆ N ₂ O ₂	N-methyl-N-nitromethanamine				[4164-28-7]
	(315–324)	69.9	(319)	DBM	[52/2][77/1]
C ₂ H ₆ N ₂ S	N-methylthiourea				[598-52-7]
		112.9±3	(298)	ME	[00/23]
		111±3.0	(381)	TE	[94/20]
C ₂ H ₆ O ₂ S	dimethyl sulfone				[67-71-0]
		77.0±2.9			[70/1][U/3]
C ₂ H ₆ O ₄	bis-hydroxymethyl peroxide				[17088-73-2]
		94.1±4.2		ME	[53/3][70/1]
C ₂ H ₈ N ₂	ethylenediamine				[107-15-3]
	(242–278)	65.6	(263)	IPM	[87/4][75/32]
C ₂ H ₈ N ₆ O ₂	1,1'-(1,2-ethanediy)bis(1-nitrosohydrazine)				[216489-95-1]
		172.4±1.3	(298)		[98/36]
C ₃ HN	cyanoacetylene				[1070-71-9]
	(247–279)	42.3	(264)		[87/4]
C ₃ H ₂ N ₂	malononitrile				[109-77-3]
	(278–299)	78.2±1.0	(298)		[90/18]
		79.1±8		ME	[67/3][70/1]
C ₃ H ₂ OS ₂	1,3-dithiol-2-one				[2314-40-1]
		73.6±0.8	(298)		[73/16][77/1]
C ₃ H ₂ OS ₃	1,3-dithiole-2-thione				[930-35-8]
		75.4±0.4	(298)		[73/16][77/1]
C ₃ H ₃ N ₃	1,3,5-triazine				[290-87-9]
	(212–229)	58.2	(222)	TE,ME	[83/7]
		54.2±0.2	(298)		[82/16]
	(283–313)	56.5±2.1			[82/9]
CH ₂ N ₄	tetrazole				[288-94-8]
	(242–264)	U 43.1	(253)	ME	[68/8]
C ₃ H ₃ N ₃ O ₂	6-azauracil				[461-89-2]
		141		LE	[74/8]
C ₃ H ₃ N ₃ O ₃	cyanuric acid				[108-80-5]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_3\text{H}_3\text{N}_5\text{O}_{10}$	(440–473)	131	(458)	ME,TE	[83/7]
		133	(298)		[83/7]
	1,1,1,2,2-pentanitropropane	77.4 ± 1.3	(298)		[62626-83-9] [99/35]
$\text{C}_3\text{H}_4\text{N}_2$	imidazole				[288-32-4]
	(292–309)	83.1 ± 0.2	(300)	ME	[87/9]
		83.1 ± 0.2	(298)	ME	[86/15]
	(288–310)	80.8	(301)	ME,TE	[83/7]
		74.5 ± 0.4	(298)	C	[80/5]
$\text{C}_3\text{H}_4\text{N}_2$		85.3	(298)		[61/3]
	pyrazole				[288-13-1]
	(268–287)	74.3 ± 0.4	(275)	ME	[87/9]
		74.0 ± 0.4	(298)		[87/9][86/15]
	(253–273)	72.7	(265)	TE,ME	[83/7]
		69.2 ± 3	(298)	C	[80/5]
		71.8			[79/11]
$\text{C}_3\text{H}_4\text{N}_2\text{O}$		67.7			[61/3]
	2-cyanoacetamide				[107-91-5]
$\text{C}_3\text{H}_4\text{N}_2\text{O}_4$	(325–348)	99.7	(336)	TE,ME	[83/7]
	3-nitro-2-isoxazoline-2-oxide				[4122-45-6]
$\text{C}_3\text{H}_4\text{OS}_2$	1,3-dithiolan-2-one	71.1 ± 8.4			[77/1][69/10]
					[2080-58-2]
$\text{C}_3\text{H}_4\text{O}_2\text{S}$	thiete sulfone (2H-thiete-1,1-dioxide)	80.3 ± 0.4			[73/16][77/1]
				B	[7285-32-7]
$\text{C}_3\text{H}_4\text{O}_3$		83.7 ± 2.5			[69/11][77/1]
	ethylene carbonate				[96-49-1]
$\text{C}_3\text{H}_4\text{O}_4$	(273–297)	78.5 ± 4.2	(285)	ME	[87/4][71/12]
					[77/1]
		73.2 ± 2.5			[70/1][58/1]
	malonic acid				[141-82-2]
$\text{C}_3\text{H}_4\text{O}_5$	(339–357)	108.9 ± 0.7	(348)	ME	[99/10]
		111.4 ± 0.7	(298)		[99/10]
	(291–320)	72.7	(306)	A	[87/4][47/6]
		105.1 ± 0.8		C	[83/26]
$\text{C}_3\text{H}_4\text{S}_3$	tartronic acid				[80-69-3]
		116.4 ± 0.3		C	[83/26]
$\text{C}_3\text{H}_4\text{S}_3$	1,3-dithiolan-2-thione				[822-38-8]
	(294–303)	81.8 ± 0.8	(298)		[67/5][70/1]
$\text{C}_3\text{H}_5\text{NO}$	acrylamide				[79-06-1]
	(303–358)	81.8	(330)		[57/6]
$\text{C}_3\text{H}_5\text{NO}$	2-azetidinone				[930-21-2]
		77.4 ± 0.3	(298)	ME	[96/21]
C_3H_6	cyclopropane				[75-19-4]
		29.2	(145)	B	[63/6]
		27.4	(298)	H	
	(115–141)	28.2	(128)	A,MS	[51/15]
$\text{C}_3\text{H}_6\text{N}_2\text{O}$	2-imadazolidinone				[120-93-4]
		83.7	(298)		[99/32]
$\text{C}_3\text{H}_6\text{N}_2\text{O}_2$	acetylurea				[591-07-1]
	(360–407)	102.4 ± 0.7	(383)	C	[88/12]
		103.1 ± 0.7	(298)		[88/12]
$\text{C}_3\text{H}_6\text{N}_2\text{O}_2$		103.1 ± 0.7	(298)	C	[85/5]
	malonamide				[108-13-4]
$\text{C}_3\text{H}_6\text{N}_4$	(397–403)	126.4 ± 0.5	(298)	C	[89/17]
	1,5-dimethyltetrazole				[5144-11-6]
$\text{C}_3\text{H}_6\text{N}_6$	(303–343)	86.2 ± 1.0		ME	[90/31]
	2,4,6-triamino-s-triazine (melamine)				[108-78-1]
	(417–531)	121.3 ± 4.2	(474)	GS	[60/6][70/1]
$\text{C}_3\text{H}_6\text{N}_6\text{O}_3$	(417–447)	123.3	(432)		[87/4]
	N,N',N''-trinitrosohexahydrotriazine				[13980-04-6]
	(343–447)	134.3 ± 0.7	(298)	ME	[78/15]
	(383–411)	112.1		ME	[74/11]
		112.1			[53/6][60/1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_3\text{H}_6\text{N}_6\text{O}_6$	N,N',N''-trinitrohexahydrotriazine	134.3	(298)		[121-82-4]
	(325–360)	112.5 ± 0.8		ME	[78/15]
	(329–371)	130.1	(350)		[74/11]
$\text{C}_3\text{H}_6\text{O}_2$	propionic acid				[69/12]
	(225–238)	74.1 ± 1	(233)	TE	[79-09-4]
		73.2 ± 1	(233)	ME	[78/16]
$(\text{C}_3\text{H}_6\text{O}_2)_2$	propionic acid dimer				[32574-16-6]
	(225–238)	81.3 ± 1	(233)	TE	[78/16]
		79.4 ± 1	(233)	ME	[78/16]
$\text{C}_3\text{H}_6\text{O}_3$	1,3,5-trioxane				[110-88-3]
	(212–231)	57.9	(223)	TE,ME	[83/7]
		55.6	(298)		[83/7]
		56.5	(298)	C	[75/17]
		56.2 ± 0.2	(298)	C	[69/8][77/1]
$\text{C}_3\text{H}_6\text{S}_3$	1,3,5-trithiane				[291-21-4]
		93.2 ± 0.2	(298)	ME	[01/14]
	(320–339)	91.5	(331)	TE,ME	[83/7]
		93.9	(298)		[83/7]
$\text{C}_3\text{H}_7\text{NO}$	acetone oxime				[127-06-0]
	(313–333)	59.6	(323)	I	[87/4][75/33]
$\text{C}_3\text{H}_7\text{NO}$	N-methyl acetamide				[79-16-3]
		70.8 ± 2.0	(298)		[96/21]
	(288–303)	54.0			[52/4][60/1]
$\text{C}_3\text{H}_7\text{NO}$	propanamide				[79-05-0]
	(283–343)	75 ± 4.0	(298)	TE	[00/1]
		79.2 ± 0.3			[75/18][77/1]
		73.3			[60/13]
	(318–346)	79.1 ± 0.4		GS	[59/3]
$\text{C}_3\text{H}_7\text{NO}_2$	L-(d)-alanine				[56-41-7]
	(407–426)	132.8 ± 1	(414)	TE,ME	[79/1][87/4]
	(413–450)	132.4 ± 1.3	(433)	C	[77/3]
		144.8 ± 4.2	(298)		[77/3]
$\text{C}_3\text{H}_7\text{NO}_2$	D-(l)-alanine				[338-69-2]
	(342–442)	$U\ 105 \pm 8$	(392)	LE	[77/2]
	(453–469)	138.3 ± 8	(461)	ME	[65/1][70/1]
$\text{C}_3\text{H}_7\text{NO}_2$	β -alanine				[64/16]
	(384–402)	133.1 ± 0.7	(393)	C	[107-95-9]
		134 ± 2	(298)	C	[83/24]
	(318–418)	$U\ 105 \pm 4$	(368)	LE	[77/2]
$\text{C}_3\text{H}_7\text{NO}_2$	ethyl carbamate				[51-79-6]
	(256–273)	77.7	(265)	TE,ME	[83/7]
		76.3	(298)		[83/7]
		71.9	(322)		[76/10]
	(292–307)	89.1 ± 0.8	(299)	GS	[59/4]
$\text{C}_3\text{H}_7\text{NO}_2$	sarcosine (N-methylglycine)				[107-97-1]
	(380–413)	146 ± 1	(298)	C	[78/4]
$\text{C}_3\text{H}_7\text{NO}_3$	DL-serine				[302-84-1]
	(354–454)	$U\ 83.7 \pm 4$	(404)	LE	[77/2]
$\text{C}_3\text{H}_7\text{NO}_2\text{S}$	L-cysteine				[52-90-4]
	(337–437)	$U\ 96.2 \pm 4.2$	(387)	LE	[77/2]
C_3H_8	propane				[74-98-6]
		28.5	(86)	B	[63/6]
$\text{C}_3\text{H}_8\text{N}_2\text{O}$	N-ethylurea				[625-52-5]
	(333–365)	91.8 ± 1.2	(354)	TE	[90/5][87/5]
		100.3 ± 0.2	(343)		[86/6][90/5]
$\text{C}_3\text{H}_8\text{N}_2\text{O}$	1,1-dimethylurea				[598-94-7]
	(326–369)	92.5 ± 1.3	(357)	TE	[90/5][87/5]
		99.1 ± 0.4	(348)		[86/6]
$\text{C}_3\text{H}_8\text{N}_2\text{O}$	1,3-dimethylurea				[96-31-1]
	(316–373)	87.2 ± 0.6	(353)	TE	[90/5][87/5]
$\text{C}_3\text{H}_8\text{N}_2\text{S}$	1,3-dimethylthiourea				[534-13-4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_3\text{H}_8\text{N}_2\text{S}$	1-ethylthiourea	111.8 ± 3	(298)	B,HA	[00/23]
		107.3 ± 4.0	(298)	B	[94/17]
		108 ± 3.0	(361)	B	[94/20]
$\text{C}_3\text{H}_8\text{O}_2\text{S}$	ethyl methyl sulfone	118.8 ± 5	(298)	ME	[625-53-6]
		77.8 ± 2.9			[00/23]
$\text{C}_3\text{H}_9\text{NO}_2\text{S}$	trimethyl amine · sulfur dioxide complex				[594-43-4]
$\text{C}_4\text{F}_6\text{NS}_2$	<i>bis</i> (trifluoromethyl)-1,3,2-dithiazol-2-yl	(292–349)	(307)		[U/3][70/1]
		60.6			[177634-55-8]
$\text{C}_4\text{F}_{12}\text{P}_4$	1,2,3,4- <i>tetrakis</i> (trifluoromethyl)tetraphosphetane	(253–283)	(268)	PG	[87/4][43/3]
		49.0 ± 1.5			[00/41]
C_4N_2	dicyanoacetylene	(292–339)	(307)		[393-02-2]
		65.3			[87/4][58/20]
$\text{C}_4\text{HF}_6\text{N}_3$	3,5- <i>bis</i> (trifluoromethyl)-1,2,4-triazole	(263–273)	(268)	I	[1071-98-3]
		44.3			[57/4][75/10]
					[87/4]
C_4H_2	butadiyne	75.6 ± 0.8	(277)	ME	[709-62-6]
		74.7 ± 0.8	(298)		[94/22]
					[94/22]
$\text{C}_4\text{H}_2\text{N}_2$	fumaronitrile	(190–232)	(211)	A	[460-12-8]
		(188–234)			[47/2]
		36.3			[33/3]
$\text{C}_4\text{H}_2\text{N}_2\text{S}$	4-cyanothiazole	(250–269)	(260)	TE,ME	[764-42-1]
		69.6	(298)		[83/7]
		68.6			[83/7]
$\text{C}_4\text{H}_2\text{O}_3$	maleic anhydride	(245–281)	(263)	ME	[67/3][70/1]
		72 ± 0.8			[1452-15-9]
					[66/5][70/1]
$\text{C}_4\text{H}_2\text{O}_4$	butyndioic acid	73.9 ± 0.4	(298)	C	[108-31-6]
					[87/4]
		85.4	(317)		[83/7]
$\text{C}_4\text{H}_2\text{O}_4$	3,4-dihydroxy-3-cyclobutene-1,2-dione	68.8	(258)	TE,ME	[83/7]
		70.0	(298)		[78/10]
		71.5 ± 5			[49/2][70/1]
$\text{C}_4\text{H}_3\text{BrN}_2\text{O}_2$	5-bromouracil	(308–325)			[110-16-7]
		NA			[72/7]
					[2892-51-5]
$\text{C}_4\text{H}_3\text{FN}_2\text{O}_2$	5-fluorouracil	152.5	(486)	ME,TE	[83/7]
		154.3	(298)		[83/7]
		83.7 ± 16.7	(298)	E	[71/5][77/1]
$\text{C}_4\text{H}_3\text{NO}_3$	2-nitrofurane			LE	[51-20-7]
		128.4			[74/8]
$\text{C}_4\text{H}_4\text{BrN}_3\text{O}$	5-bromocytosine				[51-21-8]
		NA			[02/1]
$\text{C}_4\text{H}_4\text{Cl}_4\text{O}_2\text{S}$	3,3,5,5-tetrachlorotetrahydrothiophene 1,1-dioxide	75.3 ± 2.1			[609-39-2]
		148.4 ± 1.5	(435)		[80/28][86/5]
$\text{C}_4\text{H}_4\text{F}_3\text{NO}_3$	N-(trifluoroacetyl)aminoacetic acid	(303–348)	(325)	ME	[2240-25-7]
		88.7			[75/14]
$\text{C}_4\text{H}_4\text{IN}_3\text{O}$	5-iodocytosine (4-amino-5-iodopyrimidinone)	(273–393)	(288)		[3737-41-5]
		98.8			[78/31]
$\text{C}_4\text{H}_4\text{N}_2$	pyrazine	(413–463)	(438)		[383-70-0]
		144 ± 1.5			[87/4][60/20]
$\text{C}_4\text{H}_4\text{N}_2\text{OS}$	succinonitrile	(288–317)	(303)	C	[1122-44-7]
		56.2			[75/14]
$\text{C}_4\text{H}_4\text{N}_2\text{OS}$	2-thiouracil	56.3 ± 0.5			[290-37-9]
		70 ± 0.3	(289)		[95/5]
$\text{C}_4\text{H}_4\text{N}_2\text{OS}$	4-thiouracil				[62/4][70/1]
		129.3		LE	[110-61-2]
CH_2N_4	tetrazole				[60/7][77/1]
		125.5		LE	[141-90-2]
					[74/8]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₄ H ₄ N ₂ O ₂	pyrazine 1,4-dioxide	116.9 ± 0.8	(298)	C	[2423-84-9] [97/25]
C ₄ H ₄ N ₂ O ₂	uracil				[66-22-8]
	(394–494)	127.0 ± 2.0	(439)	TE	[00/2]
	(452–587)	130.6 ± 4.0	(519)	ME,TE	[80/18]
	(452–587)	131 ± 5	(298)	TE,GS	[80/18]
	(378–428)	120.5 ± 1.3	(403)	QR	[80/19]
		121.7	(425)	MS	[79/28]
	(500–545)	133.9 ± 8	(523)	HSA	[78/17]
		126.5 ± 2.2	(440)	C	[77/13]
	(393–458)	120.5 ± 5.2	(426)	LE	[75/16][74/8]
		115.5 ± 2.1		ME	[72/32][00/2]
	U 83.7		(485)	MS	[65/2]
C ₄ H ₄ N ₂ O ₂ S	thiobarbituric acid				[504-17-6]
	(400–461)	110 ± 4.0	(430)	TE	[99/6]
C ₄ H ₄ N ₂ O ₃	barbituric acid				[67-52-7]
	(294–438)	111.3 ± 0.3		GS	[99/42]
	(392–493)	113 ± 4.0	(442)	TE	[99/6]
	(404–479)	123.3 ± 1.7	(440)	ME	[90/32]
C ₄ H ₄ N ₂ S ₂	2,4-dithiouracil				[2001-93-6]
	(393–443)	119.7 ± 2.4	(418)		[75/14]
C ₄ H ₄ N ₆	8-azaadenine				[1123-54-2]
	(418–463)	128.4 ± 1.3	(440)		[75/14]
C ₄ H ₄ N ₈ O ₁₃	bis-(2,2,2-trinitroethyl)-N-nitrosoamine				[34882-73-0]
	(333–354)	97.9 ± 0.8	(343)	ME	[73/1]
C ₄ H ₄ N ₈ O ₁₄	bis-(2,2,2-trinitroethyl)-N-nitroamine				[19836-28-3]
	(340–356)	117.6 ± 0.8	(348)	ME	[73/1]
C ₄ H ₄ O	furan				[110-00-9]
	(173–187)	39.2	(180)		[53/15]
C ₄ H ₄ O ₂	cyclobutane-1,2-dione				[33689-28-0]
	(251–289)	69.1 ± 3.5	(270)	HSA	[U/5]
	(295–335)	54.8	(315)		[85/2]
C ₄ H ₄ O ₂	cyclobutane-1,3-dione				[15506-53-3]
	(274–322)	73.6 ± 3.7	(298)	HSA	[78/14]
C ₄ H ₄ O ₃	succinic anhydride				[108-30-5]
	(298–320)	80.5 ± 1.6	(309)	ME	[90/4]
		80.7 ± 1.6	(298)	C	[90/4]
	(290–311)	82.2	(302)	ME,TE	[83/7]
C ₄ H ₄ O ₄	cis-butenedioic acid				[110-16-7]
	(348–389)	105.4 ± 1.7	(368)	ME	[74/5]
	(357–367)	110 ± 2.5			[38/1][60/1]
					[70/1]
	(356–371)	109 ± 4.2			[34/1]
C ₄ H ₄ O ₄	trans-butenedioic acid				[110-17-8]
	(371–391)	123.6 ± 2.0	(381)	TE,ME	[77/4]
		136 ± 6.3	(365)	QF	[38/1][35/1]
					[60/1]
	(358–371)	134.3 ± 4.2			[34/1]
C ₄ H ₄ O ₄	diglycolic anhydride				[4480-83-5]
	(382–303)	84.2 ± 1.1	(294)	ME,TE	[83/7]
C ₄ H ₄ S	thiophene				[110-02-1]
	(195–228)	46.8	(213)		[87/4][56/8]
	(192–213)	49.0	(203)		[44/1]
C ₄ H ₅ N	pyrrole				[109-97-7]
		NA			[41/3]
C ₄ H ₅ NO ₂	succinimide				[123-56-8]
	(317–340)	83.1 ± 1.5	(329)	ME	[90/4]
		83.6 ± 1.5	(298)		[90/4]
		88.0		B	[89/27]
C ₄ H ₅ N ₃ O	cytosine				[71-30-7]
	(505–525)	151.7 ± 0.7		GS	[98/37]
	(423–483)	147.2 ± 2.6	(453)	ME	[84/12]
		155.0 ± 3.0	(298)		[84/12]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
		167 ± 10	(298)	TE	[80/9]
	(450–470)	176 ± 10	(298)	C	[80/21]
		NA			[77/15]
$\text{C}_4\text{H}_5\text{N}_3\text{O}_2$	5-aminouracil	150.6		ME	[74/8][75/16]
					[932-52-5]
		145.6		LE	[74/8]
$\text{C}_4\text{H}_5\text{N}_3\text{O}_2$	6-azathymine				[932-53-6]
	(358–403)	112.5 ± 2.3	(380)		[74/14]
$\text{C}_4\text{H}_5\text{N}_3\text{S}$	2-thiocytosine				[333-49-3]
	(408–458)	158 ± 1.6	(433)		[75/14]
$\text{C}_4\text{H}_5\text{N}_7\text{O}_{12}$	bis-(2,2,2-trinitroethyl)amine				[34880-53-0]
	(338–349)	80.8 ± 0.4		ME	[73/1][77/1]
C_4H_6	2-butyne				[503-17-3]
	(200–239)	37.4	(220)	A	[47/2]
$\text{C}_4\text{H}_6\text{N}_2$	2-methylimidazole				[693-98-1]
	(301–318)	88.2 ± 0.7	(309)	ME	[92/25]
		88.4 ± 0.7	(298)		[92/25]
$\text{C}_4\text{H}_6\text{N}_2\text{O}$	3-amino-5-methylisoxazole				[1072-67-9]
		81.6 ± 2.5			[73/20][77/1]
$\text{C}_4\text{H}_6\text{N}_2\text{O}_2$	2,5-piperazinedione				[106-57-0]
	(413–450)	103.8 ± 2.1	(428)	ME	[87/4][56/7]
$\text{C}_4\text{H}_6\text{N}_4\text{O}$	2,4-diamino-6-hydroxypyrimidine				[56-06-4]
	(423–471)	147.6 ± 0.2		GS	[99/42]
$\text{C}_4\text{H}_6\text{N}_4\text{O}_8$	1,1,3,3-tetranitrobutane				[3759-60-2]
		87.9 ± 0.8	(298)		[99/35]
$\text{C}_4\text{H}_6\text{N}_4\text{O}_8$	2,2,3,3-tetranitrobutane				[20919-97-5]
		78.2 ± 0.8	(298)		[99/35]
$\text{C}_4\text{H}_6\text{N}_4\text{O}_8$	1,1,1,3-tetranitro-2-methylpropane				[42216-58-0]
		91.2	(298)		[99/35]
$\text{C}_4\text{H}_6\text{N}_4\text{O}_8$	1,1,1,4-tetranitrobutane				[20919-96-4]
		99.6	(298)		[99/35]
$\text{C}_4\text{H}_6\text{O}_4$	dimethyl oxalate				[553-90-2]
		74.6 ± 0.7	(298)	C	[96/10]
	(268–298)	75.6 ± 1.6	(283)	HSA	[96/10]
		75.3 ± 1.6	(298)		[96/10]
		74.9 ± 0.6		B	[96/10]
	(289–306)	47.4 ± 0.5		BG	[76/5][75/13]
$\text{C}_4\text{H}_6\text{O}_4$	succinic acid				[110-15-6]
	(356–376)	120.5	(368)	TE,ME	[83/7]
		123.1	(298)		[83/7]
	(372–401)	118.1 ± 3.3	(386)	ME	[70/1][60/4]
		120.3 ± 4.4	(298)		[70/1][60/4]
		121.8 ± 3.3	(298)		[60/4][99/10]
	(292–320)	73.6	(306)	A	[47/6]
$\text{C}_4\text{H}_6\text{O}_6$	meso tartaric acid				[147-73-9]
		156.9			[83/23]
$\text{C}_4\text{H}_6\text{O}_4$	methylmalonic acid				[516-05-2]
	(341–354)	116.2 ± 0.9	(348)	ME	[00/22]
		117.4 ± 1.9	(298)	ME	[00/22]
		113.2 ± 0.4		C	[83/26]
$\text{C}_4\text{H}_6\text{S}_3$	1,3-dithian-2-thione				[1748-15-8]
	(321–348)	88.6	(335)		[67/5]
		91.4 ± 2.5	(298)		[67/5][70/1]
$\text{C}_4\text{H}_7\text{NO}$	cis 2-butenic acid amide				[31110-30-2]
	(353–387)	68	(368)		[87/4]
$\text{C}_4\text{H}_7\text{NO}$	trans 2-butenic acid amide				[625-37-6]
	(363–413)	80	(378)		[87/4]
$\text{C}_4\text{H}_7\text{NO}_2$	diacetamide				[625-77-4]
		73.2 ± 0.8	(298)	C	[65/8][71/23]
$\text{C}_4\text{H}_7\text{NO}_3$	N-acetylglycine				[543-24-8]
	(383–400)	127.0 ± 1.0	(389)	TE,ME	[79/1]
$\text{C}_4\text{H}_7\text{NO}_4$	L-aspartic acid				[56-84-8]
	(370–470)	$U\ 96 \pm 4.2$	(420)	LE	[77/2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₄ H ₈	cyclobutane				[287-23-0]
		36.4	(145)	B	[63/6]
C ₄ H ₈ Cl ₂ S	<i>bis</i> (2-chloroethyl)sulfide (263–287)	77.2 84.5	(275)	[505-60-2] B	[87/4] [63/6][47/4]
C ₄ H ₈ Cl ₃ O ₄ P	(1-hydroxy-2,2,2-trichloroethyl)phosphonic acid dimethyl ester (293–357)	107	(308)		[52-68-6] [87/4]
C ₄ H ₈ N ₂ O	tetrahydro-2-pyrimidone	89.3	(298)		[1852-17-1] [99/32]
C ₄ H ₈ N ₂ O ₂	1,2-diacetylhydrazine (347–358)	103.1 ± 1.7	(352.5)		[3148-73-0] [87/4][59/12]
C ₄ H ₈ N ₂ O ₂	dimethylglyoxime (331–352)	96.8 ± 1.3	(341.5)	ME	[95-45-4] [87/4][56/7]
	(331–352)	97.1 ± 1.3		ME	[56/7][70/1] [60/1]
C ₄ H ₈ N ₂ O ₂	N-acetyl glycine amide				[2620-63-5]
		123.5 ± 1.7	(376)	C	[99/12]
		126.3 ± 2.3			[99/12]
		140.2 ± 2.3	(298)	C	[95/33]
	(378–406)	135 ± 3	(392)	TE	[88/6][86/16]
C ₄ H ₈ N ₂ O ₂	N-nitrosomorpholine				[59-82-2] [88/20]
		81.6			[140-79-4]
C ₄ H ₈ N ₄ O ₂	1,4-dinitrosopiperazine (325–360)	101.3 ± 8	(343)		[74/11][77/1]
C ₄ H ₈ N ₄ O ₄	1,4-dinitropiperazine (325–360)	111.3 ± 8	(343)		[4164-37-8] [74/11][77/1]
C ₄ H ₈ N ₈ O ₈	1,3,5,7-tetranitro-1,3,5,7-tetrazacyclooctane δ -form (461–487)	161.9	(474)		[2691-41-0] [76/8]
δ form	(415–479)	161 ± 0.3	(447)		[78/15]
β form	(371–403)	175.2	(385)		[69/12]
C ₄ H ₈ O ₂	butanoic acid (238–255)	76.0 ± 1.5	(248)	TE,ME	[107-92-6] [78/16]
(C ₄ H ₈ O ₂) ₂	butanoic acid dimer (238–255)	85 ± 1.5	(248)	TE,ME	[19496-06-1] [78/16]
C ₄ H ₈ O ₂	1,4-dioxane (237–272)	35.6	(255)	A	[123-91-1] [47/2]
C ₄ H ₈ O ₄	1,3,5,7-tetroxane				[293-30-1]
		79.6 ± 0.2	(298)	C	[77/1][69/8]
		79.5		C	[75/17]
C ₄ H ₈ S ₂	1,3-dithiane (266–279)	62.9 ± 0.7 69.9 ± 0.4	(298) (298)	ME GC	[505-23-7] [99/3] [89/30]
	(250–271)	72.6	(263)	TE,ME	[83/7]
		52.3 ± 0.8	(298)	C	[71/23]
C ₄ H ₈ S ₂	1,4-dithiane				[505-29-3]
		63.0	(298)		[99/15]
		68.9	(298)		[89/30]
	(253–276)	72.4	(268)	E	[83/7]
C ₄ H ₉ I	2-iodo-2-methylpropane (202–223)	49.8	(212.5)	MG	[558-17-8] [87/4][44/2]
C ₄ H ₉ Li	butyl lithium (333–368)	109.7	(350.5)		[109-72-8] [87/4][62/12]
C ₄ H ₉ ONa	sodium <i>tert</i> -butoxide	NA			[865-48-5] [90/22]
C ₄ H ₉ NO	butanamide (298–347)	82 ± 4.0 86.4 ± 0.4	(298)	TE	[541-35-5] [00/1] [75/18][77/1]
	(292–304)	85.4 ± 1.7	(298)	ME	[73/19][77/1]
	(353–373)	87	(363)		[60/1]
	(336–382)	86.4 ± 0.4	(359)	GS	[59/3]
	(298–341)	87.0 ± 0.8	(320)	ME	[59/3]
		79.9			[60/13]
C ₄ H ₉ NO	2-methylpropanamide				[563-83-7]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
	(288–354)	82 ± 4	(298)	TE	[00/01]
	(285–302)	86.1 ± 0.2	(294)	ME	[89/6]
		86.0 ± 0.2	(298)		[89/6]
C ₄ H ₉ NO ₂	2-aminoisobutyric acid				[62-57-7]
	(439–462)	125.8	(450.5)		[87/4][65/1]
	(403–424)	134.2 ± 1.0	(413.5)	TE,ME	[87/4][79/1]
	(439–469)	125.9 ± 0.4	(455)	ME	[65/1][64/16]
C ₄ H ₉ NO ₂	DL-2-aminobutanoic acid				[2835-81-6]
	(400–418)	132 ± 2	(409)	TE,ME	[79/1]
C ₄ H ₉ NO ₂	L-2-aminobutanoic acid				[1492-24-6]
		162.8 ± 0.8	(455)	ME	[65/1][64/16]
C ₄ H ₉ NO ₂	4-aminobutanoic acid				[65-12-2]
	(384–407)	138.9 ± 0.6	(395)	C	[83/24]
		140 ± 2	(298)	C	[83/24]
C ₄ H ₉ NO ₃	DL-threonine				[80-68-2]
	(341–441)	U 96 ± 8	(391)	LE	[77/2]
C ₄ H ₉ NO ₃	2-methyl-2-nitro-1-propanol				[76-39-1]
	plastic phase	59.5 ± 3.0	(319)	C	[94/27]
	crystalline phase	73.2 ± 3.7	(311)	C	[94/27]
C ₄ H ₉ NO ₄	2-methyl-2-nitro-1,3-propanediol				[77-49-6]
	plastic phase	79.3 ± 4.0	(368)	C	[94/27]
	crystalline phase	102.0 ± 5.1	(339)	C	[94/27]
C ₄ H ₉ NO ₅	2-hydroxymethyl-2-nitro-1,3-propanediol				[126-11-4]
	plastic phase	77.3 ± 3.9	(368)	C	[94/27]
C ₄ H ₉ N ₃ O ₂	1-[2-(ethenyoxy)ethyl]-1-nitrosohydrazine				[216489-98-4]
		112.1 ± 1.9	(298)		[98/36]
C ₄ H ₁₀	<i>n</i> -butane				[106-97-8]
		35.9	(107)	B	[66/3]
C ₄ H ₁₀ N ₂	piperazine				[110-85-0]
		72.1	(298)		[98/18]
		65.2	(385)	B	[97/39]
	(279–321)	73.1	(294)		[87/4]
C ₄ H ₁₀ N ₂	trimethylammonium cyanide				
	(219–236)	45	(227.5)		[87/4]
C ₄ H ₁₀ N ₂ O	N-propylurea				[927-67-3]
	(332–373)	90.7 ± 1.0	(366)		[90/5][87/5]
C ₄ H ₁₀ N ₂ O	N-isopropylurea				[691-60-1]
	(368–411)	100.6 ± 1.3	(389)		[90/5]
		99.7 ± 0.4	(352)		[90/5][86/6]
C ₄ H ₁₀ N ₂ S	trimethylthiourea				[2489-77-2]
		83 ± 3.0	(333)	TE	[94/20]
C ₄ H ₁₀ O	<i>tert</i> -butyl alcohol				[75-65-0]
	(253–298)	51.3	(275)	A	[47/2]
C ₄ H ₁₀ O ₂ S	diethyl sulfone				[597-35-3]
		86.2 ± 2.5			[U/3][70/1]
C ₄ H ₁₀ O ₄	<i>meso</i> -erythritol				[149-32-6]
		157	(298)	B	[90/7]
		135.1 ± 2.2			[50/1][60/1]
					[70/1]
C ₄ H ₁₁ NO	2-methyl-2-amino-1-propanol				[124-68-5]
	plastic phase	86.5 ± 4.3	(368)	C	[94/27]
	crystalline phase	114.5 ± 5.7	(339)	C	[94/27]
C ₄ H ₁₁ NO ₂	diethanolamine				[111-42-2]
		105.9 ± 2.	(298)	C	[82/5]
C ₅ Cl ₆	hexachlorocyclopentadiene				[77-47-4]
		73.6	(283)	B	[63/6][58/5]
C ₅ F ₁₀	decafluorocyclopentane				[376-77-2]
	(229–281)	32.1	(266)		[87/4][67/21]
		38.2	(115)		[63/6][51/8]
					[56/9]
C ₅ F ₁₂	<i>n</i> -dodecafluoropentane				[678-26-2]
		43.7	(145)		[63/6][51/8]
					[56/9]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₅ N ₄	tetracyanomethane	61.1 ± 8.8	(298)	DSC	[24331-09-7] [73/3]
C ₅ H ₂ F ₆ N ₂	3,5-bis(trifluoromethyl)pyrazole	69.0 ± 0.6	(266)	ME	[14704-41-7] [91/21]
C ₅ H ₂ N ₄ O ₆	2,4,6-trinitropyridine (335–357)	101.7 ± 2.9			[78013-51-1] [95/31]
C ₅ H ₂ N ₄ O ₇	2,4,6-trinitropyridine N-oxide (377–403)	106.3 ± 2.9			[25242-76-6] [95/31]
C ₅ H ₃ Br ₂ N	2,5-dibromopyridine	82.1 ± 2.2	(298)	C	[624-28-2] [97/16]
C ₅ H ₃ Br ₂ N	2,6-dibromopyridine	85.6 ± 3.0	(298)	C	[626-05-1] [97/16]
C ₅ H ₃ Cl ₂ N	2,3-dichloropyridine	73.5 ± 3.1	(298)	C	[2402-77-9] [97/17]
C ₅ H ₃ Cl ₂ N	2,5-dichloropyridine	67.1 ± 2.0	(298)	C	[16110-09-1] [97/17]
C ₅ H ₃ Cl ₂ N	2,6-dichloropyridine	72.0 ± 1.6	(298)	C	[2402-78-0] [97/17]
C ₅ H ₃ Cl ₂ N	3,5-dichloropyridine	67.3 ± 1.9	(298)	C	[2457-47-8] [97/17]
C ₅ H ₃ NO ₃	5-nitro-2-furancarboxaldehyde	75.3 ± 2.1			[698-63-5] [80/28][86/5]
C ₅ H ₃ N ₃	2,2-dicyanopropionitrile (293–333)	73.9 ± 0.5	(313)	T	[10359-20-3] [94/19]
C ₅ H ₄ N ₂ O ₃	4-nitropyridine-N-oxide (311–335)	108.9 ± 0.3 89.1 ± 2.5	(298)	C	[1124-33-0] [95/3] [95/31]
C ₅ H ₄ N ₄	purine	NA			[120-73-0] [74/7]
C ₅ H ₄ N ₄	1,2,4-triazolo[1,5a]pyrimidine	86.9	(419)		[275-02-5] [97/39]
C ₅ H ₄ N ₄ O	hypoxanthine (423–473)	158.1 ± 1.6	(448)		[68-94-0] [75/14]
C ₅ H ₄ N ₄ S	6-mercaptopurine (413–458)	148.5 ± 1.5	(435)		[6112-76-1] [75/14]
C ₅ H ₄ O ₂ S	2-thenoic acid (315–323)	97.1	(319)	E	[527-72-0] [53/12][60/1]
C ₅ H ₄ O ₃	2-furoic acid (317–328)	108.4 ± 2.2		ME	[88-14-2] [53/1][60/1] [70/1]
C ₅ H ₅ ClN ₂ O ₂	1-methyl-6-chlorouracil (417–465)	108.8 ± 8		HSA	[31737-09-4] [78/17]
C ₅ H ₅ ClN ₂ O ₂	3-methyl-6-chlorouracil (444–493)	104.6 ± 6		HSA	[4318-56-3] [78/17]
C ₅ H ₅ FN ₂ O ₂	1-methyl-5-fluorouracil (381–423)	116.5 ± 1.9	(402)	TE	[155-16-8] [02/1]
C ₅ H ₅ FN ₂ O ₂	(480–515)	125.5 ± 8		HSA	[78/17]
C ₅ H ₅ FN ₂ O ₂	3-methyl-5-fluorouracil (465–487)	79.5 ± 17		HSA	[4840-69-1] [78/17]
C ₅ H ₅ F ₃ N ₂	3(5)-trifluoromethyl-5(3)-methylpyrazole	78.2 ± 0.8	(297)	ME	[10010-93-2] [91/21]
C ₅ H ₅ NO	2-hydroxypyridine	86.6 ± 1.3	(298)	C	[142-08-5] [82/15][86/5]
C ₅ H ₅ NO	3-hydroxypyridine	88.3 ± 1.3	(298)	C	[109-00-2] [82/15][86/5]
C ₅ H ₅ NO	4-hydroxypyridine	118.6 ± 5.2	(298)	C	[626-64-2] [92/3]
C ₅ H ₅ NO	pyridine N-oxide	103.8 ± 1.7	(298)	C	[82/15][86/5] [694-59-7]
C ₅ H ₅ NO		79.3 ± 1.0	(298)		[88/19]
C ₅ H ₅ NO ₂	3-hydroxypyridine N-oxide (345–392)	121.8 ± 4.4	(298)	ME	[6602-28-4] [98/12]
C ₅ H ₅ NO ₂	pyrrole-2-carboxylic acid				[634-97-9]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_5\text{H}_5\text{NO}_2$	(350–354)	126.8	(352)	ME	[53/12][60/1]
	N-methylmaleimide				[930-88-1]
	(276–289)	75.3 ± 0.5	(282)	ME	[97/12]
$\text{C}_5\text{H}_5\text{N}_3\text{O}$		73.3 ± 0.5	(298)		[97/12]
	pyrazine carboxamide				[98-96-4]
	(353–383)	87.9	(368)	ME	[87/4][60/16]
$\text{C}_5\text{H}_5\text{N}_5$					[59/13]
	adenine				[73-24-5]
	(400–438)	140.4		ME	[00/17]
	(448–473)	109.2	(460.5)		[87/4]
		126.3		LE	[75/16][74/8]
		127.2		QR	[84/38][00/17]
$\text{C}_5\text{H}_5\text{N}_5\text{O}$		108.7 ± 8		ME	[65/2][70/1]
	guanine				[73-40-5]
		186.2		LE	[75/16][74/8]
$\text{C}_5\text{H}_5\text{N}_7\text{O}_{14}$	1,1,1,3,5,5,5-heptanitropentane				[20919-99-7]
$\text{C}_5\text{H}_6\text{F}_3\text{NO}_3$		111.7	(298)		[99/35]
	glycine, N-(trifluoroacetyl) methyl ester				[383-72-2]
	(293–463)	57.3	(308)		[87/4][60/20]
$\text{C}_5\text{H}_6\text{N}_2$	dimethylmalonodinitrile				[7321-55-3]
$\text{C}_5\text{H}_6\text{N}_2$		62.0 ± 0.7	(298)		[90/28]
	2-aminopyridine				[504-29-0]
		76.5 ± 0.4	(298)	C	[98/17]
		38.6 ± 1.9		DSC	[85/13]
$\text{C}_5\text{H}_6\text{N}_2$		78.7 ± 0.8	(298)	C	[84/6]
	3-aminopyridine				[462-08-8]
		80.7 ± 0.3	(298)	C	[98/17]
		84.0 ± 1.4	(298)	C	[84/6]
$\text{C}_5\text{H}_6\text{N}_2$					[504-24-5]
	4-aminopyridine				
		87.1 ± 0.4	(298)	C	[98/17]
		53.8 ± 0.8		DSC	[85/13]
$\text{C}_5\text{H}_6\text{N}_2\text{O}_2$		88.1 ± 1.1	(298)	C	[84/6]
	1-methyluracil				[615-77-0]
	(343–428)	121.7 ± 4.0	(439)	TE	[00/2]
	(378–418)	112.5 ± 2.6	(398)	QR	[80/19]
$\text{C}_5\text{H}_6\text{N}_2\text{O}_2$		104.6 ± 8	(457)	HSA	[78/17]
	3-methyluracil				[608-34-4]
	(344–419)	118.8 ± 3.0	(382)	TE	[00/2]
	(438–498)	75.3 ± 8	(463)	HSA	[78/17]
$\text{C}_5\text{H}_6\text{N}_2\text{O}_2$		69.5 ± 1.2		ME	[72/32][00/2]
	5-methyluracil (thymine)				[65-71-4]
	(383–438)	125.7 ± 3.6	(411)	ME	[84/12]
		131.3 ± 4.0	(298)		[84/12]
	(378–428)	124.4 ± 1.3	(403)	QR	[80/19]
		138 ± 10	(298)	TE	[80/9]
		134.1 ± 4.2	(298)	C	[77/13]
$\text{C}_5\text{H}_6\text{N}_2\text{O}_2$		124.3		LE	[75/16][74/8]
	6-methyluracil				[626-48-2]
	(426–503)	131	(298)		[80/13]
	glutaric anhydride				[108-55-4]
$\text{C}_5\text{H}_6\text{O}_3$		85.9 ± 1.6	(309)	ME	[90/4]
	(298–320)	86.1 ± 1.6	(298)		[90/4]
$\text{C}_5\text{H}_7\text{N}$	N-methylpyrrole				[96-54-8]
$\text{C}_5\text{H}_7\text{NO}_2$		NA			[41/3]
	glutarimide				[1121-89-7]
	(317–340)	93.6 ± 1.6	(329)	ME	[90/4]
$\text{C}_5\text{H}_7\text{NO}_2$		94.1 ± 1.6	(298)		[90/4]
	N-methylsuccinimide				[1121-07-9]
	(280–298)	80.6 ± 0.3	(289)	ME	[97/12]
$\text{C}_5\text{H}_7\text{NO}_3$		80.1 ± 0.3	(298)		[97/12]
	(dl)-5-oxoproline				[149-87-1]
$\text{C}_5\text{H}_7\text{N}_3\text{O}$		133.2 ± 1	(405)	TE,ME	[79/1]
	1-methylcytosine				[1122-47-0]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_5\text{H}_7\text{N}_3\text{O}$	(455–487)	141.2 ± 0.6		GS	[98/37]
	(423–443)	141.8 ± 8.8	(433)	ME	[84/12]
		149.1 ± 9.0	(298)		[84/12]
$\text{C}_5\text{H}_7\text{N}_3\text{O}$	3-methylcytosine				[4776-08-3]
$\text{C}_5\text{H}_7\text{N}_3\text{O}$	(487–526)	150.6		HSA	[65/2]
$\text{C}_5\text{H}_7\text{N}_3\text{O}_2$	3,5-dimethyl-4-nitrosopyrazole				[1122-04-9]
		102.9 ± 3.0	(298)	C	[01/4]
$\text{C}_5\text{H}_7\text{N}_3\text{O}_2$	1-methyl-N-hydroxycytosine				[20541-50-8]
		126.7 ± 1.5			[98/37]
$\text{C}_5\text{H}_8\text{Br}_4$	pentaerythrityl tetrabromide				[3229-00-3]
	(384–434)	84	(399)	GSM	[87/4][41/1]
$\text{C}_5\text{H}_8\text{NO}_2$	5-amino-3,4-dimethylisoxazole				[19947-75-2]
		87.9 ± 2.5			[73/21][77/1]
$\text{C}_5\text{H}_8\text{N}_2$	2,3-diazabicyclo[2.2.1]hept-2-ene				[2721-32-6]
		43.9 ± 2.1			[74/17][77/1]
		55.3 ± 0.6	(298)		[76/12]
$\text{C}_5\text{H}_8\text{N}_2$	3,5-dimethylpyrazole				[67-51-6]
		83.4 ± 2.4	(298)	C	[01/4]
		83.3 ± 0.2	(301)	ME	[91/21]
$\text{C}_5\text{H}_8\text{N}_2$	2-ethylimidazole				[1072-62-4]
	(303–321)	89.2 ± 0.4	(312)	ME	[92/25]
		89.6 ± 0.4	(298)		[92/25]
$\text{C}_5\text{H}_8\text{N}_4\text{O}_{12}$	pentaerythritol tetranitrate				[78-11-5]
	(328–405)	150.4 ± 1.3	(298)	ME	[78/15]
		146 ± 12			[71/34][78/15]
		U121.3		ME	[69/3]
	(370–411)	151.9 ± 2.1			[53/6][60/1]
$\text{C}_5\text{H}_8\text{OS}$	tetrahydro-4H-thiopyran-4-one				[70/1]
		71.7 ± 1.7	(317)	I	[1072-72-6]
		72.6 ± 1.7	(298)		[72/15]
$\text{C}_5\text{H}_8\text{O}_2$	methyl methacrylate				[72/15][77/1]
	(194–223)	60.7	(205)		[80-62-6]
$\text{C}_5\text{H}_8\text{O}_2\text{S}$	2,5-dihydro-2-methyl-thiophene-1,1-dioxide				[52/5][60/1]
		60.7 ± 2.5			[6007-71-2]
$\text{C}_5\text{H}_8\text{O}_2\text{S}$	2,5-dihydro-3-methyl-thiophene-1,1-dioxide				[69/11][77/1]
		64.0 ± 2.5			[1193-10-8]
$\text{C}_5\text{H}_8\text{O}_4$	1,5-pentanedioic acid (glutaric acid)				[69/11][77/1]
	(348–363)	117.0 ± 1.2	(356)	ME	[110-94-1]
		119.8 ± 1.2	(298)		[99/10]
	(292–320)	U52.6	(306)	A	[99/10]
$\text{C}_5\text{H}_8\text{O}_4$	dimethylmalonic acid				[47/6]
	(347–363)	110.2 ± 1.0	(355)	ME	[595-46-0]
		111.7 ± 2.1	(298)	ME	[00/22]
$\text{C}_5\text{H}_8\text{O}_4$	ethylmalonic acid				[00/22]
	(347–362)	111.2 ± 1.2	(355)	ME	[601-75-2]
		112.8 ± 2.2	(298)	ME	[00/22]
		105.5 ± 0.5		C	[83/26]
$\text{C}_5\text{H}_9\text{NO}$	cis 2-pentenoic acid amide				[15856-96-9]
	(323–333)	106.5	(328)		[87/4]
$\text{C}_5\text{H}_9\text{NO}$	trans 2-pentenoic acid amide				[15856-96-9]
	(353–383)	57.9	(368)		[87/4]
$\text{C}_5\text{H}_9\text{NO}$	δ -valerolactam				[675-20-7]
	(293–312)	74.5	(303)		[53/5][60/1]
					[60/21]
$\text{C}_5\text{H}_9\text{NO}_2$	L-(l)-proline				[147-85-3]
	(396–416)	127.4 ± 1	(406)	TE,ME	[79/1]
	(380–420)	149 ± 4	(400)	C	[78/4]
	(323–423)	U 50 $\pm$ 8	(373)	LE	[77/2]
$\text{C}_5\text{H}_9\text{NO}_3$	trans 4-hydroxy-L-proline				[51-35-4]
	(461–481)	162.6 ± 2	(471)	TE,ME	[79/1]
$\text{C}_5\text{H}_9\text{NO}_4$	L-glutamic acid				[56-86-0]
	(353–453)	U 121 $\pm$ 34	(403)	LE	[77/2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₅ H ₁₀	cyclopentane	42.6	(122)	B	[287-92-3] [63/6]
C ₅ H ₁₀ N ₂ O ₂	N-acetyl glycine, N-methylamide (348–363)	97.8	(355.5)		[7606-79-3] [87/4][55/7]
C ₅ H ₁₀ N ₂ O ₂	N-acetyl-L-alanine amide	115.0 ± 1.2 118.1 ± 1.6 115 ± 3	(376) (298) (388)	C TE	[15962-47-7] [99/12] [99/12] [88/6][86/16]
C ₅ H ₁₀ O ₂	2,2-dimethylpropanoic acid (278–303)	62.3 ± 0.6 62.1 ± 0.6	(291) (298)	GS GS	[75-98-9] [00/20] [00/20]
C ₅ H ₁₀ O ₅	1,3,5,7,9-pentoxecane	87.9 ± 0.5	(298)	C	[16528-92-0] [74/16]
C ₅ H ₁₀ O ₅	D-xylose (370–395)	158.0 ± 3.1	(382)	ME	[58-86-6] [99/1]
C ₅ H ₁₁ NO	pentanamide (333–374) (353–373)	89.3 ± 0.4 89.1		GS	[626-97-1] [59/3][70/1] [60/1]
C ₅ H ₁₁ NO	2,2-dimethylpropanamide (298–359) (288–306)	89 ± 2.0 86.6 ± 0.4	(298) (298)	TE ME	[759-10-9] [00/1] [89/6]
C ₅ H ₁₁ NO ₂	(dl) 2-aminopentanoic acid (DL-norvaline) (439–461)	120 121.1 ± 0.4	(450) (455)	ME ME	[760-78-1] [87/4][65/1] [65/1][64/16]
C ₅ H ₁₁ NO ₂	butyl carbamate (292–316)	94.1 ± 8		GS	[592-35-8] [59/4]
C ₅ H ₁₁ NO ₂	DL-valine (320–420)	U 79.5 ± 8	(370)	LE	[516-06-3] [77/2]
C ₅ H ₁₁ NO ₂	L-valine	162.8 ± 8	(455)	ME	[72-18-4] [65/1][64/16]
C ₅ H ₁₁ NO ₂	5-aminopentanoic acid (384–394)	141.8 ± 0.5 144 ± 3	(389) (289)	C C	[660-88-8] [83/24] [83/24]
C ₅ H ₁₁ NO ₂ S	DL-methionine (363–463)	U 134 ± 8	(413)	LE	[59-51-8] [77/2]
C ₅ H ₁₁ NO ₂ S	L-(d)-methionine (463–485)	125 ± 0.8	(474)	ME	[63-68-3] [87/4][65/1] [64/16]
C ₅ H ₁₂	2,2-dimethylpropane (223–256)	164 ± 4 28.2 33.2	(298) (241)	C	[81/5] [463-82-1] [87/4] [63/6][36/1]
C ₅ H ₁₂	(171–249)	23.9 22.0 22.8	(210) (298) (241)	A H A	[47/2]
C ₅ H ₁₂	n-pentane	42.0	(143)	B	[33/6] [109-66-0] [63/6]
C ₅ H ₁₂ N ₂ O	1,3-diethylurea (321–379) (384–590)	96.8 ± 0.9 NA	(361)	TE ME	[623-76-7] [90/5][87/5] [86/19]
C ₅ H ₁₂ N ₂ O	N-butylurea	99 ± 4 103.2 ± 0.8	(352)		[592-31-4] [87/6] [86/6][90/5]
C ₅ H ₁₂ N ₂ O	N-isobutylurea	101.1 ± 1.1 106.0 ± 0.5	(377) (355)	TE	[592-17-6] [90/5] [90/5][86/6]
C ₅ H ₁₂ N ₂ O	N-tert-butylurea	101.6 ± 0.7 100.7 ± 0.3	(379) (352)	TE	[1118-12-3] [90/5] [90/7][86/6]
C ₅ H ₁₂ N ₂ S	diethylthiourea	121.7 ± 3 120.2 ± 3.0	(298) (298)	B,HA B	[26914-14-7] [00/23] [94/17]
C ₅ H ₁₂ N ₂ S	tetramethylthiourea				[2782-91-4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_m/\text{kJ mol}^{-1}$	(T_m/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_5\text{H}_{12}\text{O}_2$	neopentyl glycol plastic phase crystalline phase	84.5 ± 3	(298)	ME	[00/23]
		84.0	(298)		[94/20]
		83.0 ± 0.5	(298)	C	[85/5]
		83.0 ± 0.2	(298)	C	[82/7]
					[126-30-7]
$\text{C}_5\text{H}_{12}\text{O}_3$	1,1,1- <i>tris</i> (hydroxymethyl)ethane plastic phase crystalline phase	75.5 ± 3.8	(368)	C	[94/3][94/27]
		87.6 ± 4.4	(350)	C	[94/3][94/27]
					[77-85-0]
$\text{C}_5\text{H}_{12}\text{O}_2\text{S}$	<i>tert</i> -butyl methyl sulfone	84.2 ± 4.2	(319)	C	[94/27]
		109.2 ± 5.5	(311)	C	[94/27]
$\text{C}_5\text{H}_{12}\text{O}_4$	pentaerythritol				[14094-12-3]
		82.4 ± 2.5			[U/3][70/1]
					[115-77-5]
					[94/27]
(tetragonal)	(418–455)	131.3 ± 6.6	(403)	C	[94/27]
		161 ± 1.0	(437)	TE	[90/7]
		163	(298)		[90/7]
		131.4		ME	[51/3][60/1]
$\text{C}_5\text{H}_{12}\text{O}_5$	(379–408) adonitol	143.9 ± 0.8		ME	[53/4][60/1]
					[488-81-3]
		161	(298)	B	[90/7]
$\text{C}_5\text{H}_{12}\text{O}_5$	D-arabitol				[488-82-4]
		160	(298)	B	[90/7]
$\text{C}_5\text{H}_{12}\text{O}_5$	xylitol				[87-99-0]
		161	(298)	B	[90/7]
$\text{C}_6\text{Cl}_4\text{O}_2$	tetrachloro-1,4-benzoquinone (333–356)				[118-75-2]
		98.7 ± 8.3		QF	[27/2][60/1]
					[70/1]
C_6Cl_6	hexachlorobenzene (258–313) (253–303)				[118-74-1]
		105			[94/39]
		77.4 ± 0.8	(278)	GS	[94/1]
		89.6 ± 0.2	(337)	C	[91/2]
		90.5 ± 0.2	(298)	C	[91/2]
		(461–506)			[89/32]
		(387–502)			[87/4]
		(314–373)	(402)		[87/4]
		(288–318)	(344)	GS	[86/18][97/40]
		(312–337)	(303)	GS	[80/36]
		(369–397)			[77/10]
		79.5 ± 12			[49/3][70/1]
C_6F_6	hexafluorobenzene (215–278) (238–268)	92 ± 8.2		RG	[392-56-3]
		49.2	(263)		[87/4][65/22]
		49.8	(253)	IPM,A	[79/33]
		46.0	(316)	B	[65/10]
					[355-68-0]
C_6F_{12}	dodecafluorocyclohexane (252–326) (293–333)	36.4	(267)		[87/4][67/21]
		36.2	(313)		[57/5]
					[16419-78-6]
C_6N_2	dicyanodiacetylene (294–335) (295–335)	(dicyanobutadiyne)			[16419-78-6]
		34.4	(309)		[87/4]
		35.9	(315)		[57/4]
C_6N_4	tetracyanoethylene (290–312) (333–371)				[670-54-2]
		84.3	(302)	TE,ME	[83/7]
		81.2 ± 5.9	(352)	MG	[63/4][70/1]
					[87/4]
$\text{C}_6\text{N}_6\text{O}_3$	benzotrifurazan (303–333)	78.0		GS	[58/14]
					[99/44]
$\text{C}_6\text{N}_6\text{O}_6$	benzotrifuoxan (363–433)	95.8 ± 3.8			[99/44]
		172.0 ± 2.5			[99/44]
$\text{C}_6\text{HCl}_3\text{O}_2$	trichloro-1,4-benzoquinone (301–327)				[634-85-5]
		88.7 ± 8.3		QF	[27/2][60/1]
C_6HCl_5	pentachlorobenzene				[70/1]
					[608-93-5]
$\text{C}_6\text{HCl}_5\text{O}$	pentachlorophenol	87.1 ± 0.4	(298)	C	[91/2]
					[87-86-5]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_6\text{HF}_5\text{O}$	pentafluorophenol (273–299)	67.4 ± 2.1 67.4 ± 1.7		GS	[U/4][70/1] [771-61-9] [69/5][70/1]
$\text{C}_6\text{H}_2\text{ClN}_3\text{O}_6$	1,3,5-trinitrochlorobenzene (341–363)	103.0	(352)	ME	[88-88-0] [50/2]
$\text{C}_6\text{H}_2\text{Cl}_2\text{O}_2$	2,6-dichloro-1,4-benzoquinone (274–315)	69.9 ± 8.3		QF	[697-91-6] [27/2][60/1] [70/1]
$\text{C}_6\text{H}_2\text{Cl}_4$	1,2,3,4-tetrachlorobenzene	78.8 ± 0.2	(298)	C	[634-66-2] [91/2]
$\text{C}_6\text{H}_2\text{Cl}_4$	1,2,4,5-tetrachlorobenzene	83.2 ± 0.3	(298)	C	[95-94-3] [91/2]
$\text{C}_6\text{H}_2\text{Cl}_4$	1,2,3,5-tetrachlorobenzene	79.6 ± 0.3	(298)	C	[634-90-2] [91/2]
$\text{C}_6\text{H}_2\text{Cl}_4\text{O}_2$	tetrachlorohydroquinone (298–359) (333–356)	89 88.7	(313)	QF	[87-87-6] [87/4] [27/2][60/1]
$\text{C}_6\text{H}_3\text{Br}_3\text{O}$	2,4,6-tribromophenol	97.6 ± 1.1			[118-79-6] [87/3]
$\text{C}_6\text{H}_3\text{ClN}_2\text{O}_2$	5-chlorobenzofurazan-1-oxide	81.2 ± 1.8	(298)	C	[17348-69-5] [96/6]
$\text{C}_6\text{H}_3\text{ClO}_2$	chlorobenzoquinone (264–289)	69.0 ± 8.3	(276)	QF	[695-99-8] [27/2][60/1] [70/1]
$\text{C}_6\text{H}_3\text{Cl}_3$	1,2,3-trichlorobenzene (258–313)	72.7 75.1 ± 0.75 65.7	(298) (296)	RG	[87-61-6] [94/39] [85/8] [49/7][60/1]
$\text{C}_6\text{H}_3\text{Cl}_3$	1,2,4-trichlorobenzene (279–298)	62.3	(289)	RG	[120-82-1] [49/7][60/1]
$\text{C}_6\text{H}_3\text{Cl}_3$	1,3,5-trichlorobenzene (282–301)	72.7 ± 0.5 56.5	(298) (291)	RG	[108-70-3] [85/8] [49/7][60/1]
$\text{C}_6\text{H}_3\text{Cl}_3\text{O}_2$	trichlorohydroquinone (298–336) (314–335)	101.5 101.3	(313) (324)	QF	[608-94-6] [87/4] [27/2][60/1]
$\text{C}_6\text{H}_3\text{N}_3\text{O}_4$	4-nitrobenzofurazan-1-oxide	97.3 ± 1.6	(298)	C	[18771-85-2] [96/6]
$\text{C}_6\text{H}_3\text{N}_3\text{O}_6$	1,3,5-trinitrobenzene (313–395) (353–395)	107.3 ± 0.6 99.6 ± 2.1	(298) (374)	ME ME	[99-35-4] [78/15] [50/2][70/1]
$\text{C}_6\text{H}_3\text{N}_3\text{O}_7$	picric acid (314–406)	105.1 ± 1.6	(298)	ME	[88-89-1] [78/15]
$\text{C}_6\text{H}_3\text{N}_3\text{O}_8$	2,4,6-trinitroresorcinol (325–436)	120.8 ± 1.1	(298)	ME	[82-71-3] [78/15]
$\text{C}_6\text{H}_4\text{BrCl}$	1,4-bromochlorobenzene	69.3 ± 0.1 69.3 ± 0.4 67.9 ± 0.8	(298) (298) (316)	DM ME,TE,DM	[106-39-8] [00/31] [98/13] [61/2]
$\text{C}_6\text{H}_4\text{BrI}$	1,4-bromoiodobenzene (279–355)	78.5 ± 0.2 78.5 ± 0.4	(298) (298)	DM ME,TE,DM	[589-87-7] [00/31] [98/13]
$\text{C}_6\text{H}_4\text{BrNO}_2$	4-bromo-1-nitrobenzene (293–303)	88.3	(303)	ME	[586-78-7] [87/4][25/3]
$\text{C}_6\text{H}_4\text{Br}_2$	1,4-dibromobenzene (298–354) (278–353) (228–347) (248–303)	74.2 ± 0.1 73.2 73.3 ± 0.4 73.8 59.8	(298) (313) (326) (288) (298)	ME ME,GS	[106-37-6] [00/31] [87/4] [61/2] [59/5] [40/1][60/1]
$\text{C}_6\text{H}_4\text{Br}_3\text{N}$	2,4,6-tribromoaniline	101.1 ± 1.1			[147-82-0] [87/3]
$\text{C}_6\text{H}_4\text{ClI}$	1,4-chloriodobenzene				[637-87-6]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_6\text{H}_4\text{ClNO}_2$	(259–320)	71.9 ± 0.2	(298)	DM	[00/31]
	(303–323)	71.9 ± 0.4	(298)	ME, TE, DM	[98/13]
	3-chloro-1-nitrobenzene	61.1 ± 0.6			[53/8][60/1]
	(275–286)	74.7 ± 1.7			[121-73-3] [35/1][38/1] [60/1]
$\text{C}_6\text{H}_4\text{ClNO}_2$	4-chloro-1-nitrobenzene				[100-00-5]
$\text{C}_6\text{H}_4\text{Cl}_2$	(283–303)	83.2	(293)	ME	[87/4][25/3]
	1,4-dichlorobenzene				[106-46-7]
	(258–313)	64.8 ± 0.2	(298)	DM	[00/31]
		53.1			[94/39]
	(303–423)	65.2 ± 2.0	(298)	C	[89/25]
		65.4	(313)	GS	[85/4]
		65.7			[81/12]
	(293–311)	64.8 ± 0.8	(303)		[61/2][70/1]
$\text{C}_6\text{H}_4\text{Cl}_2\text{O}$	(311–325)	63 ± 0.4	(318)		[61/2]
	(248–303)	56.9	(275)	ME	[40/1][60/1]
	2,3-dichlorophenol				[576-24-9]
		71.7 ± 2.2	(298)	C	[94/5]
$\text{C}_6\text{H}_4\text{Cl}_2\text{O}$	2,4-dichlorophenol				[120-83-2]
$\text{C}_6\text{H}_4\text{Cl}_2\text{O}$		70.1 ± 1.1	(298)	C	[94/5]
	2,5-dichlorophenol				[583-78-8]
$\text{C}_6\text{H}_4\text{Cl}_2\text{O}$		73.6 ± 2.1	(298)	C	[94/5]
	2,6-dichlorophenol				[87-65-0]
$\text{C}_6\text{H}_4\text{Cl}_2\text{O}$		75.8 ± 1.1	(298)	C	[94/5]
	3,4-dichlorophenol				[95-77-2]
$\text{C}_6\text{H}_4\text{Cl}_2\text{O}$		81.3 ± 2.3	(298)	C	[94/5]
	3,5-dichlorophenol				[591-35-5]
$\text{C}_6\text{H}_4\text{Cl}_2\text{O}_2$		82.8 ± 1.1	(298)	C	[94/5]
	(273–295)	71.8	(284)		[87/4]
	2,6-dichlorohydroquinone				[20103-10-0]
	(324–345)	92.0 ± 8.3		QF	[27/2][60/1] [70/1]
$\text{C}_6\text{H}_4\text{INO}_2$	3-iodo-1-nitrobenzene				[645-00-1]
	(295–306)	83.2 ± 1.2	(300)		[35/1][38/1] [60/1]
$\text{C}_6\text{H}_4\text{I}_2$	1,4-diiodobenzene				[624-38-4]
$\text{C}_6\text{H}_4\text{N}_2$	(372–401)	63.4	(386.5)		[87/4]
	2-cyanopyridine				[100-70-9]
$\text{C}_6\text{H}_4\text{N}_2$		70.7 ± 1.2	(298)	C	[84/6]
	3-cyanopyridine				[100-54-9]
$\text{C}_6\text{H}_4\text{N}_2$		72.1 ± 1.8	(298)	C	[84/6]
		79.0		DSC	[89/22]
	4-cyanopyridine				[100-48-1]
		73.2 ± 0.6	(298)	C	[84/6]
$\text{C}_6\text{H}_4\text{N}_2\text{O}$		75.6		DSC	[89/22]
	3-cyanopyridine N-oxide				[14906-64-0]
$\text{C}_6\text{H}_4\text{N}_2\text{O}$	(345–392)	101.9 ± 2.0	(298)	ME	[98/12]
	4-cyanopyridine N-oxide				[14906-59-3]
$\text{C}_6\text{H}_4\text{N}_2\text{O}$	(345–392)	104.4 ± 4.3	(298)	ME	[98/12]
	benzofurazan				[273-09-6]
$\text{C}_6\text{H}_4\text{N}_2\text{O}_2$		64.4 ± 1.6	(298)	C	[90/29]
		64.9	(298)		[80/6]
	benzofurazan N-oxide				[480-96-6]
		79.6 ± 1.7	(298)	C	[90/29]
$\text{C}_6\text{H}_4\text{N}_2\text{O}_3$	1-nitro-2-nitrosobenzene (dimer)				[612-29-3]
	(323–343)	95.5	(333)		[87/4][74/34]
$\text{C}_6\text{H}_4\text{N}_2\text{O}_4$	1,2-dinitrobenzene				[528-29-0]
		95.5 ± 0.9	(298)		[97/30]
	(343–377)	82.9	(358)	TE	[87/4][76/1]
	(343–397)	81.8 ± 2.3	(370)	TE	[76/1]
	(343–397)	87.9 ± 2.1	(298)	TE	[76/1]
	(328–338)	86.6 ± 1.2	(333)		[35/1][38/1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_6\text{H}_4\text{N}_2\text{O}_4$	1,3-dinitrobenzene				[60/1]
	(335–356)	76.1	(345.5)		[99-65-0]
	(332–383)	84.2 ± 1.9	(357)	TE	[87/4]
	(332–383)	87.0 ± 0.8	(298)	TE	[76/1]
	(315–329)	81.1 ± 1.7	(323)		[76/1]
$\text{C}_6\text{H}_4\text{N}_2\text{O}_4$	1,4-dinitrobenzene				[35/1][38/1]
					[60/1][70/1]
					[100-25-4]
	(339–398)	94.3 ± 0.7	(298)		[97/30]
	(339–398)	93.1 ± 2.3	(368)	TE	[76/1]
$\text{C}_6\text{H}_4\text{N}_2\text{O}_5$	2,3-dinitrophenol				[76/1]
	(303–343)	96.6	(323)		[35/1][38/1]
	2,4-dinitrophenol				[60/1]
	(293–333)	104.6 ± 4.2	(313)		[66-56-8]
	2,5-dinitrophenol				[58/1]
$\text{C}_6\text{H}_4\text{N}_2\text{O}_5$	(278–333)	93.4	(306)		[51-28-5]
	2,6-dinitrophenol				[58/1][70/1]
	(293–333)	112.1 ± 4.2	(313)		[329-71-5]
	3,4-dinitrophenol				[58/1]
	(328–383)	123.5	(355)		[577-71-9]
$\text{C}_6\text{H}_4\text{N}_2\text{S}$	2,1,3-benzothiadiazole				[58/1]
$\text{C}_6\text{H}_4\text{N}_4\text{O}_6$		70.73 ± 0.2	(298)	C	[273-13-2]
	2,4,6-trinitroaniline				[98/2]
	(328–371)	115.9	(343)	LE	[489-98-5]
$\text{C}_6\text{H}_4\text{O}_2$		125.3 ± 0.8	(298)	ME	[87/4][69/12]
	1,4-benzoquinone				[78/15]
		68.0 ± 0.5	(262)	ME,TE	[106-51-4]
$(\text{C}_6\text{H}_4\text{O}_2) - (\text{C}_6\text{H}_6\text{O}_2)$		62.8 ± 3.3			[81/4]
		68.5 ± 0.6			[56/5][77/1]
	(260–278)	62.8	(269)	QF	[53/10]
	quinhydrone (quinone-hydroquinone)				[27/2]
	(317–334)	89.1	(325.5)		[106-34-3]
$\text{C}_6\text{H}_4\text{O}_5$		88.6 ± 1	(313)	ME,TE	[87/4]
		U 181.2			[81/4]
		NA			[53/10][60/1]
	furan-2,5-dicarboxylic acid				[51/6]
	(378–402)	121.3	(391)	TE,ME	[3238-40-2]
$\text{C}_6\text{H}_4\text{S}_4$	tetrathiofulvalene				[83/5]
		61.0			[31366-25-3]
		95.3 ± 1	(345)	TGA	[95/35]
	(341–361)	92 ± 6.3	(351)	TE,ME	[80/22]
	(tetrathiofulvalene)-(7,7,8,8-tetracyanoquinodimethane)			HSA	[79/7]
$(\text{C}_6\text{H}_4\text{S}_4) - (\text{C}_{12}\text{H}_4\text{N}_4)$	(TTF–TCNQ)				[40210-84-2]
		130 ± 2	(410)	TE,ME	[80/22]
	1-bromo-4-chlorobenzene				[106-39-8]
	(250–335)	69.3 ± 0.4	(298)	TE,ME,DM	[98/13]
		69.1 ± 0.2	(298)		[98/13]
$\text{C}_6\text{H}_5\text{BrO}$	4-bromophenol				[106-41-2]
	(260–302)	87.3 ± 0.4	(298)	ME	[71/8]
	3-chlorophenol				[108-43-0]
$\text{C}_6\text{H}_5\text{ClO}$		53.1			[38/1][60/1]
	4-chlorophenol				[70/1]
	(252–293)	60.8	(278)		[106-48-9]
		51.9			[87/4]
					[38/1][60/1]
$\text{C}_6\text{H}_5\text{ClO}_2$	chlorohydroquinone				[70/1]
	(306–334)	102.9 ± 8.3	(320)	QF	[615-67-8]
					[27/2][60/1]
					[70/1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₆ H ₅ I	iodobenzene				[591-50-4]
	(243–255)	43.1			[60/1]
C ₆ H ₅ NO	(248–303)	40.0	(275)	ME	[40/1]
	nitrosobenzene (dimer)				[586-96-9]
	(297–339)	85.1 80.8	(312)		[87/4][74/34] [30/3]
C ₆ H ₅ NO ₂	2-pyridinecarboxylic acid	91.0±0.5	(329)	C	[98-98-6] [99/9]
		92.7±0.5	(298)		[99/9]
		98.0±2.3	(298)	ME	[98/12]
C ₆ H ₅ NO ₂	3-pyridinecarboxylic acid				[59-67-6]
		(345–392)			[00/3]
		123.9±3.7	(298)	ME	[99/9]
		(352–360)			[99/9]
C ₆ H ₅ NO ₂	4-pyridinecarboxylic acid	101.1±0.6	(362)	C	[84/6]
		105.2±0.6	(298)		[55-22-1]
		123.4±1.2	(298)	C	[99/9]
					[99/9]
					[98/12]
C ₆ H ₅ NO ₃	2-nitrophenol	113.9±4.7	(298)	ME	[88-75-5]
					[94/28]
		(273–292)			[87/4]
		(298–310)	(282.5)		[35/1][38/1]
C ₆ H ₅ NO ₃	3-nitrophenol	73.3	(298)	C	[60/1]
		54.8			[554-84-7]
		73.2±1.3			[94/28]
					[92/13]
					[92/13]
C ₆ H ₅ NO ₃	4-nitrophenol				[87/4]
					[35/1][38/1]
					[60/1]
					[100-02-7]
					[94/28]
C ₆ H ₅ NO ₃		92.4	(298)	C	[71/8]
		(305–352)	(298)	ME	[35/1][38/1]
		(339–351)			[60/1]
C ₆ H ₅ NO ₃	pyridine-2-carboxylic acid N-oxide				[824-40-8]
C ₆ H ₅ NO ₃	pyridine-3-carboxylic acid N-oxide				[98/12]
		(345–392)	(298)	ME	[2398-81-4]
C ₆ H ₅ NO ₃	pyridine-4-carboxylic acid N-oxide	152.3±1.9	(298)	ME	[95/3][95/11]
					[13602-12-5]
C ₆ H ₅ NO ₄	2-nitro-1,3-dihydroxybenzene	136.1±1.2	(298)	ME	[98/12]
		(253–293)			[601-89-8]
C ₆ H ₅ NO ₄	4-nitrocatechol	74.5	(273)		[58/1]
					[3316-09-4]
C ₆ H ₅ NO ₅	methyl 5-nitro-2-furancarboxylate	121.1±1.4		C	[86/3]
		104.2±2.1			[1874-23-3]
C ₆ H ₅ N ₃	1- <i>H</i> -benzotriazole				[80/28][86/5]
					[95-14-7]
		98.2±0.7	(298)	C	[99/8]
		(327–345)	(336)	ME	[89/8]
C ₆ H ₅ N ₅ O ₆	1,3-diamino-2,4,6-trinitrobenzene	99.0±0.5	(298)	ME	[89/8]
		97.9		ME	[61/3]
		140	(350)	LE	[28930-29-2]
		143.5	(298)		[87/4][69/12]
C ₆ H ₆	benzene				[78/15]
		(258–273)			[71-43-2]
		(223–279)			[94/39]
		41.7			[87/4][76/23]
		45.2	(264)	BG	
		44.6	(298)	H	
		45.1	(278)		[84/21]
		44.8	(298)	H	
		(183–197)	(298)	TE,ME	[80/1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C_6H_6	(221–268)	53.9 ± 0.8	(193)	MM	[77/4]
		49.4 ± 0.4	(193)		[77/4]
		45.6	(279)		[74/22]
		44.1	(261)		[60/1]
		43.1	(229)		[60/1]
	(263–270)	44.6	(279)	A	[56/8]
		46.6	(282)		[47/2]
		44.6	(273)		[36/3][74/22]
		U33.2	(192)		[33/4]
	(184–200)	43.3	(226)	A	[13/1]
	(214–238)				[2809-69-0]
	2,4-hexadiyne			MM	[82/11]
	(282–333)	47 ± 2	(307)		[82/11]
	solid phase transition	1.0	(118)		
$\text{C}_6\text{H}_6\text{ClN}$	4-chloroaniline				[106-47-8]
$\text{C}_6\text{H}_6\text{Cl}_6$	(283–303)	90.7	(293)	ME	[87/4][25/3]
	α -hexachlorocyclohexane (melting point 160 °C)				[319-84-6]
	(313–363)	95.7	(328)		[87/4][60/1]
$\text{C}_6\text{H}_6\text{Cl}_6$	(324–344)	92.9	(334)	TE	[47/1]
	β -hexachlorocyclohexane (melting point 314 °C)				[319-85-7]
	(506–551)	103.7			[89/32]
	(313–363)	107	(328)		[87/4][60/1]
$\text{C}_6\text{H}_6\text{Cl}_6$	(368–390)	102.9	(379)	TE	[47/1]
	γ -hexachlorocyclohexane (melting point 114 °C)				[58-89-9]
	(310–384)	92.4 ± 4.0	(298)	ME,TE	[98/15]
	(292–326)	97.7 ± 0.6	(308)	ME	[96/12]
	(243–303)	106.6 ± 0.9	(273)	GS	[94/1]
		90.1 ± 0.7	(338)	C	[91/5]
		90.8 ± 0.7	(298)	C	[91/5]
	(313–358)	70.5	(335)		[90/26]
	(313–363)	99.2	(328)		[87/4][60/1]
	(293–313)	88.9	(303)	GS	[83/5][70/8]
	(293–313)	101.2	(303)		[70/8]
	(313–343)	89.7	(328)		[60/12]
	(333–365)	115.5		TE	[47/1]
	δ -hexachlorocyclohexane (melting point 142 °C)				[319-86-8]
	(313–363)	97.3	(328)		[87/4][60/1]
	(328–358)	97.5			[47/1]
$\text{C}_6\text{H}_6\text{F}_8\text{O}_2$	2,2,3,3,4,4,5,5-octafluoro-1,6-hexanediol				[355-74-8]
$\text{C}_6\text{H}_6\text{N}_2\text{O}$		89.2 ± 8.4			[74/18][77/1]
	2-pyridinecarboxamide				[1452-77-3]
		93.1 ± 3.3	(298)	C	[01/1]
$\text{C}_6\text{H}_6\text{N}_2\text{O}$	(323–373)	93.1	(338)	ME	[87/4][60/16]
					[59/13]
	3-pyridinecarboxamide				[98-92-0]
$\text{C}_6\text{H}_6\text{N}_2\text{O}$		121.2 ± 3.3	(298)	C	[01/1]
	(363–393)	111.8	(378)	ME	[87/4][60/16]
					[59/13]
$\text{C}_6\text{H}_6\text{N}_2\text{O}$	4-pyridinecarboxamide				[1453-82-3]
		116.1 ± 1.5	(298)	C	[01/1]
	(383–412)	99.9	(397.5)	ME	[87/4][60/16]
$\text{C}_6\text{H}_6\text{N}_2\text{O}_2$					[59/13]
	3-pyridinecarboxamide N-oxide				[1986-81-8]
$\text{C}_6\text{H}_6\text{N}_2\text{O}_2$	(413–430)	119.2 ± 2.3	(298)	ME	[01/1]
	4-pyridinecarboxamide N-oxide				[38557-82-3]
$\text{C}_6\text{H}_6\text{N}_2\text{O}_2$	(409–430)	125.3 ± 1.8	(298)	ME	[01/1]
	2-nitroaniline				[88-74-4]
$\text{C}_6\text{H}_6\text{N}_2\text{O}_2$		90 ± 3.0		ME,TE	[85/7]
		82.4 ± 2	(313)		[38/1][60/1]
					[35/1]
		90 ± 4.2			[58/1][70/1]
	(310–319)	79.9 ± 1.7			[34/1]
$\text{C}_6\text{H}_6\text{N}_2\text{O}_2$		89.0 ± 0.7	(298)		[97/30]
	2-methyl-5-pyrazine carboxylic acid				[5521-55-1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_6\text{H}_6\text{N}_2\text{O}_2$	3-nitroaniline	100.9 ± 1.5	(298)	C	[97/25]
					[99-09-2]
		108.3 ± 3		ME,TE	[85/7]
	(320–384)	93.6 ± 0.7	(351)	ME	[73/5]
	(320–384)	94.6 ± 0.3	(351)	C	[73/5]
		96.5 ± 0.3	(298)	C	[73/5]
	(288–343)	97.6	(316)	ME	[58/1][70/1]
	(332–341)	88.3 ± 1.7		TE	[34/1]
$\text{C}_6\text{H}_6\text{N}_2\text{O}_2$	4-nitroaniline	88.7 ± 2.5			[38/1][60/1]
					[35/1]
					[100-01-6]
		101.4 ± 1.3	(298)	ME	[90/2]
		101.5 ± 1.7	(298)	TE	[90/2]
		94.6		GS	[87/19][91/18]
		107 ± 3		ME,TE	[85/7]
		100.4 ± 2.1	(298)	ME	[77/28][90/2]
	(351–417)	100.9 ± 0.6	(298)	ME	[73/5]
	(303–363)	109.3	(333)	ME	[58/1][70/1]
		99.3 ± 1.7	(298)	ME	[56/2]
	(346–366)	97.5 ± 1.7	(356)	ME	[56/2]
		100.7 ± 2.5	(298)	TE	[38/1]
		98.7 ± 2.5	(361)	TE	[38/1][60/1]
	(357–367)	103.3 ± 1.7	(362)		[34/1]
$\text{C}_6\text{H}_6\text{N}_2\text{O}_3$	3-methyl-4-nitropyridine N-oxide				[1074-98-2]
	(345–392)	106.7 ± 2.0	(298)	ME	[98/12]
$\text{C}_6\text{H}_6\text{N}_4\text{O}$	7-methylhypoxanthine				[1006-08-02]
		100.4 ± 13			[78/17]
$\text{C}_6\text{H}_6\text{N}_4\text{O}$	9-methylhypoxanthine				[875-31-0]
	(500–552)	84		HSA	[65/2]
$\text{C}_6\text{H}_6\text{N}_6\text{O}_6$	2,4,6-trinitro-1,3,5-benzenetriamine				[3058-38-6]
	(402–451)	168.2	(417)	LE	[87/4][69/12]
$\text{C}_6\text{H}_6\text{O}$	phenol				[108-95-2]
	(263–298)	65.3 ± 3.3	(280)	HSA	[75/3]
	(230–273)	69.7 ± 0.9	(298)	ME	[71/8]
	(282–313)	68.7 ± 0.5		GS	[60/3][70/1]
	(283–303)	68.2	(293)	ME	[58/18]
	(270–313)	68.1	(292)		[48/5]
	(278–305)	67.8		TE	[47/1][60/1]
$\text{C}_6\text{H}_6\text{O}_2$	1,2-dihydroxybenzene				[120-80-9]
		87.5 ± 0.3	(298)	C	[91/7]
		86.6 ± 1.6	(298)	C	[84/20]
		80.8			[38/1][60/1]
					[35/1]
$\text{C}_6\text{H}_6\text{O}_2$	1,3-dihydroxybenzene				[108-46-3]
		85.3 ± 0.5	(334)	C	[91/7]
		87.5 ± 0.5	(298)	C	[91/7]
	(328–379)	92.7	(353)	GS	[83/3]
	(324–335)	93.3 ± 2.1			[68/4]
	(283–323)	93.4	(303)		[58/1]
		95.4 ± 1.7			[38/1][60/1]
					[35/1]
$\text{C}_6\text{H}_6\text{O}_2$	1,4-dihydroxybenzene				[123-31-9]
		94.1 ± 0.5	(298)	C	[91/7]
		93.7 ± 0.5	(334)	C	[91/7]
	(341–400)	101.7	(370)	GS	[83/3]
		103.9 ± 1	(342)	ME,TE	[81/4]
	(298–346)	103.8	(313)		[56/5]
		90.1 ± 0.8			[53/10]
	(326–345)	103.8		QF	[27/2]
($\text{C}_6\text{H}_6\text{O}_2$)– ($\text{C}_{10}\text{H}_8\text{O}_2$)	1,4-hydroquinone-1,4-naphthoquinone				[60706-28-7]
		98.7 ± 1	(324)	TE,ME	[81/4]
$\text{C}_6\text{H}_6\text{O}_3$	1,2,3-trihydroxybenzene				[87-66-1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_6\text{H}_6\text{O}_3$	(377–398)	116.9±0.6	(298)	C	[86/3]
	1,2,4-trihydroxybenzene	89.1	(387)		[34/2]
$\text{C}_6\text{H}_6\text{O}_3$		119.8±1.6	(298)	C	[533-73-3]
	1,3,5-trihydroxybenzene				[86/3]
$\text{C}_6\text{H}_7\text{Cl}_2\text{N}$		131.7±1.0	(298)	C	[108-73-6]
	(383–406)	127.9		TE,ME	[86/3]
$\text{C}_6\text{H}_7\text{Cl}_2\text{N}$	2-chloroaniline hydrochloride				[83/23]
	(373–473)	77.6	(388)		[137-04-2]
$\text{C}_6\text{H}_7\text{Cl}_2\text{N}$	3-chloroaniline hydrochloride				[87/4][75/31]
	(383–473)	71.3	(398)		[141-85-5]
$\text{C}_6\text{H}_7\text{Cl}_2\text{N}$	4-chloroaniline hydrochloride				[87/4][75/31]
	(373–483)	77.8	(388)		[20265-96-7]
$\text{C}_6\text{H}_7\text{FN}_2\text{O}_2$	1,3-dimethyl-5-fluorouracil				[87/4][75/31]
$\text{C}_6\text{H}_7\text{F}_3\text{N}_2\text{O}_4$		120.4±3.8	(356)	TE	[02/1]
	N-[N-(trifluoroacetyl)glycyl]glycine				[400-58-8]
$\text{C}_6\text{H}_7\text{N}$	(273–423)	67	(288)		[87/4][60/20]
	3-methylpyridine				[108-99-6]
$\text{C}_6\text{H}_7\text{N}$	(225–255)	62.2	(240)	RG,ME	[87/4][51/5]
	4-methylpyridine				[108-89-4]
$(\text{C}_6\text{H}_7\text{N})-(\text{SO}_2)$	(230–257)	62.7	(243)	RG,ME	[87/4][51/5]
	aniline–sulfur dioxide complex				
$\text{C}_6\text{H}_7\text{NO}$	(277–323)	82.1	(300)		[31/2]
	2-aminophenol				[95-55-6]
$\text{C}_6\text{H}_7\text{NO}$		93.5±0.8	(332)	C	[96/18]
		95.3±0.7	(337)	C	[96/18]
		96.9±0.6	(298)	C	[96/18]
		103.9±0.9	(298)	C	[86/11]
$\text{C}_6\text{H}_7\text{NO}$	3-aminophenol				[591-27-5]
		98.8±0.9	(335)	C	[96/18]
		101.6±0.9	(298)	C	[96/18]
		104.7±1.2	(298)	C	[86/11]
$\text{C}_6\text{H}_7\text{NO}$	4-aminophenol				[123-30-8]
		101.1±0.7	(335)	C	[96/18]
		103.6±0.7	(298)	C	[96/18]
	(423–459)	111	(438)		[87/4]
$\text{C}_6\text{H}_7\text{NO}$		109.1±1.4	(298)	C	[86/11]
	(403–430)	92.1	(417)	I	[54/8][60/1]
	2-methyl-3-hydroxypyridine				[1121-25-1]
		89.3±1.3	(298)	C	[82/15][86/5]
$\text{C}_6\text{H}_7\text{NO}$	2-methyl-4-hydroxypyridine				[18617-86-6]
$\text{C}_6\text{H}_7\text{NO}$		113.0±1.3	(298)	C	[82/15][86/5]
$\text{C}_6\text{H}_7\text{NO}$	2-methyl-5-hydroxypyridine				[1121-78-4]
$\text{C}_6\text{H}_7\text{NO}$		96.2±2.1	(298)	C	[82/15][86/5]
$\text{C}_6\text{H}_7\text{NO}$	2-methyl-6-hydroxypyridine				[3279-76-3]
$\text{C}_6\text{H}_7\text{NO}$		92.0±1.3	(298)	C	[82/15][86/5]
$\text{C}_6\text{H}_7\text{NO}$	2-methylpyridine N-oxide				[931-19-1]
$\text{C}_6\text{H}_7\text{NO}$		78.2±2.2	(298)	C	[95/3]
$\text{C}_6\text{H}_7\text{NO}$	3-methylpyridine N-oxide				[1003-73-2]
$\text{C}_6\text{H}_7\text{NO}$		82.2±2.4	(298)	C	[95/3]
$\text{C}_6\text{H}_7\text{NO}$	4-methylpyridine N-oxide				[1003-67-4]
$\text{C}_6\text{H}_7\text{NO}_3\text{S}$	(345–392)	85.3±2.6	(298)	ME	[98/12]
	(316–341)	79.1±1.3			[95/31]
$\text{C}_6\text{H}_7\text{NS}$	sulfanilic acid (4-aminobenzene sulfonic acid)				[121-57-3]
		66.9			[38/1][60/1]
$\text{C}_6\text{H}_7\text{NS}$	4-methylthiopyridine				[22581-72-2]
$\text{C}_6\text{H}_7\text{NS}$	(347–383)	75.3±3.8	(365)	B	[74/21]
	1-methyl-4-thiopyridone				[6887-59-8]
$\text{C}_6\text{H}_7\text{N}_5$	(440–465)	188.3±9.2	(452)	B	[74/21]
	1-methyladenine				[5142-22-3]
$\text{C}_6\text{H}_7\text{N}_5$		138.2		ME	[00/17]
$\text{C}_6\text{H}_7\text{N}_5$	2-methyladenine				[1445-08-5]
		121.7		ME	[00/17]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_m/\text{kJ mol}^{-1}$	(T_m/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₆ H ₇ N ₅	3-methyladenine	117.5		ME	[5142-23-4]
		83.7±9		HSA	[00/17]
C ₆ H ₇ N ₅	8-methyladenine				[78/17]
		103.2		ME	[22387-37-7]
C ₆ H ₇ N ₅	9-methyladenine				[00/17]
	(413–458)	121.7	(428)	HSA	[700-00-5]
		92±8		HSA	[87/4][65/2]
C ₆ H ₈ ClN	aniline hydrochloride				[78/17]
	(383–471)	87.5	(398)		[142-04-1]
C ₆ H ₈ N ₂	1,2-diaminobenzene				[87/4][75/31]
		85.5±0.3	(298)	C	[95-54-5]
C ₆ H ₈ N ₂	1,3-diaminobenzene				[97/26]
		90.4±0.4	(298)	C	[108-45-2]
C ₆ H ₈ N ₂	1,4-diaminobenzene				[97/26]
		92.2±0.2	(298)	C	[106-50-3]
C ₆ H ₈ N ₂ O ₂	1,3-dimethyluracil				[97/26]
	(311–367)	115.8±3.0	(338)	TE	[874-14-6]
	(340–369)	96.9±1.2	(298)	C	[00/2]
		96.4±1.4	(298)	C	[89/17]
	(313–363)	101.7±2.1	(338)	QR	[85/5]
	(400–454)	46±4.2	(426)	HSA	[80/19]
		77.0±1.2		ME	[78/17]
	(344–370)	92	(352)	HSA	[72/32][00/2]
C ₆ H ₈ N ₂ O ₂	1-methylthymine				[65/2]
	(378–428)	124.4±1.3	(398)	QR	[4160-72-9]
C ₆ H ₈ O ₂	1,3-cyclohexanedione				[80/19]
		89.8±1.1	(298)	C	[504-02-9]
C ₆ H ₈ O ₂	1,4-cyclohexanedione				[93/23]
		75.0±1.0	(298)	C	[637-88-7]
		84.4	(289)	TE,ME	[93/23]
		84.2	(298)		[83/7]
C ₆ H ₈ O ₄	dimethyl fumarate				[83/7]
		NA			[624-49-7]
		84.5±1.7			[72/7]
C ₆ H ₈ O ₄	dimethyl maleate				[34/1]
		44.8			[624-48-6]
					[38/1][60/1]
	(317–341)	41.8±4.2			[35/1]
C ₆ H ₈ O ₄	cyclobutane-1,1-dicarboxylic acid				[34/1]
		112.2±0.7		C	[5445-51-2]
C ₆ H ₈ O ₄	cyclobutane-1,2-dicarboxylic acid				[83/26]
		120.0±0.9		C	[3396-14-3]
C ₆ H ₉ N ₃ O	1,3,5-trimethyl-4-nitrosopyrazole				[83/26]
		88.0±2.0	(298)	C	[7171-70-2]
C ₆ H ₉ N ₃ O	1,5-dimethylcytosine				[01/4]
	(390–437)	132.8±0.6		GS	[17634-60-5]
C ₆ H ₉ N ₃ O	1,N-dimethylcytosine				[98/37]
	(401–426)	122.2±0.3		GS	[6220-49-1]
C ₆ H ₉ N ₃ O ₂	L-histidine				[98/37]
	(392–492)	142±8	(442)	LE	[71-00-1]
C ₆ H ₉ N ₃ O ₂	1-methyl-4N-methoxycytosine				[77/2]
	(316–325)	107.6±0.3		ME	[99/42]
	(320–357)	106.9±0.4		GS	[99/42]
		106.4±0.8			[98/37]
C ₆ H ₉ N ₃ O ₂	1,5-dimethyl-N-hydroxycytosine				[6220-53-7]
	(357–394)	115.2±0.6		GS	[98/37]
C ₆ H ₉ N ₃ O ₃	trimethyl isocyanurate				[827-16-7]
	(330–346)	86.6±1.3	(338)	C	[88/12]
		88.2±1.3	(298)	C	[88/12]
		88.2±1.3	(298)	C	[89/17][85/5]
C ₆ H ₉ N ₃ O ₃	trimethyl cyanurate				[877-89-4]
	(331–360)	90.3±1.0	(298)	C	[89/17][85/5]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_6\text{H}_{10}\text{N}_6\text{O}_9$	N-(2,2-dinitropropyl)-2,2-dinitro-N-nitroso-1-propanamine (323–336)	110.9 ± 8		ME	[28464-26-8] [73/12][77/1]
$\text{C}_6\text{H}_{10}\text{N}_6\text{O}_{10}$	N-(2,2-dinitropropyl)-2,2-dinitro-N-nitro-1-propanamine (398–423)	99.2 ± 0.8		ME	[28464-24-6] [73/12]
$\text{C}_6\text{H}_{10}\text{O}$	1-methylcyclopentanol				[1462-03-9]
crystal III	(253–281)	73.7 ± 0.4	(267)	ME	[97/4]
crystal III		67.0 ± 0.2	(298)		[97/4]
crystal II		67.4 ± 0.2	(291)	C	[97/4]
$\text{C}_6\text{H}_{10}\text{O}$	cyclohexanone				[108-94-1]
	(243–265)	49.3	(254)		[48/5]
$\text{C}_6\text{H}_{10}\text{O}_3$	4-methyl-2,6,7-trioxabicyclo[2.2.2]octane				[766-32-5]
		67.4	(298)		[95/28]
$\text{C}_6\text{H}_{10}\text{O}_4$	1,6-hexanedioic acid (adipic acid)				[124-04-9]
		NA			[01/19]
	(295–318)	140		TPTD	[01/15]
		133.6 ± 1.3	(298)	ME	[99/10][60/4]
	(359–406)	129.3 ± 2.5	(383)	ME	[50/4][60/1]
					[70/1]
	(292–320)	U37.2	(306)	A	[47/6]
$\text{C}_6\text{H}_{10}\text{O}_4$	<i>cis</i> -1,3,5,7-tetraoxadecalin				[75096-35-4] or [54933-94-7]
		94.9	(298)		[98/19]
$\text{C}_6\text{H}_{10}\text{O}_4$	<i>trans</i> -1,3,5,7-tetraoxadecalin				[75096-35-4] or [54933-94-7]
		81.5	(298)		[98/19]
$\text{C}_6\text{H}_{10}\text{O}_5$	1,6-anhydro- β -D-glucose				[498-07-7]
	(344–386)	125.1 ± 1.0	(365)	ME	[99/1]
	(386–405)	100.3 ± 5.9	(395)	ME	[99/1]
$\text{C}_6\text{H}_{10}\text{O}_6$	(<i>dl</i>)-dimethyl tartrate				[608-69-5]
	(314–339)	112 ± 5.6	(326)	HSA	[81/8]
	(315–358)	113.8	(336)	ME	[54/5][77/1]
		U 95.0			[38/1][60/1]
		U 92.5			[37/6]
$\text{C}_6\text{H}_{10}\text{O}_6$	(<i>d</i>)-dimethyl tartrate				[5057-96-5]
	(310–320)	77.4 ± 8	(315)	HSA	[81/8]
	(308–317)	U 113	(312)		[54/5][77/1]
		88.3			[38/1][60/1]
		85.8			[37/6]
$\text{C}_6\text{H}_{10}\text{O}_6$	<i>meso</i> -dimethyl tartrate				[38/1][60/1]
		98.3			[37/6]
		95.8			[502-44-3]
$\text{C}_6\text{H}_{11}\text{NO}$	ϵ -caprolactam				[502-44-3]
	(330–340)	89.3 ± 0.8	(335)	ME	[92/12]
		86.3 ± 0.2	(338)	C	[92/12]
		87.3 ± 0.2	(298)		[92/12]
	(258–308)	77.5	(273)		[87/4]
	(294–314)	83.3 ± 0.8			[53/5][60/1]
					[70/1][60/21]
$\text{C}_6\text{H}_{11}\text{NO}$	cyclohexanone oxime				[100-64-1]
		74.0 ± 0.3	(354)	C	[92/6]
	(288–348)	79.9 ± 0.7	(317)	ME	[92/6]
$\text{C}_6\text{H}_{11}\text{NO}$	<i>cis</i> 2-hexenoic acid amide				[820-99-5]
	(323–333)	80	(328)		[87/4]
$\text{C}_6\text{H}_{11}\text{NO}$	<i>trans</i> 2-hexenoic acid amide				[820-99-5]
	(353–393)	55.8	(368)		[87/4]
$\text{C}_6\text{H}_{11}\text{NO}_2$	1-amino-1-cyclopentanecarboxylic acid				[52-52-8]
		123.4 ± 4	(455)	ME	[65/1][64/16]
$\text{C}_6\text{H}_{11}\text{N}_3\text{O}_6$	2,3,3-trinitro-2-methylpentane				[62154-78-3]
		90.8	(298)		[99/35]
$\text{C}_6\text{H}_{11}\text{N}_5\text{O}_8$	N-(2,2-dinitropropyl)-2,2-dinitro-1-propanamine				[1924-47-6]
		105.4 ± 4.2			[73/1][77/1]
$\text{C}_6\text{H}_{11}\text{NS}$	1-methyl-2-thiopiperidone				[13070-07-0]
	(363–370)	81.2 ± 2.9	(366)	B	[74/21]
C_6H_{12}	cyclohexane				[110-82-7]
	(223–280)	37.6	(265)		[87/4][76/23]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
		36.4 ± 0.7	(279.9)	B	[74/22]
		46.6	(186)	B	[63/6]
	(268–278)	37.2	(273)		[60/1]
	(228–268)	37.7	(248)	A	[47/2]
	(269–279)	36.5	(274)	A	[34/4]
	solid phase transition	6.73	(186)		[84/25]
C ₆ H ₁₂ N ₂	1,4-diazabicyclo[2.2.2]octane				[280-57-9]
	(324–351)	61.9 ± 3.3	(338)		[60/5][70/1]
	(353–369)	52.3 ± 3.3	(361)		[60/5][70/1]
	(323–373)	54.4	(348)		[63/6]
C ₆ H ₁₂ N ₂	3,3,4,4-tetramethyl- Δ^1 -1,2-diazetine				
		62.3 ± 1.0	(298)	C	[78/34]
C ₆ H ₁₂ N ₂ O ₄	2,3-dinitro-2,3-dimethylbutane				[3964-18-9]
		79.5 ± 0.8	(298)		[99/35]
C ₆ H ₁₂ N ₂ O ₆	2,5-hexanediol dinitrate				[42730-17-6]
	(293–313)	119	(303)		[87/4][57/7]
C ₆ H ₁₂ N ₄	1,3,5,7-tetraazatricyclo[3.3.1.1 ^{3,7}]-decane				[100-97-0]
	(298–453)	76.8	(313)		[87/4]
	(302–328)	78.8	(316)	TE,ME	[83/7]
		74.9 ± 2.9	(298)		[60/5][70/2]
	(281–298)	74.1 ± 0.8	(289)	TE	[60/2]
		75.3			[58/6]
C ₆ H ₁₂ O	cyclohexanol				[108-93-0]
	(272–298)	60.7	(285)		[87/4][48/5]
C ₆ H ₁₂ O ₂	cis 1,2-cyclohexanediol				[1792-81-0]
		89.0			[99/30]
crystal I		70 ± 3.0	(366)	C	[95/17]
crystal III		88.0 ± 1.9	(343)	C	[95/17]
	(289–320)	43.7	(304)	ME	[87/4][40/1]
C ₆ H ₁₂ O ₂	trans 1,2-cyclohexanediol				[1460-57-7]
		85.9 ± 1.4	(343)	C	[95/17]
	(289–320)	42.5	(304)	ME	[87/4][40/1]
C ₆ H ₁₂ O ₆	myo-inositol				[87-89-8]
		181			[99/30]
		154.7 ± 1.4	(477)	TE	[90/7]
		161	(298)		[90/7]
		178	(298)	B	[90/7]
		168			[83/23]
C ₆ H ₁₃ ClO ₂ S	1-hexanesulfonyl chloride				[14532-24-2]
	(273–303)	60.7 (liq)	(288)		[87/4][63/11]
C ₆ H ₁₃ NO	tert-butylacetamide				[762-84-5]
	(278–295)	78.3 ± 0.3	(287)	ME	[83/13]
		77.9 ± 0.4	(298)		[83/13]
C ₆ H ₁₃ NO	hexanamide				[628-02-4]
	(301–371)	85 ± 4.0	(298)	TE	[00/1]
	(293–303)	98.7 ± 1.7	(298)		[73/19][77/1]
	(338–368)	95.1 ± 4	(353)	GS	[59/3][70/1]
					[87/4]
C ₆ H ₁₃ NO ₂	(dl)-2-aminohexanoic acid				[616-06-8]
	(435–469)	114.5 ± 0.4	(450)	ME	[65/1][64/16]
					[87/4]
C ₆ H ₁₃ NO ₂	2-amino-3-methylpentanoic acid				[3107-04-8]
		120.1 ± 0.8	(455)	ME	[65/1][64/16]
C ₆ H ₁₃ NO ₂	L-(d)-2-amino-4-methylpentanoic acid (L-(d)-leucine)				[328-38-1]
	(323–423)	U 83.7 ± 4	(373)	LE	[77/2]
C ₆ H ₁₃ NO ₂	D-(l)-leucine				[61-90-5]
	(446–464)	150.6 ± 0.8	(455)	ME	[65/1][70/1]
					[64/16]
C ₆ H ₁₃ NO ₂	6-aminohexanoic acid				[60-32-2]
	(388–407)	153.3 ± 0.8	(398)	C	[83/24]
		155 ± 3	(298)	C	[83/24]
C ₆ H ₁₄	n-hexane				[110-54-3]
		50.8	(178)	B	[63/6]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₆ H ₁₄ N ₂ O ₂	L-lysine (397–497)	U 88 ± 8	(447)	LE	[56-87-1] [77/2]
C ₆ H ₁₄ N ₄ O ₂	L-arginine (441–541)	U 134 ± 8	(491)	LE	[74-79-3] [77/2]
C ₆ H ₁₄ O ₂	1,6-hexanediol	112.0 ± 0.4 83.3 ± 1.7	(298)	C	[629-11-8] [90/14] [72/12][77/1] [34008-94-1]
C ₆ H ₁₄ O ₂ S	<i>tert</i> -butyl ethyl sulfone	86.6 ± 2.5			[U/3][70/1] [69-65-8]
C ₆ H ₁₄ O ₆	D-mannitol	202	(298)	B	[90/7]
C ₆ H ₁₄ O ₆	D-sorbitol	186	(298)	B	[3615-56-3] [90/7]
C ₆ H ₁₄ O ₆	D-galactitol	205	(298)	B	[608-66-2] [90/7]
C ₆ H ₁₆ N ₂ O ₂	N,N'-bis(2-hydroxyethyl)ethylenediamine	142.7	(373)	B	[4439-20-7] [97/39]
C ₆ H ₁₆ N ₂ O ₂	diisopropyl ammonium nitrite (288–299)	39	(293.5)		[3129-93-9] [87/4][65/19]
C ₇ F ₁₄	perfluoromethylcyclohexane	51.6	(234)	B	[355-02-2] [63/6][57/5]
C ₇ F ₁₆	<i>n</i> -hexadecafluoroheptane	57.7		B	[335-57-9] [63/6][51/7]
C ₇ HF ₅ O ₂	pentafluorobenzoic acid (335–359)	91.6 ± 4.2		GS	[602-94-8] [69/5][70/1] [3354-42-5]
C ₇ H ₄ S ₃	4,5-benzo-1,2-dithiole-3-thione (350–361)	102.6 ± 0.4 107 ± 0.4	(355) (298)		[72/16] [72/16]
C ₇ H ₄ S ₃	4,5-benzo-1,3-dithiole-2-thione	118.8 ± 0.4	(298)		[934-36-1] [73/16][77/1]
C ₇ H ₅ BrO ₂	2-bromobenzoic acid	95.9 ± 0.4 110.9 ± 1.1 83.3 ± 0.4	(298) (298)	C C DSC	[88-65-3] [94/4] [87/8] [83/8]
C ₇ H ₅ BrO ₂	3-bromobenzoic acid	99.2 ± 0.2 105.0 ± 1.1	(298) (298)	C C	[585-76-2] [94/4] [87/8]
C ₇ H ₅ BrO ₂	4-bromobenzoic acid	103.1 ± 0.6 107.6 ± 1.1	(298) (298)	C C	[586-76-5] [94/4] [87/8]
C ₇ H ₅ ClO ₂	2-chlorobenzoic acid	100.9 ± 0.5 116.2 ± 0.6 79.5 ± 3.3	(298)	C DSC	[118-91-2] [95/24] [83/8] [38/1][60/1] [70/1]
C ₇ H ₅ ClO ₂	3-chlorobenzoic acid	101.4 ± 0.4 99.6 105.8 80.8 ± 3.3	(298) (413) (298)	C C C	[535-80-8] [95/24] [75/20] [75/20] [38/1][60/1] [70/1]
C ₇ H ₅ ClO ₂	4-chlorobenzoic acid	102.5 ± 0.4 101.9 107.9 87.9 ± 3.3	(298) (413) (298)	C C C	[74-11-3] [95/24] [75/20] [75/20] [38/1][60/1] [70/1]
C ₇ H ₅ FO ₂	2-fluorobenzoic acid (309–323)	93.9 ± 0.5 94.4 ± 0.8	(316) (298)	ME	[445-29-4] [00/5] [00/5]
C ₇ H ₅ FO ₂	3-fluorobenzoic acid (303–317)	93.3 ± 0.5 93.6 ± 0.6	(310) (298)	ME	[455-38-9] [00/5] [00/5]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_7\text{H}_5\text{FO}_2$	4-fluorobenzoic acid (358–382)	91.2 ± 1.3	(370)	GS	[456-22-4] [69/5][70/1]
					[87/4]
$\text{C}_7\text{H}_5\text{IO}_2$	2-iodobenzoic acid (345–359)	93.1 ± 3.8	(298)	ME	[69/5][00/5] [88-67-5]
		111.4 ± 0.8	(352)		[00/5]
		112.8 ± 2.0	(298)	C	[00/5]
		92.6 ± 0.2	(298)		[94/30][95/24]
$\text{C}_7\text{H}_5\text{IO}_2$	3-iodobenzoic acid (347–363)	103.0 ± 0.4	(298)	DSC	[83/8]
					[618-51-9]
		109.6 ± 0.5	(355)	ME	[00/5]
		111.1 ± 1.9	(298)		[00/5]
$\text{C}_7\text{H}_5\text{IO}_2$	4-iodobenzoic acid (363–379)	96.4 ± 0.3	(298)	C	[94/30][95/24]
					[619-58-9]
		111.0 ± 0.4	(372)	ME	[00/5]
		112.9 ± 2.5	(298)		[00/5]
$\text{C}_7\text{H}_5\text{NO}$	benzoxazole	99.3 ± 0.4	(298)	C	[94/30][95/24]
					[273-53-0]
$\text{C}_7\text{H}_5\text{NS}$	benzothiazole	69.5 ± 0.4	(298)	C	[98/20]
					[95-16-9]
$\text{C}_7\text{H}_5\text{NO}_4$	3-(5-nitro-2-furyl)-2-propenal	72.9 ± 0.6	(298)	B	[98/20]
					[1874-22-2]
$\text{C}_7\text{H}_5\text{NO}_4$	2-nitrobenzoic acid (346–356)	97.9 ± 2.1			[80/28][86/5]
					[552-16-9]
$\text{C}_7\text{H}_5\text{NO}_4$	3-nitrobenzoic acid (347–361)	115.8 ± 0.5	(356)	ME	[99/31]
		118.7 ± 0.5	(298)	ME	[99/31]
					[121-92-6]
		107.2 ± 0.4	(354)	ME	[99/31]
$\text{C}_7\text{H}_5\text{NO}_4$	4-nitrobenzoic acid (367–381)	110.0 ± 0.4	(298)	ME	[99/31]
					[62-23-7]
		115.4 ± 0.6	(374)	ME	[99/31]
		119.7 ± 0.6	(298)	ME	[99/31]
$\text{C}_7\text{H}_5\text{N}_3\text{O}_6$	2,4,6-trinitrotoluene (293–353) (301–349) (297–330) (327–349) (323–353) (340–353)		(308)		[118-96-7]
		112.4	(308)		[87/4]
		113.2 ± 1.5	(298)	ME	[78/15]
		99.2 ± 2		GS	[76/3][77/5]
		104.6 ± 1.7	(298)	ME	[71/4]
		103.3 ± 2.5	(338)		[70/3]
		U122-132		TGA	[70/26][78/15]
		118.4 ± 4.2		ME	[50/1][60/1]
					[70/1]
		102.2	(346)	ME	[50/2]
$\text{C}_7\text{H}_5\text{N}_3\text{O}_7$	2,4,6-trinitroanisole (334–342)	132.4 ± 2.1	(338)	ME	[606-35-9]
					[87/4][70/1]
$\text{C}_7\text{H}_5\text{N}_3\text{O}_7$	3-hydroxy-2,4,6-trinitrotoluene (310–365) (325–350)				[50/2]
					[602-99-3]
		111.2 ± 2.1	(298)		[78/15]
		103.3	(337)		[70/3]
$\text{C}_7\text{H}_5\text{N}_5\text{O}_8$	2,4,6-N-tetranitro-N-methylaniline (335–416)	104.6	(298)		[70/3]
					[479-45-8]
$\text{C}_7\text{H}_6\text{N}_2$	benzimidazole (340–359)	133.8 ± 1.6	(298)	ME	[78/15]
					[51-17-2]
		90.2 ± 0.6	(363)	C	[98/6]
		94.3 ± 0.6	(298)		[98/6]
		101.8 ± 0.4	(350)	ME	[87/9]
		102.2 ± 0.4	(298)	ME	[87/9]
$\text{C}_7\text{H}_6\text{N}_2$	indazole (308–317)	98.9 ± 0.4	(298)		[86/15]
					[271-44-3]
		90.9 ± 0.2	(318)	ME	[87/9]
		91.1 ± 0.2	(298)		[87/9][86/15]
		87.7 ± 0.9			[85/6]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_2$	5-methoxybenzofurazan	97.1			[61/3]
					[4413-48-3]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_7\text{H}_6\text{N}_2\text{O}_2$	5-methylbenzofurazan-1-oxide	89.2 ± 0.7	(298)	C	[96/6]
					[19164-41-1]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_3$	5-methoxybenzofurazan-1-oxide	92.2 ± 1.2	(298)	C	[96/6]
					[7791-49-3]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_3$	2-nitrobenzaloxime	96.0 ± 1.6	(298)	C	[96/6]
<i>anti</i>					[6635-41-2]
<i>syn</i>		U 26.4 ± 1.7		MS	[83/9]
		U 40.2 ± 1.7		MS	[83/9]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_3$	3-nitrobenzaloxime				[3431-62-7]
<i>anti</i>		U 41.0 ± 1.7		MS	[83/9]
<i>syn</i>		U 42.7 ± 1.7		MS	[83/9]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_3$	4-nitrobenzaloxime				[1129-37-9]
<i>anti</i>		U 56.4 ± 1.7		MS	[83/9]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_4$	1,1-dinitrophenylmethane				[25321-14-6]
	(312–323)	76.1 ± 0.8		ME	[72/6]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_4$	2,4-dinitrotoluene				[121-14-2]
	(332–342)	98.3 ± 2.5	(337)	ME	[77/1][70/3]
		99.6 ± 2.5	(298)		[77/1][70/3]
	(277–344)	95.8 ± 1.25	(310)	GS	[76/3][77/5]
		99.6 ± 1.3		ME	[70/1][71/4]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_4$	2,6-dinitrotoluene				[606-20-2]
	(277–323)	$98.3 \pm .8$	(300)	GS	[76/3][77/5]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_4$	(dinitromethyl)benzene				[611-38-1]
	(312–323)	76.1	(317.5)		[87/4]
$\text{C}_7\text{H}_6\text{N}_2\text{O}_5$	3,5-dinitro- <i>o</i> -cresol				[534-52-1]
	(290–324)	103.3		TE	[47/1][60/1]
$\text{C}_7\text{H}_6\text{O}_2$	benzoic acid				[65-85-0]
		88.3 ± 0.5	(298)	C	[01/5]
	(323–394)	90.5 ± 0.3		GS	[99/42]
		89 ± 6		TGA	[99/36]
		87.5 ± 0.4			[98/38]
	(313–343)	86.7		TGA	[97/36]
	(307–314)	88.7 ± 0.9	(311)	ME	[90/3]
		89.3 ± 0.9	(298)		[90/3]
		87.5 ± 0.3	(335)	C	[88/4]
		89.2 ± 1.0	(298)		[88/4]
		95.1 ± 1.8	(294)		[85/9]
	(293–319)	90.8 ± 0.6	(306)	QR	[85/10]
		89.5 ± 0.4		DSC	[83/8]
	(320–370)	89.1 ± 0.2		C	[82/3]
	(316–391)	89.5 ± 0.5	(353)	DM	[82/12]
	(293–313)	90.6 ± 2		ME	[82/1]
		93.45 ± 1		GS	[81/3]
	(328–398)	U 133.5 ± 4.5		C	[80/20]
	(344–395)	85 ± 2	(369)	SG	[80/30]
	(281–323)	88.3 ± 2.9		LE	[78/17]
		88.5 ± 0.8		C	[76/7]
	(294–331)	92.5 ± 4		ME	[75/6]
	(293–318)	88.5 ± 1.6		TE	[75/5]
	(273–318)	92.9 ± 0.2	(296)	ME	[74/5]
	(293–311)	88.1 ± 0.2		TCM	[73/1]
	(338–383)	89.0 ± 0.4		ME	[73/2]
	(338–383)	89.3 ± 0.4		C	[73/2]
	(290–315)	86.6 ± 1.3		ME,C	[72/9]
	(293–308)	$90. \pm 0.3$		ME	[72/3]
		89.5 ± 0.2	(298)	C	[72/1][71/6]
	(299–329)	89.1	(314)		[71/17]
	(290–315)	86.6 ± 1.7	(303)	ME	[70/19][99/42]
	(324–392)	90.4 ± 0.8	(367)	HSA	[70/9]
		89.7 ± 0.6	(298)	C	[69/2]
	(348–378)	88.9 ± 0.5	(363)	GS	[68/9]
	(291–307)	90.9	(299)	ME	[65/6]
	(243–387)	91.5 ± 0.5	(298)	GS	[54/1][70/1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_m/\text{kJ mol}^{-1}$	(T_m/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
					[60/1]
		84.2±0.8	(318)	TE	[38/1]
	(333–389)	85.8	(383)	T	[34/2]
	(377–394)	84.5±0.5	(364)	I	[27/3]
C ₇ H ₆ O ₂	4-hydroxybenzaldehyde				[123-08-0]
	(303–336)	98.2±1.3	(298)		[87/4][71/8]
	(312–336)	91.2	(324)		[60/14]
C ₇ H ₆ O ₂	tropolone				[533-75-5]
	(273–333)	84.1±0.4		ME	[71/3]
		83.7±0.8	(298)		[51/4][70/1]
C ₇ H ₆ O ₂	3-(2-furyl)-2-propenal				[623-30-3]
		76.1±2.1			[80/28][86/5]
C ₇ H ₆ O ₃	2-hydroxybenzoic acid				[69-72-7]
		95.1±0.5	(333)	C	[93/22]
		96.3±0.5	(298)		[93/22]
	(307–324)	95.7±0.8	(315)	ME	[80/2][81/11]
	(312–332)	94.9±0.4	(322)	TE	[77/4]
	(312–332)	93.22±0.8	(322)	ME	[77/4]
	(298–328)	99.2±2	(313)	ME	[74/5]
	(368–408)	94.8±0.4			[73/2]
	(368–408)	95.1±0.4		GS	[54/1][70/1]
					[60/1]
C ₇ H ₆ O ₃	3-hydroxybenzoic acid				[99-06-9]
		123.5±0.74	(363)	C	[93/22]
		125.0±0.74	(298)		[93/22]
C ₇ H ₆ O ₃	4-hydroxybenzoic acid				[99-96-7]
		112.4±0.7	(363)	C	[93/22]
		114.1±0.7	(298)		[93/22]
	(398–433)	116.3		GS	[54/1][60/1]
C ₇ H ₆ O ₄	2,3-dihydroxybenzoic acid				[303-38-8]
		116±4		TGA	[99/36]
C ₇ H ₆ O ₄	2,4-dihydroxybenzoic acid				[89-86-1]
		126±6		TGA	[99/36]
C ₇ H ₆ O ₄	2,5-dihydroxybenzoic acid				[490-79-9]
		109±3		TGA	[99/36]
C ₇ H ₆ O ₄	2,6-dihydroxybenzoic acid				[303-07-1]
		111±7		TGA	[99/36]
C ₇ H ₆ O ₄	3,4-dihydroxybenzoic acid				[99-50-3]
		153±9		TGA	[99/36]
C ₇ H ₆ O ₄	3,5-dihydroxybenzoic acid				[99-10-5]
		135±6		TGA	[99/36]
C ₇ H ₆ O ₅	3,4,5-trihydroxybenzoic acid				[149-91-7]
	(391–421)	75.1	(406)		[34/2]
C ₇ H ₇ NO	2-aminotropone				[6264-93-3]
	(273–333)	71.13±0.4		ME	[71/3]
C ₇ H ₇ NO	benzamide				[55-21-0]
	(325–342)	96.9	(333.5)		[87/4][60/21]
	(323–349)	101.7±1	(298)	C	[82/2]
C ₇ H ₇ NO	formanilide				[103-70-8]
	(298–318)	77.8	(308)		[87/4][60/21]
C ₇ H ₇ NO ₂	2-aminobenzoic acid (I)				[118-92-3]
		111.6±1.7	(298)		[72/4]
C ₇ H ₇ NO ₂	2-aminobenzoic acid (II)				
	(331–349)	100±1	(338)	TE,ME	[79/1]
		99.6±0.5	(378)	C	[74/1]
		104.9±1	(298)	C	[74/1]
C ₇ H ₇ NO ₂	3-aminobenzoic acid				[99-05-8]
	(366–385)	122±1	(375)	TE,ME	[79/1]
	(367–389)	122.3±3		C	[74/1]
		128±3.2	(298)	C	[74/1][77/13]
C ₇ H ₇ NO ₂	4-aminobenzoic acid				[150-13-0]
	(364–384)	112.3±1	(373)	TE,ME	[79/1]
	(367–389)	114±3.5		C	[74/1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
		116 ± 3.7	(298)	C	[74/1][77/13]
		U 142			[38/1][60/1]
$\text{C}_7\text{H}_7\text{NO}_2$	2-hydroxybenzaloxime				[94-67-7]
(melting point 330 K)	(423–513)	96.7 ± 9.4	(468)	DSC	[84/12]
		105.2 ± 10	(298)		[84/12]
<i>anti</i>		U 51 ± 1.7		MS	[83/9]
<i>syn</i>		U 65.3 ± 1.7		MS	[83/9]
$\text{C}_7\text{H}_7\text{NO}_2$	3-hydroxybenzaloxime				[22241-18-5]
<i>anti</i>		U 52.7 ± 1.7		MS	[83/9]
<i>syn</i>		U 57.3 ± 1.7		MS	[83/9]
$\text{C}_7\text{H}_7\text{NO}_2$	4-hydroxybenzaloxime				[699-06-9]
<i>anti</i>		U 54.4 ± 1.7		MS	[83/9]
$\text{C}_7\text{H}_7\text{NO}_2$	4-nitrotoluene				[99-99-0]
		79.1 ± 2.5	(298)	ME	[71/4]
	(298–310)	79.1	(298)		[70/3]
$\text{C}_7\text{H}_7\text{NO}_3$	2,4-dihydroxybenzaloxime				[5399-68-8]
<i>anti</i>		U 76.2 ± 1.7		MS	[83/9]
<i>syn</i>		U 93.7 ± 1.7		MS	[83/9]
$\text{C}_7\text{H}_7\text{NS}$	thiobenzamide				[2227-79-4]
		103.4 ± 2.2	(298)	C	[89/11]
		97.2 ± 0.6	(298)	C	[82/17]
C_7H_8	toluene				[108-88-3]
		43.1	(298)	B	[70/3]
$\text{C}_7\text{H}_8\text{N}_2\text{O}$	monophenylurea				[64-10-8]
	(392–412)	136 ± 6	(406)	TE	[87/2]
$\text{C}_7\text{H}_8\text{N}_2\text{O}$	2-aminobenzaloxime				[3398-07-0]
<i>anti</i>		U 33.9 ± 1.7		MS	[83/9]
<i>syn</i>		U 63.6 ± 1.7		MS	[83/9]
$\text{C}_7\text{H}_8\text{N}_2\text{O}$	(2-pyridyl)acetamide				[5231-96-9]
		103.8	(298)	B,E	[79/12]
$\text{C}_7\text{H}_8\text{N}_2$	1-amino-7-imino-1,3,5-cycloheptatriene				[33496-46-7]
	(273–333)	49.4 ± 0.4		ME	[71/3]
$\text{C}_7\text{H}_8\text{N}_4\text{O}$	9-ethylhypoxanthine				[31010-51-2]
		108.8 ± 13		HSA	[78/17]
		U 83.7		HSA	[65/2]
$\text{C}_7\text{H}_8\text{N}_4\text{O}$	1,9-dimethylhypoxanthine				[20535-82-4]
		75.3 ± 13		HSA	[78/17]
$\text{C}_7\text{H}_8\text{N}_4\text{O}_2$	1,3-dimethylxanthine (theophylline)				[58-55-9]
form I	(413–453)	132.0 ± 0.3	(433)	T	[99/40]
form I		142	(298)		[99/40]
form II	(413–453)	134.2 ± 0.3	(433)	T	[99/40]
form II		144	(298)		[99/40]
		126	(421)	ME,TE	[83/17]
		135	(298)		[83/17][99/40]
$\text{C}_7\text{H}_8\text{O}$	2-methylphenol				[95-48-7]
	(273–303)	74.8	(288)		[87/4]
	(273–303)	76.0 ± 0.8	(288)		[60/3][70/1]
$\text{C}_7\text{H}_8\text{O}$	3-methylphenol				[108-39-4]
	(273–285)	56.1	(279)		[87/4]
	(284–313)	61.7 ± 1.0 (liq)		GS	[60/3]
$\text{C}_7\text{H}_8\text{O}$	4-methylphenol				[106-44-5]
	(273–307)	73.9 ± 1.5	(290)		[60/3][70/1]
$\text{C}_7\text{H}_8\text{O}_2$	4-methoxyphenol				[150-76-5]
	(278–300)	88.7	(289)		[87/4][60/14]
$\text{C}_7\text{H}_8\text{O}_2$	3-methyl-1,2-dihydroxybenzene				[488-17-5]
		93.2 ± 1.0	(298)	C	[84/20]
$\text{C}_7\text{H}_8\text{O}_2$	4-methyl-1,2-dihydroxybenzene				[452-86-8]
		94.9 ± 1.0	(298)	C	[84/20]
$\text{C}_7\text{H}_8\text{O}_2$	2-methyl-1,4-dihydroxybenzene				[95-71-6]
	(333–368)	97.2 ± 1.4	(351)	GS	[99/28]
		100.4 ± 1.4	(298)		[99/28]
$\text{C}_7\text{H}_8\text{O}_2\text{S}$	4-methoxy-6-methyl-2-thiopyrone				[52911-98-5]
	(402–415)	130.5 ± 5.9	(408)	B	[74/21]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₇ H ₈ O ₂ S	6-methyl-2-methylthio-4-pyrone (388–433)	87.4 ± 3.8	(410)	B	[52911-99-6] [74/21]
C ₇ H ₈ O ₂ S	methyl phenyl sulfone	92 ± 2.9			[3112-85-4] [U/3][70/1]
C ₇ H ₈ O ₃	3-methoxycatechol	91.7 ± 0.8	(298)		[934-00-9] [86/3]
C ₇ H ₈ S ₃	4,5-tetramethylene-1,3-dithiole-2-thione (340–352)	98.3 102.1 ± 2.9	(346)		[698-42-0] [67/5][70/1] [67/5][70/1]
C ₇ H ₈ S ₃	4,5-tetramethylene-1,2-dithiole-3-thione (335–350)	101.6 105.3	(342) (298)		[14085-34-8] [72/16] [72/16]
C ₇ H ₉ F ₃ N ₂ O ₄	N-[N-(trifluoroacetyl)glycyl]glycine methyl ester (323–419)	127.9	(338)		[433-33-0] [87/4][60/20]
C ₇ H ₉ N	<i>p</i> -toluidine	78.8 ± 0.5	(298)		[106-49-0] [90/24]
C ₇ H ₉ N ₅	8,9-dimethyladenine (369–374)	105.8 ± 0.8	(361)	ME	[87578-82-3] [87/7]
C ₇ H ₉ N ₅	2,9-dimethyladenine (359–364)	123.5	(371)		[76470-20-7] [92/5]
C ₇ H ₁₀	norbornene	37.8 ± 0.14 37.7 ± 0.9 38.7 ± 0.5 33.6 ± 0.08	(298) (298) (298)	C BG C	[498-66-8] [82/4] [78/5] [76/4]
C ₇ H ₁₀	solid phase transition nortricyclene	4.37	(130)		[73/27] [92/17] [279-19-6]
C ₇ H ₁₀ N ₂	α - <i>tert</i> -butylmalononitrile (293–323)	38.7 ± 0.7 39.2 ± 1.1	(298) (298)	BG C	[78/5] [76/4] [4210-60-0] [90/18]
C ₇ H ₁₀ N ₂ O ₂	1,3-dimethylthymine (313–363)	59.8 ± 0.7	(298)		[4401-71-2] [80/19]
C ₇ H ₁₀ N ₂ O ₂	1,3,5-trimethyluracil (321–331)	109.2 ± 2.1	(338)	QR	[4401-71-2] [96/5]
C ₇ H ₁₀ N ₂ O ₂	1,3,6-trimethyluracil (300–340)	103.5 ± 1.5 106.7 ± 2.5	(326) (320)	ME QR	[13509-52-9] [80/19]
C ₇ H ₁₀ O	bicyclo[2.2.1]heptan-2-one (300–340)	49.0 ± 1.7	(298)	BG	[497-38-1] [78/1]
C ₇ H ₁₀ O	bicyclo[2.2.1]heptan-7-one (300–340)	47.3 ± 2.2	(298)	BG	[10218-02-7] [78/1]
C ₇ H ₁₀ O ₂	2-oxabicyclo[2.2.2]octan-3-one	69.6 ± 2.1			[4350-84-9] [80/17]
C ₇ H ₁₀ O ₃	2,4,10-trioxaadamantane	74.4 ± 0.4	(298)	C	[281-32-3] [74/16]
C ₇ H ₁₀ O ₃	tetramethylsuccinic anhydride	74.1 ± 4.2			[281-32-3] [54/3][70/1]
C ₇ H ₁₀ S ₃	4,5-tetramethylene-1,3-dithiolan-2-thione (353–369)	99.0 103.9 ± 2.9	(360) (298)		[2164-87-6] [67/5][70/1] [67/5][70/1]
C ₇ H ₁₁ N ₃ O	1,N,N-trimethylcytosine	110.9 ± 1.7			[2228-27-5] [98/37]
C ₇ H ₁₁ N ₃ O	1,5,N-trimethylcytosine (396–431)	108.0 ± 2.0		GS	[25307-94-2] [98/37]
C ₇ H ₁₁ N ₃ O ₂	1,5-dimethyl-N-methoxycytosine (327–365)	95.6 ± 0.7		GS	[98/37]
C ₇ H ₁₁ N ₅ O ₁₀	1,1,1,4,4-pentanitro-2,2-dimethylpentane	103.8	(298)		[242800-94-8] [99/35]
C ₇ H ₁₂	bicyclo[2.2.1]heptane	40.0 ± 0.1 40.3 ± 0.32 40.4 ± 0.8 39.33 ± 0.13	(298) (298)	C C	[279-23-2] [87/1] [82/4] [78/5] [73/27]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_7\text{H}_{12}\text{ClN}_5$	(284–326)	40.1 ± 0.8		BG	[71/1][77/1]
	2-chloro-4,6-bis(ethylamino)-s-triazine (Simazin)	40.0 ± 0.8	(305)	TSGC	[75/2]
	(323–403)	130.8	(338)	GS-GC	[122-34-9] [87/4][64/14]
$\text{C}_7\text{H}_{12}\text{O}_3$	1,4-dimethyl-2,6,7-trioxabicyclo[2.2.2]octane	74.9	(298)		[27761-61-1] [95/28]
$\text{C}_7\text{H}_{12}\text{O}_4$	1,7-heptanedioic acid (pimelic acid)				[111-16-0]
	(288–308)	178		TPTD	[01/15]
	(358–371)	136.6 ± 1.0	(365)	ME	[99/10]
$\text{C}_7\text{H}_{12}\text{O}_4$	butylmalonic acid	139.9 ± 1.0	(298)		[99/10]
	(348–362)	122.5 ± 1.4	(355)	ME	[534-59-8] [00/22]
		124.6 ± 2.3	(298)	ME	[00/22]
$\text{C}_7\text{H}_{13}\text{N}$	1-azabicyclooctane				[100-76-5]
		50.8 ± 0.4			[71/10][77/1]
	(273–362)	50.8 ± 0.2	(298)		[48/3][70/1] [60/1]
$\text{C}_7\text{H}_{13}\text{NO}$	<i>trans</i> -6-heptenoic acid amide				
	(362–393)	97.2	(377)		[87/4]
C_7H_{14}	cycloheptane				[291-64-5]
$\text{C}_7\text{H}_{14}\text{N}_2$	3,3,5,5-tetramethyl-1-pyrazoline	53.5	(134)		[63/6]
					[2721-31-5]
$\text{C}_7\text{H}_{14}\text{N}_2\text{O}_2$	N-acetyl L-valinamide	61.6 ± 0.2	(298)		[76/12]
		129.8 ± 1.9	(376)	C	[37933-88-3] [99/12]
		133.1 ± 2.2	(298)		[99/12]
$\text{C}_7\text{H}_{14}\text{N}_2\text{O}_2\text{S}$	(391–425)	126 ± 2.0	(418)		[90/13]
	2-methyl-2(methylthio)propanal, O-[(methylamino)carbonyl]oxime				[116-06-3]
	(298–323)	80	(310)	ME	[87/4][76/11]
$\text{C}_7\text{H}_{14}\text{O}$	1-methylcyclohexanol				[590-67-0]
$\text{C}_7\text{H}_{15}\text{NO}$		75.9 ± 0.4	(291)	C	[98/34]
	heptanamide (enanthamide)				[628-62-6]
	(345–365)	99.6			[59/3][60/1] [87/4]
$\text{C}_7\text{H}_{15}\text{NO}_2$	hexyl carbamate				[2114-20-7]
	(291–314)	96.2 ± 0.8		GS	[59/4]
C_7H_{16}	<i>n</i> -heptane				[142-82-5]
$\text{C}_7\text{H}_{16}\text{N}_2\text{S}$		57.9	(183)	B	[63/6]
	1,3-di- <i>n</i> -propylthiourea				[26536-60-7]
		134.9 ± 3	(298)	B,HA	[00/23]
$\text{C}_8\text{Cl}_4\text{N}_2$		132.5 ± 3.0	(298)	C	[94/17]
	2,4,5,6-tetrachloro-1,3-benzenedicarbonitrile				[1897-45-6]
	(363–418)	109.1	(378)	ME, GS	[87/4][80/37]
$\text{C}_8\text{H}_2\text{Cl}_4\text{N}_2$	2,3,6,7-tetrachloroquinoxaline				[25983-14-6]
	(347–361)	106.2 ± 0.3	(354)	ME	[00/26]
		108.2 ± 1.9	(298)	ME	[00/26]
$\text{C}_8\text{H}_4\text{Cl}_2\text{N}_2$	2,3-dichloroquinoxaline				[2213-63-0]
	(313–329)	92.4 ± 0.4	(321)	ME	[00/26]
		93.1 ± 0.9	(298)	ME	[00/26]
$\text{C}_8\text{H}_4\text{N}_2$	1,2-dicyanobenzene				[91-15-6]
$\text{C}_8\text{H}_4\text{N}_2$		86.9 ± 1.5	(298)	GS	[80/10]
	1,3-dicyanobenzene				[626-17-5]
$\text{C}_8\text{H}_4\text{N}_2$		90.1 ± 1.5	(298)	GS	[80/10]
	1,4-dicyanobenzene				[623-26-7]
$\text{C}_8\text{H}_4\text{N}_2\text{O}_2$		89.7 ± 1.8	(298)	ME	[92/24]
		88.8 ± 1.5	(298)	GS	[80/10]
	1,4-dicyanobenzene di-N-oxide				[3729-34-8]
$\text{C}_8\text{H}_4\text{O}_2$		73.0 ± 2.0	(298)	ME	[92/24]
	benzocyclobutenedione				[6383-11-5]
	(304–367)	U 89.5	(336)		[89/9]
$\text{C}_8\text{H}_4\text{O}_3$	phthalic anhydride				[85-44-9]
	(313–383)	87.9	(348)		[87/4][72/25]
	(333–403)	84.4 ± 1.2	(388)	GS	[79/6]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
		81 ± 1		C	[71/6]
	(303–333)	88.4 ± 1.2	(318)		[46/2][70/1]
					[60/1]
C ₈ H ₅ F ₃ OS ₂	1,1,1-trifluoro-4-(2-thienyl)-4-mercapto-3-buten-2-one	95.1 ± 3.7	(298)	C	[4552-64-1]
					[97/19]
C ₈ H ₅ F ₃ O ₂ S	1,1,1-trifluoro-4-(2-thienyl)-4-hydroxy-3-buten-2-one	86.2 ± 0.6	(298)	C	[15788-02-0]
		86.2 ± 0.6	(298)	ME	[97/19]
					[92/28]
C ₈ H ₅ F ₃ O ₃	4,4,4-trifluoro-1-(2-furanyl)-butane-1,3-dione	70 ± 10	(298)		[326-90-9]
					[97/33]
C ₈ H ₅ NO	α -oxo-benzeneacetonitrile				[613-90-1]
	(292–304)	78.7 ± 4.2	(298)		[69/9][77/1]
					[87/4]
C ₈ H ₅ NO ₂	phthalimide				[85-41-6]
	(378–418)	82.8	(393)	RG	[87/4][56/13]
C ₈ H ₅ N ₃	pyridinium dicyanomethylide				[27032-01-5]
	(403–433)	125.4	(418)		[87/4]
	(403–433)	125.5 ± 1.3		ME	[67/3][70/1]
C ₈ H ₆ ClNO ₃	2-nitrobenzeneacetyl chloride				[22751-23-1]
	(296–327)	103.6	(311)	TE	[87/4][47/1]
					[60/1]
C ₈ H ₆ ClNO ₃	3-nitrobenzeneacetyl chloride				[99-47-8]
	(299–343)	109.1	(314)	TE	[87/4][47/1]
					[60/1]
C ₈ H ₆ N ₂	phthalazine				[253-52-1]
		82.3 ± 2.3	(298)	C	[95/25]
		81.1 ± 0.4	(298)	C	[98/7][93/19]
		96.7		ME	[72/10]
C ₈ H ₆ N ₂	quinoxaline				[91-19-0]
		66.6 ± 2.0	(298)	C	[95/25]
		69.4 ± 0.6	(298)	C	[93/19]
C ₈ H ₆ N ₂	quinazoline				[253-82-7]
		77.6 ± 0.5	(298)	C	[98/7][93/19]
		76.6 ± 1.4	(298)	C	[95/25]
C ₈ H ₆ N ₂ O	2-hydroxyquinoxaline				[1196-57-2]
	(383–399)	116.1 ± 0.6	(391)	ME	[00/26]
		118.5 ± 3.1	(298)	ME	[00/26]
		125.8 ± 4.0	(298)	C	[00/15]
C ₈ H ₆ N ₂ O ₂	2,3-dihydroxyquinoxaline				[15804-19-0]
		156.3 ± 5.5	(298)	C	[00/15]
C ₈ H ₆ N ₂ O ₂	quinoxaline-1,4-dioxide				[2433-66-7]
		112.0 ± 1.9	(298)	C	[97/25]
C ₈ H ₆ N ₂ O ₂	3-aminophthalimide				[2518-24-3]
	(386–459)	108.3	(401)	RG	[87/4][56/13]
C ₈ H ₆ N ₂ O ₂	4-aminophthalimide				[3676-85-5]
	(444–498)	135.3	(459)		[87/4]
C ₈ H ₆ N ₄	monobenzo-1,3 α ,4,6 α -tetraazapentalene				
	(323–373)	74.9 ± 2.9	(348)		[67/6]
C ₈ H ₆ N ₄	monobenzo-1,3 α ,6,6 α -tetraazapentalene				
	(323–383)	63.6 ± 2.9	(350)		[67/6]
C ₈ H ₆ O ₃	piperonal				[120-57-0]
	(293–353)	90.8	(323)		[53/7][60/1]
					[87/4]
C ₈ H ₆ O ₄	phthalic acid				[88-99-3]
		129.8 ± 0.6	(298)	C	[99/16]
C ₈ H ₆ O ₄	isophthalic acid				[121-91-5]
		142.0 ± 0.7	(298)	C	[99/16]
	(493–563)	114.2	(508)		[87/4][62/2]
	(493–563)	106.7 ± 2.2	(523)	GS	[62/2][70/1]
C ₈ H ₆ O ₄	terephthalic acid				[100-21-0]
		146.6 ± 0.5	(298)	C	[99/16]
	(523–633)	139.2	(538)		[87/4][62/2]
	(523–633)	131.	(573)	GS	[62/2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)		CAS registry number
Polymorph	Temperature range (K)			Method	Reference
$\text{C}_8\text{H}_6\text{S}$	(392–425)	98.24 ± 2.5	(408)		[34/2][70/1]
	2,3-benzothiophene				[95-15-8]
	(273–403)	61.1	(288)	GS	[81/14]
$\text{C}_8\text{H}_7\text{ClO}$		65.7 ± 0.2	(298)	C	[79/2]
	2-chloroacetophenone				[532-27-4]
	(278–323)	90.7	(293)	TE	[87/4][47/1]
$\text{C}_8\text{H}_7\text{N}$					[60/1]
	indole				[120-72-9]
	(291–319)	75	(305)		[87/4]
	(275–303)	77.8 ± 1.6	(289)	ME	[74/5]
	(283–301)	70.0	(292)		[55/10]
$\text{C}_8\text{H}_7\text{NO}_3$	(283–328)	74.9	(305)		[54/4][60/1]
	3-nitroacetophenone				[121-89-1]
$\text{C}_8\text{H}_7\text{NO}_4$	(293–343)	110	(308)		[87/4]
	2-methyl-3-nitrobenzoic acid				[1975-50-4]
	(357–371)	117.3 ± 0.6	(364)	ME	[01/7]
$\text{C}_8\text{H}_7\text{NO}_4$		119.5 ± 2.3	(298)	ME	[01/7]
	2-methyl-6-nitrobenzoic acid				[13506-76-8]
	(355–369)	117.8 ± 0.6	(362)	ME	[01/7]
$\text{C}_8\text{H}_7\text{NO}_4$		120.0 ± 2.2	(298)	ME	[01/7]
	3-methyl-2-nitrobenzoic acid				[5437-38-7]
	(371–385)	121.7 ± 0.5	(378)	ME	[01/7]
$\text{C}_8\text{H}_7\text{NO}_4$		124.4 ± 2.7	(298)	ME	[01/7]
	3-methyl-4-nitrobenzoic acid				[3113-71-1]
	(363–379)	116.8 ± 0.6	(371)	ME	[01/7]
$\text{C}_8\text{H}_7\text{NO}_4$		119.3 ± 2.5	(298)	ME	[01/7]
	4-methyl-3-nitrobenzoic acid				[96-98-0]
	(363–377)	116.1 ± 0.6	(370)	ME	[01/7]
$\text{C}_8\text{H}_7\text{NO}_4$		118.6 ± 2.5	(298)	ME	[01/7]
	5-methyl-2-nitrobenzoic acid				[3113-72-2]
	(355–371)	116.5 ± 0.5	(363)	ME	[01/7]
$\text{C}_8\text{H}_7\text{NO}_5$		118.7 ± 2.2	(298)	ME	[01/7]
	3-methoxy-2-nitrobenzoic acid				[4920-80-3]
	(398–410)	136.6 ± 1.3	(404)	ME	[99/31]
$\text{C}_8\text{H}_7\text{NO}_5$		141.9 ± 1.3	(298)	ME	[99/31]
	4-methoxy-3-nitrobenzoic acid				[89-41-8]
	(387–401)	126.5 ± 0.8	(394)	ME	[99/31]
$\text{C}_8\text{H}_7\text{NO}_5$		131.2 ± 0.8	(298)	ME	[99/31]
	3-methoxy-4-nitrobenzoic acid				[5081-36-7]
	(388–402)	126.1 ± 1.1	(395)	ME	[99/31]
$\text{C}_8\text{H}_7\text{N}_3\text{O}_2$		131.0 ± 1.1	(298)	ME	[99/31]
	3,6-diaminophthalimide				[1660-15-7]
	(461–508)	98.5	(476)	RG	[87/4][56/13]
$\text{C}_8\text{H}_7\text{N}_3\text{O}_6$					[38677-56-4]
	2,2,2-trinitro-1-phenylethane				[72/6][77/1]
	(293–308)	84.1 ± 0.4	(301)	ME	[87/4]
$\text{C}_8\text{H}_7\text{N}_3\text{O}_6$					[632-92-8]
	3-methyl-2,4,6-trinitrotoluene				[87/4][78/15]
	(319–411)	129.8 ± 1.1	(365)	ME	[4732-14-3]
$\text{C}_8\text{H}_7\text{N}_3\text{O}_7$					[72/6][77/1]
	2,4,6-trinitrophenetole				[50/2]
	(352–364)	79 (liq)	(358)	ME	[87/4][50/2]
C_8H_8		120.5 ± 2.1	(347)	ME	[50/5][70/1]
	(342–352)				[50/2]
	cubane				[277-10-1]
C_8H_8	(239–262)	80.3 ± 1.6	(298)	ME	[66/2][70/1]
					[87/4]
	cyclooctatetraene				[629-20-9]
$\text{C}_8\text{H}_8\text{N}_2\text{O}_2$		54.4		B	[49/9]
	1,3-benzenedicarboxamide				[1740-57-4]
		54.4 ± 4.2		ME	[71/15][77/1]
$\text{C}_8\text{H}_8\text{N}_2\text{O}_2$					[3010-82-0]
	1,4-benzenedicarboxamide				[72/14][77/1]
	(373–498)	57.3 ± 4.2			[103-82-2]
$\text{C}_8\text{H}_8\text{O}$					[01/10]
	2-phenylacetic acid				
	(305–321)	98.6 ± 0.4	(313)	ME	

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_8\text{H}_8\text{O}_2$	2-methylbenzoic acid (297–337)	99.0 ± 0.6	(298)	ME	[01/10]
		95.9 ± 0.1	(298)	ME	[118-90-1] [86/4]
		137.7 ± 0.5		DSC	[83/8]
$\text{C}_8\text{H}_8\text{O}_2$	3-methylbenzoic acid (303–323)	97.0 ± 0.3	(298)	ME	[99-04-7] [86/4]
$\text{C}_8\text{H}_8\text{O}_2$	4-methylbenzoic acid (318–337)	98.8 ± 0.3	(298)	ME	[99-94-5] [86/4]
$\text{C}_8\text{H}_8\text{O}_2$	2,5-dimethyl-1,4-benzoquinone (273–293)	77.0	(283)	QF	[137-18-8] [27/2][60/1] [87/4]
$\text{C}_8\text{H}_8\text{O}_2$	4-hydroxyacetophenone (320–349)	95.7	(335)		[99-93-4] [87/4][60/14]
$\text{C}_8\text{H}_8\text{O}_2\text{S}$	phenyl vinyl sulfone	82 ± 2.5		B	[5535-48-8] [69/11][77/1] [579-75-9]
$\text{C}_8\text{H}_8\text{O}_3$	2-methoxybenzoic acid (318–353) (319–324) (353–368) (353–368)	101.2	(333)		[87/4]
		104.7 ± 0.3	(322)	ME	[78/7]
		90.8 ± 0.4	(360)	GS	[73/2][87/4]
		90.9	(360)	GS	[54/1][60/1]
					[586-38-9]
$\text{C}_8\text{H}_8\text{O}_3$	3-methoxybenzoic acid (318–326)	107.5 ± 0.4	(322)	ME	[87/4][78/7]
$\text{C}_8\text{H}_8\text{O}_3$	4-methoxybenzoic acid (334–344)	109.8 ± 0.6	(339)	ME	[100-09-4] [87/4][78/7]
$\text{C}_8\text{H}_8\text{O}_3$	2-hydroxy-3-methoxybenzaldehyde (282–303)	54.1	(292.5)		[148-53-8] [87/4]
$\text{C}_8\text{H}_8\text{O}_3$	4-hydroxy-3-methoxybenzaldehyde (vanillin) (293–353)	88.7	(323)		[121-33-5] [53/7][60/1]
$\text{C}_8\text{H}_9\text{NO}$	acetanilide (303–324) (317–336)	80.6	(313.5)		[103-84-4] [87/4][55/3]
		87.2	(326.5)		[87/4][60/21]
					[41977-54-2]
$\text{C}_8\text{H}_9\text{NO}$ <i>anti</i>	3-methylbenzaldoxime	U 31 ± 1.7		MS	[83/9]
$\text{C}_8\text{H}_9\text{NO}$ <i>anti</i>	4-methylbenzaldoxime	U 36 ± 1.7		MS	[3235-02-7] [83/9]
$\text{C}_8\text{H}_9\text{NO}$	2-phenylacetamide (329–352)	96.4	(340.5)		[103-81-1] [87/4][60/21]
$\text{C}_8\text{H}_9\text{NO}$	4-aminoacetophenone (314–338)	92.7	(326)		[99-92-3] [87/4][60/21]
$\text{C}_8\text{H}_9\text{NO}$	N-methylbenzamide (297–321) (307–329)	75	(309)		[613-93-4] [87/4][55/3]
		85.7	(318)		[87/4][60/21]
					[4389-45-1]
$\text{C}_8\text{H}_9\text{NO}_2$	2-amino-3-methylbenzoic acid (343–357)	105.8 ± 0.6	(350)	ME	[01/20]
		107.3 ± 1.8	(298)	ME	[01/20]
$\text{C}_8\text{H}_9\text{NO}_2$	2-amino-5-methylbenzoic acid (345–361)	108.9 ± 0.5	(353)	ME	[2941-78-8] [01/20]
		110.6 ± 1.9	(298)	ME	[01/20]
					[4389-50-8]
$\text{C}_8\text{H}_9\text{NO}_2$	2-amino-6-methylbenzoic acid (339–355)	114.7 ± 1.2	(347)	ME	[01/20]
		116.1 ± 2.0	(298)	ME	[01/20]
					[52130-17-3]
$\text{C}_8\text{H}_9\text{NO}_2$	3-amino-2-methylbenzoic acid (367–381)	125.6 ± 0.8	(374)	ME	[01/20]
		127.8 ± 2.6	(298)	ME	[01/20]
					[2458-12-0]
$\text{C}_8\text{H}_9\text{NO}_2$	3-amino-4-methylbenzoic acid (363–377)	117.3 ± 0.9	(370)	ME	[01/20]
		119.4 ± 2.5	(298)	ME	[01/20]
					[2486-70-6]
$\text{C}_8\text{H}_9\text{NO}_2$	4-amino-3-methylbenzoic acid (367–383)	119.8 ± 0.7	(375)	ME	[01/20]
		122.0 ± 2.6	(298)	ME	[01/20]
					[103-01-5]
$\text{C}_8\text{H}_9\text{NO}_2$	N-phenylglycine	114.1 ± 1.0	(365.4)	C	[80/29]
		128.0 ± 2.0	(298)		[80/29]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_8\text{H}_9\text{NO}_2$	D α -phenylglycine	148.9 ± 2.2 165.0 ± 6.0	(443) (298)	C C	[875-74-1] [80/29] [80/29]
$\text{C}_8\text{H}_9\text{NO}_2$ <i>anti</i> <i>syn</i>	2-methoxybenzaloxime	$U \ 20.1 \pm 1.7$ $U \ 32.6 \pm 1.7$		MS MS	[29577-53-5] [83/9] [83/9]
$\text{C}_8\text{H}_9\text{NO}_2$ <i>anti</i>	4-methoxybenzaloxime	$U \ 67.3 \pm 1.7$		MS	[5235-04-9] [83/9]
$\text{C}_8\text{H}_9\text{NO}_2$	methyl 2-aminobenzoate (287–298)	78.4	(292.5)	ME	[134-20-3] [87/4][54/4] [60/1]
$\text{C}_8\text{H}_9\text{NO}_7$	methyl 5-nitro-2-acetoxy-2,5-dihydro-2-furancarboxylate	89.1 ± 2.1			[22401-53-2] [80/28][86/5]
C_8H_{10}	1,2-dimethylbenzene	60.1	(248)	B	[95-47-6] [86/12]
C_8H_{10}	1,4-dimethylbenzene (247–286)	59.4 60.8	(271) (286)		[106-42-3] [87/4][74/32] [86/12]
$\text{C}_8\text{H}_{10}\text{NO}_5\text{PS}$	methyl parathion (O,O-dimethyl-O-4-nitro-phenylthiophosphate) (298–308) (278–288)	125.1 108.7	(303) (283)	GS,A GS	[298-00-0] [84/33] [83/5][79/10]
$\text{C}_8\text{H}_{10}\text{N}_2\text{O}$	4-N,N-dimethylaminonitrosobenzene (323–334)	82.0 ± 1.7	(298)	ME	[138-89-6] [94/29]
$\text{C}_8\text{H}_{10}\text{N}_2\text{O}_2$	N,N-dimethyl-3-nitroaniline	92.7 ± 0.3	(298)	C	[619-31-8] [85/5]
$\text{C}_8\text{H}_{10}\text{N}_2\text{O}_2$	N,N-dimethyl-4-nitroaniline (344–366) (372–393)	102.7 ± 1.1 98.7 ± 1.7 101.3 ± 2.0	(298) (355) (298)	C ME ME	[100-23-2] [85/5] [87/4][56/2] [94/29]
$\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$ form I form I form II form II	caffeine (1,3,7-trimethylxanthine) (413–463) (413–463)	104.8 ± 0.2 115 113.6 ± 0.2 119	(438) (298) (369) (298)	T T	[58-08-2] [99/40] [99/40] [99/40]
		105.1 ± 0.7 103.6		ME UV	[85/22] [84/37]
		100.0 ± 0.6 110	(423) (298)	MM	[79/17] [79/17][99/40]
		110.7 ± 0.7 114	(362) (298)	MM	[79/17] [79/17][99/40]
$\text{C}_8\text{H}_{10}\text{O}$	4-ethylphenol (278–317)	80.3 ± 0.5		GS	[123-07-9] [63/3][70/1] [87/4]
$\text{C}_8\text{H}_{10}\text{O}$	2,3-dimethylphenol (283–323)	$84. \pm 1.0$		GS	[526-75-0] [60/3][70/1] [87/4]
$\text{C}_8\text{H}_{10}\text{O}$	2,4-dimethylphenol (282–318)	65.9 ± 0.2		GS	[105-67-9] [60/3][70/1] [95-87-4]
$\text{C}_8\text{H}_{10}\text{O}$	2,5-dimethylphenol (282–323)	85.0 ± 0.25		GS	[60/3][70/1] [87/4]
$\text{C}_8\text{H}_{10}\text{O}$	2,6-dimethylphenol (277–313)	75.6 ± 0.17		GS	[576-26-1] [60/3][70/1] [87/4]
$\text{C}_8\text{H}_{10}\text{O}$	3,4-dimethylphenol (282–323)	85.7 ± 0.1		GS	[95-65-8] [60/3][70/1] [87/4]
$\text{C}_8\text{H}_{10}\text{O}$	3,5-dimethylphenol (282–323)	82.8 ± 0.3		GS	[108-68-9] [60/3][70/3] [87/4]
$\text{C}_8\text{H}_{10}\text{O}_2$	2,5-dimethylhydroquinone (332–361)	100.8		QF	[615-90-7] [27/2][60/1]
$\text{C}_8\text{H}_{10}\text{O}_2$	1,4-dimethoxybenzene				[150-78-7]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_8\text{H}_{10}\text{O}_2\text{S}$	methyl- <i>p</i> -tolyl sulfone	84.1 ± 2.3	(298)		[00/24]
		100 ± 3.3			[3185-99-7]
$\text{C}_8\text{H}_{10}\text{O}_2\text{S}$	methyl benzyl sulfone	99.2 ± 3			[U/3][70/1]
					[3112-90-1]
$\text{C}_8\text{H}_{11}\text{N}$	1-norbornylisocyanide	60.6 ± 0.5	(298)		[U/3][70/1]
					[103434-09-9]
$\text{C}_8\text{H}_{11}\text{N}_5$	8-ethyl-9-methyladenine	127.1 ± 0.7			[87/17]
		115.2 ± 1.0	(368)	ME	[116988-56-8]
	(365–370)				[94/6]
$\text{C}_8\text{H}_{11}\text{N}_5$	6,8,9-trimethyladenine	98.6 ± 0.2	(338)	ME	[87/7]
	(334–342)				[139909-51-6]
C_8H_{12}	bicyclo[2.2.2]octene	43.8 ± 0.4		C	[94/6]
					[931-64-6]
$\text{C}_8\text{H}_{12}\text{N}_2$	tetramethylsuccinonitrile	81.2 ± 1.7			[70/5][77/1]
					[71/10]
$\text{C}_8\text{H}_{12}\text{N}_2$	tetramethylpyrazine	94.6 ± 4.0	(298)	C	[3333-52-6]
					[73/8][77/1]
$\text{C}_8\text{H}_{12}\text{N}_2\text{O}_2$	1,3-dimethyl-5-ethyluracil	98.7 ± 1.7	(316)	ME	[1124-11-4]
	(312–321)	99.3 ± 0.2	(308)	ME	[96/28]
	(300–316)	110 ± 1.2	(330)	QR	[31703-08-9]
	(319–340)				[96/5]
$\text{C}_8\text{H}_{12}\text{O}_2$	4,4-dimethyl-1,3-cyclohexanedione	99.2 ± 2.1	(298)	ME	[83/14]
					[562-46-9]
$\text{C}_8\text{H}_{12}\text{O}_2$	5,5-dimethyl-1,3-cyclohexanedione	99.8 ± 1.1	(298)	ME	[93/23]
					[126-81-8]
$\text{C}_8\text{H}_{12}\text{O}_2$	2,2,4,4-tetramethyl-1,3-cyclobutanedione	70.3 ± 3.5		HSA	[93/23]
		72.2 ± 0.6			[933-52-8]
		72.4 ± 0.6		C	[75/3]
C_8H_{14}	bicyclo[2.2.2]octane	46.3 ± 0.8			[71/5]
	(323–363)				[71/23]
		47.7 ± 0.8	(298)	HA	[280-33-1]
		48.0 ± 2		C	[71/1][77/1]
					[87/4]
C_8H_{14}	solid phase transition	4.6	(164)		[71/1][77/1]
	3,4-dimethyl-2,4-hexadiene	53.1			[70/5][71/1]
					[77/1]
$\text{C}_8\text{H}_{14}\text{ClN}_5$	2-chloro-4-ethylamino-6-isopropylamino-1,3,5-triazine (atrazine)	114.6	(339)	GS	[84/25]
	(324–354)	113.8	(338)	GS–GC	[56/7][60/1]
	(323–403)				[1912-24-9]
$\text{C}_8\text{H}_{14}\text{N}_2$	1,4-dimethyl-2,3-diazabicyclo[2.2.2]octane	72.0 ± 0.5	(298)	C	[82/23]
					[64/14][87/4]
$\text{C}_8\text{H}_{14}\text{N}_2\text{O}_2$	α -acetylproline N-methylamide	69.1	(313)		[49570-30-1]
	(308–318)				[76/12]
$\text{C}_8\text{H}_{14}\text{N}_2\text{O}_2$	β -acetylproline N-methylamide	60.7	(327)		[19701-85-0]
	(319–335)				[87/4][55/7]
$\text{C}_8\text{H}_{14}\text{O}$	3-oxabicyclo[3.2.2]nonane	53.1 ± 0.5			[283-27-2]
					[71/10][77/1]
$\text{C}_8\text{H}_{14}\text{O}_4$	1,8-octanedioic acid (suberic acid)	148	(298)	TPTD	[505-8-6]
	(310–320)	147.8 ± 3.8	(393)	M	[01/15]
	(379–407)	143.1 ± 3.8			[99/10][60/4]
					[60/4][70/1]
					[87/4]
$\text{C}_8\text{H}_{14}\text{O}_6$	(<i>d</i>)-dimethoxysuccinic acid dimethyl ester	53.1			[37/6]
$\text{C}_8\text{H}_{14}\text{O}_6$	(<i>dl</i>)-dimethoxysuccinic acid dimethyl ester	57.7			[37/6]
$\text{C}_8\text{H}_{14}\text{O}_6$	<i>meso</i> dimethoxysuccinic acid dimethyl ester	74.1			[37/6]
$\text{C}_8\text{H}_{15}\text{N}$	3-azabicyclo[3.2.2]nonane				[283-24-9]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_8\text{H}_{15}\text{NO}$	<i>trans</i> 2-octenoic acid amide (373–393)	57.8 ± 1.3 73.5	(298) (383)	C	[70/5] [87/4]
$\text{C}_8\text{H}_{15}\text{N}_5\text{O}$	2-methoxy-4,6- <i>bis</i> (ethylamino)-1,3,5-triazine (323–403)	98.2	(338)	GS–GC	[673-04-1] [87/4][64/14]
$\text{C}_8\text{H}_{15}\text{N}_5\text{S}$	2-methylthio-4,6- <i>bis</i> (ethylamino)-1,3,5-triazine (323–355)	101.3	(338)	GS–GC	[1014-70-6] [87/4][64/14]
$\text{C}_8\text{H}_{15}\text{N}_5\text{S}$	2-methylthio-4-methylamino-6-isopropyl-1,3,5-triazine (323–357)	101.5	(338)	GS–GC	[1014-69-3] [87/4][64/14]
$\text{C}_8\text{H}_{15}\text{N}_7\text{O}_2\text{S}_3$	3-[[[2-[(aminoiminomethyl)aminol]-4-thiazolyl]methyl]thio]-N-(aminosulfonyl)propanimidamide (famotidine)	207.0		TGA	[76824-35-6] [97/36]
C_8H_{16}	cyclooctane	58.7	(166)	B	[292-64-8] [63/6]
$\text{C}_8\text{H}_{16}\text{N}_2\text{O}_2$	N-acetyl L-leucine amide	115.6 ± 1.4 119.8 ± 1.5	(376) (298)	C	[28529-34-2] [99/12] [99/12]
$\text{C}_8\text{H}_{16}\text{N}_2\text{O}_2$	N-acetyl D-leucine amide	114.8 ± 0.3 120.4 ± 0.4 101 ± 3	(393) (298) (388)	C TE	[16624-68-3] [99/12] [99/12] [88/6][86/16]
$\text{C}_8\text{H}_{16}\text{N}_2\text{O}_2$	N-acetyl L-isoleucine amide	142.7 ± 0.2 147.4 ± 0.3	(390) (298)	C	[56711-06-9] [99/12] [99/12]
$\text{C}_8\text{H}_{17}\text{NO}$	octanamide (325–374)	110.5 ± 2.9		GS,ME	[629-01-6] [59/3][87/4]
$\text{C}_8\text{H}_{17}\text{NO}_2$	8-aminooctanoic acid (391–402)	166.2 ± 0.9 170 ± 4	(397) (298)	C C	[1002-57-9] [83/24] [83/24]
C_8H_{18}	<i>n</i> -octane	68.1	(216)	B	[111-65-9] [63/6]
C_8H_{18}	2,2,3,3-tetramethylbutane (286–377) (273–338)	43.6 43.4 ± 0.2 42.9 ± 0.9	(301) (298) (298)		[594-82-1] [87/4] [52/1][70/1] [47/8]
	(263–279) solid phase transition	56.2 2.0		A,MG	[31/4] [84/25]
$\text{C}_8\text{H}_{18}\text{N}_2\text{O}_2$	1,4- <i>bis</i> -(2-hydroxyethyl)piperazine (334–356)	104.1			[122-96-3] [84/30]
$\text{C}_8\text{H}_{18}\text{O}$	1-octanol	105.4 ± 2.5	(244)		[111-87-5] [65/6]
$\text{C}_8\text{H}_{18}\text{O}_2$	1,8-octanediol	139.3 ± 0.9	(298)	C	[629-41-4] [90/14]
$\text{C}_8\text{H}_{18}\text{O}_2\text{S}$	di- <i>n</i> -butyl sulfone	100.4 ± 2.5			[598-04-9] [U/3][70/1]
$\text{C}_8\text{H}_{18}\text{O}_2\text{S}$	di- <i>tert</i> -butyl sulfone	94.1 ± 2.9			[1886-75-7] [U/3][70/1]
$\text{C}_9\text{H}_5\text{BrClINO}$	7-bromo-5-chloro-8-hydroxyquinoline (353–368)	110.1 ± 0.8 113.2 ± 0.8	(361) (298)	ME	[7640-33-7] [92/20] [92/20]
$\text{C}_9\text{H}_5\text{Br}_2\text{NO}$	5,7-dibromo-8-hydroxyquinoline (365–380)	113.6 ± 1.3 117.3 ± 1.3 94.1	(372) (298)	ME	[521-74-4] [92/20] [92/20] [63/5]
$\text{C}_9\text{H}_5\text{ClINO}$	5-chloro-7-iodo-8-hydroxyquinoline (359–378)	111.3 ± 0.4 114.8 ± 0.4	(368) (298)	ME	[130-26-7] [92/20] [92/20]
$\text{C}_9\text{H}_5\text{ClINO}$	5-iodo-7-chloro-8-hydroxyquinoline (383–414)	131.0			[35048-13-6] [63/5]
$\text{C}_9\text{H}_5\text{Cl}_2\text{NO}$	5,7-dichloro-8-hydroxyquinoline (351–366)	106.3 ± 0.7 109.3 ± 0.7 92.9	(358) (298)	ME	[773-76-2] [92/20] [92/20] [63/5]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_9\text{H}_5\text{I}_2\text{NO}$	5,7-diiodo-8-hydroxyquinoline (389–404)	121.9 ± 0.8	(396)	ME	[83-73-8]
		126.8 ± 0.8	(298)		[92/20]
	(403–423)	110.9			[92/20]
$\text{C}_9\text{H}_6\text{ClNO}$	5-chloro-8-hydroxyquinoline (317–327)	97.5 ± 0.9	(322)	ME	[63/5]
		98.7 ± 0.9	(298)		[130-16-5]
					[92/20]
$\text{C}_9\text{H}_6\text{INO}$	5-iodo-8-hydroxyquinoline (363–393)	118.8		ME	[13207-63-1]
$\text{C}_9\text{H}_6\text{N}_2\text{O}_2$	5-nitroquinoline (310–324)	93.2 ± 0.7	(317)	ME	[63/5]
		94.2 ± 0.7	(298)		[607-34-1]
					[97/2]
$\text{C}_9\text{H}_6\text{N}_2\text{O}_2$	6-nitroquinoline (336–350)	101.5 ± 1.0	(343)	ME	[613-50-3]
		103.8 ± 1.0	(298)		[97/2]
					[97/2]
$\text{C}_9\text{H}_6\text{N}_2\text{O}_2$	8-nitroquinoline (338–352)	104.3 ± 0.9	(345)	ME	[607-35-2]
		106.7 ± 0.9	(298)		[97/2]
					[97/2]
$\text{C}_9\text{H}_6\text{N}_2\text{O}_3$	5-nitro-8-hydroxyquinoline (352–362)	114.1 ± 2.2	(298)	ME	[4008-48-4]
		111.2 ± 3.0	(298)		[89/2]
					[89/2]
$\text{C}_9\text{H}_6\text{O}_2$	coumarin			C	[91-64-5]
		83.1	(298)		[91/12]
	(293–353)	86.2	(323)		[53/7][60/1]
$\text{C}_9\text{H}_6\text{O}_2$	chromone			C	[87/4]
		81.3 ± 0.2	(298)		[491-38-3]
					[88/13]
$\text{C}_9\text{H}_6\text{O}_6$	1,3,5-benzenetricarboxylic acid (553–593)	159.4	(573)	GS	[554-95-0]
					[87/4][62/2]
					[3445-76-9]
$\text{C}_9\text{H}_6\text{S}_3$	5-phenyl-1,2-dithiole-3-thione (363–373)	117.4 ± 0.4	(368)		[72/16]
		123.3 ± 0.4	(298)		[72/16]
					[59-31-4]
$\text{C}_9\text{H}_7\text{NO}$	2-hydroxyquinoline (375–390)	115.2 ± 0.6	(383)	ME	[90/1]
		119.4 ± 0.6	(298)		[90/1]
					[611-36-9]
$\text{C}_9\text{H}_7\text{NO}$	4-hydroxyquinoline (415–433)	128.8 ± 1.1	(424)	ME	[90/1]
		135.1 ± 1.1	(298)		[90/1]
					[90/1]
$\text{C}_9\text{H}_7\text{NO}$	8-hydroxyquinoline (293–303)	89.5 ± 0.9	(298)	ME	[148-24-3]
		89.0 ± 1.4	(298)		[89/2]
	(308–328)	108.8 ± 1.7			[89/2]
$\text{C}_9\text{H}_7\text{N}_3\text{O}_2$	5-amino-6-nitroquinoline (400–424)	130.7 ± 0.8	(412)	ME	[63/5][70/1]
		136.4 ± 0.8	(298)		[87/4]
					[35975-00-9]
$\text{C}_9\text{H}_7\text{NO}$	Ω -cyanoacetophenone (318–333)	99.8	(325.5)	ME	[98/11]
		92.5 ± 4.2			[614-16-4]
					[87/4]
$\text{C}_9\text{H}_7\text{NO}_2$	N-methylphthalimide (298–316)	91.1 ± 0.5	(307)	ME	[69/9][77/1]
		91.1 ± 0.5	(298)		[550-44-7]
					[97/12]
$\text{C}_9\text{H}_8\text{N}_2$	3-aminoquinoline (329–345)	101.1 ± 0.9	(337)	ME	[97/12]
		103.1 ± 0.9	(298)		[580-17-6]
		104.8 ± 4.8	(298)		[93/10]
$\text{C}_9\text{H}_8\text{N}_2$	5-aminoquinoline (329–349)	102.9 ± 0.7	(339)	ME	[93/10]
		105.0 ± 0.7	(298)		[611-34-7]
		103.3 ± 3.4	(298)		[93/10]
$\text{C}_9\text{H}_8\text{N}_2$	6-aminoquinoline (333–349)	103.6 ± 1.0	(341)	ME	[93/10]
		105.7 ± 1.0	(298)		[580-15-4]
					[93/10]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_9\text{H}_8\text{N}_2$	8-aminoquinoline				[578-66-5]
	(329–314)	93.0	(305)	ME	[93/10]
$\text{C}_9\text{H}_8\text{N}_2\text{O}$	2-methyl-3-hydroxyquinoxaline (375–391)	93.3 ± 0.5	(298)		[93/10]
		117.2 ± 0.4	(383)	ME	[14003-34-0]
		119.7 ± 2.8	(298)	ME	[00/26]
		123.0 ± 4.4	(298)	C	[00/15]
$\text{C}_9\text{H}_8\text{N}_2\text{O}_2$	2-methylquinoxaline-1,4-dioxide	107.0 ± 6.2	(298)	C	[6639-86-7]
$\text{C}_9\text{H}_8\text{N}_2\text{O}_2$	3-methylaminophthalimide (402–450)	104.9	(417)	RG	[5972-09-8]
$\text{C}_9\text{H}_8\text{O}$	1-indanone (288–308)	83.5 ± 0.7	(298)	GS	[87/4][56/13]
$\text{C}_9\text{H}_8\text{O}_2$	<i>trans</i> -cinnamic acid (333–347)	105.0 ± 0.8	(340)	ME	[83-33-0]
	(333–347)	107.1 ± 0.8	(298)	ME	[98/21]
$\text{C}_9\text{H}_8\text{O}_2\text{S}$	phenyl propadienyl sulfone	105.4 ± 2.5			[140-10-3]
$\text{C}_9\text{H}_8\text{O}_2\text{S}$	phenyl prop-1-ynyl sulfone	95.4 ± 2.5		B	[99/27]
$\text{C}_9\text{H}_8\text{O}_2\text{S}$	phenyl prop-2-ynyl sulfone	105. ± 2.5		B	[99/27]
$\text{C}_9\text{H}_8\text{O}_3$	<i>endo</i> -5-norbornene-2,3-dicarboxylic anhydride	97 ± 4.2	(298)	MG	[2525-42-0]
$\text{C}_9\text{H}_8\text{O}_4$	monomethyl terephthalate (433–493)	72.1	(448)		[69/13][70/1]
	(433–493)	82.8	(473)	GS	[2525-41-9]
		130.4			[69/13][70/1]
$\text{C}_9\text{H}_9\text{N}$	2,6-dimethylbenzonitrile	83.9 ± 2.8	(298)	C	[2525-40-8]
$\text{C}_9\text{H}_9\text{N}$	3-methylindole (288–333)	83.3	(303)		[69/13][70/1]
$\text{C}_9\text{H}_9\text{NO}_2$	4-acetomidobenzaldehyde (328–346)	99	(337)		[62/2]
$\text{C}_9\text{H}_9\text{N}_3\text{O}_6$	2,4,6-trinitromesitylene (319–397)	103.6 ± 1.2		ME	[98/28]
$\text{C}_9\text{H}_{10}\text{Cl}_2\text{N}_2\text{O}$	3-(3,4-dichlorophenyl)-1,1-dimethylurea (diuron)	119 ± 0.6	(393)	C	[21789-36-6]
		133.9 ± 0.7	(298)		[91/3]
$\text{C}_9\text{H}_{10}\text{N}_2\text{O}$	1-phenyl-3-pyrazolidinone (327–348)	84.3	(337.5)		[83-34-1]
$\text{C}_9\text{H}_{10}\text{O}$	<i>trans</i> 3-phenyl-2-propen-1-ol (288–307)	109.6	(297.5)		[87/4]
		69.5		ME	[4407-36-7]
$\text{C}_9\text{H}_{10}\text{O}_2$	2-ethylbenzoic acid (298–313)	100.5	(305.5)	ME	[54/4]
		101.1 ± 0.4	(298)	ME	[612-19-1]
	(298–313)	100.7 ± 2.5	(298)	ME	[87/4][76/6]
$\text{C}_9\text{H}_{10}\text{O}_2$	3-ethylbenzoic acid (300–318)	99.1	(309)	ME	[84/1]
		99.7 ± 0.4	(298)	ME	[76/6]
	(300–318)	99.1 ± 2.5	(298)	ME	[619-20-5]
$\text{C}_9\text{H}_{10}\text{O}_2$	4-ethylbenzoic acid (310–329)	98.2	(319.5)	ME	[87/4][76/6]
		98.9 ± 0.2	(298)	ME	[84/1]
	(311–330)	97.6 ± 0.2	(321)	ME	[84/1]
	(320–329)	97.5 ± 2.5	(298)	ME	[76/6]
$\text{C}_9\text{H}_{10}\text{O}_2$	4-acetylanisole (276–300)	77.7		V	[619-64-7]
	(283–333)	93.7	(308)		[87/4][76/6]
$\text{C}_9\text{H}_{10}\text{O}_2$	2,3-dimethylbenzoic acid (316–337)	102.3 ± 0.4	(326)	ME	[100-06-1]
					[59/2]
					[54/4][60/1]
					[87/4]
					[603-79-2]
					[84/9]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_9\text{H}_{10}\text{O}_2$	2,4-dimethylbenzoic acid (312–331)	104.6 ± 0.4	(298)	ME	[84/9]
		102.7 ± 0.3	(321)	ME	[611-01-8]
		103.5 ± 0.3	(298)	ME	[84/9]
$\text{C}_9\text{H}_{10}\text{O}_2$	2,5-dimethylbenzoic acid (315–334)	103.6 ± 0.6	(324)	ME	[84/9]
		105.0 ± 0.6	(298)	ME	[610-72-0]
					[84/9]
$\text{C}_9\text{H}_{10}\text{O}_2$	2,6-dimethylbenzoic acid (309–324)	98.2 ± 0.2	(317)	ME	[632-46-2]
		99.1 ± 0.2	(298)	ME	[84/9]
					[84/9]
$\text{C}_9\text{H}_{10}\text{O}_2$	3,4-dimethylbenzoic acid (325–347)	104.5 ± 0.3	(336)	ME	[619-04-5]
		106.4 ± 0.3	(298)	ME	[84/9]
					[84/9]
$\text{C}_9\text{H}_{10}\text{O}_2$	3,5-dimethylbenzoic acid (322–341)	100.8 ± 0.3	(332)	ME	[499-06-9]
		102.3 ± 0.3	(298)	ME	[84/9]
					[84/9]
$\text{C}_9\text{H}_{10}\text{O}_2$	3-methylphenyl acetate (274–317)	60.7	(295)	TE	[122-46-3]
$\text{C}_9\text{H}_{10}\text{O}_2\text{S}$	<i>p</i> -tolyl vinyl sulfone	82.4 ± 2.5		B	[47/1][60/1]
$\text{C}_9\text{H}_{10}\text{O}_3$	3-ethoxy-4-hydroxybenzaldehyde (296–338)				[5535-52-4]
		101.5	(311)		[69/11][77/1]
					[121-32-4]
$\text{C}_9\text{H}_{10}\text{O}_4$	2,3-dimethoxybenzoic acid (336–356)	115.1 ± 0.3	(346)	ME	[87/4][57/3]
		116.6 ± 0.3	(298)	ME	[60/1]
					[1521-38-6]
$\text{C}_9\text{H}_{10}\text{O}_4$	2,4-dimethoxybenzoic acid (346–367)	120.5 ± 0.4	(357)	ME	[85/1]
		123.4 ± 0.4	(298)	ME	[85/1]
					[91-52-1]
$\text{C}_9\text{H}_{10}\text{O}_4$	2,6-dimethoxybenzoic acid (335–378)	118.4 ± 0.4	(367)	ME	[85/1]
		121.7 ± 0.4	(298)	ME	[85/1]
					[1466-76-8]
$\text{C}_9\text{H}_{10}\text{O}_4$	3,4-dimethoxybenzoic acid (359–378)	126.1 ± 0.6	(369)	ME	[85/1]
		129.8 ± 0.6	(298)	ME	[85/1]
					[93-07-2]
$\text{C}_9\text{H}_{10}\text{O}_4$	2,5-dimethoxybenzoic acid (324–342)	113.3 ± 0.7	(333)	ME	[2785-98-0]
		116.1 ± 0.7	(298)		[96/15]
					[96/15]
$\text{C}_9\text{H}_{10}\text{O}_4$	3,5-dimethoxybenzoic acid (356–376)	124.5 ± 0.6	(369)	ME	[1132-21-4]
		127.1 ± 0.6	(298)	ME	[85/1]
					[85/1]
$\text{C}_9\text{H}_{10}\text{O}_5$	2-(diacetoxymethyl)furan	109.6 ± 2.5			[613-75-2]
$\text{C}_9\text{H}_{11}\text{ClN}_2\text{O}$	3-(4-chlorophenyl)-1,1-dimethylurea (303–379)	114.6 ± 4.9	(341)	ME,C	[80/28][86/5]
					[150-68-5]
					[87/4][72/9]
$\text{C}_9\text{H}_{11}\text{NO}$	N-(2-methylphenyl)acetamide (315–340)	96.8	(327.5)		[120-66-1]
$\text{C}_9\text{H}_{11}\text{NO}$	N-(4-methylphenyl)acetamide (331–350)	99.0	(341)		[87/4][60/21]
$\text{C}_9\text{H}_{11}\text{NO}$	N,N-dimethylbenzamide (289–305)	89.7 ± 0.3	(298)		[103-89-9]
$\text{C}_9\text{H}_{11}\text{NO}_2$	2,4,6-trimethylnitrobenzene	78.6 ± 1.0	(298)	C	[60/21]
$\text{C}_9\text{H}_{11}\text{NO}_2$	L-(<i>l</i>)-phenylalanine (342–442)	$U\ 90 \pm 6.3$	(392)	LE	[611-74-5]
		154 ± 8	(455)	ME	[95/11]
					[603-71-4]
$\text{C}_9\text{H}_{11}\text{NO}_3$	L-tyrosine (412–512)	101 ± 8	(462)	LE	[93/3][93/21]
					[63-91-2]
					[77/2]
$\text{C}_9\text{H}_{11}\text{NS}$	N,N-dimethylthiobenzamide	94.8 ± 2.0	(298)	C	[65/1][70/1]
$\text{C}_9\text{H}_{13}\text{ClN}_6$	2-[(4-chloro-6-ethylamino-s-triazin-2-yl)amino]-2-methylpropionitrile (cyanazine)				[87/4][64/16]
					[60-18-4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
	(339–365)	90.7	(352)	GS	[82/23]
$\text{C}_9\text{H}_{12}\text{F}_3\text{N}_3\text{O}_5$	N-[N-(N-(trifluoroacetyl)glycyl)glycyl]glycine methyl ester (343–433)	133.4	(358)		[651-18-3] [87/4][60/20]
$\text{C}_9\text{H}_{12}\text{N}_2$	N-methyl-7-(methylimino)-1,3,5-cycloheptatrienylamine	49.4 ± 4			[1502-10-9] [71/3][77/1]
$\text{C}_9\text{H}_{12}\text{N}_2\text{O}_2$	3-ethoxyphenylurea	75.3 ± 8.3		E	[13142-86-4] [54/12][70/1]
$\text{C}_9\text{H}_{12}\text{N}_2\text{O}_2$	4-ethoxyphenylurea	83.7 ± 8.3		E	[150-69-6] [54/12][70/1]
$\text{C}_9\text{H}_{12}\text{O}$	2,3,6-trimethylphenol	86.7 ± 0.6	(298)	GS	[2416-94-6] [99/17]
$\text{C}_9\text{H}_{12}\text{O}$	2,4,6-trimethylphenol	82.8 ± 0.3	(298)	GS	[527-60-6] [99/17]
		95.0	(298)	C	[71/24][99/17]
$\text{C}_9\text{H}_{12}\text{O}$	α,α -dimethylbenzyl alcohol (276–302)	82.8 ± 0.7	(289)	GS	[617-94-7] [99/18]
		82.3 ± 0.7	(298)		[99/18]
$\text{C}_9\text{H}_{12}\text{O}_2$	3-isopropyl-1,2-dihydroxybenzene	97.8 ± 1.7	(298)	C	[2138-48-9] [84/10]
$\text{C}_9\text{H}_{12}\text{O}_2$	1,2,3-trimethoxybenzene (375)	98.0 ± 0.3	(298)	C	[634-36-6] [00/24]
$\text{C}_9\text{H}_{12}\text{O}_2$	1,3,5-trimethoxybenzene (375)	100.6 ± 1.9	(298)	C	[621-23-8] [00/24]
$\text{C}_9\text{H}_{13}\text{N}_5$	6,9-dimethyl-8-ethyladenine (345–351)	94.1 ± 0.1	(348)	ME	[139909-52-7] [94/6]
$\text{C}_9\text{H}_{13}\text{N}_5$	8-propyl-9-methyladenine (364–370)	124.2 ± 0.8	(367)	ME	[117954-97-9] [87/7]
C_9H_{14}	bicyclo[3.2.2]non-6-ene	48 ± 1.0	(298)	C	[7124-86-9] [82/4]
C_9H_{14}	bicyclo[3.3.1]non-2-ene	48.2 ± 0.4	(298)	C	[6671-66-5] [82/4]
C_9H_{14}	bicyclo[4.2.1]non-3-ene	49.7 ± 0.8	(298)	C	[1456-33-0] [82/4]
$\text{C}_9\text{H}_{14}\text{N}_2\text{O}_2$	1,3-dimethyl-5-propyluracil (317–327)	111.0 ± 1.6	(322)	ME	[82413-39-6] [96/5]
$\text{C}_9\text{H}_{14}\text{N}_2\text{O}_2$	1,3-dimethyl-5-isopropyluracil (316–328)	102.9 ± 1.6	(322)	ME	[175412-48-3] [96/5]
$\text{C}_9\text{H}_{14}\text{N}_2\text{O}_2$	1,3-diethylthymine	89.8 ± 0.4	(298)	C	[21472-93-5] [80/7]
	(307–325)	95.0 ± 2.1	(317)	QR	[80/19]
C_9H_{16}	bicyclo[3.3.1]nonane	50.6 ± 2	(298)	TSGC	[260-65-9] [77/9]
$\text{C}_9\text{H}_{16}\text{ClN}_5$	2-chloro-4,6-bis(isopropylamino)-1,3,5-triazine (323–403)	125.1	(338)	GS–GC	[139-40-2] [87/4][64/14]
$\text{C}_9\text{H}_{16}\text{NO}_2$	2,2,6,6-tetramethyl-4-oxopiperidine-1-oxyl	83.3 ± 1.7		ME	[2896-70-0] [65/7][70/1]
					[87/4]
$\text{C}_9\text{H}_{16}\text{N}_2$	2-methyl-2-piperidinopropionitrile	80.3 ± 0.5	(298)		[2273-41-8] [97/28]
$\text{C}_9\text{H}_{16}\text{O}_4$	nonanedioic acid (367–377)	156.2 ± 0.5	(372)	ME	[123-99-9] [99/10]
		159.9 ± 1.0	(298)		[99/10]
$\text{C}_9\text{H}_{17}\text{NO}$	2,2,6,6-tetramethyl-4-oxopiperidine	60.8 ± 2.7		ME	[2896-70-0] [66/4][70/1]
$\text{C}_9\text{H}_{17}\text{NO}$	<i>trans</i> 2-nonenic acid amide (383–393)	111.9	(388)		[14952-05-7] [87/4]
$\text{C}_9\text{H}_{17}\text{NO}_2$	2,2,6,6-tetramethyl-1-hydroxy-4-oxopiperidine (288–328)	80	(303)		[3637-11-4] [87/4]
		80.1 ± 4.6		ME	[65/7][70/1]
$\text{C}_9\text{H}_{17}\text{N}_5\text{O}$	2-methoxy-4-ethylamino-6-isopropylamino-1,3,5-triazine (323–403)	94.4	(338)	GS–GC	[1610-17-9] [87/4][64/14]
$\text{C}_9\text{H}_{17}\text{N}_5\text{S}$	2-methylthio-4-ethylamino-6-isopropylamino-1,3,5-triazine (323–403)	100.9	(338)	GS–GC	[834-12-8] [87/4][64/14]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_9\text{H}_{18}\text{NO}_2$	2,2,6,6-tetramethyl-4-hydroxypiperidine-1-oxyl (293–318)	101.5 ± 5.2	(306)	ME	[2226-96-2] [66/4][70/1]
$\text{C}_9\text{H}_{18}\text{N}_2\text{OS}$	N,N-diethyl-N'-isobutanoylthiourea (363)	120.8 ± 2.5	(298)	C	[01/9]
$\text{C}_9\text{H}_{18}\text{N}_2\text{O}_2\text{S}$	3,3-dimethyl-1-(methylthio)-2-butanone O-((methylamino)carboxyl)-oxime (298–328)	93.5 ± 6	(308)	ME	[39196-18-4] [87/4][76/11]
$\text{C}_9\text{H}_{18}\text{O}_3$	di- <i>tert</i> -butylcarbonate	65.4 ± 0.2	(298)	C	[34619-03-9] [85/17]
$\text{C}_9\text{H}_{19}\text{NO}$	nonamide (353–370)	114.6 ± 3.3		ME	[1120-07-6] [59/3]
$\text{C}_9\text{H}_{19}\text{NO}_2$	2,2,6,6-tetramethyl-1,4-dihydroxypiperidine (318–348)	100.4 ± 0.6	(328)	ME	[3637-10-3] [66/4][70/1]
C_9H_{20}	<i>n</i> -nonane	74.6	(219)	B	[111-84-2] [63/6]
		71.4	(298)	H	[63/6][93/16]
$\text{C}_9\text{H}_{20}\text{N}_2\text{S}$	1,3-dibutylthiourea	141.0 ± 2	(298)	B,HA	[109-46-6] [00/23]
		137 ± 3.0	(298)	C	[94/17]
$\text{C}_9\text{H}_{20}\text{N}_2\text{S}_2$	diethylammonium diethyldithiocarbamate	111.8 ± 3.0			[1518-58-7] [79/14]
$\text{C}_9\text{H}_{20}\text{O}$	di- <i>tert</i> -butylmethanol	62.7 ± 0.9	(298)		[14609-79-1] [98/22]
$\text{C}_9\text{H}_{20}\text{O}_2$	1,9-nonanediol	148.7			[3937-56-2] [90/14]
C_{10}F_8	octafluoronaphthalene (293–323)	79.4 ± 2.5	(308)	ME	[313-72-4] [87/4][74/9]
$\text{C}_{10}\text{H}_2\text{O}_6$	1,2,4,5-benzenetetracarboxylic dianhydride (pyromellitic dianhydride)	100.4	(559)		[89-32-7] [75/11]
		88.4			[67/15]
$\text{C}_{10}\text{H}_6\text{BrNO}_2$	1-(4-bromophenyl)-1 <i>H</i> -pyrrole-2,5-dione (350–370)	105.9 ± 0.7		C	[13380-67-1] [98/25]
$\text{C}_{10}\text{H}_6\text{Cl}_4\text{O}_4$	dimethyltetrachloroterephthalane (chlorthal) (348–433)	104.9 ± 1.4	(390)	ME,GS	[1861-32-1] [81/20]
$\text{C}_{10}\text{H}_6\text{N}_2$	2-cyanoquinoline	89.3 ± 3.3	(298)	C	[1436-43-7] [95/14]
		93.4 ± 0.7	(319)	ME	[95/14]
		94.4 ± 0.7	(298)		[95/14]
$\text{C}_{10}\text{H}_6\text{N}_2$	3-cyanoquinoline	91.3 ± 1.8	(298)	C	[34846-64-5] [95/14]
		93.4 ± 0.7	(319)	ME	[95/14]
		93.2 ± 0.8	(298)		[95/14]
$\text{C}_{10}\text{H}_6\text{N}_2\text{O}_4$	1-(3-nitrophenyl)-1 <i>H</i> -pyrrole-2,5-dione (350–370)	115.7 ± 0.9		C	[7300-93-8] [98/25]
$\text{C}_{10}\text{H}_6\text{N}_2\text{O}_4$	1-(4-nitrophenyl)-1 <i>H</i> -pyrrole-2,5-dione (350–370)	117.3 ± 1.2		C	[4338-06-1] [98/25]
$\text{C}_{10}\text{H}_6\text{O}_2$	1,4-naphthoquinone	91.0 ± 0.8	(298)	C	[130-15-4] [89/21]
		90.7 ± 2	(313)	TE,ME	[81/4]
		72.4 ± 3.8			[56/5][70/1]
$(\text{C}_{10}\text{H}_6\text{O}_2)-$ $(\text{C}_{10}\text{H}_8\text{O}_2)$	(1,4-naphthoquinone)-(1,4-naphthohydroquinone) (330–351)	102.3 ± 2	(342.4)	ME,TE	[81/4] [66653-77-8]
$2(\text{C}_{10}\text{H}_6\text{O}_2)-$ $(\text{C}_{10}\text{H}_8\text{O}_2)$	2(1,4-naphthoquinone)-(1,4-naphthohydroquinone) (315–340)	88.7 ± 3	(328.5)	ME,TE	[81/4]
$\text{C}_{10}\text{H}_7\text{Br}$	2-bromonaphthalene	81.2 ± 1.0	(298)		[580-13-2] [93/13]
		64 ± 5	(298)	TE,ME	[81/6]
$\text{C}_{10}\text{H}_7\text{Cl}_7$	dihydroheptachlor (333–353)	83.8	(343)	ME	[2589-15-3] [87/4][74/36]
$\text{C}_{10}\text{H}_7\text{F}_3\text{O}_2$	benzoyltrifluoroacetone	87.1 ± 0.9	(298)	ME	[326-06-7] [92/28]
$\text{C}_{10}\text{H}_7\text{I}$	2-iodonaphthalene				[612-55-5]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{10}\text{H}_7\text{NO}_2$	1-nitronaphthalene	90.8			[56/10]
	(309–326)	68.5 ± 1.9	(318)	ME	[86-57-7]
	(325–332)	106.9 ± 2.1	(328.5)	ME	[87/4][74/9]
$\text{C}_{10}\text{H}_7\text{NO}_2$	1-nitroso-2-naphthol				[87/4][50/2]
$\text{C}_{10}\text{H}_7\text{NO}_2$	2-nitroso-1-naphthol	86.6 ± 4.2		ME	[70/1]
$\text{C}_{10}\text{H}_7\text{NO}_2$	4-nitroso-1-naphthol	56.5 ± 4.2		ME	[131-91-9]
$\text{C}_{10}\text{H}_7\text{NO}_2$	1-phenyl-1 <i>H</i> -pyrrole-2,5-dione	87.4 ± 4.2		ME	[68/6][77/1]
C_{10}H_8	azulene	98.1 ± 1		C	[132-53-6]
C_{10}H_8	(283–326)	78.4 ± 1.3	(303)	HSA	[68/6][77/1]
C_{10}H_8	(290–372)	72.7	(298)	CGC–DSC	[605-60-7]
C_{10}H_8	(253–293)	82.8	(305)		[68/6][77/1]
C_{10}H_8	(253–293)	82.9	(298)		[941-69-5]
C_{10}H_8	(253–293)	75.8	(273)		[98/25]
C_{10}H_8	(253–293)	75.3	(298)		[275-51-4]
C_{10}H_8	(253–293)	76.8 ± 0.2		C	[98/5]
C_{10}H_8	(293–323)	95.4 ± 0.4	(298)	ME	[98/5]
C_{10}H_8	naphthalene	67.6			[87/4]
C_{10}H_8	(313–353)	70.4	(298)	CGC–DSC	[87/4][93/16]
C_{10}H_8	(243–273)	71.7	(333)	GS	[58/1]
C_{10}H_8	(337–352)	73.7 ± 1.0	(258)	GS	[58/1][93/16]
C_{10}H_8	(337–352)	78.2 ± 1		GC	[72/1]
C_{10}H_8	(337–352)	70.9 ± 0.4	(323)	DSC	[62/1][70/1]
C_{10}H_8	(337–352)	72.3 ± 0.4	(298)	DSC	[47/7]
C_{10}H_8	(283–323)	75.8 ± 1.1	(303)	GS	[91-20-3]
C_{10}H_8	(283–323)	72.6 ± 0.4		DSC	[98/5]
C_{10}H_8	(283–323)	72.6 ± 0.1	(298)	TE,ME,DM	[95/7]
C_{10}H_8	(302–352)	72.8	(327)	GS	[83/1][81/1]
C_{10}H_8	(271–285)	72.8 ± 0.3		ME	[82/23]
C_{10}H_8	(271–285)	72.4 ± 0.7	(298)	C	[82/1]
C_{10}H_8	(274–353)	72.5 ± 0.1		DM	[82/3]
C_{10}H_8	(253–273)	72.6 ± 0.6		TE	[81/1]
C_{10}H_8	(280–305)	71.3	(293)	GS	[80/1]
C_{10}H_8	(253–273)	74.77 ± 0.4		TE	[79/27]
C_{10}H_8	(253–273)	73.9 ± 0.2		ME	[77/4]
C_{10}H_8	(303–329)	74.35 ± 1.7		TSGC	[77/4]
C_{10}H_8	(303–329)	72.3 ± 0.4		C	[75/1]
C_{10}H_8	(263–343)	72.5 ± 0.3		DM	[76/1]
C_{10}H_8	(263–298)	67.8 ± 3.5	(280)	HSA	[75/4]
C_{10}H_8	(263–298)	72.5	(298)	GS	[75/3]
C_{10}H_8	(281–297)	72.7 ± 1.7	(289)	ME	[74/19]
C_{10}H_8	(281–290)	$64 \pm .5$		LE	[74/9]
C_{10}H_8	(281–290)	72.1 ± 0.25	(298)	C	[73/15]
C_{10}H_8	(281–290)	73.0 ± 0.3	(298)	C	[72/1]
C_{10}H_8	(283–323)	72.7		ME	[72/2]
C_{10}H_8	(283–323)	66.5			[71/31]
C_{10}H_8	(230–260)	72.7 ± 0.3		KG	[68/1]
C_{10}H_8	(276–283)	66.3		V	[63/1][70/1]
C_{10}H_8	(283–303)	65.8	(293)	ME	[59/2]
C_{10}H_8	(253–283)	69.2	(268)		[58/18]
C_{10}H_8	(273–311)	72.1	(292)		[58/1]
C_{10}H_8	(279–294)	72.4			[57/9]
C_{10}H_8	(279–294)				[53/1][60/1]
C_{10}H_8	(279–294)				[54/10]
C_{10}H_8	(288–306)	64.0	(298)	ME	[51/12]
C_{10}H_8	(288–306)	65.7	(297)	ME	[40/1]
C_{10}H_8	(288–306)	66.5 ± 1.7	(298)	QF	[38/1]
C_{10}H_8	(237–276)	76.6			[26/2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)		CAS registry number
Polymorph	Temperature range (K)			Method	Reference
C_{10}D_8	(283–303)	82.0	(293)	ME	[25/3]
	naphthalene- d_8				[1146-65-2]
$\text{C}_{10}\text{H}_8\text{Br}_2\text{N}_2$	(282–323)	70.6 ± 0.5	(303)	GS	[83/11]
	2,3-bis(bromomethyl)quinoxaline				[3138-86-1]
	(351–365)	111.7 ± 0.5	(358)	ME	[00/26]
		114.0 ± 2.0	(298)	ME	[00/26]
$\text{C}_{10}\text{H}_8\text{NO}_2$	indole-3-acetic acid				[87-51-4]
$\text{C}_{10}\text{H}_8\text{N}_2$	(313–423)	64.0 ± 1.5	(368)	ME	[88/4]
	2,2'-bipyridine				[366-18-7]
		81.8 ± 2.3	(298)	C	[95/26]
		75.0 ± 5.0	(298)	B	[96/24]
$\text{C}_{10}\text{H}_8\text{N}_2$	2,4'-bipyridine	81.9 ± 0.3			[85/6]
					[581-47-5]
$\text{C}_{10}\text{H}_8\text{N}_2$	4,4'-bipyridine	87.9 ± 1.7	(298)	C	[95/26]
					[553-26-4]
$\text{C}_{10}\text{H}_8\text{N}_2\text{O}_2$	8-nitroquinaldine	106.3 ± 2.8	(298)	C	[95/26]
					[553-26-4]
					[95/26]
$\text{C}_{10}\text{H}_8\text{N}_2\text{O}_3$	3-acetamidophthalimide				[881-07-2]
					[97/2]
$\text{C}_{10}\text{H}_8\text{O}$	(346–360)	108.3 ± 0.8	(353)	ME	[97/2]
		111.0 ± 0.8	(298)		[97/2]
	1-naphthol				[6118-65-6]
	(296–313)	91.2 ± 0.4	(443)	RG	[87/4][56/13]
	(279–328)	89.1 ± 1.7	(304)	ME	[90-15-3]
$\text{C}_{10}\text{H}_8\text{O}$	1-naphthol (α form)	91.5 ± 3.8	(298)	B	[74/3]
					[74/5]
					[26/1][70/1]
					[27/1]
					[27/1]
$\text{C}_{10}\text{H}_8\text{O}$	1-naphthol (β form)				
$\text{C}_{10}\text{H}_8\text{O}$	(298–312)	93.3	(305)		[87/4][60/14]
	(314–324)	84.3	(319)		[87/4][60/14]
	2-naphthol				[135-19-3]
	(305–323)	94.2 ± 0.5		ME	[74/3]
	(277–324)	87.4 ± 2.5	(300)	ME	[74/5]
$\text{C}_{10}\text{H}_8\text{O}$	(283–323)	78.7 ± 0.8	(298)		[68/1][77/1]
					[87/4]
					[26/1][27/1]
					[70/1]
		83.0 ± 3.8	(298)	B	
$\text{C}_{10}\text{H}_8\text{O}$	2-naphthol (α form)				
$\text{C}_{10}\text{H}_8\text{O}$	(298–312)	97.8	(305)		[87/4][60/14]
	2-naphthol (β form)				
$\text{C}_{10}\text{H}_8\text{O}$	(314–332)	87.8	(323)		[87/4][60/14]
	1,6-oxido[10]annulene				[4759-11-9]
$\text{C}_{10}\text{H}_8\text{O}_2$	1,4-naphthohydroquinone	80.4 ± 8.4		B	[69/7][77/1]
					[571-60-8]
$\text{C}_{10}\text{H}_8\text{O}_2$	1,2-dihydroxynaphthalene	119 ± 1	(381)	ME,TE	[81/4]
					[574-00-5]
$\text{C}_{10}\text{H}_8\text{O}_2$	1,3-dihydroxynaphthalene	109.3 ± 0.9	(298)	C	[88/9]
					[132-86-5]
$\text{C}_{10}\text{H}_8\text{O}_2$	2,3-dihydroxynaphthalene	116.0 ± 1.1	(298)	C	[88/9]
					[92-44-4]
$\text{C}_{10}\text{H}_9\text{N}$	1-naphthylamine	109.6 ± 1.0	(298)	C	[88/9]
		109.4 ± 0.5	(350)	ME	[79/5]
$\text{C}_{10}\text{H}_9\text{N}$	2-naphthylamine				[134-32-7]
					[47/1][70/1]
$\text{C}_{10}\text{H}_9\text{NO}$	β -cyanopropiophenone	90.0 ± 4.2		TE	[91-59-8]
					[87/4]
$\text{C}_{10}\text{H}_9\text{NO}$	2-methyl-8-hydroxyquinoline	73.9	(298)		[68/1][77/1]
		74.1 ± 1.7			[47/2][70/1]
$\text{C}_{10}\text{H}_9\text{NO}$	2-methyl-8-hydroxyquinoline	88.3 ± 4.2			[5343-98-6]
					[69/9][77/1]
$\text{C}_{10}\text{H}_9\text{NO}$	2-methyl-8-hydroxyquinoline	101.7 ± 4.2		ME	[87/4]
		108.5	(325.5)		[826-81-3]
$\text{C}_{10}\text{H}_9\text{NO}$	2-methyl-8-hydroxyquinoline				[90/1]
		132.2 ± 1.0	(433)	ME	

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{10}\text{H}_9\text{NO}$	(296–307)	139.0 ± 1.0	(298)		[90/1]
		90.4 ± 0.7		ME	[89/2]
		87.2 ± 1.9		C	[89/2]
	(308–333)	87.9		ME	[87/4][63/5]
	4-methyl-2-hydroxyquinoline				[607-66-9]
$\text{C}_{10}\text{H}_9\text{NO}_2$	(391–405)	123.1 ± 1.6	(398)	ME	[90/1]
		128.1 ± 1.6	(298)		[90/1]
$\text{C}_{10}\text{H}_9\text{NO}_2$	indole-3-acetic acid				[87-51-4]
$\text{C}_{10}\text{H}_{10}$	(313–423)	64.0 ± 1.4 U	(368)	ME	[88/14]
	bullvalene				[1005-51-2]
$\text{C}_{10}\text{H}_{10}$		71.8	(298)	C	[81/2]
	1,4-dihydronaphthalene				[612-17-9]
$\text{C}_{10}\text{H}_{10}\text{N}_2$		63.6 ± 1.6	(298)		[99/19]
	2,3-dimethylquinoxaline				[2379-55-7]
	(294–308)	87.7 ± 0.4	(301)	ME	[00/26]
		87.8 ± 0.4	(298)	ME	[00/26]
$\text{C}_{10}\text{H}_{10}\text{N}_2$		85.8 ± 1.8	(298)	C	[96/28]
	4-aminoquinaldine				[6628-04-2]
	(352–373)	112.1 ± 0.8	(363)	ME	[98/11]
$\text{C}_{10}\text{H}_{10}\text{N}_2$		115.3 ± 0.8	(298)		[98/11]
	1-benzylimidazole				[4238-71-5]
$\text{C}_{10}\text{H}_{10}\text{N}_2\text{O}_2$		102.1 ± 0.4	(298)	ME	[99/20]
	3-dimethylaminophthalimide				
$\text{C}_{10}\text{H}_{10}\text{N}_2\text{O}_2$	(392–431)	90.9	(407)	RG	[87/4][56/13]
	1-phenyl-1,3-butanedione				[93-91-4]
$\text{C}_{10}\text{H}_{10}\text{O}_2$		91.0 ± 0.6	(298)	ME	[92/28]
	(278–300)	83.7	(289)	V	[87/4][59/2]
	<i>p</i> -tolyl propadienyl sulfone				[16192-08-8]
$\text{C}_{10}\text{H}_{10}\text{O}_2\text{S}$		113 ± 2.5		B	[69/13][70/1]
	<i>p</i> -tolyl prop-1-ynyl sulfone				[14027-53-3]
$\text{C}_{10}\text{H}_{10}\text{O}_2\text{S}$		103.3 ± 2.5		B	[69/13][70/1]
	<i>p</i> -tolyl prop-2-ynyl sulfone				[16192-07-7]
$\text{C}_{10}\text{H}_{10}\text{O}_3$		107.5 ± 2.5		B	[69/13][70/1]
	<i>cis</i> -2-methoxycinnamic acid				[14737-91-8]
	(339–352)	119.3 ± 0.6	(346)	ME	[99/27]
$\text{C}_{10}\text{H}_{10}\text{O}_3$	(339–352)	121.7 ± 0.6	(298)	ME	[99/27]
	<i>trans</i> -2-methoxycinnamic acid				[6099-03-2]
$\text{C}_{10}\text{H}_{10}\text{O}_3$	(368–382)	124.9 ± 0.6	(375)	ME	[99/27]
	(368–382)	128.8 ± 0.6	(298)	ME	[99/27]
$\text{C}_{10}\text{H}_{10}\text{O}_3$	<i>trans</i> -3-methoxycinnamic acid				[6099-04-3]
	(353–367)	120.9 ± 0.9	(360)	ME	[99/27]
	(353–367)	124.0 ± 0.9	(298)	ME	[99/27]
$\text{C}_{10}\text{H}_{10}\text{O}_3$	<i>trans</i> -4-methoxycinnamic acid				[830-09-1]
	(369–383)	130.2 ± 1.0	(376)	ME	[99/27]
	(369–383)	134.0 ± 1.0	(298)	ME	[99/27]
$\text{C}_{10}\text{H}_{10}\text{O}_4$	dimethyl isophthalate				[120-61-6]
	(295–309)	100.7 ± 0.2	(302)	ME	[98/1]
		100.9 ± 0.2	(298)		[98/1]
		100.7	(298)		[98/28]
$\text{C}_{10}\text{H}_{10}\text{O}_4$	dimethyl terephthalate				[120-61-6]
	(311–330)	103.8 ± 0.3	(321)	ME	[98/1]
		104.6 ± 0.3	(298)		[98/1]
	(373–413)	94.4	(388)		[87/4]
	(373–413)	88.3	(393)	GS	[62/2]
$\text{C}_{10}\text{H}_{11}\text{N}$		105.3			[98/28]
	2,4,6-trimethylbenzonitrile				[2571-52-0]
$\text{C}_{10}\text{H}_{11}\text{NO}$		82.9 ± 1.6	(298)	C	[91/3]
	2,4,6-trimethylbenzonitrile N-oxide				[2904-59-8]
		87.5 ± 0.5	(314)	C	[93/12]
		87.9 ± 1.9	(298)		[93/12]
	(300–338)	84.7 ± 1.8	(319)	ME	[92/14]
$\text{C}_{10}\text{H}_{11}\text{NO}$		77.5 ± 3.7	(298)	C	[91/3]
	3-amino-1-phenyl-but-2-enone				[1128-85-4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
		109.4 ± 2.1	(298)	C	[93/24]
		91.9 ± 1.9	(298)	C	[91/3]
C ₁₀ H ₁₁ NO ₂	N-phenyldiacetamide				[1563-87-7]
C ₁₀ H ₁₁ NO ₃	2,4,6-trimethoxybenzonitrile	90. ± 0.8	(298)	C	[65/8][70/1]
					[2571-54-2]
C ₁₀ H ₁₁ NO ₄	2,4,6-trimethoxybenzonitrile N-oxide	112.6 ± 2.0	(298)	C	[91/3]
					[2904-59-8]
C ₁₀ H ₁₁ N ₃ O ₂	3-dimethylamino-6-aminophthalimide (434–459)	100.4 ± 2.2	(298)	C	[01/4]
C ₁₀ H ₁₂	cyclodeca-1,2,6,7-tetraene				[10495-38-2]
		108.8	(446)	RG	[87/4][56/13]
		73.0 ± 0.4	(298)	C	[2451-55-6]
C ₁₀ H ₁₂ N ₂ O ₂	acetylglycine anilide (362–365)				[91/20]
		122.1	(363.5)		[87/4][55/7]
C ₁₀ H ₁₂ O	4-phenylbutyric acid (309–323)				[1821-12-1]
		112.4 ± 0.8	(316)	ME	[01/10]
		113.0 ± 1.0	(298)	ME	[01/10]
C ₁₀ H ₁₂ O	4-methoxy- α -methylstyrene				[1712-69-2]
		81.2 ± 0.4	(298)		[99/19]
C ₁₀ H ₁₂ O ₂	2-phenyl-2-methyl-1,3-dioxolane (293–324)				[3674-77-9]
		81.9 ± 0.5	(308)	T	[95/13]
C ₁₀ H ₁₂ O ₂	2,3,6-trimethylbenzoic acid				[2529-36-4]
		104.4 ± 0.2	(298)	ME	[87/15]
	(314–336)	103.6 ± 0.2	(325)	ME	[87/15]
C ₁₀ H ₁₂ O ₂	2,4,6-trimethylbenzoic acid				[480-63-7]
		103.6 ± 0.3	(298)	ME	[87/15]
	(316–340)	102.5 ± 0.3	(328)	ME	[87/15]
C ₁₀ H ₁₂ O ₂	2,3,4-trimethylbenzoic acid				[1076-47-7]
		109.3 ± 0.3	(298)	ME	[87/15]
	(329–351)	108.2 ± 0.3	(340)	ME	[87/15]
C ₁₀ H ₁₂ O ₂	2,3,5-trimethylbenzoic acid				[2437-66-3]
		106.7 ± 0.3	(298)	ME	[87/15]
	(320–338)	105.7 ± 0.3	(329)	ME	[87/15]
C ₁₀ H ₁₂ O ₂	2,4,5-trimethylbenzoic acid				[528-90-5]
		109.6 ± 0.5	(298)	ME	[87/15]
	(324–346)	108.3 ± 0.5	(335)	ME	[87/15]
C ₁₀ H ₁₂ O ₂	3,4,5-trimethylbenzoic acid				[1076-88-6]
		111.0 ± 0.5	(298)	ME	[87/15]
	(340–359)	109.3 ± 0.5	(350)	ME	[87/15]
C ₁₀ H ₁₂ O ₂	2-isopropylbenzoic acid (300–320)				[2438-04-2]
		100.2 ± 0.4	(310)	ME	[87/11]
		101.0 ± 0.4	(298)		[87/11]
C ₁₀ H ₁₂ O ₂	3-isopropylbenzoic acid (300–316)				[5651-47-8]
		103.3 ± 0.3	(308)	ME	[87/11]
		104.1 ± 0.3	(298)		[87/11]
C ₁₀ H ₁₂ O ₂	4-isopropylbenzoic acid (316–334)				[536-66-3]
		99.0 ± 0.3	(324)	ME	[87/11]
		101.1 ± 0.3	(298)		[87/11]
C ₁₀ H ₁₂ O ₂ S	<i>p</i> -tolyl <i>trans</i> -prop-1-enyl sulfone				[32228-15-2]
		83.7 ± 2.1		B	[69/13][70/1]
C ₁₀ H ₁₂ O ₂ S	<i>p</i> -tolyl prop-2-enyl sulfone				[3112-87-6]
		95.8 ± 2.9		B	[69/13][70/1]
C ₁₀ H ₁₂ O ₂ S	<i>p</i> -tolyl isopropenyl sulfone				[67605-02-1]
		88.7 ± 2.5		B	[69/11][77/1]
C ₁₀ H ₁₂ O ₅	3,4,5-trimethoxybenzoic acid (354–372)				[118-41-2]
		127.9 ± 0.8	(363)	ME	[01/16]
		131.2 ± 0.8	(298)	ME	[01/16]
C ₁₀ H ₁₃ NO	N,N-dimethyl-3-toluamide (373–405)				[6935-65-5]
		29.9	(388)	I	[87/4][69/25]
C ₁₀ H ₁₃ NO ₂	<i>p</i> -phenacetin (4-ethoxyphenylacetamide) (312–387)				[62-44-2]
		115.5 ± 2.4		C,ME	[72/9][87/4]
C ₁₀ H ₁₄	1,2,4,5-tetramethylbenzene				[95-93-2]
		71.7 ± 0.3	(298)	C	[94/13]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{10}\text{H}_{14}$	(263–277)	74.6 ± 0.3	(298)	ME	[89/18]
	(318–348)	71.3	(333)	A	[47/4]
	1,2,3,4-tetramethylbenzene	72.4	(298)	H	[47/4][93/16]
					[488-23-3]
$\text{C}_{10}\text{H}_{14}$	1,2,3,5-tetramethylbenzene	52.6 ± 0.2 (vap)	(298)	C	[94/13]
		55.6	(298)		[94/13][90/23]
					[527-53-7]
$\text{C}_{10}\text{H}_{14}\text{NO}_5\text{PS}$	ethyl parathion (O,O-diethyl-O-4-nitrophenylthiophosphate)	52.0 ± 0.2 (vap)	(298)	C	[94/13]
		55.2	(298)		[94/13][90/23]
$\text{C}_{10}\text{H}_{14}\text{N}_2\text{O}$	4-diethylaminonitrosobenzene	(298–318)	(308)		[56-38-2]
		100.6			[79/10][83/5]
$\text{C}_{10}\text{H}_{14}\text{N}_2\text{O}_4$	2,2-dinitroadamantane	107.9 $\pm$ 3.7	(298)	C	[120-22-9]
					[98/23]
$\text{C}_{10}\text{H}_{14}\text{O}$	3- <i>tert</i> -butylphenol	96.4 ± 1.4	(298)	T	[88381-75-3]
					[90/10]
					[585-34-2]
$\text{C}_{10}\text{H}_{14}\text{O}$	4- <i>tert</i> -butylphenol	88.9 ± 0.5	(298)	C	[99/21]
		86.0 ± 0.5	(298)	GS	[99/13]
		70.7	(281)		[87/4]
					[98-54-4]
$\text{C}_{10}\text{H}_{14}\text{O}$	thymol	89.4 ± 2.5	(298)	C	[99/21]
		85.0 ± 0.5	(313)	GS	[99/13]
		85.9 ± 0.5	(298)		[99/13]
		84.3	(292)		[87/4][60/14]
					[89-83-8]
					[87/4][60/14]
					[75/3]
$\text{C}_{10}\text{H}_{14}\text{O}$	adamantan-2-one	75.1	(284)	HSA	[71/17]
		89.1 ± 4.5	(303)	TGA	[70/1][60/1]
		U 69.0	(270)	TE	[47/1]
		91.2 ± 4.1			[57/9][87/4]
					[700-58-3]
$\text{C}_{10}\text{H}_{14}\text{O}_2$	4- <i>tert</i> -butyl-1,2-dihydroxybenzene	91.5	(298)		[78/8]
		79.7 ± 2.1	(345)	BG	[78/8]
		80.3 ± 2.5	(298)	BG	[98-29-3]
$\text{C}_{10}\text{H}_{14}\text{O}_2$	2- <i>tert</i> -butyl-1,4-dihydroxybenzene	98.7 ± 0.9	(313)	GS	[00/21]
		99.2 ± 0.9	(298)	GS	[00/21]
		99.3 ± 1.4	(298)	C	[84/20]
					[1948-33-0]
$\text{C}_{10}\text{H}_{14}\text{O}_2$	3- <i>tert</i> -butyl-1,2-dihydroxybenzene	101.2 ± 1.3	(351)	GS	[99/28]
		104.4 ± 1.3	(298)		[99/28]
					[4026-05-5]
$\text{C}_{10}\text{H}_{14}\text{O}_2$	6-methyl-3-isopropyl-1,2-dihydroxybenzene	70.1 ± 0.8 (liq)	(359)	GS	[00/21]
		73.5 ± 0.8 (liq)	(298)	GS	[00/21]
					[490-06-2]
$\text{C}_{10}\text{H}_{15}\text{NO}$	(dl)-carvoxime	96.6 ± 0.9	(298)	C	[84/20]
					[55658-55-4]
$\text{C}_{10}\text{H}_{15}\text{NO}$	(d)-carvoxime	101.6 ± 5	(334)	HSA	[81/8]
					[80124-30-7]
$\text{C}_{10}\text{H}_{15}\text{NO}_2$	1-nitroadamantane	90.8 ± 4.5	(334)	HSA	[81/8]
					[7575-82-8]
$\text{C}_{10}\text{H}_{15}\text{NO}_2$	2-nitroadamantane	63.6 ± 1.0	(339)	T	[90/10]
					[54654-31-7]
$\text{C}_{10}\text{H}_{15}\text{N}_5$	6,9-dimethyl-8-propyladenine	58.0 ± 2.3	(350)	T	[90/10]
					[153495-35-3]
$\text{C}_{10}\text{H}_{15}\text{N}_5$	8-butyl-9-methyladenine	129.0 ± 0.1	(347)	ME	[94/6]
					[117954-98-0]
$\text{C}_{10}\text{H}_{16}$	adamantane (tricyclo[3.3.1.1 ^{2,6}]decane)	135.1 ± 1.2	(366)	ME	[87/7]
					[281-23-2]
		59.1	(298)		[00/10]
		58.3	(308)		[00/4]
		52.6	(298)	CGC–DSC	[98/5]
	(278–368)	59.7	(293)		[87/4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_m/\text{kJ mol}^{-1}$	(T_m/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
	(328–373)	55.3	(343)		[87/4]
	(343–483)	54.3	(358)		[87/4][68/18]
		58.45	(298)	C	[82/4]
	(278–443)	59.5	(300)		[75/30]
	(310–336)	59.7±0.8	(326)	TSGC	[75/2]
		58.6	(298)		[75/2][93/16]
	(310–336)	59.3±0.2	(326)	BG	[71/1]
		60.5±1.3	(298)		[71/1][93/16]
	(312–366)	53.6	(332)	I	[71/7]
		54.8	(298)		[71/7][93/16]
		59.3±0.16	(298)	C	[70/2]
		59.5	(298)		[70/16]
	(313–353)	58.6±0.6	(333)	DBM	[67/1]
		59.6	(298)		[67/1][93/16]
		62.3	(298)		[67/1]
C ₁₀ H ₁₆	2,2-dimethyl-3-methylenebicyclo[2.2.1]heptane (camphene)				[79-92-5]
		46.8		C	[77/12]
C ₁₀ H ₁₆	tricyclo[4.3.1.0 ^{3,8}]decane				[53130-19-1]
	(310–335)	64.9±1.8	(323)	TSGC	[75/2]
		65.6	(298)		[75/2][93/16]
C ₁₀ H ₁₆	tricyclo[5.2.1.0 ^{2,6}]decane				[6004-38-2]
	(359–443)	52.9±1.3	(298)	BG	[71/1][77/1]
C ₁₀ H ₁₆	tricyclo[5.2.1.0 ^{4,10}]decane hexahydrotriquinacene				[17760-91-7]
		56.6±1.3	(307)	TSGC	[79/13]
C ₁₀ H ₁₆ NOS	S-2,3,3-trichloroallyl N,N-diisopropylthiocarbonate (triallate)				[2303-17-5]
	(293–318)	84			[83/5][78/18]
C ₁₀ H ₁₆ N ₂	methyl(1,1,1-trimethylpropyl)propanedinitrile				[85688-96-6]
		62.0±0.7	(298)		[90/28]
C ₁₀ H ₁₆ N ₂	(1,1-dimethylpropyl)ethylpropanedinitrile				[85688-95-5]
		76.2±0.8	(298)		[90/28]
C ₁₀ H ₁₆ N ₂ O ₂	1,3-dimethyl-5-butyluracil				[82413-40-9]
	(306–311)	106.3±1.3	(309)	ME	[96/5]
C ₁₀ H ₁₆ O	(dl)-camphor				[21368-68-3]
		51.8±0.8			[77/6]
	(273–293)	51.5±2.6	(283)	HSA	[75/3]
	(273–298)	U 65.8			[60/1][40/1]
		50.7			[60/1][37/2]
	(273–453)	53.6	(363)		[60/1][10/1]
	(285–318)	54.7	(301)		[57/9]
C ₁₀ H ₁₆ O	(d)-3-bornanone				[13854-85-8]
	(273–408)	54.2	(288)		[87/4]
	(323–339)	55	(331)		[87/4]
	(408–451)	49.8	(423)		[87/4]
C ₁₀ H ₁₆ O	cis-8-methyl-2-hydrindanone				[13351-29-6]
		60.9±0.2	(298)	C	[70/6][77/1]
C ₁₀ H ₁₆ O	adamantan-1-ol				[768-95-6]
	(320–370)	86.6±2.5	(298)	BG	[78/8]
C ₁₀ H ₁₆ O	adamantan-2-ol				[700-57-2]
	(320–370)	87.9±2.1	(345)	BG	[78/8]
		88.7±2.5	(298)	BG	[78/8]
C ₁₀ H ₁₆ S	1,7,7-trimethylbicyclo[2.2.1]heptan-2-thione				[53402-10-1]
	(262–282)	62.2±0.9	(272)	ME	[99/26]
		61.7±0.9	(298)		[99/26]
C ₁₀ H ₁₆ S ₄	1,3,5,7-tetramethyl-2,4,6,8-tetrathiaadamantane				[7000-79-5]
		117.1±4.1	(298)	TE	[78/3]
C ₁₀ H ₁₇ NOS	carbamothioic acid, N-butyl-N-(2-propynyl), S-ethyl ester				[59300-35-5]
	(298–313)	82.1	(305.5)	ME	[87/4][76/11]
C ₁₀ H ₁₇ NOS	carbamothioic acid, N,N-dipropyl S-(2-propynyl) ester				[59300-36-6]
	(298–313)	92.4	(305.5)	ME	[87/4][76/11]
C ₁₀ H ₁₇ NOS	carbamothioic acid, N-2-methylpropyl-N-(2-propynyl), S-ethyl ester				[59300-34-4]
	(298–313)	74	(305.5)	ME	[87/4][76/11]
C ₁₀ H ₁₈	bicyclo[3.3.2]decane				[283-50-1]
		58.2±2	(298)		[77/9]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{10}\text{H}_{18}$	<i>cis</i> -decahydronaphthalene	64.8 62.5	(230) (298)	B	[493-01-6] [63/6] [63/6][93/16]
$\text{C}_{10}\text{H}_{18}$	<i>trans</i> -decahydronaphthalene	66.2 64.3	(241) (298)	B	[493-02-7] [63/6] [63/6][93/16]
$\text{C}_{10}\text{H}_{18}\text{O}$	α -terpineol (283–328)	80.3	(305)	ME	[10482-56-1] [54/4][60/1]
$\text{C}_{10}\text{H}_{18}\text{O}$	(<i>dl</i>)- α -terpineol (287–308)	80.1	(297)		[98-55-5] [87/4]
$\text{C}_{10}\text{H}_{18}\text{O}$	(<i>dl</i>)-borneol (350–475)	69.3	(365)		[6627-72-1] [87/4]
$\text{C}_{10}\text{H}_{18}\text{O}$	(<i>dl</i>)-isoborneol (373–457)	41.1	(388)		[124-76-5] [87/4][36/4]
$\text{C}_{10}\text{H}_{18}\text{O}_4$	sebacic acid	165.3 $\pm$ 2.9 160.7 $\pm$ 2.5	 (389)	ME	[111-20-6] [99/10][60/4] [60/4][70/1] [87/4]
$\text{C}_{10}\text{H}_{19}\text{N}_5\text{O}$	2-methoxy-4,6- <i>bis</i> (isopropylamino)-1,3,5-triazine (323–365)	92.2	(338)	GS–GC	[1610-18-0] [87/4][64/14]
$\text{C}_{10}\text{H}_{19}\text{N}_5\text{S}$	2-methylthio-4,6- <i>bis</i> (isopropylamino)-1,3,5-triazine (323–393)	100	(338)	GS–GC	[7287-19-6] [87/4][64/14]
$\text{C}_{10}\text{H}_{20}\text{N}_2\text{OS}$	N,N-diethyl-N'-isovalerylthiourea (363)	121.5 $\pm$ 3.2	(298)	C	[01/9]
$\text{C}_{10}\text{H}_{20}\text{N}_2\text{OS}$	N,N-diethyl-N'-pivaloylthiourea (366)	114.9 $\pm$ 2.7	(298)	C	[01/9]
$\text{C}_{10}\text{H}_{20}\text{O}$	citronellol (283–333)	66.1	(308)		[106-22-9] [54/4][60/1]
$\text{C}_{10}\text{H}_{20}\text{O}$	cyclodecanol (287–292)	100.5 $\pm$ 0.5	(288)	TM	[1502-05-2] [55/5]
$\text{C}_{10}\text{H}_{20}\text{O}$	(<i>dl</i>)-menthol (279–299)	78.6 $\pm$ 4	(289)	HSA	[89-78-1] [81/8]
$\text{C}_{10}\text{H}_{20}\text{O}$	(<i>l</i>)-menthol (279–299)	95.8 $\pm$ 4.8	(289)	HSA	[2216-51-5] [81/8]
$\text{C}_{10}\text{H}_{20}\text{O}_2$	decanoic acid (293–303)	118.8 $\pm$ 2.2	(298)	ME	[334-48-5] [68/2][70/1] [87/4]
	(290–301)	117.1 $\pm$ 1.7	(295)	ME	[61/1]
$\text{C}_{10}\text{H}_{20}\text{O}_3$	peroxydecanoic acid (293–303)	117.1 $\pm$ 0.8	(298)	ME	[14156-10-6] [80/23]
$\text{C}_{10}\text{H}_{21}\text{NO}$	decanamide (353–370)	125.9 $\pm$ 1.3	(361.5)	ME	[2319-29-1] [59/3]
$\text{C}_{10}\text{H}_{22}$	decane	80.3 84.8 82.4	(298) (243) (298)	B B	[124-18-5] [80/23] [63/6] [63/6][93/16]
$\text{C}_{10}\text{H}_{22}\text{O}$	decanol (264–273)	115.5 $\pm$ 6.3 112.5 $\pm$ 6.3	(268) (298)	ME	[112-30-1] [65/5][87/4] [65/5]
$\text{C}_{10}\text{H}_{22}\text{O}_2$	1,10-decanediol	155.8 $\pm$ 0.9	(298)	C	[112-47-0] [90/14]
$\text{C}_{10}\text{H}_{24}\text{N}_4$	1,4,8,11-tetraazacyclotetradecane (352–372)	133.9 $\pm$ 2.5	(362)	TE	[295-37-4] [83/4]
$\text{C}_{11}\text{H}_6\text{N}_4$	bicyclo[2.2.1]hept-5-ene-2,2,3,3-tetracarbonitrile	117.2 $\pm$ 5.4	(408)	MG	[6343-21-1] [72/13][77/1]
$\text{C}_{11}\text{H}_7\text{N}_2$	2,2-dicyano-3-phenylpropionitrile (318–388)	96.2 $\pm$ 0.4	(353)	T	[6023-46-7] [94/44]
$\text{C}_{11}\text{H}_8\text{N}_4$	3-methyl-1,1,2,2-tetracyanocyclohex-4-ene	82 $\pm$ 2.1	(350)	MG	[13358-02-6] [71/14][77/1]
$\text{C}_{11}\text{H}_8\text{O}_2$	1-naphthoic acid	117.6 $\pm$ 0.4 110.4 $\pm$ 0.2	 (355)	DSC ME	[86-55-5] [83/8] [74/2][77/1] [87/4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{11}\text{H}_8\text{O}_2$	2-naphthoic acid	113.6	(298)	C	[74/28]
		119.5 $\pm$ 0.6		DSC	[93-09-4]
	(347–363)	113.6 $\pm$ 0.8	(365)	ME	[83/8] [74/2][77/1] [87/4]
$\text{C}_{11}\text{H}_9\text{N}$	4-phenylpyridine	117.2	(298)	C	[74/28]
		81.4 $\pm$ 1.6	(298)	BE	[939-23-1] [00/7]
$\text{C}_{11}\text{H}_9\text{NO}_2$	1-(4-methylphenyl)-1 <i>H</i> -pyrrole-2,5-dione	104.6 $\pm$ 0.8		C	[1631-28-3] [98/25]
$\text{C}_{11}\text{H}_9\text{NO}_3$	1-(4-methoxyphenyl)-1 <i>H</i> -pyrrole-2,5-dione	121.1 $\pm$ 0.8		C	[1081-17-0] [98/25]
$\text{C}_{11}\text{H}_{10}$	2-methylnaphthalene	65.7 $\pm$ 0.85		C	[91-57-6] [74/10]
		61.7 $\pm$ 1.7			[68/1][77/1] [13297-17-1]
$\text{C}_{11}\text{H}_{10}\text{N}_2\text{O}_3$	2-methyl-3-acetylquinoxaline-1,4-dioxide	117.0 $\pm$ 2.4	(298)	ME	[97/25]
$\text{C}_{11}\text{H}_{10}\text{N}_2\text{O}_3$	2-methyl-3-carboxymethoxyquinoxaline-1,4-dioxide	118.3 $\pm$ 2.6	(298)	C	[40016-70-4] [97/25]
$\text{C}_{11}\text{H}_{10}\text{O}_2$	pentacyclo[5.4.0 ^{2.6} 0 ^{3.10} 0 ^{5.9}]undecane-8,11-dione	92.6 $\pm$ 1.0	(298)	ME	[2985-72-7] [99/4]
$\text{C}_{11}\text{H}_{11}\text{N}$	2,6-dimethylquinoline	84.5 $\pm$ 1.5	(298)	C	[877-43-0] [95/27]
$\text{C}_{11}\text{H}_{11}\text{N}$	2,7-dimethylquinoline	87.5 $\pm$ 1.5	(298)	C	[93-37-8] [95/27]
$\text{C}_{11}\text{H}_{11}\text{N}_3\text{O}$	3,5-dimethyl-1-phenyl-4-nitrosopyrazole	100.4 $\pm$ 2.2	(298)	C	[5809-38-1] [01/4]
$\text{C}_{11}\text{H}_{12}\text{N}_2\text{O}_3$	4-[(4-nitrophenyl)amino]pent-3-ene-2-one	121.9 $\pm$ 3.9	(298)	C	[20771-72-6] [93/24]
$\text{C}_{11}\text{H}_{12}\text{O}_2$	1-phenyl-4,7-dioxaspiro[2.4]heptane	91.8 $\pm$ 0.8	(298)		[39522-76-4] [98/26]
$\text{C}_{11}\text{H}_{12}\text{O}_4$	<i>trans</i> -2,3-dimethoxycinnamic acid	136.6 $\pm$ 0.9	(386)	ME	[7345-82-6] [99/27]
	(380–392)	141.0 $\pm$ 0.9	(298)	ME	[99/27]
$\text{C}_{11}\text{H}_{12}\text{O}_4$	<i>trans</i> -2,4-dimethoxycinnamic acid	144.2 $\pm$ 1.3	(398)	ME	[16909-09-4] [99/27]
	(391–404)	149.2 $\pm$ 1.3	(298)	ME	[99/27]
$\text{C}_{11}\text{H}_{12}\text{O}_4$	<i>trans</i> -2,5-dimethoxycinnamic acid	134.5 $\pm$ 1.1	(384)	ME	[10538-51-9] [99/27]
	(376–391)	138.8 $\pm$ 1.1	(298)	ME	[99/27]
$\text{C}_{11}\text{H}_{12}\text{O}_4$	<i>trans</i> -3,4-dimethoxycinnamic acid	144.9 $\pm$ 0.8	(397)	ME	[2316-26-9] [99/27]
	(390–404)	149.9 $\pm$ 0.8	(298)	ME	[99/27]
$\text{C}_{11}\text{H}_{12}\text{O}_4$	<i>trans</i> -3,5-dimethoxycinnamic acid	136.7 $\pm$ 0.5	(391)	ME	[16909-11-8] [99/27]
	(385–397)	141.4 $\pm$ 0.5	(298)	ME	[99/27]
$\text{C}_{11}\text{H}_{12}\text{N}_2\text{O}_2$	L-tryptophane	87.9 $\pm$ 8 U	(390)	LE	[73-22-3] [77/2]
	(E)-3-(methylamino)-1-phenyl-but-2-en-1-one	99.2 $\pm$ 4.2	(298)	C	[14091-93-1] [93/24]
$\text{C}_{11}\text{H}_{13}\text{NO}$	4-phenylaminopent-3-ene-2-one	89.9 $\pm$ 3.8	(298)	C	[7294-89-5] [93/24]
$\text{C}_{11}\text{H}_{14}$	pentacyclo[5.4.0 ^{2.6} 0 ^{3.10} 0 ^{5.9}]undecane	54.7 $\pm$ 0.9	(337)	C	[4421-32-3] [95/10]
	(273–323)	54.9 $\pm$ 1.1	(298)	ME	[95/10]
$\text{C}_{11}\text{H}_{14}\text{N}_2\text{O}_2$	4-nitrobenzylidene <i>tert</i> -butylamine	91.1 $\pm$ 3.1	(298)	C	[718-36-5] [89/15]
$\text{C}_{11}\text{H}_{14}\text{N}_2\text{O}_2$	2-cyano-2-nitroadamantane	70.0 $\pm$ 1.9	(338)	T	[128478-71-7] [90/10]
$\text{C}_{11}\text{H}_{14}\text{N}_2\text{O}_3$	4-nitrobenzylidene <i>tert</i> -butylamine N-oxide	116.5 $\pm$ 3.1	(298)	C	[3585-88-4] [89/15]
$\text{C}_{11}\text{H}_{14}\text{O}$	5-phenylvaleric acid	118.5 $\pm$ 0.8	(321)	ME	[2270-20-4] [01/10]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{11}\text{H}_{14}\text{O}_2$	2- <i>tert</i> -butylbenzoic acid	119.4 ± 1.1	(298)	ME	[01/10]
	(306–322)				[1077-58-3]
$\text{C}_{11}\text{H}_{14}\text{O}_2$	3- <i>tert</i> -butylbenzoic acid	99.8 ± 0.4	(315)	ME	[79/4]
	(318–335)	103 ± 0.5	(327)	ME	[7498-54-6]
$\text{C}_{11}\text{H}_{14}\text{O}_2$	4- <i>tert</i> -butylbenzoic acid				[79/4]
	(325–343)	103.8 ± 0.4	(334)	ME	[98-73-7]
$\text{C}_{11}\text{H}_{14}\text{O}_2$	2,3,4,5-tetramethylbenzoic acid				[2529-39-7]
	(337–360)	113.4 ± 0.6	(348)	ME	[88/10]
$\text{C}_{11}\text{H}_{14}\text{O}_2$	2,3,4,6-tetramethylbenzoic acid	115.9 ± 0.6	(298)		[88/10]
					[2408-38-0]
$\text{C}_{11}\text{H}_{14}\text{O}_2$	(330–351)	106.9 ± 0.5	(341)	ME	[88/10]
		109.7 ± 0.5	(298)		[88/10]
$\text{C}_{11}\text{H}_{14}\text{O}_2$	2,3,5,6-tetramethylbenzoic acid				[2604-45-7]
		(330–351)	(341)	ME	[88/10]
$\text{C}_{11}\text{H}_{14}\text{O}_2$	3,5-diethylbenzoic acid	104.6 ± 0.8	(341)	ME	[88/10]
		(325–343)	(298)		[88/10]
$\text{C}_{11}\text{H}_{14}\text{O}_2$	(325–343)	106.1 ± 0.8			[3854-90-8]
			(334)		[74/15][77/1]
$\text{C}_{11}\text{H}_{14}\text{O}_2\text{S}$	<i>p</i> -tolyl but-1-enyl sulfone				[87/4]
$\text{C}_{11}\text{H}_{14}\text{O}_2\text{S}$	<i>p</i> -tolyl but-2-enyl sulfone	$111.895-49-9$		B	
					[69/11][70/1]
$\text{C}_{11}\text{H}_{14}\text{O}_2\text{S}$	<i>p</i> -tolyl but-3-enyl sulfone	107.5 ± 2.5		B	[24931-66-6]
					[69/13][70/1]
$\text{C}_{11}\text{H}_{14}\text{O}_2\text{S}$	<i>p</i> -tolyl-isobutenyl sulfone	113.4 ± 2.9		B	[17482-19-8]
					[69/13][70/1]
$\text{C}_{11}\text{H}_{14}\text{O}_2\text{S}$	<i>p</i> -tolyl 2-methylprop-2-enyl sulfone	$161.92-03-3$		B	
					[69/11][77/1]
$\text{C}_{11}\text{H}_{15}\text{N}$	1-adamantyl-1-carbonitrile	102.1 ± 2.5			[16192-04-4]
					[69/13][70/1]
$\text{C}_{11}\text{H}_{15}\text{NO}$	benzylidene <i>tert</i> -butylamine N-oxide	106.7 ± 2.9			[23074-42-2]
					[92/26]
$\text{C}_{11}\text{H}_{15}\text{NS}$	N,N-diethylthiobenzamide	67.1 ± 0.8	(303)	ME	[92/26]
		(294–312)	(298)		[3376-24-7]
$\text{C}_{11}\text{H}_{16}$	pentamethylbenzene	86.8 ± 0.9	(298)	C	[89/15]
					[18775-06-9]
$\text{C}_{11}\text{H}_{16}\text{N}_2\text{O}_2$	1,3-dimethyl-5,6-pentamethyleneuracil	91.4 ± 3.2	(298)	C	[89/11]
					[700-12-9]
$\text{C}_{11}\text{H}_{16}\text{O}$	2- <i>tert</i> -butyl-4-methylphenol	71.6 ± 0.1	(298)	C	[94/13]
		(288–318)	(298)	ME	[89/18]
$\text{C}_{11}\text{H}_{16}\text{O}$	(277–294)	77.4 ± 0.4	(298)		[82413-41-0]
					[83/14]
$\text{C}_{11}\text{H}_{16}\text{O}$	2- <i>tert</i> -butyl-5-methylphenol	111.9 ± 0.2	(346)	ME	[80/19][83/14]
		(335–358)	(330)	QR	[80/19][83/14]
$\text{C}_{11}\text{H}_{16}\text{O}$	(288–318)	108.8 ± 5	(355)	MS	[80/19][83/14]
		(274–294)			[2409-55-4]
$\text{C}_{11}\text{H}_{16}\text{O}$	4- <i>tert</i> -amylphenol	82.6 ± 0.5	(303)	GS	[99/13]
		(293–333)	(298)		[99/13]
$\text{C}_{11}\text{H}_{16}\text{O}$	(282–313)	82.9 ± 0.5	(284)		[87/4][60/14]
					[88-60-8]
$\text{C}_{11}\text{H}_{16}\text{O}$	1-(2,4,6-trimethylphenyl)ethanol	77.4	(287)	GS	[99/13]
		(336–354)	(298)		[99/13]
$\text{C}_{11}\text{H}_{17}\text{NO}$	1-adamantyl carboxamide	80.4 ± 1.3	(287)		[80-46-6]
		(336–354)	(313)	GS	[99/13]
$\text{C}_{11}\text{H}_{17}\text{N}_5$	6,9-dimethyl-8-butyladenine	88.3 ± 0.5	(298)		[31108-34-6]
					[87/4]
$\text{C}_{11}\text{H}_{18}$	1-methyladamantane	$U\ 5.7$	(297)		[5511-18-2]
					[89/6]
$\text{C}_{11}\text{H}_{18}$	(300–342)	105.9 ± 0.5	(345)	ME	[89/6]
		(306–336)	(298)		[153495-36-4]
$\text{C}_{11}\text{H}_{18}$	(306–336)	108.0 ± 0.5	(351)	ME	[94/6]
					[768-91-2]
$\text{C}_{11}\text{H}_{18}$	(306–336)	67.8 ± 1.3	(298)	BG	[77/11]
			(321)		[75/2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₁₁ H ₁₈	2-methyladamantane (310–330)	67.5 ± 2.1	(320)		[700-56-1] [75/2]
	(300–340)	68.2 ± 1.3	(298)		[77/11]
C ₁₁ H ₂₀	bicyclo[3.3.3]undecane	63.6 ± 0.8	(298)	C	[180-43-8] [75/7][77/1]
C ₁₁ H ₂₀ O ₄	undecanedioic acid (371–381)	158.6 ± 1.9 162.5 ± 1.9	(376) (298)	ME	[1852-04-6] [99/10] [99/10]
C ₁₁ H ₂₂ O ₂	undecanoic acid (303–308)	121.3 ± 1.3	(298)	ME	[112-37-8] [68/2][70/1]
C ₁₁ H ₂₂ O ₃	peroxyundecanoic acid (293–303)	125.9 ± 3.4	(298)	ME	[676-08-4] [80/23]
C ₁₁ H ₂₃ NO	N-methyl decanamide (303–325)	102.8 ± 0.8	(314)	GS	[59/4][87/4]
C ₁₁ H ₂₄	undecane	91.5	(236)	B	[1120-21-4] [63/6]
C ₁₂ Cl ₈ O	octachlorodibenzofuran (373–474)	149.4	(423)	T	[39001-02-2] [89/20][86/7]
C ₁₂ Cl ₈ O ₂	octachlorodibenzo[b,e][1,4]dioxin (393–573)	149.8	(483)	T	[3268-87-9] [89/20][86/7]
C ₁₂ F ₁₀	decafluorobiphenyl (298–323)	85.3 ± 2.3	(310)	ME	[434-90-2] [74/9][87/4]
C ₁₂ F ₁₈	hexakis(trifluoromethyl)tetracyclo[2.2.0.0 ^{2,6} .0 ^{2,5}]hexane (293–306)	49.2	(299.5)	I	[22736-20-5] [87/4][70/24]
C ₁₂ Cl ₁₀	decachlorobiphenyl (324–363)	121.8	(343)	GS	[2051-24-3] [84/26]
C ₁₂ H ₂ Cl ₈	2,2',3,3',5,5',6,6'-octachlorobiphenyl (302–334)	101.7	(318)	GS	[2136-99-4] [84/26]
C ₁₂ H ₄ Cl ₄ O	1,2,3,4-tetrachlorodibenzofuran (333–393)	118.5	(363)	T	[24478-72-6] [89/20][86/7]
C ₁₂ H ₄ Cl ₄ O	2,3,7,8-tetrachlorodibenzofuran (303–344)	124.0	(323)	T	[51207-31-9] [89/20][86/7]
C ₁₂ H ₄ Cl ₄ O ₂	2,3,7,8-tetrachlorodibenzo[b,e][1,4]dioxin	124.0	(578)		[1746-01-6] [85/19]
C ₁₂ H ₄ Cl ₆	2,2',4,4',6,6'-hexachlorobiphenyl (263–303)	103.4 ± 2.3	(283)	GS	[33976-03-2] [94/1]
C ₁₂ H ₄ N ₄	7,7,8,8-tetracyanoquinodimethane	79.0		TGA	[1518-16-7] [95/35]
	(452–553)	108 ± 2	(500)	T	[84/23]
	(382–464)	122 ± 2	(423)	ME	[84/23]
		126.1 ± 1	(413)	ME, TE	[80/22]
	(433–499)	104.8 ± 9.2	(465)	MG	[63/4][70/1] [80/23][87/4]
C ₁₂ H ₅ Cl ₃ O ₂	1,3,7-trichlorodibenzo[b,e][1,4]dioxin (310–373)	116.2	(342)	T	[67028-17-5] [89/20][86/7]
C ₁₂ H ₅ Cl ₃ O ₂	1,2,4-trichlorodibenzo[b,e][1,4]dioxin (310–374)	118.8	(342)	T	[39227-58-2] [89/20][86/7]
C ₁₂ H ₅ Cl ₅	2,2',4,5,5'-pentachlorobiphenyl (303–313)	92.7	(308)	GS	[37680-73-2] [81/19]
C ₁₂ H ₆ Cl ₂ O	3,6-dichlorodibenzofuran (305–374)	110.9	(340)	T	[74919-40-4] [89/20][86/7]
C ₁₂ H ₆ Cl ₂ O ₂	2,3-dichlorodibenzo[b,e][1,4]dioxin	108.6 ± 1.0 107.2 ± 0.8 108.6 ± 1.0 106.2	(298) (358) (298) (340)	C C T	[29446-15-9] [99/38] [98/10] [98/10] [89/20][86/7]
C ₁₂ H ₆ Cl ₂ O ₂	2,7-dichlorodibenzo[b,e][1,4]dioxin (314–374)	105.5	(344)	T	[33857-26-0] [89/20][86/7]
C ₁₂ H ₆ Cl ₂ O ₂	2,8-dichlorodibenzo[b,e][1,4]dioxin (305–363)	109.0	(334)	T	[38964-22-6] [89/20][86/7]
C ₁₂ H ₆ Cl ₄	2,2',5,5'-tetrachlorobiphenyl (303–312)	94.6	(308)	GS	[35693-99-3] [81/19]
C ₁₂ H ₆ Cl ₄	2,3,4,5-tetrachlorobiphenyl				[33284-53-6]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{12}\text{H}_7\text{ClO}_2$	(253–393)	88.7 ± 1.2	(273)	GS	[94/1]
	1-chlorodibenzo[b,e][1,4]dioxin	95.2 ± 1.1	(298)	C	[39227-53-7]
		95.2			[99/38]
$\text{C}_{12}\text{H}_7\text{ClO}_2$	(303–338)	98.6	(321)	T	[98/27]
	2-chlorodibenzo[b,e][1,4]dioxin				[89/20][86/7]
					[39227-54-8]
C_{12}H_8		97.2	(298)	C	[99/38]
		97.2 ± 0.6	(298)	C	[96/13]
	(305–348)	97.2	(327)	T	[89/20][86/7]
C_{12}H_8	acenaphthylene				[208-96-8]
		70.0	(298)	CGC–DSC	[98/5]
	(313–453)	77.2	(383)	GS	[95/7]
C_{12}H_8	(238–323)	73.2 ± 0.5	(303)	GS	[83/11]
		73.0 ± 0.4	(298)	C	[72/1]
	(286–318)	71.1 ± 1.3			[70/1][87/4]
C_{12}H_8	biphenylene				[65/3]
	(313–453)	82.7	(383)	GS	[259-79-0]
	(309–336)	U 104.5	(319)		[95/7]
$\text{C}_{12}\text{H}_8\text{Cl}_2$		87.3 ± 0.3	(298)	B	[89/9]
		$83.8 \pm .3$		C	[80/27]
	(371–381)	U 128.9 ± 2	(376)		[72/1]
$\text{C}_{12}\text{H}_8\text{Cl}_2$	2,2'-dichlorobiphenyl				[55/1][70/1]
	(310–328)	96.1	(314)	ME	[87/4]
	(310–328)	96.2 ± 4.2	(298)	ME	[13029-08-8]
$\text{C}_{12}\text{H}_8\text{Cl}_2$	4,4'-dichlorobiphenyl				[64/8]
	(263–303)	95.3 ± 1.3	(283)	GS	[64/8][70/1]
	(303–360)	103.7	(331)	ME	[87/4]
$\text{C}_{12}\text{H}_8\text{Cl}_6$	(303–360)	103.8 ± 4.2	(298)	ME	[2050-68-2]
	1,2,3,4,10,10-hexachloro-1,4,4a,5,8,8a-hexahydro- <i>endo-exo</i> -1,4:5,8-dimethylnaphthalene (aldrin)				[94/1]
	(309–343)	91.8	(326)	GS	[64/8][87/4]
$\text{C}_{12}\text{H}_8\text{Cl}_6\text{O}$	1,2,3,4,10,10-hexachloro-6,7-epoxy-1,4,4a,5,6,7,8,8a-octahydro-1,4- <i>endo-exo</i> -5,8-dimethanonaphthalene (dieldrin)				[60-57-1]
	(308–348)	93.8	(328)	GS	[82/23]
	(293–313)	98.7	(303)	GS	[69/4]
$\text{C}_{12}\text{H}_8\text{F}_2$	2,2'-difluorobiphenyl				[388-82-9]
	(301–319)	95.1	(310)		[87/4][64/8]
	(301–318)	95 ± 4.2	(298)	ME	[64/8][70/1]
$\text{C}_{12}\text{H}_8\text{F}_2$	4,4'-difluorobiphenyl				[2050-68-2]
	(294–318)	91.4	(306)	ME	[64/8]
	(294–318)	91.2 ± 4.2	(298)	ME	[64/8][70/1]
$\text{C}_{12}\text{H}_8\text{N}_2$	1,10-phenanthroline				[66-71-7]
		98.3		ME	[72/10]
					[92-82-0]
$\text{C}_{12}\text{H}_8\text{N}_2$	phenazine				[91/4]
		92.7 ± 0.4	(354)		[91/4]
		97.0 ± 0.4	(298)		[91/4]
$\text{C}_{12}\text{H}_8\text{N}_2$		91.8 ± 2.1	(298)	C	[90/9]
	(280–318)	92.4	(295)		[87/4]
		99.9 ± 2.5		ME,GS	[80/6]
$\text{C}_{12}\text{H}_8\text{N}_2\text{O}$	(303–328)	90.4 ± 2.5	(298)	TE	[75/5]
	(303–323)	90.0 ± 1.5	(298)	TCM	[U/1][75/5]
	(281–293)	90.4 ± 1.7		LE	[75/1]
$\text{C}_{12}\text{H}_8\text{N}_2$	benzo[c]cinnoline				[46/1]
	(320–360)	$101.7 \pm .2$	(340)	ME	[230-17-1]
		113		ME	[77/14][87/4]
$\text{C}_{12}\text{H}_8\text{N}_2\text{O}$	phenazine-N-oxide				[72/10]
		$100. \pm 1.3$	(298)	C	[304-81-4]
					[90/9]
$\text{C}_{12}\text{H}_8\text{N}_2\text{O}_4$	4,4'-dinitrobiphenyl				[1528-74-1]
	(441–428)	104.6 ± 1.8	(420)	ME	[53/10][60/1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{12}\text{H}_8\text{N}_4$	bicyclo[2.2.2]oct-5-ene-2,2,3,3-tetracarbonitrile	111.7 ± 5.4	(433)		[1017-93-2] [72/13][77/1]
$\text{C}_{12}\text{H}_8\text{N}_4$	dibenzo-1,3a,4,6a-tetraazapentalene (363–433)	70.3 ± 1.7	(400)		[67/6]
$\text{C}_{12}\text{H}_8\text{N}_4$	dibenzo-1,3a,6,6a-tetraazapentalene (363–443)	42.3 ± 3.4	(403)		[67/6]
$\text{C}_{12}\text{H}_8\text{O}$	dibenzofuran	84.4 ± 0.7	(298)		[132-64-9] [90/6]
		76.5 ± 0.2	(298)		[87/10]
	(304–343)	85.6	(324)	T	[89/20][86/7]
	(303–343)	79.1	(323)	GS	[86/8]
		88.7 ± 2.1			[58/2]
$\text{C}_{12}\text{H}_8\text{O}_2$	dibenzo[b,e][1,4]dioxin	89.6 ± 0.7	(298)	C	[262-12-4] [99/38]
		89.6 ± 0.7	(318)	C	[97/37]
	(303–333)	92.3	(318)	T	[89/20][86/7]
$\text{C}_{12}\text{H}_8\text{S}$	dibenzothiophene	85.1 ± 0.4	(298)	C	[132-65-0] [87/10][79/2]
	(303–348)	91.2	(325)	GS	[86/8]
	(336–366)	90.7			[81/14]
$\text{C}_{12}\text{H}_8\text{S}_2$	thianthrene	103.6 ± 0.4	(350)	IPM	[92-85-3] [93/1]
	(338–368)	98.6 ± 0.5	(353)		[89/4]
		99.4 ± 0.6	(298)		[89/4]
	(358–426)	98.0	(393)	GS	[81/14]
	(338–368)	97.5 ± 6.3	(353)	HSA	[79/7]
$\text{C}_{12}\text{H}_9\text{Cl}$	4-chlorobiphenyl	86.0 ± 0.9	(278)	GS	[2051-62-9] [94/1]
	(253–303)	73.7 ± 0.7	(326)	TE,ME	[83/25]
$\text{C}_{12}\text{H}_9\text{N}$	carbazole	101.2 ± 1.1	(355)	ME	[86-74-8] [90/8]
	(346–364)	103.3 ± 1.1	(298)	ME	[90/8]
		97.7 ± 0.3	(298)	C	[87/10]
		$84.5 \pm .8$			[55/2][70/1]
$\text{C}_{12}\text{H}_9\text{NS}$	phenothiazine	86	(351)		[92-84-2] [87/4][42/1]
$\text{C}_{12}\text{H}_9\text{N}_3\text{O}_2$	4-nitroazobenzene	110.0		GS	[2491-52-3] [87/19][91/18]
$\text{C}_{12}\text{H}_9\text{N}_3\text{O}_3$	4-hydroxy-4'-nitroazobenzene	140.1		GS	[1435-60-5] [87/19][91/18]
	(417–444)	143.8 ± 1.3	(430.5)	TE	[87/4][70/4]
		144 ± 2.5		TE,ME	[70/4]
		136.8			[68/10][88/24]
$\text{C}_{12}\text{H}_9\text{N}_3\text{O}_4$	N-(2,4-dinitrophenyl)-N-phenylamine	147.6 ± 1.7	(411)	TE	[961-68-2] [87/4][70/4]
	(402–420)	149 ± 3.0		TE,ME	[70/4]
		131.8			[68/10][88/24]
$\text{C}_{12}\text{H}_9\text{N}_3\text{O}_5$	N-(2,4-dinitrophenyl)-N-(4-hydroxyphenyl)amine	155.6 ± 4.2	(455)	TE,ME	[119-15-3] [70/4][87/4]
	(440–470)	154.0			[68/10][88/24]
$\text{C}_{12}\text{H}_9\text{N}_4\text{O}_4$	N-(4-aminophenyl)-N-(2,4-dinitrophenyl)amine	156.5 ± 2.1	(448.5)	TE	[961-68-2] [87/4][70/4]
	(437–460)	154 ± 2.9		TE,ME	[70/4]
		139.3			[68/10][88/24]
$\text{C}_{12}\text{H}_{10}$	acenaphthene	84.6	(298)	CGC–DSC	[83-32-9] [98/5]
	(313–453)	83.2	(383)	GS	[95/7]
	(283–323)	86.8 ± 0.9	(303)	GS	[83/11]
		83.4 ± 1.0	(298)		[75/8][77/7]
		82.4	(366)	B,IPM	[75/8]
	(327–356)	84.7 ± 2.7	(341)	ME	[74/9]
	(290–340)	86.2 ± 0.8		ME	[65/3][70/1]
	(291–310)	82.1 ± 0.4	(300)	V	[59/2][87/4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{12}\text{H}_{10}$	(258–308)	81.6			[58/1]
	biphenyl				[92-52-4]
		82.9	(298)	CGC–DSC	[98/5]
	(313–453)	81.8	(383)	GS	[95/7]
	(283–338)	83.4	(311)	EM	[89/1]
	(303–333)	U 113.3	(318)		[89/9]
		81.5 ± 0.2	(298)		[89/5]
		77.9 ± 0.3	(298)	C	[79/3]
		81.8 ± 0.2	(298)	C	[78/34]
	(306–332)	80.4 ± 1.6	(319)	TSGC	[75/2]
	(273–313)	76.0 ± 4.0		HSA	[75/3]
	(298–323)	83.6 ± 2.5	(310)	ME	[74/9]
	(298–318)	75.2		ME	[74/6]
		81.8 ± 0.4	(298)	C	[72/1]
	(279–299)	75.8 ± 0.6	(289)		[55/3]
		81.6 ± 2			[53/1][70/1]
					[60/1]
	(287–307)	75.1 ± 1.7	(297)		[53/10]
	(288–314)	81.6 ± 1.7	(301)		[53/13]
	(278–307)	72.8 ± 3	(302)	ME	[51/1]
$\text{C}_{12}\text{H}_{10}\text{N}_2$	<i>cis</i> -azobenzene	68.6 ± 0.8	(292)	QF	[38/1]
					[1080-16-6]
	(273–323)	92.9	(288)		[87/4]
	(298–357)	92.9 ± 1.2	(328)	ME	[77/14]
$\text{C}_{12}\text{H}_{10}\text{N}_2$	<i>trans</i> -azobenzene	U 74.9	(318)	ME	[50/3][60/1]
					[17082-12-1]
		94.1 ± 0.8	(298)	B	[96/11]
	(298–302)	93.6 ± 1.9	(298)	ME	[92/4]
	(298–341)	92.1 ± 0.9	(319)	TE,ME	[84/18]
	(299–317)	96.9 ± 0.8	(308)	TE	[77/4]
	(299–317)	94.9 ± 0.8	(308)	ME	[77/4]
	(298–347)	93.8 ± 1.2	(323)	ME	[77/14]
	(303–333)	U74.9	(318)		[50/3][60/1]
					[87/4]
$\text{C}_{12}\text{H}_{10}\text{N}_2\text{O}$	<i>trans</i> -diphenyldiazene-N-oxide	98.6 ± 0.9	(298)	C	[21650-65-7]
$\text{C}_{12}\text{H}_{10}\text{N}_2\text{O}_2$	N-(2-nitrophenyl)-N-phenylamine				[86/10]
	(335–346)	100.9 ± 2.1	(340.5)	TE	[119-75-5]
		101.9 ± 1.7		TE,ME	[87/4][70/4]
		108.4			[70/4]
$\text{C}_{12}\text{H}_{10}\text{N}_2\text{O}_2$	N-(4-nitrophenyl)-N-phenylamine				[68/10][88/24]
	(382–403)	130.6 ± 1.3	(392.5)	TE	[836-30-6]
		126.2 ± 1.6		TE,ME	[87/4][70/4]
		120.9			[70/4]
$\text{C}_{12}\text{H}_{10}\text{N}_4$	4,5-dimethyl-1,1,2,2-tetracyanocyclohex-4-ene				[68/10][88/24]
$\text{C}_{12}\text{H}_{10}\text{N}_4\text{O}_2$		107.9 ± 4.2	(378)		[69155-29-9]
					[72/13][77/1]
$\text{C}_{12}\text{H}_{10}\text{N}_4\text{O}_2$	4-amino-4'-nitroazobenzene				[730-40-5]
	(403–465)	123	(434)	GS	[89/29]
		140.1		GS	[87/19][91/18]
		127.6		UV	[84/39][84/40]
		136.4		ME	[80/25][91/18]
		140.2 ± 1.2		TE,ME	[70/4]
		134.3		ME	[68/10][88/24]
	(404–424)	137.7 ± 0.8	(414)	TE	[67/7][87/4]
					[70/4]
	(404–423)	136.4 ± 5.0	(413)	ME	[67/7][66/18]
$\text{C}_{12}\text{H}_{10}\text{O}$	2-acetylnaphthalene				[93-08-3]
$\text{C}_{12}\text{H}_{10}\text{O}$		87.9 ± 0.4	(305)	V	[59/2][87/4]
	diphenyl ether				[101-84-8]
$\text{C}_{12}\text{H}_{10}\text{O}$		82 ± 2.1		E	[58/2][70/1]
					[90-43-7]
	2-phenylphenol				[98/9]
$\text{C}_{12}\text{H}_{10}\text{O}$	(301–328)	87.6 ± 0.9	(314)	T	[98/9]
		88.5 ± 0.9	(298)		[98/9]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_m/\text{kJ mol}^{-1}$	(T_m/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{12}\text{H}_{10}\text{O}$	(292–314)	82.9	(303)		[87/4][60/14]
	4-phenylphenol				[92-69-3]
	(333–368)	106.6 ± 1.0	(351)	T	[98/9]
		109.8 ± 1.0	(298)		[98/9]
$\text{C}_{12}\text{H}_{10}\text{O}_2$	(327–348)	97.0	(337.5)		[87/4][60/14]
	2,2'-dihydroxybiphenyl				[1806-29-7]
	(334–363)	111.4 ± 1.2	(349)	T	[98/9]
		114.4 ± 1.2	(298)		[98/9]
$\text{C}_{12}\text{H}_{10}\text{O}_2$	4,4'-dihydroxybiphenyl				[92-88-6]
	(354–388)	138.6 ± 2.0	(371)	T	[98/9]
		143.0 ± 2.0	(298)		[98/9]
$\text{C}_{12}\text{H}_{10}\text{O}_2\text{S}$	diphenyl sulfone				[127-63-9]
$\text{C}_{12}\text{H}_{10}\text{O}_4$		106.3 ± 2.9			[U/3][70/1]
	quinhydrone (quinone–hydroquinone)				[106-34-3]
	(317–334)	89.1	(325.5)		[87/4]
	(300–325)	88.6 ± 1	(313)	ME,TE	[81/4]
		U 181.2			[53/10][60/1]
$\text{C}_{12}\text{H}_{10}\text{S}_2$		NA			[51/6]
	diphenyldisulfide				[882-33-7]
		95. ± 3.0		E	[62/5][70/1]
$\text{C}_{12}\text{H}_{10}\text{S}_2\text{O}_4$	diphenyldisulfone				[10409-06-0]
		91.7		E	[64/5]
$\text{C}_{12}\text{H}_{11}\text{N}$	diphenylamine				[122-39-4]
		96.7 ± 2.5		TE,ME	[70/4]
		99.2			[68/10][88/24]
		96.7 ± 2.5	(310)	QF	[53/5][70/1]
$\text{C}_{12}\text{H}_{11}\text{NO}$	(298–323)				[575-36-0]
	N-acetyl-1-naphthylamine				[87/4][60/21]
$\text{C}_{12}\text{H}_{11}\text{N}_3$	(337–360)	94.1	(348.5)		[60-09-3]
	4-aminoazobenzene				[87/19][91/18]
		106.3		GS	[84/40]
$\text{C}_{12}\text{H}_{12}$		109.4			[56/2][87/4]
	(356–373)	110.9 ± 1.7	(364)	ME	[569-4-5]
	1,8-dimethylnaphthalene				[87/4][75/8]
	(328–336)	77.9	(332)	IPM	[75/8][79/5]
		79.6	(336)	B,IPM	[74/14][77/1]
$\text{C}_{12}\text{H}_{12}$		82.7 ± 0.3	(298)	C	[581-40-8]
	2,3-dimethylnaphthalene				[87/4][75/8]
	(333–373)	82.8	(348)	IPM	[79/5]
	(287–300)	82.2 ± 0.4	(294)	ME	[75/8]
		81.0		B,IPM	[59/2][87/4]
$\text{C}_{12}\text{H}_{12}$	(278–301)	79.9 ± 0.4	(290)	V	[581-42-0]
	2,6-dimethylnaphthalene				[77/7][75/8]
	(350–383)	84.4 ± 1.9	(366)	IPM	[87/4]
		82.5	(383)	B,IPM	[75/8]
	(279–304)	84.1	(291)	V	[59/2][87/4]
$\text{C}_{12}\text{H}_{12}$	2,7-dimethylnaphthalene				[582-16-1]
	(333–369)	83.8 ± 1	(345)	IPM	[77/7][75/8]
		83.2	(369)	B,IPM	[75/8]
	(333–368)	84.6	(348)	IPM	[75/8][87/4]
$\text{C}_{12}\text{H}_{12}\text{N}_2$	4,4'-dimethyl-2,2'-bipyridyl				[1134-35-6]
$\text{C}_{12}\text{H}_{12}\text{N}_2\text{O}$		99.7 ± 2.3	(298)	C	[97/27]
	4,4'-diaminodiphenyl oxide				[101-80-4]
$\text{C}_{12}\text{H}_{12}\text{N}_2\text{O}_2$		62.8			[75/11]
	1-(4-dimethylaminophenyl)-1 <i>H</i> -pyrrole-2,5-dione				[6953-81-7]
	(350–370)	122.6 ± 0.9		C	[98/25]
$\text{C}_{12}\text{H}_{12}\text{O}_6$	1,3,5-trimethoxycarbonylbenzene				[2672-58-4]
	(350–368)	115.9 ± 0.4	(359)	ME	[95/6]
		118.9 ± 0.4	(298)		[95/6]
		117.5 ± 0.8	(298)		[67/8][95/6]
$\text{C}_{12}\text{H}_{14}\text{O}_2$	ethyl <i>trans</i> -2-phenylcyclopropanecarboxylate				[946-39-4]
$\text{C}_{12}\text{H}_{14}\text{O}_4$		96.9 ± 0.4	(298)	C	[98/24]
	1,1-diacetoxy-1-phenylethane				[28153-24-4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{12}\text{H}_{15}\text{N}_3\text{O}_2$	(308–338)	94.4 ± 2.2	(318)	GS	[96/14]
	3,6-bis(dimethylamino)phthalimide				[5972-07-6]
	(400–457)	105	(415)		[87/4]
$\text{C}_{12}\text{H}_{15}\text{N}_3\text{O}_6$		135.3		RG	[58/4]
	2,4,6-trinitro-1,3-dimethyl-5- <i>tert</i> -butylbenzene				[81-15-2]
	(312–348)	100.4	(327)		[87/4][56/3]
$\text{C}_{12}\text{H}_{16}\text{N}_2\text{O}_2$	N-benzoyl-N',N'-diethylurea	132.2 ± 2.8	(298)	C	[00/28]
$\text{C}_{12}\text{H}_{16}\text{N}_2\text{O}_5$	1-methyl-4- <i>tert</i> -butyl-3-methoxy-2,6-dinitrobenzene				[83-66-9]
	(293–353)	102.9			[53/7][60/1]
$\text{C}_{12}\text{H}_{16}\text{N}_3\text{O}_3\text{PS}_2$	azinphos-ethyl				[2642-71-9]
$\text{C}_{12}\text{H}_{16}\text{O}_2$	(326–420)	86.8	(341)		[87/4]
	pentamethylbenzoic acid				[2243-32-5]
	(347–363)	111.5 ± 1.7	(355)	ME	[88/10]
$\text{C}_{12}\text{H}_{16}\text{O}_4$		113.4 ± 1.8	(298)		[88/10]
	benzo-12-crown-4				[14174-08-4]
		104.3 ± 2.6	(298)	CGC–DSC	[00/11]
$\text{C}_{12}\text{H}_{17}\text{NO}_2$	2,6-diisopropylnitrobenzene				
$\text{C}_{12}\text{H}_{18}$		81.0 ± 1.0	(286)	GS	[00/25]
		80.6 ± 1.0	(298)	GS	[00/25]
	hexamethylbenzene				[87-85-4]
		80		TGA	[97/29]
		81.4 ± 0.1	(298)	C	[94/13]
	(288–304)	85.0 ± 0.2	(298)	ME	[89/18]
		74.9 ± 0.6		DSC	[84/2]
	(303–338)	85.2	(320)	A	[76/2]
		86.1	(298)		[76/2][93/16]
	(314–364)	83.2	(329)	A	[69/24]
		74.7 ± 2		ME	[65/4][70/1]
		80.8			[57/1][60/1]
		80.8			[49/8]
	($\text{C}_{12}\text{H}_{18}$)–($\text{C}_6\text{H}_3\text{N}_2\text{ClO}_4$)				
	(hexamethylbenzene)-(picryl chloride)				
$\text{C}_{12}\text{H}_{18}$		93.7			[49/8]
	<i>E,E,E</i> -1,5,9-cyclododecatriene				[676-22-2]
	(273–307)	75.2	(288)		[87/4]
$\text{C}_{12}\text{H}_{18}\text{O}$		74.7 ± 0.8			[73/16][77/1]
	1-adamantyl methyl ketone				[1660-04-4]
	(287–305)	84.2 ± 0.6	(298)	ME	[92/2]
$\text{C}_{12}\text{H}_{18}\text{O}$	<i>exo</i> -4-hydroxy- <i>endo</i> - <i>endo</i> -tetracyclo[6.2.1.1. ^{3,6} .0 ^{2,7}]dodecane				[7273-98-5]
$\text{C}_{12}\text{H}_{18}\text{O}$	(303–343)	79.0 ± 2.5	(298)	TSGC	[80/16]
	<i>exo</i> -4-hydroxy- <i>exo</i> - <i>endo</i> -tetracyclo[6.2.1.1. ^{3,6} .0 ^{2,7}]dodecane				[107133-43-7]
	(323–353)	74.3 ± 1.8		TSGC	[80/16]
$\text{C}_{12}\text{H}_{18}\text{O}$		76.3 ± 2.0	(298)		[80/16]
	<i>exo</i> -4-hydroxy- <i>exo</i> - <i>exo</i> -tetracyclo[6.2.1.1. ^{3,6} .0 ^{2,7}]dodecane				[74007-11-7]
	(313–353)	73.9 ± 2		TSGC	[80/16]
$\text{C}_{12}\text{H}_{18}\text{O}_2$		75.9 ± 2.2	(298)		[80/16]
	1-adamantyl-1-carboxylic acid methyl ester				[711-01-3]
	(267–283)	84.3 ± 0.6	(275)	ME	[92/26]
$\text{C}_{12}\text{H}_{18}\text{O}_2$		82.4 ± 0.6	(298)		[92/26]
	<i>trans</i> - <i>syn</i> - <i>trans</i> decahydro-3-hydroxy-2-naphthalene acetic γ -lactone				
	(240–310)	NA		ME	[57/10]
$\text{C}_{12}\text{H}_{18}\text{O}_2$	<i>trans</i> - <i>anti</i> - <i>trans</i> decahydro-3-hydroxy-2-naphthalene acetic γ -lactone				
	(240–310)	NA		ME	[57/10]
$\text{C}_{12}\text{H}_{19}\text{F}_3\text{N}_2\text{O}_4$	N[(N-trifluoroacetyl)valyl]alanine ethyl ester				
	(323–424)	115.5	(338)		[87/4][60/20]
$\text{C}_{12}\text{H}_{20}$	2,2-dimethyladamantane				[19740-34-2]
$\text{C}_{12}\text{H}_{20}$	(300–360)	73.6 ± 1.3	(298)	BG	[77/11]
	1,3-dimethyladamantane				[702-79-4]
		67.8 ± 1.3 (liq)	(298)	EB	[77/11]
$\text{C}_{12}\text{H}_{20}\text{N}_2$	1-(1-piperidinyl)cyclohexanecarbonitrile				[3867-15-0]
$\text{C}_{12}\text{H}_{20}\text{N}_2\text{O}_2$		87.8 ± 0.6	(298)		[97/28]
	N,N'-ethylenebis(4-aminopent-3-ene-2-one)				[6310-76-5]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
	(358–374)	128.2 ± 0.7	(366)	ME	[95/12]
		131.6	(298)		[95/12]
C ₁₂ H ₂₀ O ₂	bicyclo[2.2.1]heptane-7-one 2,2-dimethylpropylene ketal	84.0 ± 0.9	(298)		[217467-40-8]
C ₁₂ H ₂₂ O	cyclododecanone				[98/26]
		83.2 ± 0.3	(298)	ME	[850-1-7]
C ₁₂ H ₂₂ O	<i>trans</i> 2-cyclohexylcyclohexanol				[96/16]
	(293–325)	98.6 ± 0.5	(320)	ME	[97/5]
C ₁₂ H ₂₂ O ₄	dodecanedioic acid				[693-23-2]
	(375–296)	153.1 ± 2.9	(386)	ME	[60/4][70/1]
C ₁₂ H ₂₂ O ₁₁	D cellobiose				[528-50-7]
	(474–488)	302 ± 44.0	(481)	ME	[99/1]
C ₁₂ H ₂₄	cyclododecane				[294-62-2]
		76.2	(298)	CGC–DSC	[98/5]
		76.4 ± 1.7			[57/1]
C ₁₂ H ₂₄ N ₂ O ₂	dicyclohexyl ammonium nitrite				[3129-91-7]
	(290–298)	99.1	(294)	TE	[87/4][65/19]
		U161.8			[85/20]
	(308–339)	105.9	(324)		[61/6]
C ₁₂ H ₂₄ O ₂	dodecanoic acid (lauric acid)				[143-07-7]
	(293–303)	127.9	(298)		[87/4]
	(293–308)	132.6	(300)	ME	[68/2]
	(296–314)	140.2 ± 3.3	(304)	ME	[61/1]
	(293–313)	117.2 ± 2.9	(303)	ME	[57/3]
C ₁₂ H ₂₄ O ₃	peroxydodecanoic acid				[2388-12-7]
	(293–303)	131.4 ± 1.7	(298)	ME	[80/23]
C ₁₂ H ₂₄ O ₆	18 crown-6				[17455-13-9]
		128.1 ± 2.3	(298)	CGC–DSC	[00/11]
C ₁₂ H ₂₅ NO	dodecanamide				[1120-16-7]
	(349–368)	152.7 ± 0.8	(358.5)	ME	[59/3][87/4]
C ₁₂ H ₂₆	<i>n</i> -dodecane				[112-40-3]
		100.2	(298)	B	[72/1]
		101.7	(263)	B	[63/6]
C ₁₂ H ₂₆ O	1-dodecanol				[112-53-8]
	(285–294)	130.1 ± 1.2	(290)	ME	[65/6][87/4]
		129.3	(298)		[65/6]
C ₁₂ H ₂₆ O	di- <i>tert</i> -butyl-isopropylmethanol				[5457-42-1]
		59.3 ± 0.8	(298)		[98/22]
C ₁₃ H ₄ Cl ₆ O	1,2,4,5,7,8-hexachloroxanthene				[38178-99-3]
	(353–449)	147	(401)	T	[86/7]
C ₁₃ H ₇ NO ₂	benz[<i>g</i>]isoquinoline-5,10-dione				[46492-08-4]
	(334–381)	108.1 ± 1.6	(358)	ME	[98/3]
C ₁₃ H ₈ O	perinaphthenone				[548-39-0]
	(326–348)	97.2 ± 2.5	(337)	ME	[98/3]
C ₁₃ H ₈ O	fluorenone				[486-25-9]
	(324–349)	91.6 ± 1.6	(336)	GS	[98/21]
		93.9 ± 1.6	(298)	GS	[98/21]
		87.6 ± 0.3	(319)	C	[88/5]
		88.4 ± 0.4	(298)	C	[88/5]
	(298–343)	92.2	(320)	GS	[86/8]
C ₁₃ H ₈ OS	thioxanthone				[492-22-8]
		114.8 ± 0.4	(298)	C	[92/27]
C ₁₃ H ₈ O ₂	xanthone				[90-47-1]
		98.57 ± 0.4	(298)	C	[88/13]
C ₁₃ H ₈ O ₂	3-hydroxy-1 <i>H</i> -phenalen-1-one				[5472-84-4]
	(402–432)	151.5 ± 4.7	(417)	ME	[98/3]
C ₁₃ H ₉ ClO ₂	5-chloro-2-hydroxybenzophenone				[85-19-8]
	(293–367)	91.9	(308)	UV	[87/4][60/24]
C ₁₃ H ₉ N	acridine				[260-94-6]
		86	(430)	TGA	[98/36]
		89.5 ± 0.2	(333)	C	[94/14]
		91.7 ± 0.4	(298)	C	[94/14]
		94.5	(298)		[89/26]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_m/\text{kJ mol}^{-1}$	(T_m/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{13}\text{H}_9\text{N}$	(280–328)	92.6	(295)		[87/4]
	(303–328)	90.8 ± 1.3	(298)	TE	[75/5]
	(303–326)	93.3 ± 0.8	(298)	TCM	[U/1][75/5]
	(281–298)	91.6 ± 2.5	(290)	LE	[75/1]
	(306–345)	92.8 ± 1.3	(298)	ME	[U/2][75/5]
		78.7		E	[46/1]
	3,4-benzoquinoline (phenanthridine)				[260-27-3]
	(288–323)	100.1 ± 10.1	(306)	ME	[98/3]
		98.6	(298)		[89/26]
	(288–323)	94.6 ± 4	(308)	ME	[75/15][87/4]
$\text{C}_{13}\text{H}_9\text{N}$	5,6-benzoquinoline				[85-02-9]
	(288–323)	83.1 ± 3.6	(308)	ME	[75/15][87/4]
		106.3		ME	[72/10]
$\text{C}_{13}\text{H}_9\text{N}$	7,8-benzoquinoline				[230-27-3]
		90.2 ± 2.0	(298)		[89/26]
	(293–323)	80.8 ± 2.5	(308)	ME	[75/15][87/4]
		100.4		ME	[72/10]
$\text{C}_{13}\text{H}_9\text{NO}$	acridone				[598-95-0]
$\text{C}_{13}\text{H}_9\text{NO}_2$	N-methyl-1,8-naphthalimide				[92/27]
	(379–398)	136.2 ± 0.5	(298)	C	[2382-08-3]
		107.4 ± 0.8	(389)	ME	[00/9]
$\text{C}_{13}\text{H}_{10}$	fluorene				[00/9]
		109.7 ± 0.8	(298)	ME	[86-73-7]
		87.6	(298)	CGC–DSC	[98/5]
	(313–453)	84.9	(383)	GS	[95/7]
	(323–363)	84.9 ± 0.4	(343)	GS	[94/2]
		85.1 ± 0.4	(298)		[94/2]
	(318–333)	87.0 ± 1.0	(318)	PG	[88/26]
		80.2 ± 0.2	(298)	C	[87/10]
	(348–388)	78.9	(363)	IPM	[87/4][75/8]
	(283–323)	88.4 ± 0.6	(303)	GS	[83/11]
	(350–388)	83.1 ± 1.3			[77/7][75/8]
		81.8	(388)	B	[75/8]
	(286–300)	80.3 ± 0.8	(293)	TE	[60/2]
	(306–323)	82.8	(315)		[53/13][87/4]
	(306–322)	82.8			[53/1][60/1]
	9-aminoacridine				[90-45-9]
		115	(520)	TGA	[98/29]
	N-phenyl 4-nitrobenzaldehyde imine				[785-80-8]
		126 ± 1.3	(298)		[97/18]
$\text{C}_{13}\text{H}_{10}\text{N}_4$	1,5-diphenyltetrazole				[7477-73-8]
	(348–363)	121.5 ± 4.2	(355)	ME	[51/3][70/1]
$\text{C}_{13}\text{H}_{10}\text{N}_4$	2,5-diphenyltetrazole				[18038-45-7]
	(333–353)	119.7 ± 4.2	(343)	ME	[51/3][70/1]
$\text{C}_{13}\text{H}_{10}\text{O}$	benzophenone				[119-61-9]
	(299–320)	92.4 ± 2.2	(309)	GS	[98/21]
		93.1 ± 2.2	(298)	GS	[98/21]
		94.7 ± 1	(321)	DM	[83/2]
		92 ± 0.83	(298)	C	[74/20][83/2]
	(295–313)	95.0 ± 0.2	(304)	ME	[80/14]
		96.1			[78/15]
		84.4 ± 1.13	(298)	C	[78/13]
	(297–317)	93.9 ± 0.5	(307)	TE,ME	[77/4]
	(293–318)	95.0 ± 1.5	(305)	TE	[75/5]
	(294–318)	92.9 ± 0.8	(306)	ME	[75/6]
	(278–311)	77.0 ± 2.5	(298)	ME	[74/5]
	(298–318)	89.96	(308)	ME	[87/4][74/6]
	(295–304)	94.6 ± 0.8	(298)	TCM	[73/1]
		93.4 ± 0.3	(298)	C	[72/1]
	(293–319)	96.1	(306)		[56/3]
		91.2			[50/2]
	(290–315)	78.2 ± 1.2	(303)		[38/1][34/1]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{13}\text{H}_{10}\text{O}$	dibenzopyran (273–320) (305–353)	95 ± 2.5	(298)	TE	[32/1][70/1] [60/1]
		91.2 ± 1.6	(298)	ME	[25/1]
		92.5	(329)	T	[229-95-8] [86/7]
		112.1 ± 2.1			[58/2][70/1] [93-99-2]
$\text{C}_{13}\text{H}_{10}\text{O}_2$	phenyl benzoate	99.0 ± 0.4	(298)		[71/2]
$\text{C}_{13}\text{H}_{10}\text{O}_3$	diphenyl carbonate	89.5 ± 4.2			[71/11][77/1]
		96.2 ± 1.7			[47/2][70/1] [102-09-0]
		90 ± 8.4	(298)	E	[71/2][77/1]
					[118-55-8]
$\text{C}_{13}\text{H}_{10}\text{O}_3$	phenyl salicylate (279–315)	109.1	(294)	UV	[87/4][60/24]
$\text{C}_{13}\text{H}_{10}\text{O}_3$	2,4-dihydroxybenzophenone (312–353)	92 ± 4.2			[47/2][70/1] [131-56-6]
		134	(327)	UV	[87/4][60/24]
$\text{C}_{13}\text{H}_{10}\text{O}_4$	2,4,4'-trihydroxybenzophenone	139.0		TGA	[1470-79-7] [99/22]
$\text{C}_{13}\text{H}_{10}\text{O}_5$	2,2',4,4'-tetrahydroxybenzophenone	178.5		B	[131-55-5] [99/22]
$\text{C}_{13}\text{H}_{11}\text{N}$	N-phenyl-benzaldehyde imine (294–326)	143.4	(378)	UV	[87/4][60/24] [538-51-2]
		97.4 ± 1.2	(309)	T	[97/18]
		98.1 ± 1.2	(298)		[97/18]
		93.7 ± 0.9	(298)	C	[86/10]
$\text{C}_{13}\text{H}_{11}\text{N}$	9-methylcarbazole (313–332)	85.5 ± 2.1	(293)		[48/2] [1484-12-4]
		95.0	(322)	ME	[90/8]
		95.5	(298)		[90/8]
$\text{C}_{13}\text{H}_{11}\text{NO}$	N-phenylmethylene benzenamine N-oxide	115.0 ± 0.8	(298)	C	[1137-98-8] [86/10]
$\text{C}_{13}\text{H}_{11}\text{NO}$	2-hydroxybenzaldehyde N-phenylimine (288–325) (348–408)	115.9	(303)		[779-84-0] [87/4]
		129.9	(378)		[58/1]
					[1689-73-2]
$\text{C}_{13}\text{H}_{11}\text{NO}$	4-hydroxybenzaldehyde N-phenylimine (348–408) (288–338)	127.9	(363)		[87/4]
		116	(313)		[58/1]
					[93-98-1]
$\text{C}_{13}\text{H}_{11}\text{NO}$	benzanilide (352–369)	99.2	(360.5)		[87/4][60/21]
$\text{C}_{13}\text{H}_{11}\text{NO}_2$	N-(2-hydroxyphenylmethylene)benzenamine N-oxide	116.5 ± 1.4	(298)	C	[20357-59-9] [86/10]
$\text{C}_{13}\text{H}_{11}\text{N}_3\text{O}$	2-(2'-hydroxy-5'-methylphenyl)benzotriazole (293–333)	125.2	(308)	UV	[2440-22-4] [87/4][60/24]
$\text{C}_{13}\text{H}_{12}$	diphenylmethane (273–295) (273–298) (276–295) (278–299)	88.5 ± 0.8	(284)	GS	[101-81-5] [99/23]
		87.6 ± 0.8	(298)		[99/23]
		71.5	(286)	EM	[89/1]
		83.3 ± 3.3	(286)	HSA	[86/1]
		82.4 ± 8		V	[59/2][70/1]
		64.0			[51/1][60/1]
		72.0 ± 0.8	(297)		[38/1]
$\text{C}_{13}\text{H}_{12}$	4-methylbiphenyl	80.2 ± 1.4	(298)	C	[644-08-6] [97/27]
$\text{C}_{13}\text{H}_{12}\text{N}_2\text{O}$	1,3-diphenylurea (445–484)	152 ± 6	(464)	TE	[102-07-8] [87/2]
$\text{C}_{13}\text{H}_{12}\text{N}_4\text{O}_2$	4'-nitro-2-methylaminoazobenzene	134.7		GS	[87/19][91/18]
$\text{C}_{13}\text{H}_{12}\text{O}$	diphenylmethanol	105.7 ± 0.7	(298)		[91-01-0] [98/22]
$\text{C}_{13}\text{H}_{12}\text{O}$	4-benzylphenol (313–335)	97.4	(324)		[101-53-1] [87/4][60/14]
$\text{C}_{13}\text{H}_{13}\text{N}$	N-phenyl benzylamine				[103-32-2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
	(293–312)	103.6 ± 1.6	(303)	T	[97/10]
		51.3			[80/8]
C ₁₃ H ₁₄ N ₂	2,2'-diaminodiphenylmethane				[6582-52-1]
	(343–403)	111.3	(358)		[87/4]
C ₁₃ H ₁₆ N ₂	α -phenyl-1-piperidineacetonitrile				[5766-79-0]
		73.2 ± 0.4	(298)		[97/28]
C ₁₃ H ₁₆ N ₂ O ₃	hexahydro-1-(3-nitrobenzoyl)-1 <i>H</i> -azepine (compound is called hexamethyleneimine m-nitrobenzoate in paper)				[37000-08-1]
	(310–321)	113.0	(315)		[72/8]
		104.6		ME	[70/22][72/8]
C ₁₃ H ₁₇ NO ₃	morpholine cinnamate				
	(298–349)	118.8	(313)		[87/4]
C ₁₃ H ₁₉ NO ₂	cyclohexyl ammonium benzoate				[3129-92-8]
	(289–298)	103.1	(293.5)		[87/4][65/19]
C ₁₃ H ₂₁ NO	N,N-dimethyl-1-adamantylcarboxamide				[1502-00-7]
	(303–322)	96.9 ± 0.3	(313)	ME	[93/18][95/11]
		97.5 ± 0.3	(298)	ME	[95/11]
C ₁₃ H ₂₁ NO ₂	N-(3-phenoxy-2-hydroxypropyl)-butylamine				[3246-04-6]
	(323–348)	133.9	(335.5)		[87/4]
C ₁₃ H ₂₂	1,3,5-trimethyladamantane				[707-35-7]
	(300–360)	77.8 ± 1.3	(298)	BG	[77/11]
C ₁₃ H ₂₂ O ₃	dicyclohexyl carbonate				[4427-97-8]
	(293–313)	66.5 ± 4.2	(303)	ME	[71/11][77/1]
C ₁₃ H ₂₆ O	7-tridecanone				[462-18-0]
	(287–293)	103.8	(290)	ME	[38/3]
C ₁₃ H ₂₆ O ₂	methyl dodecanoate				[111-82-0]
	(262–273)	121.8 ± 2.1	(267)	ME	[65/6][87/4]
C ₁₃ H ₂₆ O ₂	tridecanoic acid				[638-53-9]
	(282–299)	170		TPTD	[01/15]
C ₁₃ H ₂₆ O ₃	peroxytridecanoic acid				[40915-96-6]
	(293–303)	142.7 ± 5	(298)	ME	[80/23]
C ₁₃ H ₂₇ NO	N-methyl dodecanamide				[27563-67-3]
	(323–337)	116.6 ± 0.8	(330)	GS	[59/4][87/4]
C ₁₃ H ₂₈	<i>n</i> -tridecane				[629-50-5]
		91.4	(298)	B	[72/1]
C ₁₃ H ₂₈	tri- <i>tert</i> -butylmethane				[35660-96-9]
		55.4	(298)	CGC–DSC	[98/5]
	(265–319)	57.0 ± 0.4	(288)	T	[97/7]
	(273–306)	57.7 ± 2.8	(290)	HSA	[95/16]
		61.1 ± 1.3			[95/16]
	(295–330)	7.7 ± 0.1	(311)		[86/2]
C ₁₃ H ₂₈ O	tri- <i>tert</i> -butylmethanol				[41902-42-5]
plastic	(278–318)	56.5 ± 1.0	(298)	TE	[83/18]
crystalline	(269–300)	63.2 ± 1.2	(298)	TE	[83/18]
C ₁₄ H ₆ Cl ₂ N ₂ O ₄	1-amino-4-nitro-5,8-dichloroanthraquinone				[66121-41-3]
		158.2			[68/10][88/24]
C ₁₄ H ₆ N ₂ O ₆	1,4-dinitroanthraquinone				[66121-37-7]
		131.0			[68/10][88/24]
C ₁₄ H ₆ N ₆ O ₁₂	1,2- <i>bis</i> (2,4,6-trinitrophenyl)ethylene				[20062-22-0]
	(434–479)	179.9	(449)	LE	[87/4][69/12]
		180.3			[68/14][66/11]
C ₁₄ H ₇ NO ₄	1-nitroanthraquinone				[82-34-8]
	(407–440)	139.7	(422)	TE	[87/4][70/4]
		108.9 ± 2.1	(396)	C	[82/3]
		137.9 ± 1.7		TE,ME	[70/4]
		115.5			[68/10][88/24]
C ₁₄ H ₈ Cl ₄	1,1-dichloro-2,2- <i>bis</i> (4-chlorophenyl)ethylene				[72-55-9]
		74.2			[95/32][89/32]
C ₁₄ H ₈ Cl ₆	1,1,1-trichloro-2-chloro-2,2- <i>bis</i> (4-chlorophenyl)ethane				[3563-45-9]
		89.4			[95/32][89/32]
C ₁₄ H ₈ O ₂	9,10-anthraquinone				[84-65-1]
		111.3		GS	[97/19][91/18]
		108.4	(402)	C	[82/3]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
	(373–453)	98.3	(413)	GS	[77/20][78/35]
		113±0.8	(298)	C	[73/10]
		107.5±0.8	(434)	ME	[73/10]
	(397–471)	107.9±0.8		ME	[73/10]
	(355–356)	U 105.9		TGA	[71/17]
	(470–590)	127.0±3.0		C	[71/6]
		136.6±3	(298)	C	[71/6]
		116.1±1.7		ME,TE	[70/4]
		115.1			[68/10][88/24]
	(343–403)	126.4	(373)	ME	[58/1][87/4]
		112.1	(298)		[56/5][70/1]
		110.9	(298)		[56/1]
		107.9	(298)		[54/3]
		104.6	(367)	ME	[52/3]
		108.0	(298)	ME	[52/3]
C ₁₄ H ₈ O ₂	9,10-phenanthraquinone				[84-11-7]
		108.1	(289)	C	[89/21]
		132	(383)		[56/5][70/1]
C ₁₄ H ₈ O ₃	1-hydroxy-9,10-anthraquinone				[129-43-1]
		113.1		GS	[87/19][91/18]
	(333–383)	120.6	(358)		[58/1][87/4]
		101.3±0.4	(407)	HSA	[56/1]
C ₁₄ H ₈ O ₃	2-hydroxy-9,10-anthraquinone				[605-32-3]
		136.8		GS	[87/19][91/18]
	(393–453)	153.1	(408)		[87/4]
C ₁₄ H ₈ O ₃	2,2'-biphenyldicarboxylic anhydride				[6050-13-1]
	(433–490)	91.4	(448)		[87/4]
C ₁₄ H ₈ O ₄	1,2-dihydroxyanthraquinone				[72-48-0]
	(368–498)	123.8	(383)		[87/4]
		121.9±0.5	(469)	C	[73/4]
	(434–505)	121.5±0.4	(469)	ME	[73/4]
		123.9	(403)	ME	[58/1]
C ₁₄ H ₈ O ₄	1,4-dihydroxyanthraquinone				[81-64-1]
		114.6		GS	[87/19][91/18]
	(353–373)	102.4±4.4	(363)		[84/35]
	(373–453)	89.1	(413)	GS	[77/20][78/35]
	(394–463)	121.9±0.8	(429)	ME	[73/4]
		121.1±4	(429)	C	[73/4]
	(324–351)	U 94.5	(338)	TGA	[71/17]
		123.5	(376)	ME	[58/1][87/4]
		103.5±1.3	(409)	HSA	[56/1]
C ₁₄ H ₈ O ₄	1,5-dihydroxyanthraquinone				[117-12-4]
		123.2±7		ME	[73/10]
	(363–433)	126.8	(398)	ME	[58/1][87/4]
		111.3	(456)		[56/1]
		117.6	(298)		[56/1]
C ₁₄ H ₈ O ₄	1,8-dihydroxyanthraquinone				[117-10-2]
		116.8		ME	[73/10]
	(333–403)	123	(368)	ME	[58/1][87/4]
	(335–356)	U 96.5	(345)	TGA	[71/17]
		105.8±8	(404)	HSA	[56/1]
		109.6±8	(298)		[56/1]
C ₁₄ H ₈ O ₄	2,6-dihydroxyanthraquinone				[84-60-6]
	(463–533)	173.8	(498)		[58/1][87/4]
C ₁₄ H ₈ O ₆	1,4,5,8-tetrahydroxyanthraquinone				[81-60-7]
	(403–473)	151.6	(438)		[58/1][87/4]
C ₁₄ H ₉ ClN ₂ O ₄	1,5-diaminochloro-4,8-dihydroxyanthraquinone (C.I. disperse blue 56)				[12217-79-7]
	(483–533)	93.3	(498)		[87/4]
C ₁₄ H ₉ Cl ₅	1,1-bis(-4-chlorophenyl)-2,2,2-trichloroethane (<i>p,p'</i> -DDT)				[50-29-3]
	(273–313)	120.2±1.0	(293)	GS	[94/1]
	(323–363)	115	(338)		[87/4]
	(293–353)	110	(304)	GS	[80/35]
	(293–313)	117.8	(303)	GS	[72/11]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{14}\text{H}_9\text{F}_3\text{O}_2$	(323–363)	117.5	(338)	GS	[56/4][60/1]
	(313–363)	84	(338)	GS	[49/10]
	(339–373)	118	(356)	TE	[47/1]
	4,4,4-trifluoro-1-(2-naphthyl)-butane-1,3-dione	108.7 ± 0.6	(298)	ME	[893-33-4] [97/33]
$\text{C}_{14}\text{H}_9\text{NO}_2$	1-aminoanthraquinone	121.8		GS	[82-45-1] [87/19][91/18]
	(413–443)	126.5 ± 1.3	(428)	TE	[87/4][70/4]
	(368–393)	116.3 ± 3.9	(380)		[84/35]
	(373–453)	103.3	(413)	GS	[77/20][78/35]
	(361–386)	U 90.9	(374)	TGA	[71/17]
		125.9 ± 2.5		TE,ME	[70/4]
		131.0			[68/10][88/24]
		113 ± 0.4	(463)	HSA	[56/1]
$\text{C}_{14}\text{H}_9\text{NO}_2$	2-aminoanthraquinone	136.8		GS	[117-79-3] [87/19][91/18]
		143.5 ± 2.9		TE,ME	[70/4]
		162.3			[68/10][88/24]
					[116-85-8]
$\text{C}_{14}\text{H}_9\text{NO}_3$	1-hydroxy-4-aminoanthraquinone	127.2		GS	[87/19][91/18]
	(418–438)	131.3 ± 1.7	(428)	TE	[87/4][70/4]
	(444–473)	144	(458.5)		[87/4]
		119.6			[84/40]
		133.5 ± 2.1		TE,ME	[70/4]
		120.1			[68/10][88/24]
$\text{C}_{14}\text{H}_9\text{N}_3\text{O}_4$	1,4-diamino-5-nitroanthraquinone (373–453, not crystalline)	U50.2	(413)	GS	[82-33-7] [77/20][78/35]
$\text{C}_{14}\text{H}_{10}$	anthracene				[120-12-7]
	(423–488)	94.5		MEM	[99/40]
		99.4	(298)	CGC–DSC	[98/5]
	(318–363)	100.0 ± 2.8	(341)	ME	[98/3]
	(343–448)	84.0 ± 3.0	(298)	TGA	[97/38]
	(313–453)	99.7	(383)	GS	[95/7]
	(318–373)	98.7	(346)	GS	[86/7]
	(313–363)	102.6	(338)	GS	[86/8]
	(353–399)	94.3	(376)	GS	[83/3]
	(283–323)	91.8 ± 0.9	(303)	GS	[83/11]
	(323–353)	91.2	(338)	GS	[82/23]
		97.4 ± 1.1		GS,C	[81/3]
		97.8 ± 0.1		HSA	[80/3]
	(337–361)	104.5 ± 1.5	(298)	TE,ME	[80/1]
	(358–393)	94.8	(376)	GS	[79/27]
	(363–448)	98.8 ± 0.4		HSA	[77/8]
	(328–372)	97.2		ME	[76/8]
		97.1		C	[75/20]
	(323–353)	102.9 ± 4.8	(298)	TE	[75/5]
	(283–323)	95.8 ± 6		LE	[73/15]
	(353–432)	101.0 ± 0.5		ME	[73/2]
		99.7	(393)	C	[73/2]
	(290–358)	84.1		ME,C	[72/9][71/6]
	(342–359)	98.3 ± 2.1			[64/3][70/1]
	(327–346)	90 ± 1.3	(337)	TE	[60/2]
		100.8			[58/1][70/1]
	(303–373)	103.4 ± 2.9			[58/1][70/1]
		100.8 ± 4.2			[58/1][70/1]
	(396–421)	97.5 ± 2		HSA	[53/2]
	(339–353)	102.1	(346)		[53/13]
	(338–353)	102.1 ± 2.1			[53/1][70/1]
		92.0 ± 2.1	(364)	ME	[52/3]
		90.4	(353)	ME	[51/12]
		95.4			[51/6]
		95.0			[50/5]
	(378–398)	97.3 ± 1.2		RG	[49/3]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{14}\text{H}_{10}$	diphenylacetylene	104.6±4.2			[49/6][70/1]
		93.3±4.2	(353)		[38/1]
		95.3	(298)	CGC–DSC	[501-65-5]
		(298–316)	(298)	ME	[98/5]
		(299–321)	(310)	HSA	[93/5]
$\text{C}_{14}\text{H}_{10}$	phenanthrene	(299–321)	(313)		[86/1]
		88.7±1.25			[38/1]
					[38/2][60/1]
		92±1		LE	[85-01-8]
		90.5	(298)	CGC–DSC	[98/38]
		(303–333)	(318)	ME	[98/5]
		(313–453)	(383)	GS	[98/3]
		87.2±1.1	(350)	DSC	[95/7]
		90.9±1.7	(298)	DSC	[88/4]
		(317–362)	(340)	TE	[88/4]
		(283–323)	(303)	GS	[83/27]
		(315–335)	(298)	TE,ME	[83/11]
		(325–364)	(345)	GS	[80/1]
		87.2	(372)	B	[79/27]
		(300–330)	(298)	TE	[75/8]
		(312–326)	(298)	TCM	[75/5]
		90.9±0.4	(298)	C	[U/1][75/5]
		(279–315)	(297)	TE	[72/1][77/1]
		(273–333)	(303)		[60/2]
		(310–323)			[58/1][70/1]
$\text{C}_{14}\text{D}_{10}$	phenanthrene- d_{10}	90.7±1.2	(315)	ME	[53/1][70/1]
		81.6	(323)	ME	[60/1]
		92.9			[52/3]
		84.1±0.8	(313)		[51/12]
					[49/6][70/1]
$\text{C}_{14}\text{H}_{10}\text{F}_4$	1,1,2,2-tetrafluoro-1,2-diphenylethane	92.2±1.1	(303)	GS	[38/1]
$\text{C}_{14}\text{H}_{10}\text{N}_2\text{O}_2$	1,4-diaminoanthraquinone	101.8	(298)		[1517-22-2]
$\text{C}_{14}\text{H}_{10}\text{N}_2\text{O}_2$	(448–474)	143.0		GS	[83/11]
		151.2±1.3	(461)		[425-32-1]
		136.0			[97/34]
		(378–403)	(390)		[128-95-0]
		(473–453)	(413)	GS	[87/19][91/18]
$\text{C}_{14}\text{H}_{10}\text{N}_2\text{O}_2$	1,5-diaminoanthraquinone	149.2±2.5		TE,ME	[87/4][70/4]
		123.4			[84/40]
		138.1		GS	[84/35]
					[77/20][78/35]
					[70/4]
$\text{C}_{14}\text{H}_{10}\text{O}$	anthrone	103.0±0.8	(350)	GS	[68/10][88/24]
$\text{C}_{14}\text{H}_{10}\text{O}_2$	benzil	106.1±0.8	(298)	GS	[67/16][91/18]
		103.3	(298)		[129-44-2]
		99.6	(354)	C	[84/35]
					[90-44-8]
					[98/21]
$\text{C}_{14}\text{H}_{10}\text{O}_3$	benzoic anhydride	98.4±1.1	(329)		[98/21]
					[91/12]
					[91/12]
					[134-81-6]
					[59/2][70/1]
$\text{C}_{14}\text{H}_{10}\text{O}_4$	benzoyl peroxide	82.8			[87/4]
					[38/1][38/2]
					[60/1]
					[93-97-0]
					[71/2][77/1]
$\text{C}_{14}\text{H}_{10}\text{O}_4$	diphenyl oxalate	96.2±4.2	(298)	B	[47/2][70/1]
		96.7±4.2			[94-36-0]
					[75/9]
					[71/11][77/1]
					[3155-16-6]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
		102.5 ± 8.4		B	[71/2][77/1]
C ₁₄ H ₁₀ O ₄	2,2'-biphenyldicarboxylic acid (433–493)	166.1	(448)		[482-05-3] [87/4]
C ₁₄ H ₁₀ O ₅	O-phenyl-O,O-benzoyl peroxy carbonate	97.9 ± 2.5 133.9 ± 4.2		E	[962-16-3] [75/9][77/1] [71/11][77/1]
C ₁₄ H ₁₁ FO ₃	2'-fluoro-2-hydroxy-4-methoxybenzophenone (307–318)	109.3	(312.5)	EV	[3119-88-8] [87/4][66/6]
C ₁₄ H ₁₁ FO ₃	3'-fluoro-2-hydroxy-4-methoxybenzophenone (322–343)	U 17.3	(332.5)	EV	[3506-35-2] [87/4][66/6]
C ₁₄ H ₁₁ FO ₃	4'-fluoro-2-hydroxy-4-methoxybenzophenone (322–343)	U 37.7	(332.5)	EV	[3602-47-9] [87/4][66/6]
C ₁₄ H ₁₁ F ₃	1,1,2-trifluoro-1,2-diphenylethane	93.1	(298)		[68936-77-6] [97/34]
C ₁₄ H ₁₁ N ₃ O ₂	1,4,5-triaminoanthraquinone (373–453)	U70.3	(413)	GS	[6407-69-8] [77/20][78/35]
C ₁₄ H ₁₂	9,10-dihydroanthracene (313–453)	93.9	(383)	GS	[613-31-0] [95/7]
	(318–379)	92.4 ± 4 94.2 ± 0.8	(298)	ME ME	[75/12][87/4] [75/12]
	(319–377)	92.2 ± 0.6 93.9 ± 0.6	(298)	C C	[75/12][87/4] [75/12]
	(279–328)	93.3 ± 4 89.5	(304) (388)		[58/1][70/1] [51/2][60/1]
C ₁₄ H ₁₂	<i>trans</i> -diphenylethane	102.0	(298)	CGC–DSC	[103-30-0] [98/5]
	(298–343)	99.6	(313)		[87/4]
		U 61.1		MS	[83/9]
	(293–338)	103.8 ± 2.5 100.7 ± 0.4	(315) (298)		[83/16] [83/1]
	(310–340)	99.6 ± 1.7 102.1 ± 0.6 99.2 ± 0.4	(298)	SRFG TE,ME,DM TE TCM	[75/5] [73/1] [72/1]
	(303–315)	86.5 ± 0.1	(309)	TM	[55/5]
C ₁₄ H ₁₂	9-methylfluorene (318–358)	82.8 ± 0.3 82.8 ± 0.3	(338) (298)	B	[2523-37-7] [94/2] [94/2]
C ₁₄ H ₁₂ F ₂	1,1-difluoro-1,2-diphenylethane	94.7 ± 0.9	(298)		[350-62-9] [97/34]
C ₁₄ H ₁₂ N ₂	N-methyl-9-acridinamine	107	(480)	TGA	[22739-29-3] [98/29]
C ₁₄ H ₁₂ N ₂	10-methyl-9-acridinimine	94	(550)	TGA	[5291-44-1] [98/29]
C ₁₄ H ₁₂ N ₂	dibenzylideneazaine	93.3 ± 2.1	(293)		[588-68-1] [48/2]
C ₁₄ H ₁₂ N ₂ O ₂	<i>cis</i> -5a,6,11a,12-tetrahydro[1,4]benzothiazino[3,2-b][1,4]benzoxazine (383–392)	122.0 129.0 ± 1.3	(387) (298)	ME	[192998-96-2] [97/1] [97/1]
C ₁₄ H ₁₂ N ₂ S ₂	<i>cis</i> -5a,6,11a,12-tetrahydro[1,4]benzothiazino[3,2-b][1,4]-benzothiazine (383–392)	118.0 123.3 ± 1.2	(387) (298)	ME	[165454-33-1] [97/1] [97/1]
C ₁₄ H ₁₂ N ₄ O ₂	1,4,5,8-tetraminoanthraquinone (373–453)	U82	(413)	GS	[2475-45-8] [77/20][78/35]
C ₁₄ H ₁₂ O	desoxybenzoin	99.3 ± 4.2			[451-40-1] [47/2][70/1]
C ₁₄ H ₁₂ O ₂ S	E-(2-phenylethenyl)sulfonylbenzene (phenyl <i>trans</i> -B-styrylsulfone)	105 ± 3.8		B	[16212-06-9] [69/11][77/1]
C ₁₄ H ₁₂ O ₃	2-hydroxy-4-methoxybenzophenone (281–337)	118.9	(296)	UV	[131-57-7] [87/4][60/24]
	(308–323)	U39.7	(315)	EV	[66/6]
C ₁₄ H ₁₂ O ₄	2,2'-dihydroxy-4-methoxybenzophenone				[131-53-3]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{14}\text{H}_{12}\text{O}_4$	(303–342)	103.8		B	[99/22]
		228	(318)	UV	[87/4][60/24]
	2,4-dihydroxy-4'-methoxybenzophenone	138.3		B	[131-53-3] [99/22]
$\text{C}_{14}\text{H}_{13}\text{N}$	N-ethylcarbazole				[86-28-2]
	(310–329)	98.4 ± 0.3	(319)	ME	[90/8]
		99.1 ± 0.3	(298)	ME	[90/8]
$\text{C}_{14}\text{H}_{13}\text{NO}$	N,N-diphenylacetamide				[519-87-9]
	(343–376)	122.7	(358)		[87/4]
$\text{C}_{14}\text{H}_{13}\text{NO}_2$	N-(4-methoxyphenylmethylene) benzenamine N-oxide				[3585-93-1]
		130.6 ± 1.2	(298)	C	[86/10]
$\text{C}_{14}\text{H}_{14}$	2,2'-dimethylbiphenyl				[605-39-0]
	(283–288)	65.7	(285)	ME	[74/6][87/4]
$\text{C}_{14}\text{H}_{14}$	3,3'-dimethylbiphenyl				[612-75-9]
	(288–308)	71.9	(298)	ME	[74/6]
$\text{C}_{14}\text{H}_{14}$	4,4'-dimethylbiphenyl				[613-33-2]
$\text{C}_{14}\text{H}_{14}$		95.1 ± 2.0	(298)	C	[97/27]
	1,2-diphenylethane				[103-29-7]
	(293–323)	92.9	(308)	EM	[89/1]
	(273–318)	91.2 ± 0.4	(295)		[83/16]
		91.5 ± 0.7	(298)	B	[80/27]
		91.4 ± 0.5	(298)	C	[72/1]
	(286–307)	84.1 ± 0.4		V	[59/2][70/1]
	(290–317)	72.4 ± 1.3	(304)	ME	[51/1]
		73.2			[38/1][60/1]
					[38/2]
$\text{C}_{14}\text{H}_{14}\text{FN}_3$	N,N-dimethyl-4-[(4-fluorophenyl)azo]benzenamine				[150-74-3]
		91.2		UV	[84/39]
$\text{C}_{14}\text{H}_{14}\text{FN}_3\text{O}_2\text{S}$	4-[[4-(dimethylamino)phenyl]azo]benzenesulfonyl chloride				[4644-89-7]
		105.6		UV	[84/39]
$\text{C}_{14}\text{H}_{14}\text{NO}_3$	bis(4-methoxyphenyl)nitrogen oxide				[2643-00-7]
	(328–363)	100.7	(343)		[87/4][65/7]
$\text{C}_{14}\text{H}_{14}\text{N}_2$	N,N'-diphenylacetamidine				[621-09-0]
	(343–383)	122.6 ± 3.8	(363)	ME	[58/12]
$\text{C}_{14}\text{H}_{14}\text{N}_2\text{O}$	<i>p</i> -azoxyanisole				[1562-94-3]
		134.8 ± 3.7	(298)	C	[93/11]
$\text{C}_{14}\text{H}_{14}\text{N}_2\text{O}_2$	4-(2-hydroxyethoxy)azobenzene				
		120.9		GS	[56/14][91/18]
$\text{C}_{14}\text{H}_{14}\text{N}_4\text{O}_2$	3-nitro-4'-(N,N-dimethylamino)azobenzene				[3837-55-6]
	(388–412)	133.9 ± 3.8	(400)	ME	[67/7]
	(392–410)	133.1 ± 3.8	(401)	TE	[67/7][87/4]
$\text{C}_{14}\text{H}_{14}\text{N}_4\text{O}_2$	4-nitro-4'-(N,N-dimethylamino)azobenzene				[2491-74-9]
	(413–425)	134.3 ± 7.5	(419)	ME	[67/7][66/18]
	(414–428)	135.1 ± 0.9	(421)	TE	[67/7][87/4]
		134.3		ME	[56/14][91/18]
$\text{C}_{14}\text{H}_{14}\text{O}$	1,1-diphenylethanol				[599-67-7]
		105.0 ± 0.8	(298)		[98/22]
$\text{C}_{14}\text{H}_{14}\text{O}_2\text{S}$	dibenzyl sulfone				[620-32-6]
		125.5 ± 2.9			[U/3][70/1]
$\text{C}_{14}\text{H}_{14}\text{O}_2\text{S}$	di- <i>p</i> -tolyl sulfone				[599-66-6]
		109.6 ± 2.9			[U/3][70/1]
$\text{C}_{14}\text{H}_{14}\text{O}_8$	1,2,4,5-tetramethoxycarbonylbenzene				[635-10-9]
	(371–391)	140.4 ± 0.8	(381)	ME	[95/6]
		143.3 ± 0.8	(298)		[95/6]
		135.9 ± 1.3	(298)		[67/8][95/6]
$\text{C}_{14}\text{H}_{14}\text{S}$	dibenzyl sulfide				[538-74-9]
		93.3 ± 5		E	[62/5][70/1]
$\text{C}_{14}\text{H}_{15}\text{N}_3$	4-(N,N-dimethylamino)azobenzene				[60-11-7]
	(346–354)	117.6 ± 1.7	(350)	ME	[67/7]
	(352–354)	115.9 ± 1.3	(353)	TE	[67/7]
		120.9 ± 1.7	(373)	ME	[56/2][87/4]
$\text{C}_{14}\text{H}_{15}\text{N}_3$	(E) 4-(N,N-dimethylamino)azobenzene				[25548-37-2]
		132 ± 8	(381)	TE	[85/22]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{14}\text{H}_{15}\text{N}_3$	2,3'-dimethyl-4-aminoazobenzene	112.5		GS	[87/19][91/18]
$\text{C}_{14}\text{H}_{16}$	1,4,5,8-tetramethylnaphthalene	99.8 ± 1.4	(298)	C	[2717-39-7] [74/14][77/1]
$\text{C}_{14}\text{H}_{16}\text{ClN}_3\text{O}_2$	{1-(4-chlorophenoxy)-3,3-dimethyl-1-(1 <i>H</i> -1,2,4-triazol-1-yl)}butanone (Triadimefon)	111.1 ± 2.2	(303)	GS	[43121-43-3] [97/6]
$\text{C}_{14}\text{H}_{16}\text{O}_5$	benzoyl(3-cyclohexyloxy)carbonyl peroxide	96.2 ± 4.2	(303)	ME	[20666-86-8] [71/11][77/1]
$\text{C}_{14}\text{H}_{18}\text{N}_2\text{O}_5$	2,6-dimethyl-3,5-dinitro-4- <i>tert</i> -butylacetophenone	107.9	(323)	ME	[81-14-1] [53/7][60/1]
$\text{C}_{14}\text{H}_{18}\text{O}$	diamantanone	$103.1 \pm .62$	(320)	TSGC	[30545-23-4] [80/4]
$\text{C}_{14}\text{H}_{18}\text{O}_2$	6,6-dimethyl-1-phenyl-4,8-dioxaspiro[2.5]octane	97.5 ± 0.3	(298)		[180988-52-7] [98/26]
$\text{C}_{14}\text{H}_{20}$	1,8-cyclotetradecadiyne				[1540-80-3]
	(315–364)	87.6 ± 1.0	(338)	HSA	[98/5]
		94.3	(298)	CGC–DSC	[98/5]
	(317–332)	166.0 ± 3.2	(325)	ME	[64/1][70/1]
$\text{C}_{14}\text{H}_{20}$	1,2,3,4,5,6,7,8-octahydroanthracene (octhracene)	82.3 ± 1.2	(298)	BG	[1079-71-6] [71/1][77/1]
$\text{C}_{14}\text{H}_{20}$	diadamantane				[2292-79-7]
	(305–333)	96.0 ± 0.8	(319)	TSGC	[75/2]
		117.2 ± 8		B	[71/9]
$\text{C}_{14}\text{H}_{20}\text{O}$	diamantan-1-ol				[30545-14-3]
	(319–349)	$118. \pm 0.6$	(334)		[80/4][75/2]
$\text{C}_{14}\text{H}_{20}\text{O}$	diamantan-3-ol				[30545-24-5]
	(323–354)	116.1 ± 4.4	(338)		[80/4][75/2]
$\text{C}_{14}\text{H}_{20}\text{O}$	diamantan-4-ol				[30651-03-7]
	(322–353)	117.8 ± 0.2	(337)		[80/4][75/2]
$\text{C}_{14}\text{H}_{20}\text{O}_5$	benzo-15-crown-5				[14098-44-3]
		123.2 ± 2.0	(298)	CGC–DSC	[00/11]
$\text{C}_{14}\text{H}_{21}\text{F}_3\text{N}_2\text{O}_4$	proline, 1-[N-(trifluoroacetyl)-1-leucyl]methyl ester				
	(313–366)	121.3	(328)		[87/4][60/20]
$\text{C}_{14}\text{H}_{21}\text{NO}$	4-isopropylbenzylidene <i>t</i> -butylamine N-oxide	101.8 ± 4.1	(298)	C	[121678-88-4] [89/15]
$\text{C}_{14}\text{H}_{22}$	1,4-di- <i>tert</i> -butylbenzene				[1012-72-2]
	(288–333)	82.1 ± 0.4	(310)	T	[98/14]
		82.8 ± 0.4	(298)		[98/14]
	(285–325)	82.8	(305)	ME, RG	[51/5][87/4]
$\text{C}_{14}\text{H}_{22}\text{O}$	2,6-di- <i>tert</i> -butylphenol				[128-39-2]
		84.6 ± 0.5	(298)	GS	[99/17]
		81.5 ± 2.3	(298)	C	[99/21]
		U110.9	(298)	C	[71/24][99/17]
$\text{C}_{14}\text{H}_{22}\text{O}$	2,4-di- <i>tert</i> -butylphenol				[96-76-4]
	(288–327)	86.1 ± 0.3	(308)	GS	[99/13]
		86.7 ± 0.3	(298)		[99/13]
		92.9 ± 2.8	(298)	C	[99/21]
$\text{C}_{14}\text{H}_{22}\text{O}$	3,5-di- <i>tert</i> -butylphenol				[1138-52-9]
		97.7 ± 3.7	(298)		[01/17]
	(302–325)	68.2	(313.5)		[87/4]
$\text{C}_{14}\text{H}_{22}\text{O}$	4- <i>tert</i> -octylphenol				[124765-79-3]
	(297–351)	96.3 ± 0.9	(324)	GS	[99/13]
		97.9 ± 0.9	(298)		[99/13]
$\text{C}_{14}\text{H}_{22}\text{O}_2$	3,5-di- <i>tert</i> -butyl-1,2-dihydroxybenzene				[1020-31-1]
	(313–346)	103.7 ± 0.5	(330)	GS	[00/21]
		104.7 ± 0.5	(298)		[00/21]
		100.1 ± 0.6	(298)	C	[84/20]
$\text{C}_{14}\text{H}_{22}\text{O}_2$	2,5-di- <i>tert</i> -butyl-1,4-dihydroxybenzene				[88-58-4]
	(333–368)	108.8 ± 1.7	(351)	GS	[99/28]
		122.4 ± 1.7	(298)		[99/28]
$\text{C}_{14}\text{H}_{22}\text{O}_6$	dicyclohexyl peroxydicarbonate				[1561-49-5]
		100.4 ± 4.2			[71/11][77/11]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_m/\text{kJ mol}^{-1}$	(T_m/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{14}\text{H}_{24}$	(293–313)	100.4 ± 8.3	(303)	ME	[62/3][70/1]
	<i>trans-anti-trans</i> perhydroanthracene				[28071-99-0]
	(269–313)	66.1	(284)		[87/4]
$\text{C}_{14}\text{H}_{24}$	(275–313)	72.7 ± 3.3	(294)	ME	[63/2][70/1]
	<i>trans-syn-trans</i> perhydroanthracene				[1755-19-7]
	(293–335)	88.1	(308)		[87/4]
$\text{C}_{14}\text{H}_{24}$	(335–393)	87.4 ± 2.4	(365)	ME	[63/2][70/1]
	1,3,5,7-tetramethyladamantane				[1687-36-1]
	(310–350)	83.7 ± 1.3	(298)	BG	[77/11]
$\text{C}_{14}\text{H}_{26}\text{O}$	(295–315)	81.1 ± 10.9	(305)	TSGC	[75/2]
	cyclotetradecanone				[295-17-0]
$\text{C}_{14}\text{H}_{28}$		80.75			[38/1][60/1]
	cyclotetradecane				[295-17-0]
		95.6	(298)	CGC–DSC	[98/5]
	(300–321)	97.9 ± 1.7	(310)	HSA	[92/1]
	(295–307)	134.8 ± 1.5	(301)	ME	[64/1][70/1]
$\text{C}_{14}\text{H}_{28}\text{O}$	(285–290)	89.3 ± 0.4	(287)	TM	[55/5]
	2-tetradecanone				[2345-27-9]
		130.9 ± 0.5	(298)	C	[79/8]
$\text{C}_{14}\text{H}_{28}\text{O}_2$	tetradecanoic acid (myristic acid)				[544-63-8]
	(282–305)	174		TPTD	[01/15]
	(312–325)	139.7 ± 3.8	(318)	ME	[61/1][70/1]
$\text{C}_{14}\text{H}_{28}\text{O}_3$	peroxytetradecanoic acid				[19816-73-0]
	(293–303)	156.0 ± 4.1		ME	[80/23]
$\text{C}_{14}\text{H}_{29}\text{NO}$	tetradecanamide				[638-58-4]
	(248–375)	167.4 ± 2.5	(352)	ME	[59/3][87/4]
$\text{C}_{14}\text{H}_{30}$	<i>n</i> -tetradecane				[629-59-4]
		117.6	(298)	B	[72/1]
$\text{C}_{14}\text{H}_{30}\text{O}$	1-tetradecanol				[112-72-1]
		126.0 ± 0.6			[77/17]
	(293–307)	143.9	(300)	ME	[65/6]
$\text{C}_{15}\text{H}_{10}\text{O}$	diphenylcyclopropenone				[886-38-4]
	(353–378)	119.7 ± 8	(365)	HSA	[85/3]
	(323–343)	141 ± 4	(333)	ME	[76/9][87/4]
$\text{C}_{15}\text{H}_{10}\text{O}_2$	α -benzoyloxyphthalide				
	(343–388)	U 125.3	(366)		[89/9]
$\text{C}_{15}\text{H}_{10}\text{O}_3$	1-methoxy-9,10-anthraquinone				[82-39-3]
		128.0		GS	[87/19][91/18]
		106.6	(385)	HSA	[56/1]
$\text{C}_{15}\text{H}_{10}\text{O}_3$	2-methoxy-9,10-anthraquinone				[3274-20-2]
		124.7		GS	[87/19][91/18]
		118.4 ± 0.4	(419)	HSA	[56/1]
$\text{C}_{15}\text{H}_{11}\text{F}_3\text{O}_3$	2-hydroxy-2'-trifluoromethyl-4-methoxybenzophenone				[3119-86-6]
	(323–363)	U 13.3	(338)	EV	[87/4][66/6]
$\text{C}_{15}\text{H}_{11}\text{F}_3\text{O}_3$	2-hydroxy-3'-trifluoromethyl-4-methoxybenzophenone				[7396-89-6]
	(313–323)	103.8	(318)	EV	[87/4][66/6]
$\text{C}_{15}\text{H}_{11}\text{F}_3\text{O}_3$	2-hydroxy-4'-trifluoromethyl-4-methoxybenzophenone				[7396-90-9]
	(313–333)	91	(323)	EV	[87/4][66/6]
$\text{C}_{15}\text{H}_{11}\text{N}$	2-phenylquinoline				[612-96-4]
	(337–351)	103.1 ± 0.8	(344)	ME	[97/14]
		105.4 ± 0.9	(298)		
$\text{C}_{15}\text{H}_{11}\text{NO}_2$	1-amino-2-methyl-9,10-anthraquinone				[82-28-0]
	(360–388)	124.6 ± 7.3	(374)		[84/35]
$\text{C}_{15}\text{H}_{11}\text{NO}_2$	1-(<i>N</i> -methylamino)-9,10-anthraquinone				[82-38-2]
		112.6			[84/40]
	(363–383)	115.9 ± 3.5	(373)		[84/35]
	(384–405)	123.8 ± 3.3	(395)	ME	[60/8][87/4]
		123.8 ± 3.2		ME	[64/11][91/18]
					[66/18]
$\text{C}_{15}\text{H}_{11}\text{NO}_4$		115.5 ± 0.4	(461)	HSA	[56/1]
		114.7 ± 3	(406)	HSA	[56/1]
	1-amino-2-methoxy-4-hydroxy-9,10-anthraquinone				
		132.0			[84/40]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₁₅ H ₁₁ N ₃ O ₂	4-hydroxy-3-(phenylazo)-2(1 <i>H</i>)-quinolinone (Disperse Yellow 4)	127.2			[6407-80-3] [68/10][88/24]
C ₁₅ H ₁₂	9-methylanthracene	98.9		RG	[779-02-2] [58/4]
C ₁₅ H ₁₂ N ₂ O ₂	1-amino-4-(<i>N</i> -methylamino)anthra-9,10-quinone	140.6		GS	[1220-94-6] [67/16][91/18]
C ₁₅ H ₁₂ N ₂ O ₃	1,4-diamino-2-methoxyanthra-9,10-quinone	147.0 151.9		GS	[2872-48-2] [84/40] [67/16][91/18]
C ₁₅ H ₁₂ O	dibenzosuberone	109.3	(298)	B	[1210-35-1] [98/21]
C ₁₅ H ₁₂ O	5,7-dihydro-6 <i>H</i> -dibenzo[<i>a,c</i>]cyclohepten-6-one (333–347)	93.4 ± 0.8 95.6 ± 0.8	(340) (298)	GS GS	[1139-82-8] [98/21] [98/21]
C ₁₅ H ₁₂ O ₂	dibenzoylmethane	115.7 ± 0.9	(298)	ME	[120-46-7] [92/28]
C ₁₅ H ₁₃ ClN ₂ O ₅	gallocyanine (C.I. Disperse Blue 95) (433–493)	88.2	(448)		[1562-85-2] [87/4]
C ₁₅ H ₁₃ NO ₂	<i>N</i> -benzoyl- <i>N</i> -methylbenzamide (246–269)	116.8 ± 0.4 120.1 ± 0.4	(356) (298)	ME	[23825-32-3] [97/12] [97/12]
C ₁₅ H ₁₄ Cl ₂ N ₄ O ₃	4-(<i>N</i> -methyl- <i>N</i> -2-hydroxyethylamino)-4'-nitro-2',6'-dichloroazobenzene	135.1			[6232-56-0] [68/10][88/24]
C ₁₅ H ₁₄ F ₃ N ₃	<i>N,N</i> -dimethyl-4-[[4-(trifluoromethyl)phenyl]azo]benzenamine	95.8		UV	[405-82-3] [84/39]
C ₁₅ H ₁₄ F ₃ N ₃ O	<i>N,N</i> -dimethyl-4-[[4-(trifluoromethoxy)phenyl]azo]benzenamine	96.8		UV	[1494-75-3] [84/39]
C ₁₅ H ₁₄ F ₃ N ₃ S	<i>N,N</i> -dimethyl-4-[[4-[(trifluoromethyl)thio]phenyl]azo]benzenamine	100.8		UV	[1494-77-5] [84/39]
C ₁₅ H ₁₄ N ₂	<i>N,N</i> -dimethyl-9-acridinamine	86	(510)	TGA	[3295-59-8] [98/29]
C ₁₅ H ₁₄ N ₂	<i>N</i> -methyl-10-methylacridinimine	72	(480)	TGA	[213623-43-9] [98/29]
C ₁₅ H ₁₄ O	4,5,6-trimethylbenzoxalene	139.7 ± 2.5			[10435-68-4] [66/3][70/1]
C ₁₅ H ₁₄ O	1,3-diphenyl-2-propanone	89.1 ± 5			[102-04-5] [54/2][77/1] [70/1]
C ₁₅ H ₁₄ O ₂	2,2-diphenyl-1,3-dioxalene	99.7 ± 1.1	(298)		[4359-34-6] [98/26]
C ₁₅ H ₁₄ O ₂ S	(<i>Z</i>)-1-methyl-4-(2-phenylethenyl)sulfonyl benzene	116.3 ± 3.8		B	[54897-33-5] [69/11][77/1]
C ₁₅ H ₁₄ O ₂ S	(<i>E</i>)-1-methyl-4-(2-phenylethenyl)sulfonyl benzene	108.4 ± 2.5		B	[16212-08-1] [69/11][77/1]
C ₁₅ H ₁₄ O ₄	2-hydroxy-4,4'-dimethoxybenzophenone	121.1		B	[631-38-0] [99/22]
C ₁₅ H ₁₄ O ₅	2,2'-dihydroxy-4,4'-dimethoxybenzophenone	130.2 147.0		B UV	[131-54-4] [99/22] [60/24]
C ₁₅ H ₁₅ N ₃ O ₂	<i>N</i> -[4-[(2-hydroxy-5-methylphenyl)azo]phenyl]acetamide (Disperse Yellow 3) (403–465)	107 140.6	(434)	GS	[2832-40-8] [89/29] [68/10][88/24]
C ₁₅ H ₁₆ N ₄ O ₂	3-methyl-3'-nitro-4- <i>N,N</i> -dimethylaminoazobenzene (368–393) (370–388)	101.7 ± 1.7 98.7 ± 2.5	(381) (379)	ME TE	[4313-14-8] [67/7] [67/7]
C ₁₅ H ₁₆ N ₄ O ₂	3-methyl-4'-nitro-4- <i>N,N</i> -dimethylaminoazobenzene (369–392) (371–390)	125.5 ± 1.3 126.4 ± 3.8	(381) (381)	TE ME	[92114-99-3] [67/7][87/4] [67/7]
C ₁₅ H ₁₆ O ₂	dimethoxydiphenylmethane	103.9 ± 1.7	(298)		[2235-01-0] [98/26]
C ₁₅ H ₁₇ NO ₂	<i>N</i> -(2-hydroxy-3-phenoxypropyl)phenylamine (323–333)	99.9	(328)		[16112-55-3] [87/4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_m/\text{kJ mol}^{-1}$	(T_m/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{15}\text{H}_{18}\text{N}_2$	4-isopropylaminodiphenylamine (323–348)	113.8 ± 2.1 120.7	(335)	GS	[76/19] [101-72-4] [71/19]
$\text{C}_{15}\text{H}_{21}\text{NO}$	2-methyl-1-phenyl-2-N-piperidinyl-1-propanone	94.8 ± 1.3		B	[13430-30-3] [94/11]
$\text{C}_{15}\text{H}_{22}$	1-methyldiamantane (310–333)	80.7 ± 0.4	(321)	TSGC	[26460-76-4] [75/2]
$\text{C}_{15}\text{H}_{22}$	3-methyldiamantane (305–327)	103.1 ± 1.0	(316)	TSGC	[38375-86-2] [75/2]
$\text{C}_{15}\text{H}_{22}$	4-methyldiamantane (310–333)	79.4 ± 1.25	(321)	TSGC	[30545-18-9] [75/2]
$\text{C}_{15}\text{H}_{22}\text{O}_2$	3,5-di- <i>tert</i> -butylbenzoic acid (339–357)	108.4 ± 4.2	(348)	ME	[16225-26-6] [74/15][87/4] [77/1]
$\text{C}_{15}\text{H}_{24}$	1,3-di- <i>tert</i> -butyl-5-methylbenzene (275–301)	82.4 ± 0.5 81.8 ± 0.5	(288) (298)	T	[15181-11-0] [98/14] [98/14]
$\text{C}_{15}\text{H}_{24}\text{O}$	2,6-di- <i>tert</i> -butyl-4-methylphenol	91.9 ± 3.2 86.8 ± 0.8 88.0 ± 0.8 87.8 U117.3	(298) (319) (298) (318) (298)	C GS GS GS C	[128-37-0] [01/17] [99/17] [99/17] [87/4][71/19] [71/24][99/17]
$\text{C}_{15}\text{H}_{28}\text{O}$	cyclopentadecanone (296–315)	86.0 ± 0.6 77.4	(305)	ME	[502-72-7] [97/13] [38/1][60/1] [70/1]
$\text{C}_{15}\text{H}_{28}\text{O}_2$	pentadecanolide (290–310)	81.3	(300)	ME	[32539-85-8] [87/4][60/1] [54/4]
$\text{C}_{15}\text{H}_{30}$	cyclopentadecane	74.6 ± 0.4			[295-48-7] [57/1][70/1]
$\text{C}_{15}\text{H}_{30}\text{O}$	2-pentadecanone	139.3 ± 1.6	(298)	C	[2345-28-0] [79/8]
$\text{C}_{15}\text{H}_{30}\text{O}_2$	methyl tetradecanoate	137.7 ± 2.1	(281)	ME	[124-10-7] [65/6]
$\text{C}_{15}\text{H}_{30}\text{O}_2$	pentadecanoic acid (283–305)	178		TPTD	[1002-84-2] [01/15]
$\text{C}_{15}\text{H}_{31}\text{NO}$	N-methyl tetradecanamide (332–347)	130.4 ± 0.8	(340)	GS	[7438-09-7] [59/4][87/4]
$\text{C}_{15}\text{H}_{32}$	<i>n</i> -pentadecane	107.8	(298)	B	[629-62-9] [72/1]
$\text{C}_{16}\text{F}_{34}$	<i>n</i> -perfluorohexadecane (288–303)	104.6	(295)	ME	[355-49-7] [51/9][87/4]
$\text{C}_{16}\text{H}_6\text{Br}_4\text{N}_2\text{O}_2$	5,7-dibromo-2-(5,7-dibromo-1,3-dihydro-3-oxo-2 <i>H</i> -indol-2-ylidene)- 1,2-dihydro-3 <i>H</i> -indol-3-one (C.I. Vat Blue 5) (519–634)	129	(577)	GS	[2475-31-2] [86/14]
$\text{C}_{16}\text{H}_9\text{BrN}_2\text{O}_2$	5-bromo-2-(1,3-dihydro-3-oxo-2 <i>H</i> -indol-2-ylidene)-1,2-dihydro-3 <i>H</i> - indol-3-one (C.I. Vat Blue 3) (519–634)	57	(577)	GS	[6492-73-5] [86/14]
$\text{C}_{16}\text{H}_{10}$	fluoranthene (313–453) (283–323)	98.3 84.6 ± 0.9 99.2 ± 0.8	(383) (303) (298)	GS GS C	[206-44-0] [95/7] [83/11] [72/1][77/1]
$\text{C}_{16}\text{H}_{10}$	(328–353) (298–358) pyrene (308–398) (313–453) (369–383) (283–323) (398–423)	102.1 ± 2 102.6 103.1 ± 6.5 97.9 100.3 ± 0.3 91.2 ± 0.5 100.2 ± 0.4 101.0 ± 0.5	(340) (328) (353) (383) (353) (303) (410)	ME ME GS PG GS IPM C	[65/3][70/1] [58/1] [129-00-0] [98/3] [95/7] [88/26] [83/11] [80/26] [74/4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
	(348–419)	100.8 ± 1.5		ME	[74/4]
		95.7		ME	[53/1][77/1]
	(298–363)	100.5	(330)	ME	[70/1]
	(345–358)	100.1 ± 1.7	(351)	ME	[58/1]
C ₁₆ H ₁₀ O	1-hydroxypyrene				[52/3]
	(369–394)	129.0 ± 3.2	(382)	ME	[5315-79-7]
C ₁₆ H ₁₀ N ₂ O ₂	2-(1,3-dihydro-3-oxo-2 <i>H</i> -indol-2-ylidene)-1,2-dihydro-3 <i>H</i> -indol-3-one (C.I. Vat Blue 1)				[98/3]
	(519–634)	136	(577)	GS	[482-89-3]
C ₁₆ H ₁₀ O	2,3,5,6-dibenzoxalene (benz[b]indeno[1,2- <i>e</i>]pyran)				[86/14]
	(375–388)	125.9	(381.5)		[243-24-3]
		129.4 ± 1.3			[87/4]
C ₁₆ H ₁₀ S	1,2-benzodiphenylene sulfide				[66/3][70/1]
	(325–373)	111.9 ± 1.2	(349)	ME	[239-35-0]
C ₁₆ H ₁₂ N ₂ O	2-hydroxy-1-phenylazonaphthalene				[98/3]
	(350–374)	116.7 ± 5.4	(362)		[842-07-9]
C ₁₆ H ₁₂ O	2,5-diphenylfuran				[84/35]
		102	(340)		[955-83-9]
C ₁₆ H ₁₂ S ₂	3,6-diphenyl-1,2-dithiin				[89/7]
		174.5 ± 2.5	(355)		[16212-85-4]
		183.1 ± 2.5	(298)		[73/23][77/1]
C ₁₆ H ₁₃ N	N-phenyl-1-naphthylamine				[73/23][77/1]
	(313–333)	96.5	(323)	GS	[90-30-2]
C ₁₆ H ₁₃ N	N-phenyl-2-naphthylamine				[87/4][71/19]
	(333–363)	115.8	(348)	GS	[135-88-6]
C ₁₆ H ₁₃ NO	9-acetamidoanthracene				[87/4][71/19]
	(446–500)	134.8	(461)	RG	[37170-96-0]
C ₁₆ H ₁₃ NO	N-(4-hydroxyphenyl)-2-naphthylamine				[58/4][87/4]
	(373–408)	126.8	(390)	GS	[93-45-8]
C ₁₆ H ₁₃ NO ₂	1-(dimethylamino)-9,10-anthraquinone				[71/19]
	(396–408)	U 3.6	(402)		[5960-55-4]
C ₁₆ H ₁₃ NO ₃	1-(2-hydroxyethylamino)-9,10-anthraquinone				[87/4]
	(403–417)	152.7 ± 3.8	(410)	ME	[4465-58-1]
C ₁₆ H ₁₃ NO ₅	1-amino-2-hydroxyethyl-4-hydroxy-9,10-anthraquinone				[60/8][66/18]
		135.2			[84/40]
C ₁₆ H ₁₄	9,10-dimethylanthracene				[781-43-1]
	(372–382)	114.6	(377)		[87/4]
	(381–434)	103.2	(396)	RG	[58/4][87/4]
C ₁₆ H ₁₄	2,7-dimethylphenanthrene				[1576-69-8]
		106.7 ± 0.8		ME	[65/5][70/1]
C ₁₆ H ₁₄	4,5-dimethylphenanthrene				[3674-69-9]
	(313–453)	85.7	(383)	GS	[95/7]
		104.6 ± 1.3		ME	[65/5][70/1]
C ₁₆ H ₁₄	9,10-dimethylphenanthrene				[604-83-1]
		119.5 ± 1.3			[66/3][70/1]
C ₁₆ H ₁₄	1-phenylnaphthalene				[605-02-7]
	(313–453)	88.6	(383)	GS	[95/7]
C ₁₆ H ₁₄	1,4-diphenylbutadiene				[866-65-7]
		87.0		RG	[58/4]
C ₁₆ H ₁₄	4,5,9,10-tetrahydropyrene				[781-17-9]
	(385–410)	90.4	(400)	IPM	[93/7]
C ₁₆ H ₁₄ Cl ₂ O ₂	1,1-dichloro-2,2-bis-(4-methoxyphenyl)ethylene				[4359-34-6]
		79.2			[95/32][89/32]
C ₁₆ H ₁₄ F ₄ N ₄ O ₂	N-methyl-N-(2,2,3,3-tetrafluoropropyl)-4-[(4-nitrophenyl)azobenzenamine				[80135-84-8]
		100.8		UV	[84/39]
C ₁₆ H ₁₄ N ₂ O ₂	1,4-bis(N-methylamino)anthra-9,10-quinone				[2475-44-7]
	(385–413)	151.8 ± 3.9	(399)		[84/35]
		150.2		GS	[67/16][91/18]
C ₁₆ H ₁₅ NO	3-anilino-1-phenylbut-2-enone				[18594-93-9]
		126.8 ± 3.0	(298)	C	[93/24]
C ₁₆ H ₁₆	[2.2]-para-cyclophane				[1633-22-3]
	(353–409)	96.4 ± 1.5		TSGC	[80/15]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{16}\text{H}_{16}$	(343–383)	96.3 $\pm$ 4.2 92.9 $\pm$ 0.84	(363)	ME	[73/13][77/1] [66/11][87/4] [70/1]
	[2.2]-meta-cyclophane (308–332)	91.6 $\pm$ 1.7	(320)	ME	[2319-97-3] [69/6][77/1] [87/4]
		92.0 $\pm$ 2	(298)	ME	[69/6][77/1]
$\text{C}_{16}\text{H}_{16}$	[2.2]-meta-para-cyclophane (311–328)	86.6	(336)	ME	[5385-36-4] [69/6][77/1] [87/4]
$\text{C}_{16}\text{H}_{16}$	1,1-bis-(4-methylphenyl)ethane	87.5 $\pm$ 0.9	(298)	ME	[69/6][77/1] [2919-20-2] [99/19]
$\text{C}_{16}\text{H}_{16}$	1,2,3,6,7,8-hexahydropyrene (390–405)	101.0 $\pm$ 1.4	(298)		[1732-13-4] [93/7]
$\text{C}_{16}\text{H}_{16}\text{O}_3$	α,α -dimethylbenzyl perbenzoate (293–313)	92.3	(398)	IPM	[7074-00-2] [71/11][77/1]
$\text{C}_{16}\text{H}_{16}\text{O}_{10}$	pentamethoxycarbonylbenzene (389–413)	43.1 $\pm$ 4.2	(303)	ME	[3327-06-8] [95/6]
$\text{C}_{16}\text{H}_{17}\text{ClN}_4\text{O}_3$		160.0 $\pm$ 0.8	(401)	ME	[67/8][95/6]
		165.1 $\pm$ 0.8	(298)		[3180-81-2] [68/10][88/24]
		142.7			[193472-70-7] [97/34]
$\text{C}_{16}\text{H}_{17}\text{F}$	2-fluoro-2-methyl-1,3-diphenylpropane	102.2 $\pm$ 1.1	(298)		[15582-77-5] [94/11]
$\text{C}_{16}\text{H}_{17}\text{NO}$	1,2-diphenyl-2-N,N-dimethylamino-1-ethanone	140.1 $\pm$ 1.9		B	[99081-88-6] [86/10]
$\text{C}_{16}\text{H}_{17}\text{NO}$	N-(4-isopropylphenylmethylene) benzenamine N-oxide	127.2 $\pm$ 1.7	(298)	C	[1520-44-1] [74/6][87/4] [5789-35-5] [84/16]
$\text{C}_{16}\text{H}_{18}$	(dl) 1,3-diphenylbutane (288–303)	73.6	(296)	ME	[3788-15-6] [87/4][65/7]
$\text{C}_{16}\text{H}_{18}$	2,3-diphenylbutane (293–348)	96.7	(326)		[4792-83-0] [93/11]
$\text{C}_{16}\text{H}_{18}\text{NO}_5$	bis(2,4-dimethoxyphenyl)nitrogen oxide (333–363)	144.1 $\pm$ 11.4	(348)		[101225-69-8] [93/3]
$\text{C}_{16}\text{H}_{18}\text{N}_2\text{O}_3$	<i>p</i> -azoxyphenetole	126.2 $\pm$ 2.7	(298)	C	[3025-52-3] [87/19][91/18]
$\text{C}_{16}\text{H}_{18}\text{N}_2\text{O}_2$	2,2',6,6'-tetramethylazobenzene-N,N-dioxide	107 $\pm$ 12	(298)	ME	[60/8] [2872-52-8]
$\text{C}_{16}\text{H}_{18}\text{N}_4\text{O}_2$	4-(N,N-diethylamino)-4'-nitroazobenzene	146.0		GS	[84/39][84/40] [68/10][88/24]
$\text{C}_{16}\text{H}_{18}\text{N}_4\text{O}_3$	(422–441)	151.5 $\pm$ 4.2	(431)	ME	[60/8]
	4-(N-ethyl-N-2-hydroxyethylamino)-4'-nitroazobenzene	136.8		UV	[60/8][66/18] [2481-94-9]
		189.5			[87/19][91/18]
$\text{C}_{16}\text{H}_{19}\text{N}_3$	(420–433)	176.6 $\pm$ 1.3	(426)	ME	[84/35] [84/39] [84/40]
	4-(N,N-diethylamino)azobenzene	132.2		GS	[199394-72-4] [97/18]
	(330–353)	91.4 $\pm$ 2.9	(342)		
$\text{C}_{16}\text{H}_{23}\text{N}$	N-cyclohexyl-(2,4,6-trimethyl)benzaldehyde imine	106.4 103.4		UV	
$\text{C}_{16}\text{H}_{24}\text{N}_2\text{O}_2$	N-benzoyl-N',N'-diisobutylurea	104.9 $\pm$ 0.8	(298)	B	
$\text{C}_{16}\text{H}_{26}\text{O}$	2,4,5-triisopropylbenzyl alcohol (313–346)	137.5 $\pm$ 4.4	(298)	C	[00/28]
$\text{C}_{16}\text{H}_{28}$	tricyclo[8.2.2.2 ^{4,7}]hexadecane (316–338)	113.0	(330)		[63/13] [283-68-1]
$\text{C}_{16}\text{H}_{30}\text{N}_2$		85.2 $\pm$ 1.3	(327)	ME	[69/6][77/1] [87/4]
	(316–338)	91.6 $\pm$ 2.1	(298)	ME	[69/6][77/1] [19219-01-3] [84/22]
	tetracyclopropylsuccinonitrile	110.2 $\pm$ 1.5			[2550-52-8]
$\text{C}_{16}\text{H}_{30}\text{O}$	cyclohexadecanone				

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{16}\text{H}_{30}\text{O}_4$	hexadecanedioic acid (377–398)	82			[38/1][60/1]
		151.0 $\pm$ 3.3	(388)	ME	[505-54-4]
		155.4 $\pm$ 3.3	(298)		[60/4][87/4]
$\text{C}_{16}\text{H}_{32}$	cyclohexadecane				[60/4][99/10]
$\text{C}_{16}\text{H}_{32}\text{O}_2$	hexadecanoic acid (palmitic acid) (294–316) (320–333)	81.8 $\pm$ 0.4			[295-65-8]
		154		TPTD	[57/1][70/1]
		154.4 $\pm$ 4.2	(326)	ME	[57-10-3]
$\text{C}_{16}\text{H}_{33}\text{NO}$	hexadecanamide (364–378)				[01/15]
$\text{C}_{16}\text{H}_{34}$	hexadecane				[61/1][70/1]
		181.6 $\pm$ 1.3	(371)	ME	[87/4]
		135.1	(298)	B	[629-54-9]
$\text{C}_{16}\text{H}_{34}\text{O}$	1-hexadecanol (323–335) (308–320)	134.9	(291)	B	[59/3][87/4]
		83.4 $\pm$ 8		ME	[544-76-3]
		109.4 (liq)	(329)		[72/1]
		167.4 $\pm$ 2.1	(314)	ME	[63/6]
		169.5 $\pm$ 2.1	(298)		[49/1]
$\text{C}_{17}\text{H}_{10}\text{O}$	benzanthrone (389–409) (353–388)				[36653-82-4]
		121.6 $\pm$ 0.6	(399)	ME	[87/4]
		126.6 $\pm$ 0.6	(298)		[65/6][87/4]
		129.7 $\pm$ 2.1	(298)	QR	[65/6]
		119.7 $\pm$ 5.4	(370)	ME	[82-05-3]
		124.6 $\pm$ 6.0	(298)		[99/11]
		114.2 $\pm$ 0.8		QR	[99/11]
$\text{C}_{17}\text{H}_{12}$	benzo[a]fluorene (313–453)	115.5	(398)		[99/24]
$\text{C}_{17}\text{H}_{12}$	benzo[b]fluorene (344–398) (313–453)	105.4	(383)	GS	[84/12]
		119.3 $\pm$ 1.3	(371)	ME	[79/28]
		111.2	(383)	GS	[52/3][60/1]
$\text{C}_{17}\text{H}_{13}\text{N}$	N-methyl-2,3,5,6-dibenzazalene				[238-84-6]
$\text{C}_{17}\text{H}_{14}\text{N}_2\text{O}_2$	1-[(2-methoxyphenyl)azo]-2-hydroxynaphthalene (374–388)	131.8 $\pm$ 1.3			[95/7]
		142.4 $\pm$ 2.2	(381)		[243-17-4]
$\text{C}_{17}\text{H}_{16}\text{F}_4\text{N}_4\text{O}_2$	N-ethyl-N-(2,2,3,3-tetrafluoropropyl)-4-[4-nitrophenyl]azo-benzenamine				[98/3]
$\text{C}_{17}\text{H}_{16}\text{F}_4\text{N}_4\text{O}_3$	2-[[4-[(4-nitrophenyl)azo]phenyl](2,2,3,3-tetrafluoropropyl)amino]-ethanol	103.0		UV	[95/7]
		103.0		UV	[66/3][70/1]
		136.0	(375)	ME	[6626-64-8]
$\text{C}_{17}\text{H}_{16}\text{OS}$	tetrahydro-2,6-diphenyl-4 <i>H</i> -thiopyran-4-one	144 $\pm$ 3	(298)	ME	[66/3][70/1]
$\text{C}_{17}\text{H}_{16}\text{O}_4$	diphenylmethylene diacetate (348–388)				[1229-55-6]
		122.1 $\pm$ 1.2	(368)	GS	[84/35]
$\text{C}_{17}\text{H}_{17}\text{N}_5\text{O}_2$	4-nitro-4'-(N-2-cyanoethyl-N-ethylamino)azobenzene	147.3			[91488-84-5]
$\text{C}_{17}\text{H}_{18}\text{O}_3$	4-(<i>tert</i> -butylphenyl)salicylate (293–336)	137.4	(308)	UV	[84/39]
$\text{C}_{17}\text{H}_{18}\text{O}_4$	2-hydroxy-4,4'-diethoxybenzophenone	134.9	(298)	B	[1543-74-4]
$\text{C}_{17}\text{H}_{20}\text{O}_2$	diethoxydiphenylmethane	97.1 $\pm$ 1.1	(298)		[84/39]
$\text{C}_{17}\text{H}_{21}\text{NO}_4$	cocaine (294–314)				[37014-01-0]
		127.2		GS	[72/15]
$\text{C}_{17}\text{H}_{32}\text{O}$	cycloheptadecanone	112.3 $\pm$ 2.8	(304)	GS	[72/15][77/1]
		75.7			[54334-63-3]
$\text{C}_{17}\text{H}_{34}$	cycloheptadecane				[96/14]
		66.1 $\pm$ 0.6			[84/40]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{17}\text{H}_{34}\text{O}_2$	methyl hexadecanoate (291–301)	152.3 ± 2	(296)	ME	[112-39-0] [65/6][87/4]
$\text{C}_{17}\text{H}_{34}\text{O}_2$	heptadecanoic acid (margaric acid) (291–316)	168		TPTD	[506-12-7] [01/15]
$\text{C}_{17}\text{H}_{35}\text{NO}$	N-methyl hexadecanamide (345–355)	144.5 ± 0.8	(350)	GS	[7388-58-1] [59/4][87/4]
$\text{C}_{17}\text{H}_{36}$	heptadecane (288–293)	125.1 131.3 ± 13	(298) (290)	ME	[629-78-7] [72/1] [49/1][60/1]
$\text{C}_{17}\text{H}_{36}\text{O}$	1-heptadecanol				[1454-85-9] [65/6][70/1]
$\text{C}_{18}\text{H}_{10}$	benzo[3,4]cyclobuta[1,2-a]biphenylene ([3]phenylene)	169.5 ± 2.2 115.1 ± 0.8			[65513-20-4] [00/18]
$\text{C}_{18}\text{H}_{10}\text{BrNO}_3$	2(4-bromo-3-hydroxy-2-quinolinyl)-1 <i>H</i> -indene-1,3(2 <i>H</i>)-dione (C. I. Disperse Yellow 64) (483–523)	130.6	(498)		[10319-14-9] [87/4]
$\text{C}_{18}\text{H}_{10}\text{Cl}_2\text{O}_2\text{S}_2$	6-chloro-2-(6-chloro-4-methyl-3-oxobenzo[b]thien-2(3 <i>H</i>)-ylidene)- 4-methyl-benzo[b]thiophen-3(2 <i>H</i>)-one (C.I. Vat Red 1) (519–634)	148	(577)	GS	[2379-74-0] [86/14]
$\text{C}_{18}\text{H}_{10}\text{Cl}_2\text{O}_2\text{S}_2$	5-chloro-2-(5-chloro-7-methyl-3-oxobenzo[b]thien-2(3 <i>H</i>)-ylidene)- 7-methyl-benzo[b]thiophen-3(2 <i>H</i>)-one (C.I. Vat Violet 2) (519–634)	93	(577)	GS	[5462-29-3] [86/14]
$\text{C}_{18}\text{H}_{10}\text{O}_2$	1,2-benzanthra-9,10-quinone	82.8 ± 4.0			[2498-66-0] [56/5][70/1]
$\text{C}_{18}\text{H}_{10}\text{O}_2$	5,12-tetracenequinone	108.8 ± 5.0			[1090-13-7] [56/5][70/1]
$\text{C}_{18}\text{H}_{10}\text{O}_4$	6,11-dihydroxy-5,12-naphthacenedione (426–446)	144.2 ± 1.4	(436)	ME	[1785-52-0] [98/3]
$\text{C}_{18}\text{H}_{11}\text{NO}_3$	2-(3-hydroxy-2-quinolinylidene)-indeno-1,3-dione (Disperse Yellow 54) (483–513)	125.2 ± 0.4 139	(498)	LE	[7576-65-0] [98/38] [73/6]
$\text{C}_{18}\text{H}_{12}$	benz[a]anthracene (1,2-benzanthracene, tetraphene) (313–453) (330–390)	115.5 113.4	(383) (345)	GS ME	[56-55-3] [95/7] [87/4][74/30]
		104 ± 2	(351)	TE	[83/27]
		U 81.3 ± 2.5	(303)	GS	[83/11]
		123.3 ± 3	(298)		[80/1]
		120.5	(405)	ME	[67/2]
		104.6 ± 4.2	(390)	ME	[64/3][87/4]
		119.7	(363)		[58/1]
		U 109.2			[51/2][60/1]
$\text{C}_{18}\text{H}_{12}$	triphenylene (313–453) (381–406) (363–468) (338–398)	114.5 126.5 ± 4 107.6 118 ± 4 107.1	(383) (298) (378) (368) (425)	GS TE,ME ME	[217-59-4] [95/7] [80/1] [87/4] [58/1][70/1] [67/2]
$\text{C}_{18}\text{H}_{12}$	naphthacene (tetracene) (386–472) (313–453) (419–446)	126.1 ± 9.0 126.5 143.7 ± 0.5 124.7 ± 4	(429) (383) (298) (422)	ME GS TE,M ME	[92-24-0] [98/3] [95/7] [80/1] [67/2][77/1] [70/1]
		128.8	(473)	HSA	[65/15]
		132.6	(468)	HSA	[64/10]
		117.2	(459)	ME	[52/3][60/1]
		U92.0	(384)	ME	[51/12]
		124.3			[51/2][60/1]
$\text{C}_{18}\text{H}_{12}$	benzo[c]phenanthrene (3,4-benzophenanthrene)	106.3 ± 4.2			[195-19-7] [51/2][70/1] [67/2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{18}\text{H}_{12}$	chrysene				[218-01-9]
	(313–453)	118.8	(383)	GS	[95/7]
		131 ± 4	(298)	TE,ME	[80/1]
		117.6 ± 4	(400)	ME	[67/2][70/1]
	(353–418)	121.4	(385)		[58/1]
$\text{C}_{18}\text{H}_{12}\text{N}_2$		117.6			[51/2][60/1]
	2,2'-biquinoline				[119-91-5]
	(393–411)	129.5 ± 0.8	(402)	ME	[97/14]
		134.7 ± 1.3	(298)		[97/14]
$\text{C}_{18}\text{H}_{12}\text{O}$	2-phenylindeno[2,1-b]pyran	96.6 ± 0.9			[85/6]
$\text{C}_{18}\text{H}_{14}$	(394–424)	132.8	(409)		[10435-67-3]
	5,12-dihydrotetracene				[87/4][66/3]
$\text{C}_{18}\text{H}_{14}$	(338–398)	115.9 ± 4	(368)		[959-02-4]
		120.5			[58/1][70/1]
$\text{C}_{18}\text{H}_{14}$	diphenylfulvene				[51/2][60/1]
$\text{C}_{18}\text{H}_{14}$		104.6 ± 8.3		E	[2175-90-8]
	<i>o</i> -terphenyl				[57/2][70/1]
$\text{C}_{18}\text{H}_{14}$		101.2 ± 0.5	(298)	B	[84-15-1]
		97 ± 1	(298)	B	[97/15]
	<i>m</i> -terphenyl				[71/8]
	(329–353)	115.5 ± 1.6	(341)	T	[92-06-8]
$\text{C}_{18}\text{H}_{14}$		118.1 ± 1.6	(298)		[97/15]
		120 ± 1	(298)		[97/15]
	(313–363)	119	(338)		[71/8]
	<i>p</i> -terphenyl				[58/1]
	(353–383)	116.2 ± 2.4	(368)	T	[92-94-4]
$\text{C}_{18}\text{H}_{14}\text{O}$		120.4 ± 2.4	(298)		[97/15]
		113 ± 2	(298)	B	[97/15]
		118.4	(397)	ME	[71/8]
	(333–393)	120.6	(363)		[67/2]
	2,6-diphenylphenol				[58/1]
$\text{C}_{18}\text{H}_{14}\text{O}$	(334–363)	116.1 ± 1.1	(348)	T	[92-69-3]
		119.1 ± 1.1	(298)		[98/9]
$\text{C}_{18}\text{H}_{15}\text{N}$	triphenylamine				[98/9]
$\text{C}_{18}\text{H}_{15}\text{NO}_2$	(322–373)	87.9 ± 1.3	(337)		[603-34-9]
	9-diacetylaminoanthracene				[78/2][87/4]
$\text{C}_{18}\text{H}_{15}\text{OP}$		106.4		RG	[3808-37-5]
	triphenylphosphine oxide				[58/4]
	(385–408)	131 ± 2	(399)	ME,TE	[791-28-6]
$\text{C}_{18}\text{H}_{15}\text{P}$		66 ± 6	(298)	B	[89/28]
	triphenylphosphine				[78/11]
		113.2 ± 3.0	(298)		[603-35-0]
		109.2 ± 1.1	(350)		[88/21]
$\text{C}_{18}\text{H}_{15}\text{PO}_4$		96.2 ± 8.4	(298)		[84/13]
	triphenyl phosphate				[82/20][60/9]
		114.4 ± 2.6	(298)	B	[115-86-6]
$\text{C}_{18}\text{H}_{15}\text{PS}$					[89/23]
	triphenylphosphine sulfide				[3878-45-3]
	(388–419)	136.8 ± 6.1	(403)	HSA	[96/9]
$\text{C}_{18}\text{H}_{16}\text{N}_4$		142.8 ± 6.8	(298)		[96/9]
	dihydrodibenzotetra-aza-annulene				[22119-35-3]
$\text{C}_{18}\text{H}_{18}$	(443–583)	81.5 ± 6.4	(513)	T	[83/29]
	2,4,5,7-tetramethylphenanthrene				[7396-38-5]
$\text{C}_{18}\text{H}_{18}$		114.2 ± 1.7		ME	[65/5][70/1]
	3,4,5,6-tetramethylphenanthrene				[7343-06-8]
$\text{C}_{18}\text{H}_{18}$		133.5 ± 3.8		ME	[65/5][70/1]
	9-butylanthracene				[1498-69-7]
$\text{C}_{18}\text{H}_{18}\text{O}_2$	(293–313)	108.1	(303)		[87/4][64/15]
	3-diphenylmethyl-2,4-pentanedione				[19672-37-8]
$\text{C}_{18}\text{H}_{18}\text{O}_4$	(348–383)	112.8 ± 0.4	(366)	T	[95/2]
	2,2'-diphenyl-bi-(1,3-dioxolan-2-yl)				[25062-95-7]
$\text{C}_{18}\text{H}_{18}\text{O}_{12}$	(320–362)	132.8 ± 2.1	(341)	T	[95/13]
	hexamethoxycarbonylbenzene				[6237-59-8]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{18}\text{H}_{20}$	(403–422)	140.7 ± 1.1	(413)	ME	[95/6]
		154.3 ± 1.2	(298)		[95/6]
	[3.3]para-cyclophane (322–343)	97.8 ± 0.8	(332)	ME	[2913-24-8] [69/6][77/1] [87/4]
$\text{C}_{18}\text{H}_{20}\text{O}_4$	(322–343)	103.3 ± 1.0	(298)	ME	[69/6][77/1]
	2-hydroxy-4-butoxy-4'-methoxybenzophenone	126.3		B	[39716-92-2] [99/22]
$\text{C}_{18}\text{H}_{21}\text{N}$	N-benzyl-pivalophenone imine	109.7 ± 3.3	(298)	B	[97/18]
$\text{C}_{18}\text{H}_{22}$	2,3-dimethyl-2,3-diphenylbutane				[1889-67-4]
	(293–348)	96.7 ± 0.8	(320)		[83/16]
$\text{C}_{18}\text{H}_{22}\text{N}_2\text{O}_2$	2,2',4,4',6,6'-hexamethylazobenzene-N,N-dioxide				[100046-00-2]
		107 ± 12	(298)	ME	[93/3]
$\text{C}_{18}\text{H}_{22}\text{N}_4$	2,3-dimethyl-2,3-bis(phenylazo)butane				[133930-64-0]
		113.8 ± 1.8		B	[93/14]
$\text{C}_{18}\text{H}_{22}\text{O}_2$	(dl)-2,3-dimethoxy-2,3-diphenylbutane				[41047-48-7]
	(322–355)	114.2 ± 6.3	(339)		[90/17]
$\text{C}_{18}\text{H}_{22}\text{O}_4$	1,2-diphenyl-1,1,2,2-tetramethoxyethane				[39787-30-9]
	(351–399)	77.6 ± 0.6	(375)	T	[95/13]
$\text{C}_{18}\text{H}_{24}$	1,2,3,4,4a,7,8,9,10,11,12,12a-dodecahydrochrysene				[1610-22-6]
	(293–313)	115.4	(303)		[87/4][64/15]
$\text{C}_{18}\text{H}_{29}\text{NO}$	2,4,6-tri- <i>tert</i> -butylnitrosobenzene				[24973-59-9]
		91.0 ± 3.2	(298)	C	[95/20]
$\text{C}_{18}\text{H}_{29}\text{NO}_2$	2,4,6-tri- <i>tert</i> -butylnitrosobenzene				[4074-25-3]
		94.8 ± 1.0	(351)	GS	[00/25]
		96.4 ± 1.0	(298)	GS	[00/25]
		81.4 ± 1.8	(298)	C	[95/20]
$\text{C}_{18}\text{H}_{30}$	1,3,5-tri- <i>tert</i> -butylbenzene				[1460-02-2]
	(298–341)	79.9 ± 0.3	(319)	T	[98/14]
		81.2 ± 0.3	(298)		[98/14]
$\text{C}_{18}\text{H}_{30}$	(273–315)	79.7 ± 0.4	(294)	ME	[65/6][87/4]
	hexaethylbenzene				[604-88-6]
	(327–352)	95.0 ± 4.0	(340)	HSA	[86/1]
$\text{C}_{18}\text{H}_{30}$		U 41.3 ± 0.9		DSC	[84/2]
	perhydrochrysene				[2090-14-4]
$\text{C}_{18}\text{H}_{30}$	(273–353)	82.3 (liq)	(288)		[87/4][64/15]
	1-phenyldodecane				[123-01-3]
$\text{C}_{18}\text{H}_{30}\text{O}$		135.1	(298)		[00/10]
	2,4,6-tri- <i>tert</i> -butylphenol				[732-26-3]
		87.5 ± 0.4	(298)	GS	[99/17]
	(295–339)	85.6 ± 0.4	(317)	ME	[65/6][87/4]
	(415–551)	63.2 (liq)	(430)		[87/4]
		U128.1	(298)	C	[71/24][99/17]
	(292–313)	83.9	(302)		[60/14]
$\text{C}_{18}\text{H}_{30}\text{O}_4$		84.2 ± 0.5	(298)	V	[60/14][99/17]
	4-diacetylbenzene diethyl ketal				[47189-08-2]
	(306–327)	112.5	(316.5)		[78/9][87/4]
$\text{C}_{18}\text{H}_{31}\text{N}$	2,4,6-tri- <i>tert</i> -butylaniline				[961-38-6]
		92.5 ± 1.1	(298)	GS	[00/6]
$\text{C}_{18}\text{H}_{34}\text{O}$	cyclooctadecanone				[6907-37-5]
		77.4			[38/1][60/1]
$\text{C}_{18}\text{H}_{36}\text{O}_2$	octadecanoic acid (stearic acid)				[57-11-4]
	(296–319)	158		TPTD	[01/15]
	(331–340)	166.5 ± 4.2	(336)	ME	[61/1][70/1]
$\text{C}_{18}\text{H}_{36}\text{O}_2$	ethyl palmitate				[628-97-7]
	(286–294)	150.8	(290)	ME	[87/4][67/18]
$\text{C}_{18}\text{H}_{37}\text{NO}$	octadecanamide				[124-26-5]
	(367–379)	195.8 ± 4.2	(373)	ME	[59/3][87/4]
$\text{C}_{18}\text{H}_{38}$	1,1,2,2-tetra- <i>tert</i> -butylethane				[62850-21-9]
	(303–366)	71.9	(341)		[84/15]
$\text{C}_{18}\text{H}_{38}$		74.3	(298)		[84/15]
	<i>n</i> -octadecane				[593-45-3]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{18}\text{H}_{38}\text{O}$		152.7	(298)	C	[72/1]
	(288–298)	153.0 ± 5	(293)	ME	[49/1][60/1]
					[70/1]
	1-octadecanol				[112-92-5]
$\text{C}_{19}\text{H}_{15}\text{N}$	(318–329)	187.4 ± 1.3	(324)	ME	[65/6][87/4]
		191.2 ± 1.3	(298)		[65/6]
	N-phenyl benzophenone imine				[574-45-8]
	(348–387)	115.5 ± 1.8	(367)	T	[97/18]
$\text{C}_{19}\text{H}_{15}\text{N}_3$		119.7 ± 1.8	(298)		[97/18]
	triphenylazidomethane				[14309-25-2]
$\text{C}_{19}\text{H}_{16}$	(335–363)	120.6	(349)		[87/4][74/31]
	triphenylmethane				[519-73-3]
		109.1	(298)	GS	[99/25]
		112.0	(298)	CGC–DSC	[98/5]
	(343–363)	113.9	(353)	EM	[89/1]
	(303–358)	106.8	(330)	GS	[86/8]
	(325–349)	100 ± 0.4	(339)	V	[59/2][70/1]
		100.7	(298)		[86/17][36/2]
$\text{C}_{19}\text{H}_{16}\text{F}_8\text{N}_4\text{O}_2$		105 ± 0.8			[74/31]
	N-ethyl-4-[(4-nitrophenyl)azo]-N-(2,2,3,3,4,4,5,5-octafluoropentyl)-benzenamine				[91488-85-6]
$\text{C}_{19}\text{H}_{16}\text{O}$		112.6		UV	[84/39]
	triphenylmethanol				[76-84-6]
		121.8 ± 1.7	(298)	GS	[98/22]
$\text{C}_{19}\text{H}_{16}\text{O}_2$	(353–373)	122	(363)		[87/4]
	2-fluorenyl-2-methyl-1,3-cyclopentandione				[160731-89-5]
	(353–388)	122.3 ± 1.6	(371)	T	[95/2]
$\text{C}_{19}\text{H}_{17}\text{NO}_2$	1-piperidinoanthraquinone				[4946-83-2]
	(383–392)	U 18.3 (387.5)			[87/4]
$\text{C}_{19}\text{H}_{18}\text{O}_2$	2-diphenylmethyl-2-methyl-1,3-cyclopentandione				[160731-87-3]
	(355–393)	120.2 ± 1.1	(374)	T	[95/2]
$\text{C}_{19}\text{H}_{20}\text{O}_2$	3-diphenylmethyl-3-methyl-2,4-pentandione				[137932-36-6]
		114.4 ± 0.6	(298)	T,B	[95/2]
$\text{C}_{19}\text{H}_{21}\text{NO}$	(±)1,2-diphenyl-2-N-piperidinyl-1-ethanone				[127529-16-2]
		147.1 ± 1		B	[94/11]
$\text{C}_{19}\text{H}_{34}$	tricyclohexylmethane				[1610-24-8]
	(301–321)	117.4	(311)		[87/4][64/15]
$\text{C}_{19}\text{H}_{36}\text{O}$	cyclononadecanone				
		82.4			[38/1][60/1]
$\text{C}_{19}\text{H}_{38}\text{O}_2$	methyl octadecanoate				[112-61-8]
	(299–310)	158.2 ± 2.5	(304)		[65/6][87/4]
$\text{C}_{19}\text{H}_{38}\text{O}_2$	nonadecanoic acid				[646-30-0]
		198.7 ± 5			[68/2][70/1]
$\text{C}_{19}\text{H}_{40}$	<i>n</i> -nonadecane				[629-92-5]
		143.6	(298)	C	[72/1]
	(288–303)	136.6	(296)		[64/15]
$\text{C}_{20}\text{H}_{12}$	perylene				[198-55-0]
	(391–424)	132.6 ± 3.6	(408)	ME	[98/3]
	(313–453)	123.2	(383)	GS	[95/7]
	(443–518)	145.2 ± 2.5	(298)	C,ME	[73/7]
		125.5 ± 4.2	(298)	ME	[67/2][70/1]
	(383–453)	139	(418)		[58/1][87/4]
		129.6 ± 2.1	(415)	ME	[52/3]
		121.3	(370)	ME	[51/12]
$\text{C}_{20}\text{H}_{12}$	benzo[b]fluoranthene				[205-99-2]
	(313–453)	119.2	(383)	GS	[95/7]
$\text{C}_{20}\text{H}_{12}$	benzo[k]fluoranthene				[207-08-9]
	(363–430)	130	(378)		[87/4]
		120 ± 10		TE	[83/28]
$\text{C}_{20}\text{H}_{12}$	benzo[a]pyrene				[50-32-8]
	(313–453)	122.5	(383)	GS	[95/7]
	(358–431)	118.3	(373)	ME	[87/4][74/30]
$\text{C}_{20}\text{H}_{12}$	benzo[e]pyrene				[192-97-2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{20}\text{H}_{12}\text{BrNO}_4$	(313–453)	117.9	(383)	GS	[95/7]
	(359–423)	119.1	(373)	ME	[87/4][74/30]
	1-amino-2-(4-bromophenoxy)-4-hydroxy-9,10-anthraquinone				[59722-76-8]
$\text{C}_{20}\text{H}_{13}\text{NO}_4$	(373–453)	163.6	(413)		[78/35]
	1-amino-4-hydroxy-2-phenoxy-9,10-anthraquinone (Disperse Red 60)				[17418-58-5]
	(359–366)	152.5	(362.5)		[87/4]
$\text{C}_{20}\text{H}_{14}$		141.8			[84/40]
	(373–453)	103.8	(413)		[78/35]
	9,10-dihydro-9,10-(1',2') benzoanthracene (tryptcene)				[477-75-8]
$\text{C}_{20}\text{H}_{14}$		104.6 ± 12.6			[73/13][77/1]
	9-phenylanthracene				[602-55-1]
	(313–453)	118.7	(383)	GS	[95/7]
$\text{C}_{20}\text{H}_{14}$	(352–395)	119.7		TE	[74/33]
	(353–426)	115.3	(368)		[58/4]
$\text{C}_{20}\text{H}_{14}$	binaphthalene				[11068-27-2]
	(313–453)	138.3	(383)	GS	[95/7]
$\text{C}_{20}\text{H}_{14}\text{N}_2\text{O}_2$	1-anilino-4-amino-9,19-anthraquinone				[4395-65-7]
		138.6			[84/40]
	(373–453)	135.1	(413)	GS	[77/20][78/35]
$\text{C}_{20}\text{H}_{14}\text{N}_2\text{O}_4$	1-amino-2-(4-aminophenoxy)-4-hydroxy-9,10-anthraquinone				[56405-27-7]
	(373–453)	U50.2	(413)		[78/35]
	resorcinol dibenzoate				[94-01-9]
$\text{C}_{20}\text{H}_{14}\text{O}_4$	(323–399)	165.8	(338)	UV	[87/4][60/24]
	1,1,1-trifluoro-2,2,2-triphenylethane				[68643-31-2]
		112.3 ± 1.0	(298)		[97/34]
$\text{C}_{20}\text{H}_{16}$	1',9-dimethyl-1,2-benzanthracene				[313-74-6]
		112.5 ± 3.3		ME	[65/5][70/1]
	3',6-dimethyl-1,2-benzanthracene				[316-51-8]
$\text{C}_{20}\text{H}_{16}$		112.5 ± 3.3		ME	[65/5][70/1]
	7,12-dimethylbenz[a]anthracene				[57-97-6]
	(379–390)	135			[87/4][64/3]
$\text{C}_{20}\text{H}_{16}$	(379–396)	107.8 (vap)			[87/4][64/3]
	5,6-dimethylchrysene				[3697-27-6]
		130 ± 1.3			[66/3][70/1]
$\text{C}_{20}\text{H}_{16}\text{O}_4\text{S}_2$		134 ± 1.3			[66/3][70/1]
	(379–408)	135 ± 2.4	(394)	ME	[64/3]
	6-ethoxy-2-(6-ethoxy-3-oxobenzob[thien-2(3H)-ylidene]benzo[b]-thiophen-3(2H)-one (C.I. Vat Orange 5)				[3263-31-8]
$\text{C}_{20}\text{H}_{17}\text{N}_3\text{O}_2$	(519–634)	65	(577)	GS	[86/14]
	N-(benzoyl)-4-(2-hydroxy-4-methylphenylazo)benzenamine				
	(Disperse Yellow 50)				
$\text{C}_{20}\text{H}_{18}$	(464–484)	69	(474)	GS	[89/29]
	1,1,1-triphenylethane				[15271-39-6]
		108.6	(298)		[99/25]
$\text{C}_{20}\text{H}_{18}\text{O}_6$	9-fluorenyl-tris(methoxycarbonyl)methane				[170464-52-5]
		132.6	(298)	GS	[95/29]
	pagodane (undecacyclo[9.9.0.0 ^{1,5} .0 ^{2,12} .0 ^{2,18} .0 ^{3,7} .0 ^{6,10} .0 ^{8,12} .0 ^{11,15} .0 ^{13,17} .0 ^{16,20}]eicosane)				[89683-62-5]
$\text{C}_{20}\text{H}_{20}\text{O}_2$	(418–473)	90.2 ± 2.3	(446)	T	[94/9]
	2-diphenylmethyl-2-ethyl-1,3-cyclopentanedione				[160731-88-4]
	(342–377)	122.8 ± 0.7	(360)	T	[95/2]
$\text{C}_{20}\text{H}_{20}\text{O}_6$	1,1,1-tris(methoxycarbonyl)-2,2-diphenylethane				[170464-52-5]
		136.0	(298)	GS	[95/29]
	N-2-[(2-chloro-6-cyano-4-nitrophenyl)azo]-5-(diethylamino)phenyl-propanamide (Disperse Blue 165)				
$\text{C}_{20}\text{H}_{22}\text{O}_2$	(464–484)	90.7	(474)	GS	[89/29]
	3-diphenylmethyl-3-ethyl-2,4-pentanedione				[160731-83-9]
	(349–387)	122.3 ± 1.5	(368)	T	[95/2]
$\text{C}_{20}\text{H}_{24}\text{O}_6$	dibenzo-18-crown-6				[14187-32-7]
		178.8 ± 6.9	(298)	CGC–DSC	[00/11]
	hexacyclopropylethane				[26902-55-6]
$\text{C}_{20}\text{H}_{30}$		109.0 ± 2.1			[84/32]
	eicosanedioic acid				[2424-92-2]
	(380–395)	165.7 ± 3.3	(388)	ME	[60/4][87/4]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)		CAS registry number
Polymorph	Temperature range (K)			Method	Reference
		170.0 ± 3.3	(298)		[60/4][99/10]
C ₂₀ H ₄₀ O ₂	eicosanoic acid (337–346)	199.6 ± 7.5	(342)	ME	[506-30-9] [61/1][70/1][87/4]
C ₂₀ H ₄₀ O ₂	ethyl stearate (297–306)	161.4	(301.5)	ME	[111-61-5] [87/4][67/18]
C ₂₀ H ₄₂	<i>n</i> -eicosane				[112-95-8]
		179.5 ± 2.0	(367)	B	[94/8]
		U152.3 ± 5.0	(298)	B	[91/1]
		170.4	(298)	C	[72/1]
C ₂₀ H ₄₂ O	1-eicosanol (327–341)				[629-96-9]
		218 ± 3.8	(332)	ME	[65/6][87/4]
		223 ± 3.8	(298)		[65/6]
C ₂₁ H ₆ N ₁₂ O ₁₈	2,4,6- <i>tris</i> (2,4,6-trinitrophenyl)-1,3,5-triazine (479–551)	167.9	(494)		[49753-54-0] [87/4]
C ₂₁ H ₁₄ N ₂ O ₃	2-phenyl-3-benzoylquinoxaline-1,4-dioxide				[13494-38-7]
		167.4 ± 4.0	(298)	ME	[97/25]
C ₂₁ H ₁₄ N ₂ O ₃	1,4-diamino-2-benzoyl-9,10-anthraquinone				
		168.5			[84/40]
C ₂₁ H ₁₅ BrN ₂ O ₂	1-amino-2-bromo-4-[(4-methylphenyl)amino]-9,10-anthraquinone (418–438)	167.0 ± 6.0	(428)		[128-83-6] [84/35]
C ₂₁ H ₁₅ NO ₃	2-hydroxy-4-[(4-methylphenyl)amino]-9,10-anthraquinone (349–378)	121.0 ± 7.6	(363)		[84/35]
	[Note: Compound is listed as the 2-hydroxy-derivative in the paper; however, it is listed as the 1-hydroxy-derivative in <i>Chem. Abstracts</i>]				
C ₂₁ H ₁₆	3-methylcholanthrene (401–425)	127.2 ± 2.4	(413)		[56-49-5] [87/4][64/3]
C ₂₁ H ₁₆ N ₂ O ₂	1-anilino-4-(<i>N</i> -methylamino)-9,10-anthraquinone				
		136.9			[84/40]
C ₂₁ H ₁₇ N ₃ O ₃	(5-cyano-3,4-diphenyl-6-oxo-1,6-dihydropyridazin-1-yl)acetate (396–414)	131.9 ± 9.3	(405)	ME	[82232-20-0] [82/24]
C ₂₁ H ₁₈ F ₂	1,1-difluoro-3,3,3-triphenylpropane				[193472-73-0]
		113.2 ± 1.7	(298)		[97/34]
C ₂₁ H ₁₉ F	1-fluoro-3,3,3-triphenylpropane				[193472-69-4]
		129.3 ± 0.6	(298)		[97/34]
C ₂₁ H ₁₉ F	2-fluoro-1,2,3-triphenylpropane				[193472-72-9]
		132.5 ± 3.0	(298)		[97/34]
C ₂₁ H ₂₀ Cl ₂ O ₃	(3-phenoxyphenyl)methyl- <i>cis</i> -3-(2,2-dichloroethenyl)-2,2-dimethyl- cyclopropanecarboxylate (<i>cis</i> -permethrin)				
		108.8	(323)	GS,A	[86/20]
C ₂₁ H ₂₃ NO ₅	diacetylmorphine (heroin) (324–339)	144.5 ± 4.0	(331)	GS	[561-27-3] [84/27]
C ₂₁ H ₂₄ O ₂	3-diphenylmethyl-3-propyl-2,4-pentanedione				[160731-84-0]
		124.7	(298)	T,B	[95/2]
C ₂₁ H ₂₆	[1,8]-para-cyclophane (354–376)	105 ± 1.3	(365)	ME	[6169-94-4] [69/6][77/1]
		110.9 ± 2.1	(298)	ME	[69/6][77/1]
C ₂₁ H ₂₆ O ₄	2-hydroxy-4,4'-dibutoxybenzophenone				[6127-74-8]
		148.0		B	[99/22]
C ₂₁ H ₃₀ O	1,1-diadamantyl ketone (362–379)	109.0 ± 1.8	(298)	ME	[38256-01-8] [92/2]
C ₂₁ H ₄₂ O ₂	methyl eicosanoate (311–318)	190.8 ± 10	(314)	ME	[1120-28-1] [65/6][87/4]
C ₂₁ H ₄₂ O ₂	ethyl nonadecanoate (302–308)	149.7	(305)	ME	[18281-04-4] [87/4][67/18]
C ₂₁ H ₄₄	heneicosane				[629-94-7]
		141.8 ± 10	(298)	B	[91/1]
C ₂₂ H ₁₀ O ₂	anthanthrone (dibenzochrysene-6,12-dione) (450–550)	152.2	(465)		[641-13-4] [87/4]
C ₂₂ H ₁₂	anthranthrene (dibenzo[def,mno]chrysene)				[191-26-4]
		135 ± 5	(479)	ME	[52/3]
C ₂₂ H ₁₂	benzo[ghi]perylene (313–453)	129.9	(383)	GS	[191-24-2] [95/7]
		127.8	(404)	ME	[87/4][74/30]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₂₂ H ₁₂ O ₂	(450–510)	135.1	(465)	ME	[87/4]
	(454–502)	125.5	(478)		[67/2]
	6,13-pentacenequinone				[3029-32-1]
C ₂₂ H ₁₄		116.3 ± 5.9	(298)	ME	[56/5][70/1]
	1,2:6,7-dibenzophenanthrene (benzo[b]chrysene)				[214-17-5]
		136.4	(417)		[67/2]
C ₂₂ H ₁₄	(398–513)	136.9	(413)	ME	[87/4]
	pentacene				[135-48-8]
	(443–483)	156.9 ± 13.6	(463)		[98/3]
C ₂₂ H ₁₄	(494–526)	154 ± 5	(512)	ME, TE	[80/1]
	(495–530)	184 ± 10	(298)		[80/1]
	(455–555)	157.7	(505)		[67/2]
C ₂₂ H ₁₄	picene			ME	[213-46-7]
	(425–488)	140.1	(456)		[67/2]
	(409–527)	140.7	(424)		[87/4]
C ₂₂ H ₁₄	1,2:3,4-dibenzanthracene (benzo[b]triphenylene)			GS	[215-58-7]
	(313–453)	135	(383)		[95/7]
	(425–452)	159 ± 6	(298)		[80/1]
C ₂₂ H ₁₄	1,2:5,6-dibenzanthracene (dibenz[a,h]anthracene)			GS	[53-70-3]
		134.1			[95/7]
	(436–462)	162 ± 6	(298)		[80/1]
C ₂₂ H ₁₆ O	(417–502)	141.8	(457)	ME	[67/2]
	3,8-dimethylnaphtho[3,2,1-kl]xanthene (3,8-dimethylceroxene)				
	(373–433)	138.2	(388)		[87/4][59/14]
C ₂₂ H ₁₇ NO ₃ S	2-(3-methoxypropyl)-1 <i>H</i> -xantheno[2,2,9-def]isoquinoline-1,3(2 <i>H</i>)-dione				[36245-88-2]
	(605–647)	111.8	(620)		[87/4]
	(647–685)	150.8	(662)		[87/4]
C ₂₂ H ₁₈ N ₂ O ₂	1-amino-2-methyl-4-[(4-methylphenyl)amino]-9,10-anthraquinone				[116-77-8]
	(418–435)	142.2 ± 2.3	(426)		[84/35]
	1-(<i>N</i> -methylamino)-4-[(3-methylphenyl)amino]-9,10-anthraquinone				[6408-50-8]
C ₂₂ H ₁₈ N ₂ O ₂	(418–434)	129.0 ± 4.7	(426)		[84/35]
	1-(<i>N</i> -methylamino)-4-[(4-methylphenyl)amino]-9,10-anthraquinone				[128-85-8]
	(403–426)	153.9 ± 3.9	(414)		[84/35]
C ₂₂ H ₂₀ N ₂ O ₄	<i>N,N'</i> -bis(2-methoxyphenyl)terephthalamide			ME	[36360-34-6]
	(456–470)	197.5 ± 4.2			[73/21][77/1]
	<i>N,N'</i> -bis(3-methoxyphenyl)terephthalamide				[6957-81-9]
C ₂₂ H ₂₀ N ₂ O ₄	(456–470)	209.2 ± 8.4		ME	[73/21][77/1]
	<i>N,N'</i> -bis(4-methoxyphenyl)terephthalamide				[7144-15-2]
		227.6 ± 8.4			[73/21][77/1]
C ₂₂ H ₂₁ F	2-benzyl-2-fluoro-1,3-diphenylpropane				[193472-71-8]
		127.5 ± 0.8	(298)		[97/34]
	1,1,1-triphenylbutane				[43044-69-5]
C ₂₂ H ₂₂		114.3	(298)		[99/25]
	<i>N,N'</i> -ethylenebis(3-amino-1-phenylbut-2-en-1-one)				[16087-30-2]
	(407–426)	192.9 ± 5.3	(415)		[95/12]
C ₂₂ H ₂₄ N ₂ O ₂		198.8 ± 5.3	(298)	ME	[95/12]
	1,1'-diphenyl-1,1'-bicyclopentyl				[59358-70-2]
		141.4			[83/16]
C ₂₂ H ₂₆ N ₂ O ₂	1,4-bis(<i>N</i> -butylamino)-9,10-anthraquinone			E, B	[17354-14-2]
	(389–398)	116.4 ± 2.3	(394)		[84/35]
	1,4-bis(<i>N</i> -isobutylamino)-9,10-anthraquinone				[10720-45-7]
C ₂₂ H ₂₆ N ₂ O ₂	(368–388)	96.4 ± 2.1	(378)	ME	[84/35]
	(4 <i>R</i> ,4' <i>R</i> ,5 <i>R</i> ,5' <i>R</i>)-5,5-diphenyl-3,3',4,4'-tetramethyl-2,2'-bioxazolidine				[145513-29-7]
	(349–358)	130.8 ± 0.8	(356)		[95/19]
C ₂₂ H ₂₈ N ₂ O ₂	(2 <i>R</i> , 3 <i>R</i> , 6 <i>R</i> , 7 <i>R</i>)-2,6-diphenyl-3,4,7,8-tetramethyl- <i>cis</i> -perhydro-[1,4]-oxazino-[3,2- <i>b</i>]-[1,4]-oxazine			ME	[145438-85-3]
		116.6 ± 1.0	(358)		[95/19]
	(2 <i>R</i> , 3 <i>S</i> , 6 <i>R</i> , 7 <i>S</i>)-2,6-diphenyl-3,4,7,8-tetramethyl- <i>cis</i> -perhydro-[1,4]-oxazino-[3,2- <i>b</i>]-[1,4]-oxazine				[145438-85-3]
C ₂₂ H ₂₈ N ₂ O ₂	(353–364)	123.1 ± 1.6	(358)	ME	[95/19]
	2,4,6-triisopropylbenzophenone				[33574-11-7]
	(353–364)	116 ± 7	(298)		[82/10]
C ₂₂ H ₂₈ O	3',5'-diisopropyl-4,4-dimethyl-3-phenyl-1,2-benzocyclobuten-3-ol			C	[33574-16-2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{22}\text{H}_{38}$	(354–364)	117.9	(298)	C	[82/10]
	<i>meso</i> 3,4-di(1-cyclohexen-1-yl)-2,2,5,5-tetramethylhexane (347–404)	117.2 $\pm$ 2.4	(376)	T	[62678-54-0] [93/6]
$\text{C}_{22}\text{H}_{31}\text{NO}_4$	N,N-bis(3-phenoxy-2-hydroxypropyl)butyl amine (363–411)	114.3	(378)		[23257-62-7] [87/4]
		146.0 $\pm$ 4.2			[76/19]
$\text{C}_{22}\text{H}_{42}$	<i>meso</i> -(E, E)-5,6-di- <i>tert</i> -butyl-2,2,9,9-tetramethyl-3,7-decadiene (297–353)	110.0 $\pm$ 1.7	(325)	T	[95/1]
$\text{C}_{22}\text{H}_{42}$	(<i>dl</i>)-(E, E)-5,6-di- <i>tert</i> -butyl-2,2,9,9-tetramethyl-3,7-decadiene (297–346)	74.4 $\pm$ 1.7	(307)	T	[95/1]
$\text{C}_{22}\text{H}_{44}\text{O}_2$	ethyl eicosanoate (307–313)	171.5	(310)	ME	[18281-05-5] [87/4][67/18]
$\text{C}_{22}\text{H}_{46}\text{O}$	1-docosanol (335–341)	206.7 $\pm$ 10	(330)	ME	[661-19-8] [65/6][87/4]
		238.5 $\pm$ 10	(298)		[65/6]
$\text{C}_{22}\text{H}_{46}$	docosane				[629-97-0]
		172.6 $\pm$ 2.0	(391)	B	[94/8]
$\text{C}_{23}\text{H}_{24}\text{O}_6$		U 151.1 $\pm$ 10	(298)	B	[91/1]
	<i>tris</i> (ethoxycarbonyl)-9-fluorenylmethane	143.2	(298)	GS	[170464-53-6] [95/29]
$\text{C}_{23}\text{H}_{25}\text{F}$	1-adamantylfluorodiphenylmethane (353–393)	125.9 $\pm$ 1.3	(373)	T	[154393-25-6] [94/10]
$\text{C}_{23}\text{H}_{26}\text{O}_6$	1,1,1- <i>tris</i> (ethoxycarbonyl)-2,2-diphenylethane	140.1	(298)	GS	[95/29]
$\text{C}_{23}\text{H}_{36}\text{N}_2\text{O}_2$	(5 α ,17 β)-N-(1,1-dimethylethyl)-3-oxo-4-azaandrost-1-ene-17-carboxamide (Finasteride) (463–488)	143.7		TGA	[98319-26-7] [97/36]
$\text{C}_{23}\text{H}_{48}$	tricosane				[638-67-5]
		U146.8 $\pm$ 10	(298)	B	[91/1]
$\text{C}_{24}\text{H}_{12}$	coronene				[191-07-1]
	(421–504)	133.1 $\pm$ 5.1	(463)	ME	[98/3]
	(313–453)	143.2	(383)	GS	[95/7]
	(427–510)	135.9	(468)	ME	[87/4][74/30]
		128.4		ME	[67/2]
	(433–513)	147	(473)		[58/1]
	(476–555)	143.2	(407)	ME	[52/3]
$\text{C}_{24}\text{H}_{12}$		148.5	(407)	ME	[51/12]
	<i>bis</i> -benzo[3,4]cyclobuta[1,2-a:1',2'-c]biphenylene ([4]phenylene)	131.0 $\pm$ 4.2			[102234-01-5] [00/18]
$\text{C}_{24}\text{H}_{12}\text{O}_2$	3,4:9,10-dibenzpyrene-5,8-quinone	112.5 $\pm$ 5.4			[3302-52-1] [56/5][70/1]
$\text{C}_{24}\text{H}_{14}$	dibenzo[a,e]pyrene				[192-65-4]
	(414–506)	146.4	(429)		[87/4]
$\text{C}_{24}\text{H}_{14}$	(434–526)	137.6	(480)	ME	[67/2]
	dibenzo[fg,op]naphthacene				[192-51-8]
	(430–555)	147.4	(445)		[87/4]
	(454–526)	146.9	(490)	ME	[67/2]
$\text{C}_{24}\text{H}_{16}\text{N}_2\text{O}_2$	(called 1,2,6,7-dibenzpyrene in paper, which we have taken to be dibenzo[fg,op]naphthacene based upon the melting point temperature reported in the paper)				
	2,2'-(1,4-phenylene) <i>bis</i> (5-phenyl)oxazole	140	(480)		[1806-34-4] [89/7]
	(600–680)	94.4	(615)		[87/4]
$\text{C}_{24}\text{H}_{18}$	1,3,5-triphenylbenzene				[612-71-5]
		150.9	(298)	CGC–DSC	[98/5]
	(364–388)	145.6 $\pm$ 0.9	(376)	T	[97/15]
		150.3 $\pm$ 0.9	(298)		[97/15]
		152 $\pm$ 0.3	(298)	C,ME	[74/4]
	(410–444)	142	(425)	ME	[74/4][87/4]
	(384–400)	142.2	(422)	ME	[67/2]
$\text{C}_{24}\text{H}_{24}\text{O}_4$		149.7 $\pm$ 4.1	(298)	ME	[58/1][70/1]
	<i>syn</i> 4,9- <i>bis</i> (methoxycarbonyl)pagodane (dimethyl undecacyclo [9.9.0.0 ^{1,5} .0 ^{2,12} .0 ^{2,18} .0 ^{3,7} .0 ^{6,10} .0 ^{8,12} .0 ^{11,15} .0 ^{13,17} .0 ^{16,20}]eicosane-4- <i>syn</i> , 9- <i>syn</i> -dicarboxylate)				[89702-41-0]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₂₄ H ₂₄ O ₄	(393–447)	146.1 ± 3.0	(420)	T	[94/9]
	1,6-bis(methoxycarbonyl)dodecahedrane (dimethyl undecacyclo-[9.9.0.0 ^{2,9} .0 ^{3,7} .0 ^{4,20} .0 ^{5,18} .0 ^{6,16} .0 ^{8,15} .0 ^{10,14} .0 ^{12,19} .0 ^{13,17}]eicosane-1,6-dicarboxylate)				[124316-65-0]
C ₂₄ H ₂₆ N ₂ O ₂	(395–450)	139.7 ± 1.3	(422)	T	[94/9]
	1,5-dipiperidylanthraquinone				[14580-70-2]
C ₂₄ H ₂₈ P ₂ O ₂	(408–458)	173.3	(428)		[58/1][87/4]
	1,4-bis(diphenylphosphino)butane				[7688-25-7]
C ₂₄ H ₃₀		171.6	(443)	B	[89/28]
	1,1'-diphenyl-1,1'-bicyclohexyl				[59358-71-3]
C ₂₄ H ₃₀ O ₄		150.2		E,B	[83/16]
	2,2'-diphenyl-bi-(5,5-dimethyl-1,3-dioxan-2-yl)				[167321-36-0]
C ₂₄ H ₃₂	(372–420)	130.2 ± 1.8	(396)	T	[95/13]
	[6.6]-para-cyclophane				[4384-23-0]
C ₂₄ H ₄₈ O ₂	(352–371)	108.8 ± 0.8	(361)	ME	[69/6][77/1]
		115.1 ± 2.1	(298)	ME	[69/6][77/1]
C ₂₄ H ₅₀	ethyl docosanoate				[5908-87-2]
	(313–318)	196.5	(315.5)	ME	[87/4][67/18]
C ₂₅ H ₅₀	tetracosane				[646-31-1]
		162 ± 12	(298)	B	[91/1]
C ₂₅ H ₂₀	tetraphenylmethane				[630-76-2]
		140.0	(298)		[99/25]
C ₂₅ H ₃₆ O ₂	(396–466)	150.6 ± 4	(298)	TE,ME	[72/5][77/1]
	(404–466)	143.3 ± 1	(419)	ME	[87/4][72/5]
C ₂₅ H ₅₀ O ₂	2,2'-methylenebis(6-tert-butyl-4-methylphenol)				[119-47-1]
	(383–403)	114.0	(393)	GS	[71/19]
C ₂₅ H ₅₂	ethyl tricosanoate				[18281-07-7]
	(316–322)	175.2	(319)	ME	[87/4][67/18]
C ₂₆ H ₁₆	pentacosane				[629-99-2]
		173.6 ± 10	(298)	B	[91/1]
C ₂₆ H ₁₈	dibenzo[g,p]chrysene				[191-68-4]
	(408–493)	142.2	(423)		[87/4]
C ₂₆ H ₁₈	(417–500)	141.8	(458)	ME	[67/2]
	9,10-diphenylanthracene				[1499-10-1]
C ₂₆ H ₁₈	(313–453)	137.5	(383)	GS	[95/7]
		116.4			[58/4]
C ₂₆ H ₁₈	(393–433)	143.6	(413)		[58/1][87/4]
	(481–502)	156.9 ± 4.2	(492)	HSA	[53/2][70/1]
C ₂₆ H ₂₀	9,9'-bifluorenyl				[1530-12-7]
	(383–408)	131.8 ± 1.1	(395)	T	[94/2]
C ₂₆ H ₂₀		132.6 ± 1.1	(298)		[94/2]
	tetraphenylethene				[632-51-9]
C ₂₆ H ₂₀ N ₂ O ₂	(343–389)	129.3 ± 0.7	(366)	T	[99/29]
		133.4 ± 0.7	(298)		[99/29]
C ₂₆ H ₂₂	2,2'-(1,4-phenylene)bis(4-methyl-5-phenyl)oxazole				[3073-87-8]
		150	(480)		[89/7]
C ₂₆ H ₂₂	1,1,2,2-tetraphenylethane				[632-50-8]
		136.8 ± 2.9	(298)	GS	[90/12]
C ₂₆ H ₂₂	(370–423)	131.4 ± 2.1	(396)	GS	[90/12]
	1,1,1,2-tetraphenylethane				[2294-94-2]
C ₂₆ H ₂₆		132.6 ± 2.1	(298)	GS	[90/12]
	(340–400)	128.7 ± 2.1	(370)	GS	[90/12]
C ₂₆ H ₂₆		126.4 ± 1.7	(434)	HSA	[56/1]
	pentacyclo[18.2.2.2(9,12).0(4,15).0(4,15).0(6,17)]hexacos-4,6(17),9,11,15,20,22,23,25-nonane (triple layered [2.2]paracyclophane)				[35117-21-6]
C ₂₆ H ₃₈	(299–412)	119.1 ± 1.5		TSGC	[80/15]
		125.9 ± 2.5	(298)	TSGC	[80/15]
C ₂₆ H ₅₄	2,3-dimethyl-2,3-bis-(4-tert-butylphenyl)-butane				[5171-91-5]
		161.9		E,B	[83/16]
C ₂₇ H ₄₈	hexacosane				[630-01-3]
		177.2 ± 10	(298)	B	[91/1]
	17-(1,5-dimethylhexyl)-10,13-dimethyl-hexahydro-1H-cyclopenta[a]-phenanthrene (5 α -cholestane)				[481-21-0]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
C ₂₇ H ₅₆	heptacosane	133.8	(298)		[00/10]
					[593-49-7]
C ₂₈ H ₁₄	phenanthro[1,10,9,8-opqra]perylene	196.0 ± 30	(298)	B	[91/1]
	(580–630)	180.5 ± 5	(605)	ME	[190-39-6]
C ₂₈ H ₁₂ Cl ₂ N ₂ O ₄	C.I. Vat Blue 6				[87/4][52/3]
	(519–634)	199	(577)	GS	[130-20-1]
C ₂₈ H ₁₄ N ₂ O ₄	C.I. Vat Blue 4				[86/14]
	(519–634)	167	(577)	GS	[81-77-6]
C ₂₈ H ₁₈	9,9'-bianthryl				[86/14]
		128.4 ± 0.2			[1055-23-8]
	(413–473)	127.9	(443)		[70/1][58/1]
		148.1			[58/1][87/4]
C ₂₈ H ₁₈	9,9'-biphenanthryl				[51/2][60/1]
		151.5			[20532-03-0]
C ₂₈ H ₂₂	9,9'-dimethyl-9,9'-bifluorenyl				[51/2][60/1]
	(368–403)	118.7 ± 1.3	(386)	T	[15300-82-0]
		119.7 ± 1.3	(298)		[94/2]
C ₂₈ H ₃₀ N ₄	2,3,7,8,12,13,17,18-octamethylporphyrin				[94/2]
	(593–653)	268 ± 11		GS	[1257-25-6]
C ₂₈ H ₃₈	1,1'-diphenyl-1,1'-bicyclooctyl				[01/12]
		174.5		E,B	[59358-73-5]
C ₂₈ H ₅₈	octacosane				[83/16]
		208.9 ± 10	(298)	B	[630-02-4]
C ₃₀ H ₁₄ O ₂	8,16-pyranthenedione (C. I. Vat Orange 9)				[91/1]
	(503–543)	197.7	(518)		[128-70-1]
		181.2	(498)	ME	[87/4]
C ₃₀ H ₁₆	pyranthrene				[51/12]
		194.5 ± 6.7	(595)	ME	[191-13-9]
C ₃₀ H ₄₆	3,4-diethyl-3,4-bis-(4-tert-butylphenyl)-hexane				[52/3]
		167.8		E,B	[85668-74-2]
C ₃₁ H ₁₅ NO ₃	C.I. Vat Green 3				[83/16]
	(519–634)	155	(577)	GS	[3271-76-9]
C ₃₂ H ₂ Br ₁₆ N ₈	hexadecabromophthalocyanine				[86/14]
	(438–493)	109.2 ± 16.3	(453)	ME	[28746-04-5]
C ₃₂ H ₂ Cl ₁₆ N ₈	hexadecachlorophthalocyanine				[87/4][70/7]
	(398–443)	141.0 ± 17.6	(413)	ME	[28888-81-5]
C ₃₂ H ₁₄	ovalene				[87/4][70/7]
		211.7 ± 7.9	(600)	ME	[190-26-1]
C ₃₂ H ₁₈ N ₈	β -29H,31H-phthalocyanine				[52/3]
	(598–698)	223.8 ± 1.3		ME	[574-93-6]
C ₃₂ H ₅₀	2,4,5,7-tetramethyl-4,5-bis-(4-tert-butylphenyl)-octane				[00/30]
		182.8		E,B	[85668-75-3]
C ₃₂ H ₅₀	4,5-diethyl-4,5-bis-(4-tert-butylphenyl)-octane				[83/16]
		182.4		E,B	[85668-73-1]
C ₃₂ H ₆₆	dotriacontane				[83/16]
		271.1 ± 2.5			[544-85-4]
C ₃₄ H ₁₆ O ₂	dibenzanthrone (violanthrone)				[70/1]
	(513–548)	208.8	(528)		[116-71-2]
		202.9	(542)	ME	[87/4]
C ₃₄ H ₁₆ O ₂	isodibenzanthrone (isoviolanthrone)				[51/12]
	(523–553)	221.1	(538)		[128-64-3]
		215.5	(537)	ME	[87/4]
C ₃₄ H ₁₈	benzo[rst]phenanthro[1,10,9-cde]pentaphene				[51/12]
	(478–603)	154.1	(493)		[190-93-2]
C ₃₄ H ₁₈	violanthrene				[87/4]
		223.8 ± 8.8	(590)		[81-31-2]
C ₃₄ H ₁₈	violanthrene A (melting point 478 °C)				[52/3][60/1]
	(562–724)	195.8	(653)	ME	[67/2]
C ₃₄ H ₁₈	violanthrene B (melting point 330 °C)				
	(555–625)	153.5	(590)	ME	[67/2]

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
	(Note: This entry is likely the original reference for benzo[<i>rst</i>]phenanthro-[1,10,9- <i>cde</i>]pentaphene listed in reference [87/4]. <i>Chemical Abstracts</i> cites [67/2] as reporting the heat of sublimation of benzo[<i>rst</i>]-phenanthro[1,10,9- <i>cde</i>]pentaphene.)				
C ₃₄ H ₁₈	isoviolanthrene A (melting point 510 °C)				[4430-29-9]
	(588–724)	218	(590)	ME	[52/3][60/1]
C ₃₄ H ₁₈	tetrabenzo[<i>de,hi,op,st</i>]pentacene				[191-79-7]
	(348–448)	118.5	(363)	ME	[87/4][67/2]
C ₃₄ H ₅₄	4,5-dipropyl-4,5- <i>bis</i> -(4- <i>tert</i> -butylphenyl)-octane				[85668-72-0]
		198.3		E,B	[83/16]
C ₃₈ H ₃₀	1-diphenylmethylene-4-triphenylmethyl-2,5-cyclohexadiene				[18909-18-7]
	(348–394)	114.6	(363)		[87/4]
C ₃₈ H ₃₀ O ₂	<i>bis</i> (triphenylmethyl)peroxide				[596-30-5]
	(392–434)	158.1	(407)		[87/4]
C ₃₈ H ₆₂	5,6-dibutyl-5,6- <i>bis</i> -(4- <i>tert</i> -butylphenyl)-decane				[85668-76-4]
		220.9		E,B	[83/16]
C ₄₂ H ₂₈	5,6,11,12-tetraphenyltetracene				[517-51-1]
	(453–523)	160.6 ± 4.2	(488)		[58/1][70/1]
C ₄₄ H ₂₆ Br ₄ N ₄	5,10,15,20- <i>tetrakis</i> (3-bromophenyl)porphine				[68772-71-4]
		204 ± 4		GS	[00/36]
C ₄₄ H ₂₆ Br ₄ N ₄	5,10,15,20- <i>tetrakis</i> (4-bromophenyl)porphine				[29162-73-0]
		135 ± 4		GS	[00/36]
C ₄₄ H ₂₆ Cl ₄ N ₄	5,10,15,20- <i>tetrakis</i> (4-chlorophenyl)porphine				[22112-77-2]
		311 ± 5		GS	[00/36]
C ₄₄ H ₂₆ F ₄ N ₄	5,10,15,20- <i>tetrakis</i> (2-fluorophenyl)porphine				[27185-62-2]
		225 ± 8		GS	[00/36]
C ₄₄ H ₂₆ F ₄ N ₄	5,10,15,20- <i>tetrakis</i> (4-fluorophenyl)porphine				[37095-43-5]
		178 ± 4		GS	[00/36]
C ₄₄ H ₃₀ N ₄	5,10,15,20-tetraphenylporphine				[917-23-7]
		240 ± 7		GS	[00/36]
form I	(540–630)	267 ± 9			[94/41]
form II	(630–670)	185 ± 10			[94/41]
	(588–678)	110.9 ± 5.0	(603)	ME	[87/4][70/7]
C ₄₈ H ₃₈ N ₄	5,10,15,20- <i>tetrakis</i> (2-methylphenyl)porphine				[37083-40-2]
		159 ± 5		GS	[00/36]
C ₄₈ H ₃₈ N ₄	5,10,15,20- <i>tetrakis</i> (3-methylphenyl)porphine				[50849-45-1]
		177 ± 5		GS	[00/36]
C ₄₈ H ₃₈ N ₄	5,10,15,20- <i>tetrakis</i> (4-methylphenyl)porphine				[14527-51-6]
		178 ± 3		GS	[00/36]
C ₆₀	buckminsterfullerene				[99685-96-8]
	(775–1033)	180 ± 2	(298)	ME	[00/37]
	(789–907)	152.8 ± 0.1	(897)	GS	[98/4]
		183.5 ± 1.0	(298)		[98/4]
		179.2 ± 3.5	(298)		[96/20][98/4]
	(730–990)	175.2 ± 2.9	(860)	ME,TE	[95/8]
		181 ± 2.0	(298)	ME,TE	[95/8]
		219.6		TGA	[95/35]
	(546–722)	180 ± 10.0	(634)	UV	[94/18]
		158 ± 3.0	(700)	ME	[94/12]
		168.5 ± 1.2	(298)	ME	[94/12][98/4]
		181.1 ± 2.6	(298)	ME	[94/25][98/4]
		181.4 ± 2.3	(700)	MS	[94/15][92/22]
		158.6	(773)	ME	[93/15]
		184.1 ± 3.1	(298)	GS	[92/21][98/4]
		183.2 ± 3.5	(298)	ME	[92/22][98/4]
		180.6 ± 1.5	(298)	ME	[92/23][98/4]
	(673–873)	159.0 ± 4.2		ME	[92/16]
		>163 (powder)		TGA	[92/10]
	(640–800)	167.8 ± 5.4	(707)	ME,MS	[91/9]
		U90.9		ME,MS	[90/20]
C ₆₀ F ₁₆	hexadecafluorobuckminsterfullerene				
		186 ± 9		ME,MS	[00/32]
C ₆₀ F ₃₆	hexatriacontylfluorobuckminsterfullerene (4 isomer average)				

TABLE 6. Reported enthalpies of sublimation of organic compounds, 1910–2001—Continued

Molecular formula	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Polymorph	Temperature range (K)				Reference
$\text{C}_{60}\text{F}_{36}$	(422–525)	134 ± 6	(473)	MS	[94/26][96/8]
	hexatriacontylfluorobuckminsterfullerene				[150180-35-1]
$\text{C}_{60}\text{F}_{42}$	(408–539)	139 ± 8	(466)	ME,MS	[00/8]
	dotetracontylfluorobuckminsterfullerene	135 ± 8.0			[99/2]
$\text{C}_{60}\text{F}_{44}$	(430–510)	110 ± 10		ME,MS	[150155-92-3]
	tetratetracontylfluorobuckminsterfullerene				[00/14]
$\text{C}_{60}\text{F}_{44}\text{O}$	(430–510)	112 ± 6		ME,MS	[147771-02-6]
	tetracontylfluorobuckminsterfullerene				[00/14]
$\text{C}_{60}\text{F}_{46}$	(430–510)	111 ± 3		ME,MS	[337371-51-4]
	hexatetracontylfluorobuckminsterfullerene				[00/14]
$\text{C}_{60}\text{F}_{48}$	(430–510)	114 ± 7		ME,MS	[143471-96-9]
	octatetracontylfluorobuckminsterfullerene				[00/14]
$\text{C}_{60}\text{H}_{16}$	(395–528)	109 ± 7.0	(476)	ME,MS	[143471-98-1]
	hexadecahydrobuckminsterfullerene				[99/2][00/34]
$\text{C}_{60}\text{H}_{36}$		≥ 186		E	[00/38][01/11]
	hexatriacontylhydrobuckminsterfullerene				
C_{70}	(560–680)	162 ± 5		MS	[00/38][01/11]
		152	(630)		[01/11]
C_{70}	Fullerene— C_{70}	175	(298)		[01/11]
	(864–1099)	199 ± 2	(298)	ME	[115383-22-7]
C_{70}	(783–904)	189.8 ± 3.1	(844)	ME	[00/37]
		200 ± 6.0	(298)		[96/1]
C_{70}		174 ± 3.0	(740)	ME	[96/1]
		193.4 ± 1.5	(750)	MS	[94/12]
C_{70}		186.6	(788)	ME	[94/15]
	(673–873)	188.3 ± 4.2		ME	[93/15]
C_{70}	(640–800)	180.0 ± 9.2	(739)	ME,MS	[92/16]
	Fullerene— C_{76}				[91/9]
C_{76}	(637–911)	190 ± 7	(764)	ME	[135113-15-4]
	(834–1069)	206 ± 4.0	(298)	TE	[98/30]
$\text{C}_{76}\text{H}_{94}\text{N}_4$	5,10,15,20-tetrakis(3,5-di- <i>tert</i> -butylphenyl)porphine				[97/21]
		209 ± 5			[89372-90-7]
C_{84}	Fullerene— C_{84}				[00/36]
	(658–980)	202 ± 4.0	(853)	ME	[135113-16-5]
					[99/5]

TABLE 7. Enthalpies of sublimation of some organometallic and inorganic compounds, 1910–2001

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
A1					
C ₅ H ₅ AlBr ₃ N	aluminum tribromide–pyridine complex				[15348-61-5]
	(501–633)	71.2 ± 0.6		T	[89/24]
C ₁₅ H ₃ AlF ₁₈ O ₆		83.3		B,E	[67/12]
	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)aluminum(III)				[17786-67-3]
	(333–363)	52		TGA	[00/35]
	(324–344)	77.6 ± 6.2	(334)	BG	[87/20][88/22]
		79.0 ± 6.5	(298)		[87/20]
C ₁₅ H ₁₂ AlF ₉ O ₆		74.1 ± 2.5			[85/23][87/20]
	(323–347)	109.6 ± 3.8	(335)		[72/17]
	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)aluminum(III)				[14354-59-7]
	(373–403)	74		TGA	[00/35]
	(363–423)	113.4 ± 1.3		GS	[85/16]
	(369–392)	102.7 ± 3.2			[78/27]
		108 ± 2.0			[77/18][88/16]
		43.1	(443)		[77/25]
	(354–396)	93.7 ± 6.7	(375)		[72/17]
		41.0			[65/11]
C ₁₅ H ₂₁ AlO ₆		40.0			[60/17]
	<i>tris</i> (2,4-pentanedionato)aluminum(III)				[13963-57-0]
	(413–443)	93		TGA	[00/35]
		120 ± 3.0	(298)	ME	[77/18][88/2]
	(432–464)	102.0 ± 3.2	(448)	BG	[88/22]
		47.1 ± 1.0			[81/13]
		118.9 ± 7.9			[80/30]
		24.3	(458)		[77/25]
		121.7 ± 4.2	(298)		[75/19]
	(383–413)	66.1 ± 3.3	(398)		[72/17]
C ₁₆ H ₄₀ Al ₂ N ₂		23.4			[65/11]
	(417–476)	20.5			[60/17]
C ₁₆ H ₁₅ Al	<i>tetramethylbis</i> [μ -[N-(1-methylethyl)-2-propanaminto]]dialuminum(III)				[115381-27-6]
		99.2		ME	[88/20]
C ₁₈ H ₁₅ Al	triphenylaluminum				[841-76-9]
	(432–477)	172 ± 5	(455)	ME,TE	[84/8]
C ₂₄ H ₁₂ AlF ₉ O ₆ S ₃	<i>tris</i> (1-(2-thenoyl)-4,4,4-trifluoro-1,3-butanedione)aluminum(III)				[14054-83-2]
		U46.4			[60/17]
C ₂₇ H ₁₈ AlN ₃ O ₃	<i>tris</i> (8-hydroxyquinolino)aluminum(III)				[2085-33-8]
		137.7		TGA	[95/35]
C ₃₀ H ₁₈ AlF ₉ O ₆	<i>tris</i> (1-phenyl-4,4,4-trifluoro-1,3-butanedione)aluminum(III)				[14323-12-7]
		U55.2			[60/17]
C ₃₀ H ₂₇ AlO ₆	<i>tris</i> (1-phenyl-1,3-butanedionato)aluminum(III)				[14376-06-8]
	(462–478)	186.8 ± 2.1	(470)	ME,TE	[95/9]
		195.2 ± 2.1	(298)		[95/9]
		326.5	(496)		[88/16]
C ₃₀ H ₃₀ F ₂₁ AlO ₆		193.7 ± 0.3	(298)		[83/20]
	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyl-4,6-octanedionato)aluminum(III)				[18716-26-2]
	(363–398)	71.1 ± 2.5	(381)		[72/17]
C ₃₂ H ₁₆ AlClN ₈	aluminum(III)–(phthalocyaninato)chloro complex				[14154-42-8]
	(588–703)	236.4 ± 1.7		ME	[00/30]
C ₃₂ H ₁₆ AlFN ₈	aluminum(III)–(phthalocyaninato)fluoro complex				[5196-93-4]
	(658–768)	266.9 ± 2.5		ME	[00/30]
C ₃₃ H ₅₇ AlO ₆	<i>tris</i> (2,2,6,6-tetramethyl-3,5-heptanedionato)aluminum(III)				[14319-08-5]
	(413–443)	88		TGA	[00/35]
Am		119 ± 3.0			[77/18][83/20]
(C ₁₅ H ₁₃ AmF ₁₈ O ₆)–2(C ₁₂ H ₂₇ O ₄ P)	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)americium(III)-2(tributylphosphate) complex				[58760-64-8]
	(425–511)	133.9 ± 1.7	(468)	TRM	[78/32]
(C ₁₅ H ₁₂ AmF ₉ O ₆)–2(C ₁₂ H ₂₇ O ₄ P)	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)americium(III)-2(tributylphosphate) complex				[75101-27-8]
	(509–545)	222.6 ± 29.2	(527)	TRM	[78/32]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
(C ₂₄ H ₃₀ AmF ₉ O ₆)–2(C ₁₂ H ₂₇ O ₄ P)	<i>tris</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)americium(III)- 2(tributylphosphate) complex (438–493)	129.7 ± 23.4	(465)	TRM	[75101-26-7] [78/32]
C ₃₃ H ₅₇ AmO ₆	<i>tris</i> (2,2,6,6-tetramethyl-3,5-heptanedionato)americium(III) (373–423)	200.8		ME	[71817-66-8] [79/30]
As	<i>tetrakis</i> (trifluoromethyl)tetraarsene (317–354)	76.6	(335)		[7547-15-1] [66/14]
C ₄ As ₄ F ₁₂	<i>tris</i> (N,N-diethyldithiocarbamate)arsenic(III) (124 ± 3)	124 ± 3	(298)		[17767-20-3] [87/16]
C ₁₅ H ₃₀ AsN ₃ S ₆	triphenylarsine 98.3 ± 8.4	98.3 ± 8.4			[603-32-7] [82/20][64/6]
C ₁₈ H ₁₅ As	triphenylarsine oxide 149.0 ± 5.4	149.0 ± 5.4			[1153-05-5] [94/31]
C ₂₁ H ₄₂ AsN ₃ S ₆	<i>tris</i> (dipropyldithiocarbamate)arsenic(III) 145.1 ± 5.3	145.1 ± 5.3	(298)	DSC,E	[86431-46-1] [99/34]
C ₂₇ H ₅₄ AsN ₃ S ₆	<i>tris</i> (N,N-dibutyldithiocarbamate)arsenic(III) 128 ± 3	128 ± 3	(298)		[48233-55-2] [94/31]
C ₂₇ H ₅₄ AsN ₃ S ₆	<i>tris</i> (N,N-diisobutyldithiocarbamate)arsenic(III) 128 ± 2	128 ± 2	(298)	DSC,E	[41582-74-5] [97/31]
Au	dimethyl(1,1,1-trifluoro-2,4-pentanedionato)gold(III) (265–300)	83.5			[00/33]
C ₇ H ₁₀ AuF ₃ O ₂					
B					
CH ₅ BO ₂	dihydroxymethylborane (293–362)	64.1	(308)		[13061-96-6] [87/4]
	(298–338)	65.2	(318)		[40/4]
(CH ₅ N)–(BH ₃)	methylamine–borane complex (273–318)	78.7 ± 4.2		ME	[1722-33-4] [59/16]
(CH ₅ N)–(C ₃ H ₉ BO ₃)	methylamine–methylborate complex 58.2	58.2			[51/13]
CH ₁₁ B ₂ NS ₁	N-methyl-N-silylaminodiborane (214–230)	36.6	(222)		[50/7]
(C ₂ H ₃ OF)–(BF ₃)	methylfluorocarbonyl–trifluoroboron complex (223–273)	26.3	(248)		[353-44-6] [57/8]
(C ₂ H ₅ B ₃)–(C ₃ H ₉ N)	1,5-dicarbopentaborane(5)–trimethylamine complex (220–253)	49.7	(236)		[72/28]
C ₂ H ₆ BF ₂ N	dimethylamino difluoroboron (308–359)	76.5	(333)		[54/13]
(C ₂ H ₇ N)–(BH ₃)	dimethylamine–borane complex (273–308)	77.4 ± 2.9		ME	[74-94-2] [69/16]
(C ₂ H ₇ N)–(C ₃ H ₉ BO ₃)	dimethylamine–methylborate complex 70.3	70.3			[51/13]
C ₂ H ₆ B ₄	1,6-dicarbahexaborane (190–209)	31.2	(194)		[20693-67-8] [87/4]
C ₂ H ₁₂ B ₁₀	1,2-dicarbododecaborane (<i>o</i> -carborane) (283–333)	50.3	(318)		[16872-09-6] [87/4]
	(333–423)	49.4	(348)		[87/4]
		65.4 ± 1.0	(298)		[82/20][76/13]
C ₂ H ₁₂ B ₁₀	1,7-dicarbododecaborane (<i>m</i> -carborane) (283–333)	67.5	(298)		[16986-24-6] [87/4]
	(333–423)	63.3	(348)		[87/4]
		58.5 ± 1.0	(298)		[82/20][76/13]
C ₂ H ₁₂ B ₁₀	1,12-dicarbododecaborane (<i>p</i> -carborane) 61.3 ± 1.0	61.3 ± 1.0	(298)		[20644-12-6] [82/20][76/13]
(C ₃ H ₇ N)–(BH ₃)	azetidine–borane complex (297–321)	67.9	(309)		[56/17]
(C ₃ H ₉ B)–(C ₂ H ₉ Nsi)	trimethylboron–silyldimethylamine complex (243–268)	51.4	(255)		[54/11]
(C ₃ H ₉ B)–(C ₇ H ₁₃ N)	trimethylboron–azabicyclo[2.2.2]octane complex (273–388)	79.6			[48/3]
(C ₃ H ₉ N)–(BF ₃)	trimethylamine–boron trifluoride complex (373–413)	68.9	(393)		[420-20-2] [87/4][43/5]
(C ₃ H ₉)–(B ₂ F ₄)	trimethylamine–diboron tetrafluoride (tetramer)				[3801-72-7]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_3\text{H}_9\text{B}_3\text{Cl}_3\text{N}_3$	(366–399) 2,4,6-trichloro-1,3,5-trimethylborazine	65.1	(382)		[58/8] [703-86-6]
$(\text{C}_3\text{H}_9\text{N})-(\text{BH}_3)$	(363–404) trimethylamine–borane complex	57.9	(383.5)		[87/4] [75-22-9]
$(\text{C}_3\text{H}_9\text{N})-(\text{C}_3\text{H}_6\text{BCl}_2\text{N})$	(273–363) trimethylamine–dimethylaminoboron dichloride complex	56.9 ± 0.8		ME	[59/16]
$\text{C}_3\text{H}_{10}\text{BN}$	(296–367) N-methylaminodimethylborane	57	(311)		[87/4][37/1]
$\text{C}_3\text{H}_{12}\text{B}_{10}\text{O}_2$	<i>o</i> -carboranecarboxylic acid	66.1 ± 1.7	(317)		[52/8] [4023-40-9]
$\text{C}_3\text{H}_{12}\text{B}_{10}\text{O}_2$	<i>m</i> -carboranecarboxylic acid	56.9 ± 0.8	(298)		[88/16][66/7] [18178-04-6]
$\text{C}_3\text{H}_{12}\text{B}_{10}\text{O}_2$	<i>p</i> -carboranecarboxylic acid	97.0 ± 1.7	(298)		[82/20][70/14] [18581-81-2]
$\text{C}_3\text{H}_{12}\text{B}_{10}\text{O}_2$		97.7 ± 0.7	(298)		[82/20][70/14] [23087-98-1]
$\text{C}_3\text{H}_{14}\text{B}_{10}$	methyl- <i>o</i> -carborane	96.3 ± 0.7	(298)		[82/20][70/14] [16872-10-9]
$\text{C}_3\text{H}_{14}\text{B}_{10}\text{O}$	hydroxymethyl- <i>o</i> -carborane	63.8 ± 0.6	(298)		[82/20][76/13] [19610-34-5]
$\text{C}_3\text{H}_{14}\text{B}_{10}\text{O}$	hydroxymethyl- <i>m</i> -carborane	77.0 ± 1.3	(298)		[82/20][76/13] [53257-04-8]
$\text{C}_3\text{H}_{14}\text{B}_{10}\text{O}$	hydroxymethyl- <i>p</i> -carborane	78.3 ± 1.3	(298)		[82/20][76/13] [35795-98-3]
$\text{C}_4\text{H}_{11}\text{BO}_2$	dihydroxy- <i>n</i> -butylborane	83.9 ± 1.3	(298)		[82/20][76/13] [4426-47-5]
$\text{C}_4\text{H}_{12}\text{B}_2\text{Br}_4\text{N}_2$	(303–340) dibromo(dimethylamino)borane dimer	69.9 ± 0.8	(321)	BG	[56/12] [25928-66-9]
$(\text{C}_4\text{H}_{12}\text{GeO})-(\text{BF}_3)$		87.4 ± 22.2		BG	[83/12]
$\text{C}_4\text{H}_{16}\text{B}_{10}$	trimethylmethoxygermane–boron trifluoride complex	59.5	(297)	SG	[61/9] [17032-21-2]
$\text{C}_4\text{H}_{18}\text{B}_4\text{N}_2$	(289–306) dimethyl- <i>o</i> -carborane	65.3 ± 0.7	(298)		[82/20][76/13]
$(\text{C}_5\text{H}_5\text{N})-(\text{BBr}_3)$	1,4-piperazinediyl bis(diborane(6))	63.9	(332)		[68/16] [3022-54-6]
$(\text{C}_5\text{H}_{11}\text{N})-(\text{BCl}_3)$	(318–346) boron tribromide–pyridine complex	65.8 ± 0.2		T	[89/24]
$(\text{C}_5\text{H}_{11}\text{N})-(\text{BH}_3)$	(523–602) piperidine–boron trichloride complex	105.5 ± 1.1	(393)	C	[89/24]
$\text{C}_5\text{H}_{21}\text{B}_3\text{N}_2\text{S}$	(448–457) piperidine–borane complex	87.8	(361)	GS	[60/19] [56/17]
$\text{C}_6\text{H}_{12}\text{BNO}_3$	(342–380) 1,2,3,3,4,4,5,5,6,6-decahydro-1,3,3,5,5-pentamethyl-2 <i>H</i> -1,3,5,2,4,6-thiadiazatriborine	57.7			[72/27] [283-56-7]
$(\text{C}_7\text{H}_9\text{N})-(\text{BH}_3)$	2,8,9-trioxa-5-aza-1-boratricyclo[3.3.3.0 ^{1,5}]undecane	111.9 ± 0.9	(418)	C	[84/14]
$\text{C}_7\text{H}_{14}\text{BNO}_3$	2,6-dimethylpyridine–borane complex	83.8	(368)	T	[56/15] [283-62-5]
$\text{C}_8\text{H}_{16}\text{BNO}_3$	(358–378) 2,9,10-trioxa-5-aza-1-boratricyclo[4.3.3.0 ^{1,6}]dodecane	105.2 ± 0.6	(390)	C	[84/14] [283-64-7]
$\text{C}_8\text{H}_{18}\text{BNO}_3$	2,10,11-trioxa-5-aza-1-boratricyclo[4.4.3.0 ^{1,6}]tridecane	102.2 ± 1.0	(390)	C	[84/14] [283-65-8]
$\text{C}_8\text{H}_{18}\text{B}_{10}\text{O}_3$	2,10,11-trioxa-5-aza-1-boratricyclo[4.4.4.0 ^{1,6}]tetradecane	97.9 ± 1.0	(418)	C	[84/14] [146959-04-8]
$\text{C}_8\text{H}_{18}\text{B}_{10}\text{O}_3$	1,2-dicarbadodecaborane(12)-1-carboperoxoic acid 1,1-dimethyl-2-propynyl ester	120.7 ± 7.4		ME	[99/43]
$\text{C}_8\text{H}_{24}\text{B}_{10}$	(329–343) 1,7-dicarbadodecaborane(12)-1-carboperoxoic acid 1,1-dimethyl-2-propynyl ester	80.1 ± 6.1		ME	[146959-05-9] [99/43]
	1-hexyl- <i>o</i> -carborane	86.2 ± 1.4	(298)		[20740-05-0] [82/20][78/23]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{11}\text{H}_{24}\text{B}_{10}\text{O}_3$	1,2-dicarbadodecaborane(12)-1-carboperoxoic acid 2-(1-methylethyl)-1,1-dimethyl-2-propynyl ester (345–362)	125.1 ± 7.0		ME	[146959-06-0] [99/43]
$\text{C}_{18}\text{H}_{15}\text{B}$	triphenylboron	103.8 ± 2.5 92.1 ± 2.5 81.6 ± 2.1	(360) (298)	TE,ME	[960-71-4] [84/4] [78/2] [82/20][67/10] [1088-01-3] [82/20][67/10] [13965-73-6] [58/8] [13703-88-3] [66/9] [933-18-6] [66/9] [55/6] [13779-24-3] [66/9] [13871-09-5] [69/22][71/30]
$\text{C}_{18}\text{H}_{33}\text{B}$	tricyclohexylboron	81.6 ± 4.2	(298)		
B_2F_4	diboron tetrafluoride (178–209.5)	35.5	(193)		
$\text{B}_3\text{Br}_3\text{H}_3\text{N}_3$	2,4,6-tribromoborazine (342–395)	86.2 ± 0.4		I	
$\text{B}_3\text{Cl}_3\text{H}_3\text{N}_3$	2,4,6-trichloroborazine (303–353) (313–357)	70.5 ± 0.4 71.1		I	
$\text{B}_3\text{F}_3\text{H}_3\text{N}_3$	2,4,6-trifluoroborazine (273–454)	63.1 ± 0.1		I	
$\text{B}_3\text{H}_{12}\text{N}_3$	hexahydroborazine (321–349)	104.6 ± 12.6		ME	
$(\text{NH}_3) - (\text{B}_3\text{H}_7)$	ammonia–triborane complex (306–328) (304–327)	71.5 ± 0.4 71.5		ME	[59/16] [59/15]
Ba ($\text{C}_{10}\text{H}_2\text{BaF}_{12}\text{O}_4$)–($\text{C}_{12}\text{H}_{24}\text{O}_6$)	bis(1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)barium(II)–18-crown-6 complex (412–468) (428–473)	104.9 ± 1.3 115 ± 2	(440) (450)	T	[143737-48-8] [95/22] [93/4] [155138-07-1] [94/40] [93/9] [16034-35-2] [96/7] [160669-81-8] [96/7]
$\text{C}_{22}\text{H}_{38}\text{BaO}_4$	bis(2,2,6,6-tetramethylheptane-3,5-dionato)barium(II) NA (413–473)	90.2	(443)		
$\text{C}_{34}\text{H}_{42}\text{BaCu}_2\text{F}_{24}\text{O}_8$	tetrakis(hexafluoroisopropoxy)bis(2,2,6,6-tetramethylheptane-3,5-dionato)barium(II)dycopper(II) (383–448)	102.7	(416)		
$\text{C}_{56}\text{H}_{80}\text{BaF}_{24}\text{O}_{12}\text{Y}_2$	tetrakis(hexafluoroisopropoxy)tetrakis(2,2,6,6-tetramethylheptane-3,5-dionato)barium(II)diyttrium(III) (360–403)	84.8	(382)		
Be $\text{C}_{10}\text{H}_2\text{BeF}_{12}\text{O}_4$	bis(1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)beryllium(II) (289–349)	66.1	(319)	BG	[19648-82-9] [87/20] [13939-10-1] [87/20][88/22] [87/20] [60/17] [10210-64-7] [77/18][88/2] [88/25] [88/22] [85/5] [83/10] [60/17] [19049-40-2] [87/4] [59/17] [55/9] [55/9]
$\text{C}_{10}\text{H}_8\text{BeF}_6\text{O}_4$	bis(1,1,1-trifluoro-2,4-pentanedionato)beryllium(II) (354–383)	85.3 ± 6.3 88.0 ± 6.5 U30.5	(368) (298)	BG	
$\text{C}_{10}\text{H}_{14}\text{BeO}_4$	bis(2,4-pentanedionato)beryllium(II)	94 ± 1.0 95.3 ± 2.0 82.3 91 ± 1.4 85.3 ± 3.5 U35.6	(298) (298) (298) (298) (298) (298)	ME BG C DSC	
$\text{C}_{12}\text{H}_{18}\text{Be}_4\text{O}_{13}$	hexakis(aceto)-oxotetraberyllium (390–451)	115.3	(420.5)		
monoclinic form I form II	(394–422) (426–446)	115.3 132.6 113.4	(408) (436)		
$\text{C}_{20}\text{H}_{12}\text{BeF}_6\text{O}_4$	bis(1-phenyl-4,4,4-trifluoro-1,3-butanedionato)beryllium(II) U35.8			I	
$\text{C}_{20}\text{H}_{18}\text{BeO}_4$	bis(benzoylacetato)beryllium(II) (416–438)	151.6 ± 1.8 158.0 ± 1.8 142.3 ± 1.4	(427) (298) (298)	TE,ME C	[14128-75-7] [95/9] [95/9] [83/20] [36915-22-7]
$\text{C}_{22}\text{H}_{38}\text{BeO}_4$	bis(2,2,6,6-tetramethylheptane-3,5-dionato)beryllium				

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
BeF ₂	beryllium fluoride (713–795)	84.2 236.4 ± 2.9 231.8 ± 1.7	(750) (755)	BG TE MS	[88/22] [7787-49-7] [65/20] [65/20]
Bi					
C ₁₅ H ₃₀ BiN ₃ S ₆	<i>tris</i> (N,N-diethyldithiocarbamate)bismuth(III)	213 ± 3	(298)		[20673-31-8] [94/31]
C ₁₈ H ₁₅ Bi	triphenylbismuth	110.9 ± 8.4	(298)		[603-33-8] [82/20][79/22]
C ₂₁ H ₄₂ BiN ₃ S ₆	<i>tris</i> (dipropyldithiocarbamate)bismuth(III)	285.2 ± 5.0		DSC,E	[57407-97-3] [99/34]
C ₂₇ H ₅₄ BiN ₃ S ₆	<i>tris</i> (N,N-dibutyldithiocarbamate)bismuth(III)	202 ± 3	(298)		[34410-99-6] [94/31]
C ₂₇ H ₅₄ BiN ₃ S ₆	<i>tris</i> (N,N-diisobutyldithiocarbamate)bismuth(III)	147 ± 3	(298)	DSC,E	[90285-80-6] [97/31]
BiCl ₃	bismuth (III) chloride (371–468) (371–468)	124.7 118.8 ± 0.4	(420)	ME ME	[7787-60-2] [66/8][59/11] [59/11]
Ca					
C ₂₂ H ₃₈ CaO ₄	<i>bis</i> (2,2,6,6-tetramethylheptane-3,5-dionato)calcium(II)	72.0		GS	[3618-89-0] [90/15]
Cd					
C ₄ H ₁₆ CdCl ₂ N ₈ S ₄	<i>trans</i> -dichloro- <i>tetrakis</i> (thiourea)cadmium(II) (377–405)	75 ± 20			[28813-21-0] [70/11]
C ₁₀ H ₁₄ CdCl ₂ N ₆ O ₂	[cadmium(1-methylcytosine) ₂ Cl ₂] (483–503)	135.3 ± 20 145 ± 20	(493) (298)	ME	[84/12] [84/12]
C ₁₀ H ₁₄ CdO ₄	<i>bis</i> (2,4-pentanedionato)cadmium(II) (438–448)	144.9 ± 22 154 ± 22	(443) (298)	ME	[14689-45-3] [84/12] [84/12]
C ₁₀ H ₂₀ CdN ₂ S ₄	<i>bis</i> (diethyldithiocarbamate)cadmium(II) (433–469)	133.2	(451)		[14239-68-0] [87/4]
C ₁₄ H ₂₈ CdN ₂ S ₄	<i>bis</i> (dipropyldithiocarbamate)cadmium(II)	199 ± 1	(298)	DSC,E	[55519-99-8] [92/19]
C ₁₈ H ₁₂ CdN ₂ O ₂	<i>bis</i> (8-hydroxyquinolino)cadmium(II)	201.7 ± 7.5 144.9 ± 22 154 ± 22	(298) (443) (298)	ME ME	[14245-29-5] [94/16] [84/12] [84/12]
C ₁₈ H ₃₆ CdN ₂ S ₄	<i>bis</i> (dibutyldithiocarbamate)cadmium(II)	123 ± 3	(298)	DSC,E	[14566-86-0] [91/15]
C ₁₈ H ₃₆ CdN ₂ S ₄	<i>bis</i> (diisobutyldithiocarbamate)cadmium(II)	281 ± 2	(298)	DSC,E	[69090-75-1] [94/33]
C ₂₀ H ₁₆ CdN ₂ O ₂	<i>bis</i> (8-hydroxy-2-methylquinolino)cadmium(II) (537–554)	190.9 ± 7.3 203.3 ± 7.3	(546) (298)	ME	[15685-78-6] [98/8] [98/8]
C ₄₄ H ₂₈ CdN ₄	5,10,15,20-tetraphenylporphine cadmium(II)	222 ± 6		GS	[14977-07-2] [00/36]
Ce					
C ₁₅ H ₁₅ Ce	<i>tris</i> (cyclopentadienyl)cerium (528–653)	104.6 ± 2.1			[1298-53-9] [73/31]
CeBr ₃	cerium(III) bromide (887–1003)	300 ± 10	(298)	TE	[14457-87-5] [00/19]
CeCl ₃	cerium(III) chloride (955–1070)	331 ± 5	(298)	TE	[7790-86-5] [00/19]
CeI ₃	cerium(III) iodide (910–1031)	295 ± 10	(298)	TE	[7790-87-6] [00/19]
Cf					
(C ₁₅ H ₃ CfF ₁₈ O ₆)–2(C ₆ H ₁₄ OS)	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)californium-249– dipropyl sulfoxide (1:2) complex (402–434)	93.6 ± 6.0		GS,TRM	[123611-97-2] [89/31]
(C ₁₅ H ₃ CfF ₁₈ O ₆)–2(C ₁₂ H ₂₇ OP)	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)californium-249– tributylphosphine oxide (1:2) complex (431–485)	130.6 ± 1.9		GS,TRM	[123628-36-4] [89/31]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
(C ₁₅ H ₃ CfF ₁₈ O ₆)–2(C ₁₂ H ₂₇ O ₄ P)	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)californium-249–tributylphosphate (1:2) complex (413–451)	133.0 ± 6.1		GS, TRM	[123712-43-6] [89/31]
Cl					
HCl	hydrogen chloride (121–133)	19.7	(127)		[90/33]
	(134–150)	19.6	(142)		[90/33]
Co					
C ₄ H ₁₆ Cl ₂ CoN ₈ S ₄	<i>trans</i> -dichloro- <i>tetrakis</i> (thiourea)cobalt(II) (356–382)	129 ± 20			[22738-43-8] [70/11]
C ₈ Co ₂ O ₈	octacarbonyldicobalt (264–278)	84.3 ± 0.5	(271)	TE	[10210-68-1] [95/36]
	(288–315)	103.8	(301.5)		[87/4][68/15]
	(207–287)	65.2 ± 3.3	(298)		[82/20][75/26]
		75.3 ± 6.3		EM	[73/26]
C ₈ H ₁₀ Cl ₂ CoN ₆ O ₂	[cobalt(cytosine) ₂ Cl ₂] (483–523)	151.8 ± 14 162 ± 14	(503) (298)	ME	[74543-51-4] [84/12] [84/12]
C ₉ CoMnO ₉	nonacarbonylcobaltmanganese	85 ± 2 72 ± 2	(308) (298)	C	[35646-82-3] [98/33] [98/33]
C ₉ CoO ₉ Re	nonacarbonylcobaltrhenium	94 ± 4 83 ± 4	(313) (298)	C	[15039-80-2] [98/33] [98/33]
C ₁₀ BrCo ₃ O ₉	(bromomethylidyne)tricobaltenneacarbonyl	99.6 ± 1.7	(298)		[19439-14-6] [82/20][75/26]
C ₁₀ ClCo ₃ O ₉	(chloromethylidyne)tricobaltenneacarbonyl	117.6 ± 2.5	(298)		[13682-02-5] [82/20][75/26]
C ₁₀ H ₈ Cl ₄ CoN ₂	[cobalt(2-chloropyridine) ₂ Cl ₂] (345–365)	101.2 ± 6.7	(355)	DSC	[14361-73-0] [82/18]
C ₁₀ H ₈ Cl ₄ CoN ₂	[cobalt(3-chloropyridine) ₂ Cl ₂] (345–365)	77.0 ± 4.2	(355)	DSC	[14361-78-5] [82/18]
C ₁₀ H ₁₀ Co	dicyclopentadienyl cobalt	72.1 ± 0.1 70.3 ± 4.2	(298)		[1277-43-6] [88/3] [82/20][75/23]
C ₁₀ H ₁₄ CoO ₄	<i>bis</i> (2,4-pentanedionato)cobalt(II) (433–463)	149		TGA	[14024-48-7] [00/35]
	(322–371)	130.1 ± 6.3 118.7 ± 2.2 81.2 U62.8	(298) (298) (370)	ME	[90/21] [85/5] [70/10] [60/17]
C ₁₂ Co ₄ O ₁₂	tetracobaltdodecacarbonyl	96.2 ± 4.2	(298)		[17786-31-1] [82/20][74/26]
C ₁₂ H ₁₄ Cl ₂ CoN ₂	[cobalt(2-methylpyridine) ₂ Cl ₂] (345–365)	86.6 ± 3.8	(355)	DSC	[13869-67-5] [82/18]
C ₁₄ H ₁₀ Br ₂ CoN ₂ S ₂	[cobalt(benzothiazole) ₂ Br ₂] (381–399)	127.7 ± 4.1	(390)	DSC	[21422-14-0] [73/28]
C ₁₅ H ₃ CoF ₁₈ O ₆	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)cobalt(III) (333–363)	73		TGA	[16702-37-7] [00/35]
C ₁₅ H ₁₂ CoF ₉ O ₆	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)cobalt(III) (373–403)	119 114 ± 4.0 108.8 ± 0.4	(298)	TGA	[16827-64-8] [00/35] [88/1]
	(383–433)			GS	[85/16]
C ₁₅ H ₂₁ CoO ₆	<i>tris</i> (2,4-pentanedionato)cobalt(III) (433–463)	138 NA		TGA	[21679-46-9] [00/35] [94/34]
	(318–382)	134.6 ± 4.0 142.6 ± 6.9 86.3 107.1 74.9 ± 4.6 U13.0	(298) (471)	ME DSC	[90/21] [87/13] [71/17] [70/10] [64/4] [61/8]
C ₁₅ H ₃₀ CoN ₃ S ₆	<i>tris</i> (diethyldithiocarbamato)cobalt(III)				[13963-60-5]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{16}\text{H}_{14}\text{Br}_2\text{CoN}_2\text{O}_2$	(448–587)	95 ± 6	(518)		[79/20]
	[cobalt(2-methylbenzoxazole) $_2\text{Br}_2$]				[22974-96-5]
$\text{C}_{16}\text{H}_{14}\text{Cl}_2\text{CoN}_2\text{O}_2$	(345–390)	111.1 ± 4.2	(368)	DSC	[82/18][74/27]
	[cobalt(2-methylbenzoxazole) $_2\text{Cl}_2$]				[52657-96-2]
$\text{C}_{16}\text{H}_{14}\text{Br}_2\text{CoN}_2\text{S}_2$	(345–390)	92.4 ± 2.5	(368)	DSC	[82/18][74/27]
	[cobalt(2-methylbenzothiazole) $_2\text{Br}_2$]				[26225-02-5]
$\text{C}_{16}\text{H}_{14}\text{Cl}_2\text{CoN}_2\text{S}_2$	(335–354)	115.1 ± 4.1	(345)	DSC	[73/28]
	[cobalt(2-methylbenzothiazole) $_2\text{Cl}_2$]				[26225-01-4]
$\text{C}_{18}\text{H}_{12}\text{CoN}_2\text{O}_2$	(332–356)	122.6 ± 1.2	(345)	DSC	[73/28]
	bis(8-hydroxyquinolino)cobalt(II)				[13978-88-6]
		205.3 ± 4.0	(298)	ME	[94/16]
	(533–569)	185.7 ± 9	(551)	ME	[84/12]
$\text{C}_{18}\text{H}_{14}\text{CoN}_4$		200 ± 10	(298)		[84/12]
	dibenzotetra-aza–annulene cobalt(II) complex				[41283-94-7]
		178.2 ± 16.7	(360)		[82/25]
$\text{C}_{18}\text{H}_{18}\text{Br}_2\text{CoN}_2\text{O}_2$	[cobalt(2,5-dimethylbenzoxazole) $_2\text{Br}_2$]				[52230-48-5]
	(345–390)	95.4 ± 4.6	(368)	DSC	[82/18][74/27]
$\text{C}_{18}\text{H}_{18}\text{Cl}_2\text{CoN}_2\text{O}_2$	[cobalt(2,5-dimethylbenzoxazole) $_2\text{Cl}_2$]				[52230-47-4]
	(345–390)	104.6 ± 5.8	(368)	DSC	[82/18][74/27]
$\text{C}_{20}\text{H}_{16}\text{CoN}_2\text{O}_2$	bis(8-hydroxy-2-methylquinolino)cobalt(II)				[17992-18-6]
	(457–473)	196.1 ± 5.9	(465)	ME	[98/8]
		204.4 ± 5.9	(298)		[98/8]
$\text{C}_{22}\text{H}_{38}\text{CoO}_4$	bis(2,2,6,6-tetramethyl-3,5-heptanedionato)cobalt(II)				[13986-53-3]
	(433–463)	143		TGA	[00/35]
$\text{C}_{24}\text{H}_{12}\text{CoF}_9\text{O}_6\text{S}_3$	tris(1-(2-thenoyl)-4,4,4-trifluoro-1,3-butanedione)cobalt(III)				[41875-84-7]
		45.6			[61/8]
$\text{C}_{24}\text{H}_{12}\text{CoF}_9\text{O}_9$	tris(2-furoyltrifluoroacetono)cobalt(III)				[64137-83-3]
		35.6			[61/8]
$\text{C}_{30}\text{H}_{18}\text{CoF}_9\text{O}_6$	tris(1-phenyl-4,4,4-trifluoro-1,3-butanedionato)cobalt(III)				[31125-84-5]
		51.0			[61/8]
$\text{C}_{30}\text{H}_{27}\text{CoO}_6$	tris(1-phenyl-1,3-butanedionato)cobalt(III)				[14524-55-1]
		39.0			[61/8]
$\text{C}_{32}\text{H}_{16}\text{CoN}_8$	cobalt (II) phthalocyanine				[3317-67-7]
		183.7 ± 13.8		ME	[70/7]
$\text{C}_{32}\text{H}_{46}\text{CoN}_2\text{O}_4$	bis(2,2,6,6-tetramethyl-3,5-heptanedionato)(2,2'-bipyridyl)cobalt(II)				[18347-53-8]
		126 ± 4.0		B	[96/24]
		130.3		UV/Vis	[96/24]
		124.4		MEM	[96/24]
$\text{C}_{33}\text{H}_{57}\text{CoO}_6$	tris(2,2,6,6-tetramethyl-3,5-heptanedionato)cobalt(III)				[14877-41-3]
	(433–463)	132		TGA	[00/35]
		126 ± 3.0	(298)		[88/1]
CoBr_2	cobalt(II) bromide				[7789-43-7]
	(764–911)	207 ± 4.0	(802)	TE	[97/20]
		216 ± 1.0	(298)		[97/20]
Cr					
C_6CrO_6	chromium hexacarbonyl				[13007-92-6]
	(266–272)	65.7	(269)	TE	[95/36]
	(323–391)	68.5 ± 1.1			[93/28]
	(288–423)	68.5	(355.5)		[87/4]
		68.9 ± 2	(298)		[84/17]
		70.0 ± 2	(298)	C	[83/20]
	(240–280)	71.6 ± 1.7	(260)	ME	[80/34][79/19]
		69.5	(298)	C	[75/20]
		72.0 ± 4.2	(298)		[82/20][75/24]
	(274–301)	71.5 ± 0.8	(288)	BG	[66/17]
	(319–411)	69.3			[52/7]
		71.9			[35/2]
	(308–408)	63.6	(358)	MM	[34/3]
	thiazole(pentacarbonyl)chromium				[55293-31-7]
	(270–301)	102.0 ± 2.7	(286)	ME	[79/19]
$\text{C}_8\text{H}_4\text{CrN}_2\text{O}_5$	pyrazole(pentacarbonyl)chromium				[71127-65-6]
	(270–303)	88.4 ± 1.8	(287)	ME	[79/19]
$\text{C}_8\text{H}_9\text{CrNO}_5$	trimethylamine(pentacarbonyl)chromium				[15228-26-9]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_8\text{H}_9\text{CrO}_5\text{P}$	(248–293)	80.2 ± 0.7	(271)	ME	[80/34]
	trimethylphosphine(pentacarbonyl)chromium	91.2 ± 1.6		ME	[26555-09-9] [80/34]
$\text{C}_8\text{H}_{12}\text{CrMoO}_8$	chromium molybdenum tetraacetate	165.0 ± 8.4			[71561-64-3] [82/20]
					[15020-15-2]
$\text{C}_8\text{H}_{12}\text{Cr}_2\text{O}_8$	tetra- μ -acetatodichromium(II) (330–340)	299.6 ± 10	(335)	ME,TE	[84/20]
		313.8 ± 27.0	(298)		[82/20]
		145		E	[79/23] [66179-02-0]
$\text{C}_9\text{H}_4\text{CrN}_2\text{O}_5$	pyrazine(pentacarbonyl)chromium	99.7		ME	[79/19]
$\text{C}_9\text{H}_5\text{ClCrO}_3$	chlorobenzenechromium tricarbonyl	102.5 ± 4.2	(298)		[12082-03-0] [82/20][75/20]
$\text{C}_9\text{H}_6\text{CrO}_3$	benzene chromium tricarbonyl	91.2	(298)	C	[12082-08-5] [75/20]
		U58.6			[61/4][73/29]
		97.9		TE	[59/6][73/29]
$\text{C}_{10}\text{H}_5\text{CrNO}_5$	pyridine(pentacarbonyl)chromium				[14740-77-3]
$\text{C}_{10}\text{H}_8\text{CrO}_3$	cycloheptatriene chromium tricarbonyl	103.2 ± 1.8	(306)	ME	[79/19]
		94.1	(298)	C	[12125-72-3] [75/20]
$\text{C}_{10}\text{H}_8\text{CrO}_3$	η^6 -toluene(tricarbonyl)chromium	93.0 ± 2.0	(298)	C	[12125-87-0] [84/17]
		94.6 ± 4.2	(298)		[82/20][75/20]
		104.2 ± 2.0	(298)	C	[12116-44-8] [84/17]
$\text{C}_{10}\text{H}_{10}\text{Cr}$	dicyclopentadienyl chromium				[1271-24-5]
		71.0	(298)		[84/24]
		62.8 ± 4.2	(298)		[82/20][75/23]
		69.9 ± 1.7			[77/24]
$\text{C}_{10}\text{H}_{14}\text{CrO}_4$	<i>bis</i> (2,4-pentanedionato)chromium(II) (330–370)	129.8 ± 8.7	(298)	ME	[14024-50-1] [90/21]
		111	(439)	T	[81/15]
$\text{C}_{10}\text{H}_{11}\text{CrNO}_5$	piperidine(pentacarbonyl)chromium				[15710-39-1]
$\text{C}_{11}\text{H}_8\text{CrO}_4$	norbornadienechromium tetracarbonyl	93.5 ± 1.9	(282)	ME	[79/19]
		89.0 ± 4.0	(298)		[12146-36-0] [82/20][77/22]
$\text{C}_{11}\text{H}_8\text{CrO}_4$	η^6 -acetophenone(tricarbonyl)chromium	107.0 ± 0.6	(298)	C	[12153-11-6] [84/17]
$\text{C}_{11}\text{H}_8\text{CrO}_5$	η^6 -methyl benzoate(tricarbonyl)chromium	114.0 ± 5.0	(298)	C	[12125-87-0] [84/17]
					[12109-10-3]
$\text{C}_{11}\text{H}_{11}\text{CrNO}_3$	η^6 -N,N-dimethylaniline(tricarbonyl)chromium	118.4 ± 10	(298)	C	[84/17]
$\text{C}_{12}\text{H}_{12}\text{Cr}$	dibenzenechromium (323–363)				[1271-54-1]
		89.4	(343)		[87/4]
		78.2 ± 6.3	(298)		[82/20][73/29]
		82.0 ± 2.1		ME	[73/22]
		90.6 ± 0.3			[69/21]
$\text{C}_{12}\text{H}_{12}\text{CrO}_3$	mesitylene chromium tricarbonyl	78.2 ± 6.2	(298)		[58/13]
		108.4	(298)	C	[12129-67-8] [75/20]
		U64.4			[61/5][77/21]
$\text{C}_{12}\text{H}_{12}\text{CrO}_3$	(1,2,4-trimethylbenzene) chromium tricarbonyl	U33.5			[32913-41-0] [61/5][77/21]
					[12110-37-1]
$\text{C}_{13}\text{H}_8\text{CrO}_3$	(1,2,3,4,4a,8a-h-naphthalene)tricarbonyl chromium	107 ± 3	(298)	C	[79/16]
$\text{C}_{15}\text{H}_3\text{F}_{18}\text{CrO}_6$	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)chromium(III) (333–363)	46		TGA	[14592-80-4] [00/35]
		112 ± 4.0	(298)		[87/12]
		123.0 ± 1.3	(335)		[72/17]
$\text{C}_{15}\text{H}_{12}\text{CrF}_9\text{O}_6$	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)chromium(III) (373–403)	71		TGA	[14592-89-2] [00/35]
		182 ± 4.0	(426)	C	[87/12]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{15}\text{H}_{18}\text{CrO}_3$	(373–438)	117 ± 4.0	(298)	GS	[87/12]
	(403–423)	115.1 ± 0.8			[85/16]
		112.5 ± 4.8			[78/29]
	(377–413)	53.6	(447)		[77/25]
	hexamethylbenzene chromium tricarbonyl	108.8 ± 1.3	(395)		[72/17]
$\text{C}_{15}\text{H}_{21}\text{CrO}_6$		123.0 ± 4.0	(298)	C	[12088-11-8]
	<i>tris</i> (2,4-pentanedionato)chromium(III)				[75/20][77/22]
	(413–443)	91		TGA	[21679-31-2]
	(350–375)	126.8 ± 4.2	(298)	ME	[00/35]
	(457–486)	113.0 ± 4.8		BG	[90/21]
		132.1 ± 1.9	(298)	C	[88/22][87/21]
		28.9	(463)		[85/5]
		112.1	(390)		[77/25]
	(363–393)	40.2 ± 1.7	(378)		[70/10]
		110.9 ± 0.8	(298)	HSA	[72/17]
$\text{C}_{18}\text{H}_{24}\text{Cr}$	<i>bis</i> (η^6 -1,3,5-trimethylbenzene)chromium				[70/9][70/17]
		104 ± 1	(298)	ME	[77/18][88/2]
$\text{C}_{20}\text{H}_{16}\text{Cr}$	<i>bis</i> (naphthalene)chromium				[67/11]
		105.0 ± 10		C	[1274-07-3]
$\text{C}_{23}\text{H}_{15}\text{CrO}_5\text{P}$	triphenylphosphine(pentacarbonyl)chromium				[79/16]
	(324–347)	170.2 ± 6.8	(336)	ME	[33085-81-3]
$\text{C}_{24}\text{H}_{24}\text{Cr}_2\text{N}_4\text{O}_4$	<i>tetrakis</i> (6-methyl-2-hydroxypyridyl)dichromium(II)				[79/16]
		150.0 ± 4.0	(298)		[14917-12-5]
$\text{C}_{24}\text{H}_{36}\text{Cr}$	<i>bis</i> (η^6 -hexamethylbenzene)chromium				[80/34]
		119 ± 4	(298)	C	[67634-82-6]
$\text{C}_{30}\text{H}_{27}\text{CrO}_6$	<i>tris</i> (1-phenyl-1,3-butanedionato)chromium(III)				[82/20][81/18]
		186 ± 2	(298)		[12156-66-0]
$\text{C}_{30}\text{H}_{30}\text{F}_{21}\text{CrO}_6$	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyl-4,6-octanedionato)chromium(III)				[79/16]
	(323–353)	37.7 ± 0.8	(338)		[16432-36-3]
$\text{C}_{33}\text{H}_{57}\text{CrO}_6$	<i>tris</i> (2,2,6,6-tetramethyl-3,5-heptanedionato)chromium(III)				[87/12]
	(413–443)	85		TGA	[17966-86-8]
CrI_2	chromium II iodide	133 ± 2	(298)		[72/17]
	(943–1054)	298.7	(298)	65	[14434-47-0]
Cs	cesium pivalate				[00/35]
$\text{C}_5\text{H}_9\text{CsO}_2$		163.5 ± 7.2			[87/12]
CsI	cesium iodide				[13478-28-9]
		195.6	(298)	GS	[56/19]
		193.1	(298)	T	[20442-70-0]
		193.1	(298)	T	[98/31]
Cu		191.1	(298)	MS	[7789-17-5]
					[98/4]
					[85/15][98/4]
					[84/29][98/4]
$\text{C}_6\text{H}_{12}\text{CuN}_2\text{S}_4$	<i>bis</i> (dimethyldithiocarbamato) copper complex				[84/28][98/4]
		156.0 ± 0.3	(298)	C	[137-29-1]
	(443–473)	147.4 ± 0.8	(458)		[95/21]
$\text{C}_8\text{H}_{12}\text{Cu}_2\text{O}_8$	<i>tetrakis</i> (acetato)dicropper(II)	149.0 ± 2.5		GC	[87/4][78/12]
	(321–360)	106.1 ± 0.9	(298)	ME, TE	[76/21]
$\text{C}_8\text{H}_{14}\text{CuN}_4\text{O}_4$	<i>bis</i> (dimethylglyoxime)copper(II)				[24411-13-0]
		93.1 ± 0.8	(298)	TE, ME	[90/2]
$\text{C}_{10}\text{H}_2\text{CuF}_{12}\text{O}_4$	<i>bis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)copper(II)				[14221-10-4]
		108 ± 6	(298)	C	[90/2]
$\text{C}_{10}\text{H}_8\text{CuF}_6\text{O}_4$	<i>bis</i> (1,1,1-trifluoro-2,4-pentanedionato)copper(II)				[14781-45-4]
	(373–403)	112		TGA	[88/7]
	(342–359)	113.3 ± 2.4	(350)	TE	[14324-82-4]
		115.9 ± 2.4	(298)		[00/35]
	(342–359)	114.4 ± 1.6	(350)	ME	[95/4]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{10}\text{H}_{14}\text{CuO}_4$		117.0 ± 1.6	(298)		[95/4]
		112 ± 3.0	(298)	C	[88/7]
	(383–463)	110.0 ± 0.8		GS	[85/16]
		U50.6			[60/17]
	<i>bis</i> (2,4-pentanedionato)copper(II)				[13395-16-9]
	(413–443)	120		TGA	[00/35]
	(377–398)	122.5 ± 1.2	(387)	TE	[95/4]
		122.5 ± 1.2	(298)		[95/4]
	(377–398)	122.6 ± 0.7	(387)	ME	[95/4]
		127.1 ± 1.2	(298)		[95/4]
		122.3 ± 1.1	(393)	ME	[95/4]
		127.0 ± 1.1	(298)		[95/4]
		116.6 ± 2.0	(298)	C	[94/7]
	(315–386)	115.1 ± 2.1	(298)		[91/17]
		142.6 ± 6.9	(471)	DSC	[87/12]
		107.1 ± 5.7	(492)		[87/13]
		127.5 ± 3.2	(298)		[85/5]
		154 ± 22	(298)		[84/12]
		109.9 ± 3.4	(298)	C	[84/11]
		57.1		TE	[81/13]
$\text{C}_{10}\text{H}_{20}\text{CuN}_2\text{S}_4$		109.6			[72/24]
		106.1		TG	[71/17]
		109 ± 6	(400)		[70/10]
		57.3		DSC	[71/18]
		62.8			[62/8]
	<i>bis</i> (diethyldithiocarbamate)copper(II)				[13681-87-3]
		162.6 ± 5	(298)		[89/10]
	(420–465)	149.1 ± 0.4	(442.5)		[87/4][78/12]
		103.8 ± 2.4			[79/20]
		116.2 ± 1.3			[79/14]
$\text{C}_{12}\text{H}_{12}\text{CuF}_6\text{O}_4$		149.0 ± 2.5			[76/21]
		87 ± 1.7		I	[69/15]
	<i>bis</i> (1,1,1-trifluorohexane-2,4-dione)copper(II)				[30133-85-8]
$\text{C}_{12}\text{H}_{18}\text{CuO}_4$		119.1 ± 1.7	(298)	ME	[98/32]
	<i>bis</i> (3-methyl-2,4-pentanedionato)copper(II)				[14781-49-8]
		130.7 ± 1	(396.7)	ME	[92/8]
$\text{C}_{14}\text{H}_{16}\text{CuF}_6\text{O}_4$		135.6 ± 1	(298)		[92/8]
		132.7 ± 2.5	(298)	C	[92/8]
	<i>bis</i> (1,1,1-trifluoro-5-methylhexane-2,4-dione)copper(II)				[33896-35-4]
$\text{C}_{14}\text{H}_{28}\text{CuN}_2\text{S}_4$		122.4 ± 0.9	(298)	ME	[98/32]
	<i>bis</i> (dipropyldithiocarbamate)copper complex				[14354-08-6]
		158.6 ± 5	(298)		[89/10]
$\text{C}_{14}\text{H}_{28}\text{CuN}_2\text{S}_4$		118.4 ± 3.3			[78/12]
	<i>bis</i> (diisopropyldithiocarbamate)copper complex				[14354-07-5]
	(440–465)	129.5 ± 2.9	(452.5)		[87/4][78/12]
$\text{C}_{16}\text{H}_8\text{CuF}_6\text{O}_6$	<i>bis</i> (4,4,4-trifluoro-1-(2-furanyl)butane-1,3-dione)copper(II)				[13928-10-4]
		161.1 ± 2.1	(298)	ME	[98/32]
	<i>bis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)copper(II)				[150026-91-8]
$\text{C}_{16}\text{H}_{20}\text{CuF}_6\text{O}_4$		120.2 ± 1.0	(298)	ME	[98/32]
	(353–379)	102 ± 3	(366)	T	[93/4]
	(381–443)	76.5 ± 2 (liq)	(412)	T	[93/4]
$\text{C}_{16}\text{H}_{20}\text{CuF}_6\text{O}_4$	<i>bis</i> (1,1,1-trifluoro-5-methylheptane-2,4-dione)copper(II)				[220869-88-5]
		122.5 ± 0.9	(298)	ME	[98/32]
	<i>bis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)copper(II)-15-crown-5 complex				
$\text{C}_{16}\text{H}_{20}\text{F}_6\text{O}_4\text{Cu}$ ($\text{C}_{10}\text{H}_{20}\text{O}_5$)	(368–443)	80.2 ± 2 (liq)	(405)	T	[93/4]
	<i>bis</i> (8-hydroxyquinolino)copper(II)				[10380-26-6]
		168.7 ± 7.3	(298)	ME	[94/16]
$\text{C}_{18}\text{H}_{14}\text{CuN}_4$		160.3 ± 3	(491)	ME	[84/12]
		170 ± 3	(298)		[84/12]
	dibenzotetra-aza-annulene copper(II) complex				[41283-96-9]
$\text{C}_{18}\text{H}_{30}\text{CuO}_4$	(493–553)	99.7 ± 8.7	(523)	T	[83/29]
	<i>bis</i> (2,2-dimethylheptane-3,5-dionato)copper(II)				[15321-96-7]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{18}\text{H}_{30}\text{CuO}_4$	(344–364)	125.0 ± 1.3	(354)	TE	[95/4]
		127.8 ± 1.3	(298)		[95/4]
	(344–364)	125.1 ± 0.5	(354)	ME	[95/4]
		127.9 ± 0.5	(298)		[95/4]
		122.8 ± 1.7	(298)		[84/11]
$\text{C}_{20}\text{H}_{12}\text{CuF}_6\text{O}_4$	<i>bis</i> (2,6-dimethylheptane-3,5-dionato)copper(II)	118.0 ± 347	(298)		[17653-77-9] [84/11]
$\text{C}_{20}\text{H}_{16}\text{CuN}_2\text{O}_2$	<i>bis</i> (4,4,4-trifluoro-1-phenylbutane-1,3-dione)copper(II)	172.1 ± 3.1	(298)	ME	[14126-89-7] [98/32]
$\text{C}_{20}\text{H}_{18}\text{CuO}_4$	<i>bis</i> (8-hydroxy-2-methylquinolate)copper(II)				[14522-43-1]
	(402–419)	166.5 ± 3.4	(410)	ME	[98/8]
		172.1 ± 3.4	(298)		[98/8]
	<i>bis</i> (1-phenylbutane-1,3-dionato)copper(II)				[14128-84-8]
	(429–450)	152.2 ± 1.7	(439)	TE	[95/4]
$\text{C}_{20}\text{H}_{20}\text{CuF}_{14}\text{O}_4$		159.3 ± 1.7	(298)		[95/4]
	(429–450)	152.2 ± 1.9	(439)	ME	[95/4]
		159.3 ± 1.9	(298)		[95/4]
		160 ± 4	(298)	C	[79/21]
	<i>bis</i> (1,1,1,2,2,3,3-hetafluoro-7,7-dimethyloctane-4,6-dionato)copper(II)				[38926-19-1]
$\text{C}_{20}\text{H}_{34}\text{CuO}_4$		122.8 ± 0.7	(298)	ME	[98/32]
	<i>bis</i> (2,2,6-trimethylheptane-3,5-dionato)copper(II)				[141752-16-3]
	(346–362)	127.4 ± 0.7	(354)	ME	[95/4]
		130.2 ± 0.7	(298)		[95/4]
	(346–362)	127.8 ± 1.5	(354)	TE	[95/4]
$\text{C}_{22}\text{H}_{24}\text{CuN}_2\text{O}_2$		130.6 ± 1.5	(298)		[95/4]
		129.0 ± 1.3	(351)	ME	[95/4]
		131.7 ± 1.3	(298)		[95/4]
		126.4 ± 1.1	(298)		[84/11]
	<i>bis</i> [(4-phenylimino-2-pentanoato)]copper(II)				[15214-38-7]
$\text{C}_{22}\text{H}_{38}\text{CuO}_4$		128.1 ± 0.8	(298)	ME, TE	[90/2]
	<i>bis</i> (2,2,6,6-tetramethylheptane-3,5-dionato)copper(II)				[14040-05-2]
	(345–377)	127.6 ± 0.4	(361)	TE	[01/6]
	(335–366)	127.2 ± 1.7	(351)	TE	[01/6]
	(433–463)	120		TGA	[00/35]
	(400–430)	74.8		TGA, DTA	[96/29]
	(392–415)	100	(404)	T	[96/4]
		124.5 ± 0.8	(372)	ME	[95/4]
		129.1 ± 0.8	(298)		[95/4]
	(362–452)	124.6	(407)		[93/9]
	(418–473)	123.6	(445)		[92/18]
		105.9		GS	[90/15]
		111.6			[88/23][93/9]
		122.8 ± 6.5	(298)	C	[84/11]
		112		C	[84/11]
$\text{C}_{28}\text{H}_{16}\text{CuF}_6\text{O}_4$	<i>bis</i> (4,4,4-trifluoro-1-(2-naphthalenyl)butane-1,3-dione)copper(II)				[30983-56-3]
$\text{C}_{32}\text{H}_{16}\text{CuN}_8$		208.4 ± 4.9	(298)	ME	[98/32]
$\text{C}_{32}\text{H}_{16}\text{CuN}_8$	copper(II) α -phthalocyanine	114.0		TGA	[95/35]
$\text{C}_{32}\text{H}_{16}\text{CuN}_8$	copper(II) β -phthalocyanine				[147-14-8]
$\text{C}_{39}\text{H}_{59}\text{F}_{12}\text{O}_8\text{CuY}$	(618–713)	231.8 ± 2.1		ME	[00/30]
		211.1		TGA	[95/35]
	(657–863)	266.1			[69/23]
	(657–722)	266.1 ± 5.1		ME	[65/15][70/7]
	<i>bis</i> (hexafluoroisopropoxy)tris(2,2,6,6-tetramethylheptane-3,5-dionato)copper(II)yttrium(III)				[160364-36-3]
$\text{C}_{44}\text{H}_{28}\text{CuN}_4$	(370–410)	81.2	(390)		[96/7]
	5,10,15,20-tetraphenylporphine copper(II)	160 ± 5		GS	[14172-91-9] [00/36]
Dy					
$\text{C}_{15}\text{H}_{15}\text{Dy}$	<i>tris</i> (cyclopentadienyl)dysprosium(III)				[12088-04-9]
$\text{C}_{30}\text{H}_{30}\text{DyF}_{21}\text{O}_6$		105.0 ± 2.1			[73/32]
	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)dysprosium(III)				[18232-98-3]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{33}\text{H}_{57}\text{O}_6\text{Dy}$	(370–385)	156.5 ± 2.9		ME	[71/25]
	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)dysprosium(III)				[15522-69-7]
	(373–388)	171.5	(380)	ME	[81/21]
	(388–413)	152.7	(400)	ME	[81/21]
DyBr_3	(410–456)	133.5	(433)	BG	[69/19]
	dysprosium tribromide				[14456-48-5]
DyCl_3	(878–1151)	289 ± 6.0	(298)	TE	[99/20]
	dysprosium trichloride				[10025-74-8]
DyI_3	(924–1214)	283 ± 5.0	(298)	TE	[99/20]
	dysprosium triiodide				[15474-63-2]
Er	(889–1157)	282 ± 4.0	(298)	TE	[99/20]
$\text{C}_{15}\text{H}_{12}\text{ErF}_9\text{O}_6$	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)erbium(III)				[70332-27-3]
$\text{C}_{15}\text{H}_{15}\text{Er}$	(473–494)	79.5 ± 11.5	(484)		[96/27]
	<i>tris</i> (cyclopentadienyl)erbium(III)				[39330-74-0]
	(503–558)	97.2 ± 3.2	(530)		[96/27]
$\text{C}_{24}\text{H}_{33}\text{Er}$		97.1 ± 3.3			[73/32]
	<i>tris</i> [(1,2,3,4,5- η)-1-(1-methylethyl)-2,4-cyclopentadien-1-yl]erbium(III)				[130521-76-5]
$\text{C}_{30}\text{H}_{30}\text{ErF}_{21}\text{O}_6$	(464–502)	78.6 ± 3.0	(483)		[96/27]
	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)erbium(III)				[17978-75-5]
$\text{C}_{33}\text{H}_{57}\text{O}_6\text{Er}$	(349–362)	154.8 ± 4.2		ME	[71/25]
	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)erbium(III)				[14319-09-6]
		130.8 ± 2.2	(298)	DSC	[99/33]
	(471–505)	93.9 ± 4.6	(488)		[96/27]
	(363–418)	154.0	(390)	ME	[81/21]
	(358–381)	149.3 ± 1.7		ME	[71/25]
Eu	(410–454)	133.2	(432)	BG	[69/19]
$\text{C}_{33}\text{H}_{57}\text{O}_6\text{Eu}$	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)europlum(III)				[15522-71-1]
Fe	(363–433)	179.9	(398)	ME	[81/21]
	(373–423)	180.0		ME	[79/30]
	(430–466)	165.4	(448)	BG	[69/19]
$\text{C}_2\text{FeN}_2\text{O}_4$	dicarbonyldinitrosyl iron				[13682-74-1]
$\text{C}_4\text{FeI}_2\text{O}_4$	(272–291)	47.2	(281.5)		[87/4]
	iron tetracarbonyl diiodide				[14878-30-9]
$\text{C}_4\text{H}_{16}\text{Cl}_2\text{FeN}_8\text{S}_4$		86.0 ± 4.0	(298)		[82/20][79/25]
	<i>trans</i> -dichloro- <i>tetrakis</i> (thiourea)iron(II)				[28813-18-5]
$\text{C}_6\text{H}_5\text{FeIO}_3$	(372–405)	110 ± 20			[70/11]
	allylirontricarboxyl iodide				[12189-10-5]
$\text{C}_8\text{H}_6\text{Fe}_2\text{O}_6\text{S}_2$		84.5 ± 4.0	(298)		[82/20][79/25]
	hexacarbonylbis(methanethiolato)diiron				[14878-96-7]
		102.8	(333)	C	[95/34]
$\text{C}_9\text{Fe}_2\text{O}_9$		109.8	(298)		[95/34]
	diiron nonacarbonyl				[15321-51-4]
	(296–314)	135.3	(305)		[87/4]
$\text{C}_9\text{H}_{12}\text{FeO}$		75.3 ± 21.0	(298)		[82/20][72/21]
	<i>bis</i> (1,3-butadiene)ironcarbonyl				
$\text{C}_{10}\text{H}_{10}\text{Fe}$		76.1 ± 4.2	(298)		[82/20][76/16]
	ferrocene				[102-54-5]
		73.3 ± 0.1	(298)	C	[01/5]
		74.3 ± 0.4	(298)	ME	[95/15]
		73.2 ± 0.7	(298)	C	[95/15]
	(292–300)	72.5 ± 1.0	(296)	ME	[90/3]
		72.4 ± 1.0	(298)		[90/3]
	(294–302)	70.3 ± 1.0	(298)	ME	[89/3][90/3]
	(278–309)	72.1 ± 0.4	(294)	ME	[88/3]
		71.9 ± 0.4	(298)		[88/3]
	(348–446)	64.6	(397)		[87/4]
		75.6 ± 0.4	(298)	TE, ME, DM	[83/6]
		74.0 ± 2	(298)	TE	[81/10]
	(328–398)	70.0 ± 2		C	[80/20]
		72.6 ± 1.4	(298)	ME	[80/24]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{10}\text{H}_{10}\text{Fe}_2\text{O}_6\text{S}_2$	(385–455)	73.6 ± 0.4	(298)		[82/20][75/23]
		74.1 ± 1.7	(298)	TCM	[73/1]
		84.0 ± 2		C	[71/6]
		72.7 ± 2	(298)	ME	[69/4]
		76.6 ± 1	(298)	ME	[62/7]
		83.3		ME	[59/6]
		70.5	(406)	BG	[52/6]
					[28829-01-8]
		105.4	(340)	C	[95/34]
		112.0	(298)		[95/34]
$\text{C}_{10}\text{H}_{14}\text{FeO}_4$	<i>bis</i> (2,4-pentanedionato)iron(II) (330–368)				[14024-17-0]
		131.2 ± 8.7	(298)	ME	[90/21]
		117.6	(385)		[70/10]
$\text{C}_{11}\text{H}_8\text{FeO}_3$	cyclooctatetraeneirontricarbonyl				[12093-05-9]
$\text{C}_{12}\text{H}_{12}\text{FeO}$	acetylferrocene (329–358)	87.0 ± 4.0	(298)		[82/20][79/25]
					[1271-55-2]
$\text{C}_{12}\text{Fe}_3\text{O}_{12}$	triiron dodecacarbonyl				[81/10]
$\text{C}_{13}\text{H}_{16}\text{FeO}$	<i>bis</i> (1,3-cyclohexadiene)ironcarbonyl	115.6 ± 2.5	(298)		[17685-52-8]
		96.0 ± 21.0	(298)		[82/20][72/21]
$\text{C}_{14}\text{H}_{14}\text{FeO}_2$	1,1'-diacetylferrocene (360–400)				[34978-83-1]
		91.9 ± 2.5	(298)		[82/20][76/16]
$\text{C}_{15}\text{H}_3\text{F}_{18}\text{FeO}_6$	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)iron(III) (333–363)				[1273-94-5]
		60		TGA	[81/10]
$\text{C}_{15}\text{H}_{12}\text{F}_9\text{FeO}_6$	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)iron(III) (373–403) (378–438)				[17786-67-3]
		96		TGA	[00/35]
		104.6 ± 0.8		GS	[14526-22-8]
					[00/35]
$\text{C}_{15}\text{H}_{21}\text{FeO}_6$	<i>tris</i> (2,4-pentanedionato)iron(III) (413–443) (369–388) (338–355) (309–360)	80.3	(433)		[85/16]
		128.9	(345)		[77/25]
		87.0			[70/10]
					[60/17]
					[14024-18-1]
		112		TGA	[00/35]
		118		TGA	[97/29]
		124.6 ± 0.9	(378)	TE,ME	[96/3]
		128.6 ± 0.9	(298)		[96/3]
		114.2 ± 1.5			[92/11]
		126.4 ± 3.1	(298)	ME	[90/21]
		138.4 ± 5.2	(298)	C	[85/5]
		100	(395)	T	[81/15]
		113.6 ± 3.8			[80/30]
		99 ± 0.8			[79/21][81/15]
$\text{C}_{15}\text{H}_{30}\text{FeN}_3\text{S}_6$	<i>tris</i> (diethyldithiocarbamato)iron(III) (413–443)				[70/17]
		114.2	(385)		[70/10]
		65.3 ± 3.3	(298)		[82/20][68/13]
		U23.4		I	[64/2]
		81.6			[60/17]
$\text{C}_{17}\text{H}_{14}\text{FeO}$	benzoylferrocene (358–382)				[34768-31-5]
		65.7 ± 1.7	(246)		[70/12]
$\text{C}_{18}\text{H}_{27}\text{FeO}_6$	<i>tris</i> (3-methylpentane-2,4-dionato)iron(III) (358–382)				[1272-44-2]
		116.3 ± 6	(298)	TE,ME	[83/15]
$\text{C}_{20}\text{H}_{30}\text{Fe}$	<i>bis</i> (η^5 -pentamethylcyclopentadienyl)iron (358–382)				[13978-46-6]
		164.5	(422)		[92/29]
$\text{C}_{24}\text{H}_{12}\text{F}_9\text{FeO}_6\text{S}_3$	<i>tris</i> (1-(2-thenoyl)-4,4,4-trifluoro-1,3-butanedione)iron(III) (358–382)				[12126-50-0]
		96.8 ± 0.6	(298)	C	[01/5]
$\text{C}_{24}\text{H}_{18}\text{FeO}_2$	1,1'-dibenzoylferrocene (358–381)				[14319-78-9]
		U46.4			[60/17]
$\text{C}_{30}\text{H}_{27}\text{FeO}_6$	<i>tris</i> (benzoylacetato)iron(III) (358–381)				[12180-80-2]
		109.3 ± 6	(298)	TE,ME	[83/15]
$\text{C}_{33}\text{H}_{57}\text{FeO}_6$	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)iron(III) (413–443)				[14323-17-2]
		U45.6		I	[64/2]
		111		TGA	[14876-47-2]
					[00/35]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{45}\text{H}_{33}\text{FeO}_6$	(360–378)	128.5 ± 0.9	(369)	TE,ME	[96/3]
		129.3 ± 1.2	(298)		[96/3]
	(316–330)	136.1 ± 1.9			[92/11]
		106.7		ME	[73/18]
	<i>tris</i> (dibenzoylmethano)iron(III)				[14405-49-3]
FeBr_2		210 ± 10			[92/11]
		U31.8		I	[64/2]
	iron(II) dibromide				[7789-46-0]
	(655–833)	197.6 ± 5	(744)	TE,ME	[96/26]
		208 ± 5	(298)		[96/26]
FeCl_2	(680–720)	196 ± 8	(700)	TE	[60/23][96/26]
	(673–962)	197 ± 2	(817)	GS	[55/12][96/26]
		210 ± 6	(298)		[55/12][96/26]
	(623–718)	197 ± 4	(670)	ME	[55/12][96/26]
	iron(II) dichloride				[7758-94-3]
FeF_2	(693–866)	198.9 ± 6	(780)	TE,ME	[96/26]
		204 ± 6	(298)		[96/26]
	(694–745)	189 ± 8	(719)	TE	[60/23][96/26]
	(621–658)	186 ± 12	(640)	MS	[58/22][96/26]
		193 ± 12	(298)		[58/22][96/26]
Ga	iron(II) difluoride				[7789-28-8]
	(958–1178)	263 ± 4	(1068)	TE,ME	[96/26]
		271 ± 4	(298)		[96/26]
	(848–1142)	263 ± 3	(995)	ME	[76/22][96/26]
		270 ± 3	(298)		[76/22][96/26]
$\text{C}_3\text{H}_9\text{Ga}$	trimethyl gallium				[1445-79-0]
$\text{C}_{15}\text{H}_3\text{F}_{18}\text{GaO}_6$	(247–257)	45.2	(252)		[87/4]
	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)gallium(III)				[19648-92-1]
$\text{C}_{15}\text{H}_{12}\text{F}_9\text{GaO}_6$	(333–363)	53		TGA	[00/35]
	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)gallium(III)				[15453-83-5]
	(373–403)	75		TGA	[00/35]
	(378–433)	118.8 ± 1.7		GS	[85/16]
$\text{C}_{15}\text{H}_{21}\text{GaO}_6$	(386–401)	89.4 ± 6.7			[78/27]
	<i>tris</i> (pentane-2,4-dionato)gallium(III)				[14405-43-7]
	(413–433)	90		TGA	[00/35]
$\text{C}_{16}\text{H}_{36}\text{Ga}_4\text{S}_4$	$[\{(\text{CH}_3)_3\text{C}\}\text{Ga}(\mu^3-\text{S})]_4$				[135283-83-9]
	(367–380)	110	(373)	TGA	[97/29]
$\text{C}_{16}\text{H}_{36}\text{Ga}_4\text{Se}_4$	$[\{(\text{CH}_3)_3\text{C}\}\text{Ga}(\mu^3-\text{Se})]_4$				[13528-84-0]
	(375–388)	119	(381)	TGA	[97/29]
$\text{C}_{16}\text{H}_{36}\text{Ga}_4\text{Te}_4$	$[\{(\text{CH}_3)_3\text{C}\}\text{Ga}(\mu^3-\text{Te})]_4$				[135258-40-1]
	(391–422)	131	(406)	TGA	[97/29]
$\text{C}_{20}\text{H}_{44}\text{Ga}_4\text{S}_4$	$[(\text{C}_2\text{H}_5(\text{CH}_3)_2\text{C})\text{Ga}(\mu^3-\text{S})]_4$				[166331-96-0]
	(369–382)	124	(375)	TGA	[97/29]
$\text{C}_{20}\text{H}_{44}\text{Ga}_4\text{Se}_4$	$[(\text{C}_2\text{H}_5(\text{CH}_3)_2\text{C})\text{Ga}(\mu^3-\text{Se})]_4$				[176100-40-6]
	(395–407)	137	(375)	TGA	[97/29]
$\text{C}_{20}\text{H}_{44}\text{Ga}_4\text{Te}_4$	$[(\text{C}_2\text{H}_5(\text{CH}_3)_2\text{C})\text{Ga}(\mu^3-\text{Te})]_4$				[176100-41-7]
	(416–432)	140	(324)	TGA	[97/29]
$\text{C}_{24}\text{H}_{52}\text{Ga}_4\text{S}_4$	$[\{(\text{C}_2\text{H}_5)_2(\text{CH}_3)\text{C}\}\text{Ga}(\mu^3-\text{S})]_4$				[166331-97-1]
	(407–420)	137	(413)	TGA	[97/29]
$\text{C}_{24}\text{H}_{52}\text{Ga}_4\text{Se}_4$	$[\{(\text{C}_2\text{H}_5)_2(\text{CH}_3)\text{C}\}\text{Ga}(\mu^3-\text{Se})]_4$				[187612-49-3]
	(388–420)	147	(404)	TGA	[97/29]
$\text{C}_{24}\text{H}_{52}\text{Ga}_4\text{Te}_4$	$[\{(\text{C}_2\text{H}_5)_2(\text{CH}_3)\text{C}\}\text{Ga}(\mu^3\text{Te})]_4$				[176100-42-8]
	(432–447)	151	(439)	TGA	[97/29]
$\text{C}_{28}\text{H}_{60}\text{Ga}_4\text{S}_4$	$[\{(\text{C}_2\text{H}_5)_3\text{C}\}\text{Ga}(\mu^3-\text{S})]_4$				[187612-47-1]
	(432–444)	149	(438)	TGA	[97/29]
$\text{C}_{28}\text{H}_{60}\text{Ga}_4\text{Se}_4$	$[\{(\text{C}_2\text{H}_5)_3\text{C}\}\text{Ga}(\mu^3-\text{Se})]_4$				[187612-51-7]
	(452–464)	156	(458)	TGA	[97/29]
$\text{C}_{28}\text{H}_{60}\text{Ga}_4\text{Te}_4$	$[\{(\text{C}_2\text{H}_5)_3\text{C}\}\text{Ga}(\mu^3-\text{Te})]_4$				[187612-52-8]
	(444–456)	156	(450)	TGA	[97/29]
$\text{C}_{33}\text{H}_{57}\text{O}_6\text{Ga}$	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)gallium(III)				[34228-15-4]
	(413–443)	87		TGA	[00/35]
		102.1		ME	[73/18]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
(GaBr ₃)–(NH ₃)	gallium tribromide–ammonia complex				[54955-92-9]
		67.4 ± 1.3			[75/27]
(GaCl ₃)–(NH ₃)	gallium trichloride–ammonia complex				[50599-24-1]
		75.6 ± 1.3			[75/27]
Gd					
C ₁₀ H ₁₀ ClGd	<i>bis</i> (cyclopentadienyl)gadolinium chloride				[11087-14-2]
	(338–438)	138.5 ± 2.1		ME	[71/32]
C ₁₅ H ₁₅ Gd	<i>tris</i> (cyclopentadienyl)gadolinium				[1272-21-5]
	(513–623)	106.7 ± 2.9			[73/31]
C ₃₀ H ₃₀ F ₂₁ GdO ₆	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)gadolinium(III)				[17631-67-3]
	(362–385)	154.8 ± 0.8		ME	[71/25]
C ₃₃ H ₅₇ O ₆ Gd	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)gadolinium(III)				[14768-15-1]
		166.1 ± 3.5	(298)	DSC	[99/33]
		78.8 ± 1.5			[96/31][00/16]
	(383–418)	181.2	(400)	ME	[81/21]
		163.6		ME	[73/18]
	(420–456)	161.3	(438)	BG	[69/19]
Ge					
C ₃ H ₉ FGe	fluorotrimethylgermane				[661-370-0]
	(250–284)	40.0	(267)	SG	[87/4][61/9]
C ₈ H ₂₄ Ge ₄ O ₄	octamethyltetragermoxane				
		68.2 ± 4.2	(298)		[82/20][72/19]
C ₁₆ H ₁₂ Ge	diethynyldiphenylgermane				[1675-59-8]
		133.9		BE	[75/28]
C ₁₆ H ₁₈ Ge	1,1-diphenylgermolane				[4514-06-1]
		104.6 ± 2.8	(298)	B	[88/31]
C ₂₀ H ₁₈ Ge	triphenyl vinylgermanium				[4049-97-2]
		98.7 ± 1.6	(298)	ME,TE	[88/8]
C ₂₄ H ₂₀ Ge	tetraphenylgermane				[1048-05-1]
	(402–480)	148.6	(441)		[87/4]
		156.9 ± 4.2	(298)		[82/20][69/17]
C ₂₆ H ₂₀ Ge	triphenyl phenylethynylgermane				[4131-49-1]
		107.5 ± 1.5	(298)	ME,TE	[88/8]
C ₂₈ H ₂₈ Ge	tetrabenzylgermane				[1048-05-1]
		168.6 ± 8.4	(298)		[82/20][70/15]
C ₃₂ H ₁₆ Cl ₂ GeN ₈	germanium phthalocyanine dichloride				[19566-97-3]
		147.4 ± 12.4			[72/31]
C ₃₆ H ₃₀ Ge ₂ O	<i>bis</i> (triphenyl germanium) oxide				[2181-40-0]
		98.0 ± 1.5	(298)	ME,TE	[88/8]
C ₃₆ H ₃₀ Ge ₂	hexaphenyldigermane				[2816-39-9]
		209.2 ± 4.2	(298)		[82/20][70/15]
GeCl ₄	germanium tetrachloride				[10038-98-9]
	(187–221)	44.6 ± 0.2		MG	[64/12]
GeF ₂	germanium difluoride				[13940-63-1]
		82.8 ± 4.2	(298)	MS	[71/26]
		93.3 ± 10.5	(298)		[71/26]
GeI ₄	germanium tetraiodide				[13573-08-5]
		87.1 ± 3	(298)		[99/14]
		86.7 ± 3	(298)		[99/14]
	(323–420)	76.5 ± 5.7	(298)	TE	[87/22]
Ha					
HaCl ₅	hahnium(V) pentachloride				[146837-09-4]
		≤120	(298)		[96/25]
HaOCl ₃	hahnium(V) oxychloride				[143928-41-0]
	(298–607)	152 ± 18	(298)		[96/25]
Hf					
C ₁₀ H ₁₀ Cl ₂ Hf	<i>bis</i> (cyclopentadienyl)hafnium dichloride				[68/19][01/3]
		107.3 ± 2.4	(298)		[12116-66-4]
		110.2 ± 2.9	(298)	ME	[01/3]
	(394–447)	100.3	(420.5)		[87/4]
		106.7 ± 2.1	(298)		[82/20][76/14]
		100.4 ± 1.3			[77/23]
C ₂₀ H ₁₆ F ₁₂ HfO ₈	<i>tetrakis</i> (1,1,1-trifluoro-2,4-pentanedionato)hafnium(IV)				[17475-68-2]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{20}\text{H}_{28}\text{HfO}_8$	(383–438)	129.7 ± 3.8		GS	[85/16]
	(383–438)	124.7 ± 3.8		GS	[85/16]
	<i>tetrakis</i> (pentane-2,4-dionato)hafnium(IV)	150.6 ± 4.2			[17475-67-1] [91/23]
HfCl_4	hafnium tetrachloride				[13499-05-3]
Hg	(398–500)	97.9 ± 1.2	(499)	T	[94/22]
	(353–433)	107.9 ± 0.8			[73/30]
CH_3BrHg	methylmercuric bromide				[506-83-2]
CH_3ClHg	(258–297)	67.6 ± 1.6	(277.5)	V	[87/4][51/14]
	methylmercuric chloride				[115-09-3]
	(278–307)	64.9 ± 1.6	(298)	V	[87/4][82/20] [50/6][51/14]
CH_3HgI	methylmercuric iodide				[143-36-2]
$\text{C}_2\text{H}_5\text{BrHg}$	(263–290)	65.3 ± 1.6	(276)	V	[51/14]
	ethylmercuric bromide				[107-26-6]
	(285–303)	76.5 ± 2.9	(294)	V	[87/4][82/20] [51/10][51/14]
$\text{C}_2\text{H}_5\text{ClHg}$	ethylmercuric chloride				[107-27-7]
$\text{C}_2\text{H}_5\text{HgI}$	(283–303)	76.2 ± 2.9	(293)	V	[87/4][82/20] [51/10][51/14]
	ethylmercuric iodide				[2440-42-8]
	(286–303)	79.7 ± 2.9	(294.5)	V	[87/4][82/20] [51/10][51/14]
$\text{C}_4\text{H}_{16}\text{Cl}_2\text{HgN}_8\text{S}_4$	<i>trans</i> -dichloro- <i>tetrakis</i> (thiourea)mercury(II)				[28813-22-1]
$\text{C}_{10}\text{H}_{14}\text{Cl}_2\text{HgN}_6\text{O}_2$		101 ± 20			[70/11]
	[mercury(1-methylcytosine) $_2\text{Cl}_2$]				
	(428–443)	150.8 ± 19	(435)	ME	[84/12]
$\text{C}_{10}\text{H}_{20}\text{HgN}_2\text{S}_4$		159 ± 19	(298)		[84/12]
	<i>bis</i> (diethyldithiocarbamate) mercury complex				[14239-51-1]
	(378–403)	47.6	(390.5)		[87/4]
$\text{C}_{12}\text{H}_{12}\text{Hg}$	diphenylmercury				[587-85-9]
$\text{C}_{14}\text{H}_{14}\text{Hg}$	(314–303)	112.8 ± 0.8	(298)	ME	[58/9]
	<i>bis</i> (benzyl)mercury				[780-24-5]
	(350–390)	88.7 ± 2.1		ME,TE	[84/19]
$\text{C}_{14}\text{H}_{28}\text{HgN}_2\text{S}_4$	<i>bis</i> (dipropyldithiocarbamate)mercury(II)				[21439-56-5]
$\text{C}_{16}\text{H}_{10}\text{Hg}$		200 ± 2	(298)	DSC,E	[92/19]
	<i>bis</i> (phenylethynyl)mercury				[6077-10-7]
	(350–390)	99.2 ± 1.4		ME,TE	[84/19]
$\text{C}_{18}\text{H}_{36}\text{HgN}_2\text{S}_4$	<i>bis</i> (dibutyldithiocarbamate)mercury(II)				[21439-58-7]
$\text{C}_{18}\text{H}_{36}\text{HgN}_2\text{S}_4$		193 ± 3	(298)	DSC,E	[91/15]
	<i>bis</i> (diisobutyldithiocarbamate)mercury(II)				[79001-48-2]
		247 ± 1	(298)	DSC,E	[94/33]
Ho					
$\text{C}_{15}\text{H}_{15}\text{Ho}$	<i>tris</i> (cyclopentadienyl)holmium(III)				[1272-22-6]
$\text{C}_{33}\text{H}_{57}\text{HoO}_6$		102.1 ± 2.1			[73/32]
	(338–348)	119.7 ± 2.1		ME	[71/32][71/33]
	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)holmium(III)				[15522-73-3]
		131.0 ± 2.9		DSC	[93/26]
	(363–418)	152.7	(390)	ME	[81/21]
	(420–458)	131.4	(439)	BG	[69/19]
In					
$\text{C}_3\text{H}_9\text{In}$	trimethyl indium				[3385-78-2]
$\text{C}_{15}\text{H}_{12}\text{F}_9\text{InO}_6$		48.5 ± 2.5	(298)		[82/20][68/11]
	(328–362)	57.7	(344)		[87/4][41/4]
	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)indium(III)				[15453-87-9]
	(378–428)	112.1 ± 1.3		GS	[85/16]
	(398–478)	77.4 ± 0.6 (liq)			[78/27]
	<i>tris</i> (diethyldithiocarbamate)indium(III)				[15741-07-8]
$\text{C}_{15}\text{H}_{30}\text{InN}_3\text{S}_6$		176.7 ± 3.3	(298)	DSC,E	[00/13]
$\text{C}_{20}\text{H}_{48}\text{In}_2\text{P}_4$	<i>bis</i> [μ -[<i>bis</i> (1,1-dimethylethyl)phosphino]]tetramethyldiindium(III)				[115381-28-7]
$\text{C}_{21}\text{H}_{42}\text{InN}_3\text{S}_6$		130.0		ME	[88/20]
	<i>tris</i> (dipropyldithiocarbamate)indium(III)				[87052-01-5]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
		372.8 ± 3.4	(298)	DSC,E	[00/13]
C ₂₁ H ₄₂ InN ₃ S ₆	<i>tris</i> (diisopropylthiocarbamate)indium(III)				[85883-33-6]
		279.5 ± 3.5	(298)	DSC,E	[00/13]
C ₂₇ H ₅₄ InN ₃ S ₆	<i>tris</i> (dibutylthiocarbamate)indium(III)				[23467-56-3]
		358.3 ± 3.2	(298)	DSC,E	[00/13]
C ₂₇ H ₅₄ InN ₃ S ₆	<i>tris</i> (diisobutylthiocarbamate)indium(III)				[85129-27-7]
		182.0 ± 3.3	(298)	DSC,E	[00/13]
C ₃₃ H ₅₇ InO ₆	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)indium(III)				[34269-03-9]
		129.3		ME	[73/18]
InBr ₃	indium(III) bromide				[13465-09-3]
		147 ± 4	(298)	TE	[97/23]
InCl ₃	indium(III) chloride				[10025-82-8]
	(495–648)	152 ± 4	(570)	TE	[98/35]
		158 ± 4	(298)		[98/35]
		150.4	(710)		[94/37]
		161.1	(298)		[94/37][98/35]
	(453–572)	151.1 ± 1.2	(489)	MS	[88/30]
		155.6 ± 1.2	(298)		[88/30][98/35]
	(478–563)	161.1 ± 1.6	(524)		[88/30]
		168.5 ± 1.6	(298)		[88/30][98/35]
	(623–773)	156.3	(698)		[74/29]
		166.6	(298)		[74/29][98/35]
InI ₃	indium(III) iodide				[13510-35-5]
	(399–479)	136 ± 5.0	(298)	TE,ME	[97/22]
Ir					
C ₇ H ₇ IrO ₄	dicarbonyl-2,4-pentanedionato iridium complex				[14023-80-4]
	(286–325)	92. ± 1.3	(306)	ME	[78/20][87/4]
C ₇ H ₁₃ Cl ₂ IrO ₂	<i>bis</i> (chloroethylene)-2,4-pentanedionato iridium complex				
	(281–298)	89.5 ± 4.2	(290)	ME	[78/20][87/4]
C ₉ H ₁₅ IrO ₂	<i>bis</i> (ethylene)-2,4-pentanedionato iridium complex				[52654-27-0]
	(283–311)	82.8 ± 4.2	(297)	ME	[78/20][87/4]
C ₁₁ H ₁₉ IrO ₂	<i>bis</i> (propylene)-2,4-pentanedionato iridium complex				[66467-05-8]
	(269–304)	90 ± 1.3	(287)	ME	[78/20][87/40]
C ₁₂ O ₁₂ Ir ₄	tetrairidiumdodecacarbonyl				[11065-24-0]
		104.6 ± 20	(298)		[82/20][74/26]
C ₁₃ H ₁₉ IrO ₆	<i>bis</i> (vinyl acetate)-2,4-pentanedionato iridium complex				[66467-07-0]
	(325–344)	120.5 ± 2.9	(333)	ME	[78/20]
C ₁₃ H ₁₉ IrO ₆	<i>bis</i> (methyl acrylate)-2,4-pentanedionato iridium complex				[66467-08-1]
	(311–335)	117.2 ± 5	(323)	ME	[78/20]
C ₁₅ H ₂₁ IrO ₆	<i>tris</i> (2,4-pentanedionato)iridium(III)				[15635-87-7]
	(423–473)	129.3 ± 0.8		GS	[00/12]
	(383–433)	130.5 ± 3.4		ME	[00/12]
	(387–513)	101.6 ± 1.8		MCV	[00/12]
	(468–518)	86.6 ± 1.7		SMZG	[00/12]
		NA			[94/34]
La					
C ₁₅ H ₁₅ La	<i>tris</i> (cyclopentadienyl)lanthanum				[1272-23-7]
		114.6 ± 4.0	(298)		[82/20][74/23]
	(548–663)	102.1 ± 2.9			[73/31]
C ₃₀ H ₃₀ F ₂₁ LaO ₆	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)lanthanum(III)				[19106-89-9]
	(387–403)	145.2 ± 2.9		ME	[71/25]
C ₃₃ H ₅₇ LaO ₆	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)lanthanum(III)				[14319-13-2]
		156.0 ± 4.6		DSC	[97/3][00/16]
		116.1 ± 8.4			[96/17]
		107.9 ± 4.6			[96/31][00/16]
	(388–423)	179.5	(405)	ME	[81/21]
		164.4		ME	[73/18]
	(450–520)	143.6	(485)	BG	[69/19]
Li					
C ₂ H ₅ Li	ethyl lithium				[811-49-4]
	(298–333)	116.6	(315.5)		[87/4][62/9]
C ₄ H ₉ Li	butyl lithium				[109-72-8]
	(333–368)	109.7	(350.5)		[87/4][62/12]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
C ₁₁ H ₁₉ LiO ₂	(2,2,6,6-tetramethylheptane-3,5-dionato)lithium	174.5		ME	[73/18]
LiF	lithium fluoride				[7789-24-4]
	(1073–1121)	268.2±4.2			[59/18][58/20]
	(957–1113)	267.8±4.2			[58/21]
Lu					
C ₁₅ H ₁₅ Lu	<i>tris</i> (cyclopentadienyl)lutetium(III)	123.0±2.9			[1272-24-8]
					[73/32]
C ₁₅ H ₂₁ LuO ₆	<i>tris</i> (2,4-pentanedionato)lutetium(III)				[17966-84-6]
	(403–433)	79±13	(418)		[83/22]
C ₃₃ H ₅₇ O ₆ Lu	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)lutetium(III)				[15497-45-2]
		135.8±2.9	(298)	DSC	[99/33]
	(363–413)	154.8	(390)	ME	[81/21]
	(420–448)	134.2	(434)	BG	[69/19]
Mg					
C ₁₀ H ₁₀ Mg	<i>bis</i> (cyclopentadienyl) magnesium				[1284-72-6]
		68.2±1.3	(298)		[82/20][67/9]
					[67/17]
C ₁₀ H ₂₂ Mg	<i>bis</i> (2,2-dimethylpropyl)magnesium				[19978-31-5]
	(318–348)	160.0±2.0	(333)	ME	[83/19]
C ₁₈ H ₁₂ MgN ₂ O ₂	<i>bis</i> (8-hydroxyquinolino)lato)magnesium(II)				[14639-28-2]
		230.2±4.0	(298)	ME	[94/16]
C ₂₀ H ₁₆ MgN ₂ O ₂	<i>bis</i> (8-hydroxy-2-methylquinolino)lato)magnesium(II)				[14406-92-9]
	(533–549)	212.2±6.5	(541)	ME	[98/8]
		224.3±6.5	(298)		[98/8]
MgF ₂	magnesium fluoride				[7783-40-6]
	(1220–1450)	359.8	(1330)	MS	[62/10]
	(1273–1513)	327.3±4.3	(1400)	TE	[64/13]
		348.2±4.3	(298)		[64/13]
Mn					
C ₄ H ₁₆ Cl ₂ MnN ₈ S ₄	<i>trans</i> -dichloro- <i>tetrakis</i> (thiourea)manganese(II)				[28813-17-4]
	(382–409)	133±20			[70/11]
C ₅ BrMnO ₅	bromo(pentacarbonyl)manganese				[14516-54-2]
		58.6±8.4	(298)		[82/20][72/21]
		88.0±2.0	(298)	C	[82/21]
C ₅ ClMnO ₅	chloro(pentacarbonyl)manganese				[14100-30-2]
		58.6±8.4	(298)		[82/20][72/21]
		91±9			[82/21]
C ₅ IMnO ₅	iodo(pentacarbonyl)manganese				[14879-42-6]
		77.4±1.4	(298)	C	[82/21]
C ₆ F ₃ MnO ₅	pentacarbonyl(trifluoromethyl)manganese				[13601-14-4]
		77.8±1.0	(298)	C	[82/21]
C ₆ H ₃ MnO ₅	methyl(pentacarbonyl)manganese				[13601-24-6]
		60.3±1.0			[82/20][74/25]
	(293–403)	60.2			[58/7]
C ₇ F ₃ MnO ₆	pentacarbonyl(trifluoroacetyl)manganese				[14099-62-8]
		79±5	(298)	C	[82/21]
C ₇ H ₃ MnO ₆	acetyl(pentacarbonyl)manganese				[13963-91-2]
		80±7	(298)	C	[82/21]
C ₈ H ₅ MnO ₃	cyclopentadienyl(tricarbonyl)manganese				[12079-65-1]
		52.4±3.1			[82/20][65/12]
C ₈ H ₁₀ Cl ₂ MnN ₆ O ₂	[manganese-(cytosine) ₂ Cl ₂]				[74543-44-5]
	(433–453)	U 146±21	(443)	ME	[84/12]
C ₁₀ MnO ₁₀ Re	decacarbonylmanganeserhenium				[14693-30-2]
		109±4	(363)	C	[98/33]
		86±4	(298)		[98/33]
	(363–440)	68.6	(401)	MM	[71/20]
C ₁₀ Mn ₂ O ₁₀	decacarbonyldimanganese				[10170-69-1]
		80.3±4.2	(298)		[82/20][58/10]
		92.3±2.1	(298)	C	[82/21]
	(351–428)	80.3±2.1	(390)	MM	[71/20]
		62.8±4.2			[60/15]
C ₁₀ H ₆ Mn ₂ O ₈ S ₂	<i>bis</i> (μ-methanethiolato)octacarbonyldimanganese				[21321-38-0]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{10}\text{H}_{10}\text{Mn}$	<i>bis</i> (cyclopentadienyl)manganese complex (298–445)	114.2 ± 0.8	(340)	C	[95/18]
		72.4	(371.5)		[1271-27-8]
		75.7 ± 1.7	(298)		[87/4]
		72.4			[82/20][71/22]
$\text{C}_{10}\text{H}_{14}\text{MnO}_4$	<i>bis</i> (2,4-pentanedionato) manganese(II) (390–440)			ME	[56/11]
		139.3 ± 2.5	(298)		[14024-58-9]
		87	(343)		[90/21]
		88.7			[81/15]
$\text{C}_{11}\text{H}_5\text{MnO}_5$	phenyl(pentacarbonyl)manganese	88.7	(400)	C	[72/24]
		84.9 ± 4.4	(298)		[70/10]
$\text{C}_{12}\text{H}_5\text{MnO}_6$	benzoyl(pentacarbonyl)manganese	123 ± 3	(298)	C	[13985-77-8]
$\text{C}_{12}\text{H}_7\text{MnO}_5$	benzyl(pentacarbonyl)manganese	84.5 ± 0.7	(298)	C	[82/21]
$\text{C}_{15}\text{H}_{12}\text{F}_9\text{MnO}_6$	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato) manganese(III) (378–413)	120.5 ± 9.2		GS	[15612-92-7]
		117.3			[82/21]
		77.8			[14049-86-6]
					[82/21]
$\text{C}_{15}\text{H}_{21}\text{MnO}_6$	<i>tris</i> (2,4-pentanedionato) manganese(III) (320–380)			ME	[14526-24-0]
		124.7 ± 3.8	(298)		[85/16]
		120 ± 10	(298)		[71/17]
		99	(392)		[64/4]
		113.0	(370)		[14284-89-0]
$\text{C}_{18}\text{H}_{12}\text{MnN}_2\text{O}_2$	<i>bis</i> (8-hydroxyquinolato) manganese(II) (615–650)	77.8 ± 0.8	(298)	ME	[90/21]
		194.6 ± 10.4	(298)		[88/28]
		208.4 ± 14	(633)		[81/15]
		226 ± 14	(298)		[70/10]
$\text{C}_{20}\text{H}_{16}\text{MnN}_2\text{O}_2$	<i>bis</i> (8-hydroxy-2-methylquinolate)manganese(II) (521–541)	199.6 ± 7.2	(531)	ME	[82/20][68/12]
		211.2 ± 7.2	(298)		[14495-13-7]
					[94/16]
$\text{C}_{30}\text{H}_{27}\text{MnO}_6$	<i>tris</i> (1-phenylbutane-1,3-dionato)manganese(III)	195 ± 10	(298)	E	[84/12]
$\text{C}_{33}\text{H}_{57}\text{MnO}_6$	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)manganese(III)	140 ± 10	(298)	E	[84/12]
$\text{C}_{44}\text{H}_{28}\text{MnN}_4$	5,10,15,20-tetraphenylporphine manganese (II)	175 ± 1		UV	[14515-78-7]
MnF_2	manganese(II) fluoride	318.4 ± 8.4	(298)	ME	[98/8]
Mo					[14376-07-9]
C_6MoO_6	molybdenum hexacarbonyl (265–300) (316–423) (240–285) (343–383) (323–403) (292–308)	77.7		ME	[88/28]
		69.1	(331)		[14324-99-3]
		76.9 ± 0.9	(263)		[88/28]
		73.8 ± 1.0			[31004-82-7]
		69.7	(363)		[93/27]
		72.5			[7782-64-1]
		72.8			[64/18]
		68.2			
$\text{C}_7\text{H}_3\text{MoNO}_5$	acetonitrile molybdenum pentacarbonyl (260–279)	105.8 ± 5.6	(298)		[13939-06-5]
$\text{C}_8\text{F}_{12}\text{Mo}_2\text{O}_8$	dimolybdenum tetratrifluoroacetate (330–370)	113.6 ± 1.7	(350)	ME, TE	[00/33]
$\text{C}_8\text{H}_{12}\text{CrMoO}_8$	chromium molybdenum tetraacetate	165.0 ± 8.4			[87/4]
$\text{C}_8\text{H}_{12}\text{Mo}_2\text{O}_8$	dimolybdenum tetraacetate (400–420)	170.5 ± 7	(410)	ME, TE	[79/19][80/34]
$\text{C}_8\text{H}_{12}\text{Mo}_2\text{O}_8$	tetra- μ -acetatodimolybdenum(II)	165.0 ± 8.4	(298)		[75/22][74/24]
$\text{C}_8\text{H}_{24}\text{MoN}_4$	<i>tetrakis</i> (dimethylamino)molybdenum	72.4 ± 6	(298)	C	[60/18]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_9\text{H}_9\text{MoN}_3\text{O}_3$	<i>tris</i> (acetonitrile) molybdenum tricarbonyl (283–308)	111.3 $\pm$ 3.0 96.0 $\pm$ 10.0	(298) (298)		[15038-48-9] [80/31] [82/20][78/25]
$\text{C}_{10}\text{H}_5\text{MoNO}_5$	pyridine(pentacarbonyl)molybdenum (283–299)	102.0 $\pm$ 2.0	(291)	ME	[14324-76-6] [79/19]
$\text{C}_{10}\text{H}_8\text{MoO}_3$	cycloheptatriene(tricarbonyl)molybdenum	88.0 $\pm$ 4.0	(298)		[12125-77-8] [82/20][77/22]
$\text{C}_{10}\text{H}_{10}\text{Cl}_2\text{Mo}$	dichlorobis(η^5 -2,4-cyclopentadien-1-yl)molybdenum	100.4 $\pm$ 4.2	(298)	E	[12184-22-4] [76/20]
$\text{C}_{10}\text{H}_{10}\text{I}_2\text{Mo}$	bis(η^5 -2,4-cyclopentadien-1-yl)diiodomolybdenum	100.4 $\pm$ 4.2	(298)	E	[12184-29-1] [76/20]
$\text{C}_{10}\text{H}_{11}\text{MoNO}_5$	piperidine(pentacarbonyl)molybdenum (275–289)	121.9 $\pm$ 5.1	(282)	ME	[19456-57-6] [79/19]
$\text{C}_{10}\text{H}_{12}\text{Mo}$	bis(η^5 -2,4-cyclopentadien-1-yl)dihydromolybdenum	81.4 $\pm$ 1.0 92.5 $\pm$ 2.1		ME	[1291-40-3] [90/30] [76/20]
$\text{C}_{11}\text{H}_8\text{MoO}_4$	norbornadienemolybdenumtetracarbonyl	92.0 $\pm$ 4.0	(298)		[12146-37-1] [82/20][77/22]
$\text{C}_{12}\text{H}_{12}\text{Mo}$	dibenzene molybdenum	94.6 $\pm$ 8			[12129-68-9] [70/1][61/4]
$\text{C}_{12}\text{H}_{16}\text{Mo}$	dimethyldicyclopentadienylmolybdenum	70.4 $\pm$ 4.2	(298)		[39333-52-3] [82/20][80/37]
$\text{C}_{12}\text{H}_{36}\text{Mo}_2\text{N}_6$	<i>hexakis</i> (dimethylamine)dimolybdenum(II)	111 $\pm$ 8	(298)	C	[51956-20-8] [79/18][81/16]
$\text{C}_{14}\text{H}_{20}\text{Mo}_2\text{O}_8$	di- μ -acetatobis(pentane-2,4-dionato)dimolybdenum(II)	163.0 $\pm$ 5.0	(298)		[82/20][79/23]
$\text{C}_{16}\text{H}_{14}\text{Mo}_2\text{N}_2\text{O}_4$	di(6-methyl-2-hydroxypyridyl)diacetatodimolybdenum(II)	161.0 $\pm$ 4.0	(298)		[82/20][81/18]
$\text{C}_{18}\text{H}_{15}\text{MoN}_3\text{O}_3$	<i>tris</i> (pyridine)tricarbonylmolybdenum	142.0 $\pm$ 10.0	(298)		[15279-79-5] [82/20][78/25]
$\text{C}_{18}\text{H}_{42}\text{Mo}_2\text{O}_6$	<i>hexakis</i> (isopropoxy)dimolybdenum	113 $\pm$ 10	(298)	C	[62521-20-4] [81/16]
$\text{C}_{24}\text{H}_{24}\text{Mo}_2\text{N}_4\text{O}_4$	<i>tetrakis</i> (6-methyl-2-hydroxypyridyl)dimolybdenum(II)	157.0 $\pm$ 3.0	(298)		[67634-80-4] [82/20][81/18]
$\text{C}_{24}\text{H}_{56}\text{Mo}_2\text{O}_8$	<i>octakis</i> (isopropoxy)dimolybdenum(II)	137.0 $\pm$ 15	(298)	C	[79376-50-4] [81/16]
N					
NH_3	ammonia (177–195)	31.2			[7664-41-7] [37/4]
NH_3O	hydroxylamine (261–280)	64.2			[7803-49-8] [65/18]
NH_4Br	ammonium bromide (273–298)	U 46.5	(285)		[41/5] [12124-97-9]
		183.7 187.9	(550) (298)	I	[71/28] [55/11]
NH_4Cl	ammonium chloride				[12125-02-9]
		168.6 176.6 $\pm$ 0.4 177.0	(550) (298)	I TE	[71/28] [61/7] [55/11]
NH_4I	ammonium iodide	182.0	(298)		[12027-06-4] [55/11]
NH_4CN	ammonium cyanide				
		84.5	(298)		[55/11]
NH_4SCN	ammonium thiocyanate				[1762-95-4] [55/11]
		133.9	(298)		[10102-43-9] [29/2]
NO	nitric oxide (94–109)	16.6	(101)		[302-01-2] [41/2][01/8]
N_2H_4	hydrazine				[7803-58-9] [97/32][59/12]
		U 46.0			[6484-52-2] [62/11]
$\text{N}_2\text{H}_4\text{O}_2\text{S}$	sulfamide (347–358)	101.5 $\pm$ 1.0			
$\text{N}_2\text{H}_4\text{O}_3$	ammonium nitrate (349–438)	178.7			

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
N ₂ O	nitrous oxide	174.9	(298)		[55/11]
	(68–80)	25.1 ± 0.4	(74)	LE	[10024-97-2]
	(148–182)	24.6	(161)		[74/13]
	(103–123)	23.6	(113)	MG	[35/5]
Na					
C ₄ H ₉ ONa	sodium <i>tert</i> -butoxide	NA			[865-48-5]
C ₆ H ₁₃ ONa	sodium methyldiethylmethoxide	NA			[90/22]
C ₇ H ₁₅ ONa	sodium triethylmethoxide	NA			[67638-48-6]
C ₃₂ H ₄₀ F ₁₂ NaO ₈ Pr	sodium <i>tetrakis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)-praseodymate	155 ± 2	(453)	T	[90/22]
C ₃₂ H ₄₀ F ₁₂ NaO ₈ Tb	sodium <i>tetrakis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)-terbate	163 ± 3	(445)	T	[53535-82-3]
C ₃₂ H ₄₀ F ₁₂ NaO ₈ Y	sodium <i>tetrakis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)-yttrate	130 ± 3	(460)	T	[90/22]
		142 ± 12	(483)		[93557-93-8]
Nb					
C ₅ H ₁₅ NbO ₅	niobium pentamethoxide	80.3 ± 10.5		ME,E	[93/4]
C ₁₀ H ₁₀ Cl ₂ Nb	<i>bis</i> (cyclopentadienyl)niobium dichloride	127.4 ± 4.4	(298)	ME	[12576-88-4]
NbBr ₅	niobium(V) pentabromide	115 ± 18	(298)		[93/4]
	(298–479)	112.5	(298)		[12793-14-5]
NbCl ₅	niobium(V) pentachloride	94.0	(298)		[01/3]
	(298–479)	95 ± 16	(298)		[13478-45-8]
NbCl ₃ O	niobium(V) oxychloride	128.5	(298)		[96/25]
	(298–607)	124 ± 16	(298)		[96/25][91/22]
Nd					
C ₁₅ H ₁₅ Nd	<i>tris</i> (cyclopentadienyl)neodymium(III)	108.8 ± 3.8			[1066-25-7]
	(533–633)	134.7 ± 2.1		ME	[72/23][77/21]
C ₃₀ H ₃₀ F ₂₁ NdO ₆	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)neodymium(III)	155.2 ± 2.9		ME	[12793-14-5]
C ₃₃ H ₅₇ O ₆ Nd	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)neodymium(III)	159.1 ± 3.4	(298)	DSC	[01/3]
		92.9 ± 2.5			[13478-45-8]
	(378–423)	177.0	(400)	ME	[96/25]
	(430–491)	158.4	(460)	BG	[96/25][91/22]
Ni					
C ₄ H ₁₆ Cl ₂ N ₈ NiS ₈	<i>trans</i> -dichloro- <i>tetrakis</i> (thiourea)nickel(II)	74 ± 20			[113597-20-1]
C ₄ NiO ₄	nickel tetracarbonyl	41.6 ± 0.5			[96/25]
C ₆ H ₁₂ N ₂ NiS ₄	<i>bis</i> (dimethyldithiocarbamato) nickel complex	139.9 ± 2.1	(463)		[113597-20-1]
	(448–478)	151.9 ± 2.1		GC	[96/25]
C ₈ F ₁₈ NiO ₂ P ₂	dicarbonyl <i>bis</i> [<i>tris</i> (trifluoromethyl)phosphine]nickel	47.2	(298)		[15492-47-4]
C ₈ F ₂₈ NiP ₄	<i>tetrakis</i> [<i>bis</i> (trifluoromethyl)phosphinous fluoride]nickel	66.6	(318)		[99/33]
C ₁₀ H ₈ F ₆ NiO ₄	<i>bis</i> (1,1,1-trifluoro-2,4-pentanedionato)nickel(II)	157.7 ± 3.3		GS	[96/31][00/16]
C ₁₀ H ₁₀ Ni	<i>bis</i> (cyclopentadienyl) nickel	71.5 ± 0.6			[81/21]
		70.2 ± 1.5	(298)		[69/19]
	(353–419)	72.4 ± 1.3	(298)	MM	[28813-19-6]
					[70/11]
					[13463-39-3]
					[53/9]
					[15521-65-0]
					[87/4][78/12]
					[76/21]
					[15188-79-1]
					[66/15]
					[14917-18-1]
					[66/15]
					[14324-83-5]
					[85/16]
					[1271-28-9]
					[88/3]
					[84/24]
					[82/20][75/23]
					[67/17]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{10}\text{H}_{14}\text{NiO}_4$	<i>bis</i> (2,4-pentanedionato)nickel(II)				[3264-82-2]
	(357–420)	126.4 ± 4.4	(298)	ME	[90/21]
		108.2 ± 5	(207)	DSC	[87/13]
		108.2 ± 4.9	(480)	DSC	[87/12]
		155 ± 80	(298)	C	[85/5]
	(378–403)	127.7 ± 10	(381)	ME	[84/12]
		132 ± 10	(298)	ME	[84/12]
		69.0		I	[71/17]
		95.4	(400)		[70/10]
		69.0			[60/17]
$\text{C}_{10}\text{H}_{20}\text{N}_2\text{NiS}_4$	<i>bis</i> (diethyldithiocarbamato)nickel(II)				[14267-17-5]
	(448–478)	157.3 ± 6.0		C	[89/16]
	(440–478)	152 ± 0.8	(459)		[87/4][78/12]
	(507–650)	98.8 ± 6	(579)	DSC	[79/20]
	(443–543)	91.9 ± 6	(493)	DSC	[79/20]
		151.9 ± 2.1			[76/21]
		61.1 ± 1.7		I	[69/15]
$\text{C}_{12}\text{H}_8\text{N}_2\text{NiO}_4$	<i>bis</i> (picolinato)nickel(II)	76.6		I	[63/7]
$\text{C}_{13}\text{H}_6\text{F}_{24}\text{N}_2\text{Ni}_2\text{O}_3\text{P}_4$	μ -carbonyldicarbonyl <i>bis</i> [μ -(methylimino) <i>bis</i> [<i>bis</i> (trifluoromethyl)phosphine]]dinickel				[14402-98-3]
	(370–390)	92.3	(380)		[68/17]
$\text{C}_{14}\text{H}_{10}\text{NiO}_4$	<i>bis</i> (salicyladehydato)nickel(II)	85.4		I	[63/7]
$\text{C}_{14}\text{H}_{12}\text{N}_2\text{NiO}_2$	<i>bis</i> (salicyliminato)nickel(II)	158.2		I	[63/7]
$\text{C}_{14}\text{H}_{12}\text{N}_2\text{NiO}_4$	<i>bis</i> (salicylaldoximate)nickel(II)				[14363-30-5]
	(403–423)	106.6 ± 29	(413)		[84/12]
		112 ± 29	(298)		[84/12]
$\text{C}_{14}\text{H}_{28}\text{N}_2\text{NiS}_4$	<i>bis</i> (dipropyldithiocarbamate)nickel complex				[14516-30-4]
		147.2 ± 5.0		C	[89/16]
		126.1 ± 0.8			[78/12]
$\text{C}_{14}\text{H}_{28}\text{N}_2\text{NiS}_4$	<i>bis</i> (diisopropyldithiocarbamate) nickel complex				[15694-55-0]
		148. ± 5.0		C	[89/16]
	(442–477)	143.4 ± 2.1	(459.5)		[87/4][78/12]
$\text{C}_{16}\text{H}_{14}\text{N}_2\text{NiO}_2$	N,N- <i>bis</i> (salicylidene)ethylenediaminonickel(II)				[14167-20-5]
	(459–545)	149.8 ± 7.0		ME	[99/39]
$\text{C}_{18}\text{H}_{12}\text{N}_2\text{NiO}_2$	<i>bis</i> (8-hydroxyquinolino)nickel(II)				[14100-15-3]
		175.4 ± 6.7	(298)	ME	[94/16]
	(468–503)	129.9 ± 6	(486)	ME	[84/12]
		139 ± 6	(298)		[84/12]
$\text{C}_{18}\text{H}_{14}\text{N}_4\text{Ni}$	dibenzotetra-aza-annulene nickel(II) complex				[39251-81-5]
	(463–553)	116.6 ± 5.5	(508)	T	[83/29]
$\text{C}_{18}\text{H}_{36}\text{N}_2\text{NiS}_4$	<i>bis</i> (dibutyldithiocarbamate)nickel(II)				[13927-77-0]
		132.6 ± 5.0		C	[89/16]
$\text{C}_{18}\text{H}_{36}\text{N}_2\text{NiS}_4$	<i>bis</i> (diisobutyldithiocarbamate)nickel(II)				[28371-07-5]
		133.6 ± 5.0		C	[89/16]
	(423–443)	152.1 ± 1.3	(433)		[87/4][78/12]
$\text{C}_{20}\text{H}_{16}\text{N}_2\text{NiO}_2$	<i>bis</i> (8-hydroxy-2-methylquinolino)nickel(II)				[15200-70-1]
	(489–505)	170.9 ± 3.7	(496)	ME	[98/8]
		180.9 ± 3.7	(298)		[98/8]
$\text{C}_{22}\text{H}_{38}\text{NiO}_4$	<i>bis</i> (2,2,6,6-tetramethylheptane-3,5-dionato)nickel(II)				[14481-08-4]
	(453–493)	111		MEM	[99/40]
		145.2 ± 10		ME	[78/22]
$\text{C}_{32}\text{H}_{16}\text{N}_8\text{Ni}$	nickel(II) phthalocyanine	144.6		TGA	[14055-02-8]
$\text{C}_{44}\text{H}_{28}\text{N}_4\text{Ni}$	5,10,15,20-tetraphenylporphine nickel(II)				[95/35]
		152 ± 4		GS	[14172-92-0]
NiBr_2	nickel(II) bromide				[13462-88-9]
	(714–969)	207 ± 4.0	(841)	TE	[97/20]
		226 ± 1.0	(298)		[97/20]
NiF_2	nickel(II) fluoride				[10028-18-9]
	(1054–1106)	332.2 ± 4.1		ME	[64/19]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
Np					
(C ₁₀ H ₂ F ₁₂ NpO ₆)–(C ₃ H ₉ OP)	<i>bis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)neptunium(IV) dioxide-trimethylphosphine oxide adduct (370–418)	90 ± 3			[106617-32-7] [88/15][87/23]
C ₂₀ H ₄ F ₂₄ NpO ₈	<i>tetrakis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)neptunium(IV) (314–375)	81 ± 3			[110900-26-0] [88/15][87/23]
(C ₂₀ H ₄ F ₂₄ NpO ₈)–(C ₃ H ₉ OP)	<i>tetrakis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)neptunium(IV)-trimethylphosphine oxide adduct (353–404)	100 ± 4			[110934-11-7] [88/15][87/23]
C ₃₂ H ₄₀ F ₁₂ NpO ₈	<i>tetrakis</i> (1,1,1-trimethyl-5,5,5-hexafluoro-2,4-pentanedionato)neptunium(IV) (374–424)	106 ± 3			[99791-99-8] [88/15]
C ₄₀ H ₄₀ F ₂₈ NpO ₈	<i>tetrakis</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)neptunium(IV) (350–368)	147.7 ± 2.9	(359)	ME	[70/23]
Os					
C ₁₀ H ₁₀ Os	<i>bis</i> (cyclopentadienyl)osmium (393–506)	72.9 ± 1.4 80.5 ± 1.7 75.3	(298)		[1273-81-0] [84/31] [84/31] [59/10]
C ₁₂ O ₁₂ Os ₃	triosmium dodecacarbonyl (423–543)	104.6 ± 20 108.4	(298) (483)		[15696-40-9] [82/20][74/26] [74/35]
P					
CCl ₃ F ₂ P	difluoro(trichloromethyl)phosphine (264–283)	36.8(liq)	(274)		[1112-03-4] [87/4]
CH ₃ Cl ₂ OP	methylphosphonic dichloride	62.3			[676-97-1] [70/1][55/4]
CH ₅ O ₃ P	methylphosphonic acid	48.1 ± 4.2			[993-13-5] [55/4][70/1]
C ₂ HF ₆ OP	phosphinous acid, <i>bis</i> (trifluoromethyl) ester (233–251)	46.6	(242)		[359-65-9] [87/4][62/13]
C ₂ HF ₆ PS ₂	phosphinodithioic acid, <i>bis</i> (trifluoromethyl) ester (273–287)	41.9	(280)		[18799-75-2] [87/4]
C ₂ H ₆ ClP	chlorodimethylphosphine (233–268)	55.5	(253)		[811-62-1] [87/4][58/16]
C ₂ H ₇ O ₃ P	ethylphosphonic acid	50.6 ± 4.2			[15845-66-6] [55/4][70/1]
C ₃ F ₉ P ₃ S ₅	2,4,5- <i>tris</i> (trifluoromethyl)-1,3,2,4,5-dithiatriphospholane-2,4,5-trisulfide (363–373)	96.6	(368)		[26349-17-7] [70/21]
C ₃ N ₃ P	tricyanophosphine (293–323)	78.3 75.3 ± 2.9	(308) (298)	ME	[1116-01-4] [87/4][76/17] [95/30][76/17]
C ₃ H ₆ F ₃ PS	dimethyl(trifluoromethyl)phosphine sulfide (300–320)	68.0	(310)		[26348-92-5] [70/20]
C ₃ H ₉ OP	trimethylphosphine oxide	50.2 ± 4.2		E	[676-96-0] [82/20][60/10]
C ₃ H ₉ PS	trimethylphosphine sulfide (366–394)	70.3	(380)		[2404-55-9] [66/13]
C ₄ F ₁₂ P ₄	1,2,3,4- <i>tetrakis</i> (trifluoromethyl)tetraphosphetane (292–339)	65.3	(307)		[393-02-2] [87/4][58/17]
C ₄ H ₈ Cl ₃ O ₄ P	(1-hydroxy-2,2,2-trichloroethyl)phosphonic acid dimethyl ester (293–357)	107	(308)		[52-68-6] [87/4]
C ₄ H ₁₃ NP ₂	<i>bis</i> (dimethylphosphino)amine (300–310)	61.7	(305)		[53/14]
C ₅ H ₁₁ P	phosphorinane (250–291) (294–345)	43.3 39.9 (liq)	(276) (309)	T T	[4743-40-2] [87/4][66/16] [87/4][66/16]
C ₆ F ₁₈ NP ₃	nitrilotris[<i>bis</i> (trifluoromethyl)phosphine] (273–309)	68.4	(291)		[65/21]
C ₉ H ₁₂ N ₃ P	<i>tris</i> (2-cyanoethyl)phosphine (397–427)	105.7 ± 2	(413)	ME, TE	[4023-54-4] [81/22]
C ₁₀ H ₁₄ NO ₅ PS	ethyl parathion (O,O-diethyl-O-4-nitrophenylthiophosphate) (298–318)	100.6	(308)		[56-38-2] [79/10][83/5]
C ₁₂ H ₁₆ N ₃ O ₃ PS ₂	azinphos-ethyl				[2642-71-9]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{15}\text{H}_{30}\text{N}_3\text{PS}_6$	(326–420)	86.8	(341)		[87/4]
	phosphorus- <i>tris</i> (N,N-diethyldithiocarbamate)				[17767-20-3]
$\text{C}_{18}\text{H}_{15}\text{OP}$	triphenylphosphine oxide	143 ± 2	(298)		[87/16]
					[791-28-6]
$\text{C}_{18}\text{H}_{15}\text{P}$	triphenylphosphine	131 ± 2	(399)	ME,TE	[89/28]
		66 ± 6	(298)	B	[78/11]
					[603-35-0]
					[88/21]
$\text{C}_{18}\text{H}_{15}\text{O}_4\text{P}$	triphenylphosphate	113.2 ± 3.0	(298)		[84/13]
		109.2 ± 1.1	(350)		[82/20][60/9]
		96.2 ± 8.4	(298)		[115-86-6]
$\text{C}_{18}\text{H}_{15}\text{PS}$	triphenylphosphine sulfide	114.4 ± 2.6	(298)	B	[89/23]
					[3878-45-3]
$\text{C}_{20}\text{H}_{20}\text{NP}$	N-ethyl triphenylphosphine imine	136.8 ± 6.1	(403)	HSA	[96/9]
		142.8 ± 6.8	(298)		[96/9]
$\text{C}_{21}\text{H}_{42}\text{N}_3\text{PS}_6$	<i>tris</i> (dipropyldithiocarbamate)phosphorous	75.3 ± 8.4	(298)		[47182-04-7]
					[82/20][60/10]
$\text{C}_{28}\text{H}_{28}\text{P}$	1,4- <i>bis</i> (diphenylphosphino)butane	127.4 ± 4.2		DSC,E	[100575-31-3]
					[99/34]
$\text{C}_{27}\text{H}_{54}\text{N}_3\text{PS}_6$	<i>tris</i> (diisobutyldithiocarbamate)phosphorous	171.6 ± 2.5	(443)	B	[7688-25-7]
		138 ± 3		DSC,E	[89/28]
PH_3	phosphine				[194281-16-8]
					[97/31]
PBr_3S	thiophosphoryl bromide	17.2		MM	[7803-51-2]
					[37/3]
$\text{P}_3\text{Cl}_6\text{N}_3$	phosphonitrilic chloride (trimer)	61.2	(291)	GSM	[3931-89-3]
					[41/1]
$\text{P}_3\text{F}_6\text{N}_3$	phosphonitrilic fluoride (trimer)	76.1			[940-71-6]
					[43/6]
$\text{P}_4\text{F}_8\text{N}_4$	phosphonitrilic fluoride (tetramer)	53.6		T	[15599-91-4]
		53.6		ME	[58/15]
Pb	phosphonitrilic fluoride (tetramer)	58.2		T	[58/19]
		57.7			[14700-00-6]
$\text{C}_{10}\text{H}_2\text{F}_{12}\text{O}_4\text{Pb}$	<i>bis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)lead(II)				[58/15]
					[97/35]
$\text{C}_{10}\text{H}_{14}\text{O}_4\text{Pb}$	<i>bis</i> (2,4-pentanedionato)lead(II)	111.7 ± 1.3		GS	[15282-88-9]
					[97/35]
$\text{C}_{10}\text{H}_{20}\text{N}_2\text{PbS}_4$	<i>bis</i> (diethyldithiocarbamate)lead complex	102.4 ± 5.0		GS	[97/35]
		87.0		LE	[94/35][97/35]
$\text{C}_{12}\text{H}_{10}\text{Br}_2\text{Pb}$	diphenyl lead dibromide	129.9 ± 2.5	(463)		[17549-30-3]
					[87/4][78/12]
$\text{C}_{16}\text{H}_{20}\text{F}_6\text{O}_4\text{Pb}$	<i>bis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)lead(II)	141.8 ± 0.8	(298)	ME	[3134-29-6]
					[88/16][76/15]
$\text{C}_{18}\text{H}_{12}\text{N}_2\text{PbO}_2$	<i>bis</i> (8-hydroxyquinolino)lead(II)	117.5 ± 2.8		GS	[21751-12-2]
					[97/35]
$\text{C}_{18}\text{H}_{15}\text{BrPb}$	triphenyl lead bromide	187.1 ± 6.2	(298)	ME	[14976-96-6]
					[94/16]
$\text{C}_{18}\text{H}_{15}\text{IPb}$	triphenyl lead iodide	134.7 ± 3.3	(298)	ME	[894-06-4]
					[88/16][76/15]
$\text{C}_{20}\text{H}_{20}\text{F}_{14}\text{O}_4\text{Pb}$	<i>bis</i> (6,6,7,7,8,8,8-heptafluoro-2,2-dimethyl-3,5-octanedionato)lead(II)	130.1 ± 0.4	(298)	ME	[894-07-5]
					[88/16][76/15]
$\text{C}_{22}\text{H}_{38}\text{O}_4\text{Pb}$	<i>bis</i> (2,2,6,6-tetramethyl-3,5-heptanedionato)lead(II)	75			[21600-78-2]
					[92/9]
		117.5 ± 2.8		GS	[21319-43-7]
					[97/35]
$\text{C}_{24}\text{H}_{20}\text{Pb}$	tetraphenyl lead	66.9		LE	[94/35][97/35]
		86			[92/9]
		74.1		ME	[73/18]
					[595-89-1]
$\text{C}_{24}\text{H}_{20}\text{Pb}$	tetraphenyl lead	151	(446)		[87/4]
		159 ± 1	(298)	ME,TE	[77/19]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{32}\text{H}_{16}\text{N}_8\text{Pb}$	(298–316)	194.6 ± 6.3	(298)	E	[82/20][72/20]
		U 80.2	(298)	ME	[62/6]
		82.8	(298)		[72/22]
	lead(II) phthalocyanine				[15187-16-3]
		156.3		TGA	[95/35]
PbF_2	(542–663)	195.7			[84/7]
	lead(II) fluoride				[7783-46-2]
PbI_2		267.8			[71/27]
	lead(II) iodide				[10101-63-0]
	(598–640)	173.1 ± 1.6	(298)	ME	[96/23]
	(474–582)	167.7 ± 1.3	(298)	MS	[96/23][85/14]
	(900–1150)	182.5 ± 1.0	(298)		[96/23][79/26]
PbSe	(563–613)	165.2 ± 1.8	(298)	ME	[96/23][64/7]
	(579–650)	166.4 ± 1.0	(298)	ME	[96/23][39/1]
	(923–1073)	165.5 ± 3.0	(298)	GS	[96/23][29/1]
	lead selenide				[12069-00-0]
	(835–1047)	226 ± 1		TE	[93/20]
Pd					
$\text{C}_{10}\text{H}_2\text{F}_{12}\text{O}_4\text{Pd}$	<i>bis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)palladium(II)				[64916-48-9]
	(293–313)	84.6 ± 1.6		ME	[00/29]
$\text{C}_8\text{H}_{10}\text{Pd}$	(cyclopentadienyl)allyl palladium				[1271-03-0]
	(291–333)	49.9	(312)		[87/4][76/18]
$\text{C}_{10}\text{H}_8\text{F}_6\text{O}_4\text{Pd}$	<i>bis</i> (1,1,1-trifluoro-2,4-pentanedionato)palladium(II)				[63742-52-9]
	(332–378)	115.2 ± 1.4		ME	[00/29]
	(423–448)	105.0 ± 0.8		GS	[85/16]
$\text{C}_{10}\text{H}_{10}\text{F}_6\text{N}_2\text{O}_2\text{Pd}$	<i>bis</i> (1,1,1-trifluoro-4-imino-2-pentanonato)palladium(II)				[203874-01-5]
	(332–386)	110.9 ± 0.7		ME	[00/29]
$\text{C}_{10}\text{H}_{14}\text{O}_4\text{Pd}$	<i>bis</i> (2,4-pentanedionato)palladium(II)				[14024-61-4]
	(347–416)	130.1 ± 2.8		ME	[00/29]
	(330–394)	122.7 ± 8.6	(298)		[91/17]
	(363–393)	127.6 ± 17	(378)	ME	[84/12]
		132 ± 17	(298)		[84/12]
$\text{C}_{10}\text{H}_{20}\text{N}_2\text{PdS}_4$	<i>bis</i> (diethyldithiocarbamate)palladium(II)				[15170-78-2]
		153.1 ± 1.9			[99/37]
$\text{C}_{16}\text{H}_{20}\text{F}_6\text{O}_4\text{Pd}$	<i>bis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)palladium(II)				[77964-87-5]
	(315–357)	131.4 ± 1.9		ME	[00/29]
$\text{C}_{16}\text{H}_{20}\text{F}_6\text{O}_6\text{Pd}$	<i>bis</i> (1,1-dimethyl-1-methoxy-5,5,5-trifluoro-2,4-pentanedionato)-palladium(II)				[301198-67-4]
	(315–369)	113.8 ± 1.2		ME	[00/29]
$\text{C}_{18}\text{H}_{12}\text{N}_2\text{O}_2\text{Pd}$	<i>bis</i> (8-hydroxyquinolato)palladium(II)				[14638-30-3]
	(483–503)	158.5 ± 4	(493)	ME	[84/12]
		168 ± 4	(298)		[84/12]
$\text{C}_{20}\text{H}_{12}\text{F}_6\text{O}_4\text{Pd}$	<i>bis</i> (4,4,4-trifluoro-1-phenyl-1,3-butanedionato)palladium(II)				[85159-01-9]
	(386–452)	148.6 ± 1.4		ME	[00/29]
$\text{C}_{20}\text{H}_{18}\text{O}_4\text{Pd}$	<i>bis</i> (1-phenyl-1,3-butanedionato)palladium(II)				[15186-07-9]
	(410–471)	152.9 ± 1.4		ME	[00/29]
$\text{C}_{22}\text{H}_{38}\text{O}_4\text{Pd}$	<i>bis</i> (2,2,6,6-tetramethyl-2,4-heptanedionato)palladium(II)				[15214-66-1]
	(343–401)	125.4 ± 1.4		ME	[00/29]
$\text{C}_{44}\text{H}_{28}\text{N}_4\text{Pd}$	5,10,15,20-tetraphenylporphine palladium(II)				[76775-77-4]
		207 ± 5		GS	[00/36]
Pm					
$\text{C}_{33}\text{H}_{57}\text{O}_6\text{Pm}$	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)promethium(III)				[67840-53-3]
	(433–463)	131.8			[79/29]
Pr					
$\text{C}_{15}\text{H}_{15}\text{Pr}$	<i>tris</i> (cyclopentadienyl)praseodymium				[11077-59-1]
		125.5 ± 3.0	(298)		[82/20][74/23]
	(533–653)	113.0 ± 1.7			[73/31]
	(338–438)	131.0 ± 2.1		ME	[71/32][71/33]
$\text{C}_{33}\text{H}_{57}\text{O}_6\text{Pr}$	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)praseodymium(III)				[15492-48-5]
		104.3 ± 2.6			[96/31][00/16]
		163.0 ± 3.6		DSC	[93/26][00/16]
	(383–423)	178.7	(403)	ME	[81/21]
	(450–495)	165.4	(473)	BG	[69/19]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{32}\text{H}_{40}\text{F}_{12}\text{O}_8\text{NaPr}$	sodium <i>tetrakis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)-praseodymate (423–483)	155 ± 2	(453)	T	[93557-93-8] [93/4]
PrBr_3	praseodymium(III) bromide	288 ± 4 306 ± 4 292	(900) (298) (298)	TE	[13536-53-3] [00/27] [00/27] [00/27]
PrCl_3	praseodymium(III) chloride	317 ± 4 340 ± 4 324	(1000) (298) (298)	TE	[10361-79-2] [00/27] [00/27] [00/27]
PrI_3	praseodymium(III) iodide	263 ± 4 280 ± 4 275	(900) (298) (298)	TE	[13813-23-5] [00/27] [00/27] [00/27]
Pt					
$\text{C}_8\text{H}_{14}\text{Pt}$	cyclopentadienyltrimethylplatinum	77.8 ± 2.0	(298)		[1271-07-4] [82/20][77/21]
$\text{C}_{10}\text{H}_{14}\text{O}_4\text{Pt}$	<i>bis</i> (2,4-pentanedionato)platinum(II) (363–383)	129.4 ± 9 133 ± 9	(373) (298)	ME	[15170-57-7] [84/12] [84/12]
$\text{C}_{10}\text{H}_{20}\text{N}_2\text{PtS}_4$	<i>bis</i> (diethyldithiocarbamate)platinum(II)	157.1 ± 2.0			[15730-38-8] [99/37]
$\text{C}_{12}\text{H}_{16}\text{Pt}$	dicyclopentadienyldimethylplatinum	83.7 ± 3.5	(298)		[42613-14-9] [82/20][77/21]
Pu					
$\text{C}_{40}\text{H}_{40}\text{F}_{28}\text{O}_8\text{Pu}$	<i>tetrakis</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)plutonium(IV) (349–363)	153.5 ± 7.9	(356)	ME	[28041-99-8] [70/23]
Rb					
$\text{C}_5\text{H}_9\text{O}_2\text{Rb}$	rubidium pivalate	167.1 ± 5.6			[70205-79-7] [98/31]
Re					
$\text{C}_4\text{H}_6\text{Br}_4\text{O}_4\text{Re}_2$	<i>bis</i> (μ -acetato)tetrabromodirhenium stereoisomer				[75027-96-2]; [75081-56-0]
<i>cis</i>	(410–510)	66.6			[84/36]
<i>trans</i>	(410–510)	59.9			[84/36]
$\text{C}_4\text{H}_6\text{Cl}_4\text{O}_4\text{Re}_2$	<i>bis</i> (μ -acetato)tetrachlorodirhenium stereoisomer				[62320-69-8]; [100495-10-1]
<i>cis</i>	(450–560)	72.8			[84/36]
<i>trans</i>	(450–560)	64.7			[84/36]
$\text{C}_5\text{BrO}_5\text{Re}$	bromopentacarbonylrhenium	92.1 ± 2		C	[14220-21-4] [83/21]
$\text{C}_5\text{ClO}_5\text{Re}$	chloropentacarbonylrhenium	110.9 ± 2		C	[14099-01-5] [83/21]
$\text{C}_5\text{HO}_5\text{Re}$	rhenium hydride pentacarbonyl complex (279–369)	45.1	(324)		[16457-30-0] [87/4]
$\text{C}_6\text{H}_3\text{O}_5\text{Re}$	rhenium methylpentacarbonyl complex (315–380)	65.2 70.0 ± 2 65.3 ± 1.0 64.9	(347.5) (298) (298)	C	[14524-92-6] [87/4][60/22] [83/21] [82/20][74/25] [58/7]
$\text{C}_{10}\text{O}_{10}\text{Re}_2$	dirhenium decacarbonyl	100.9 ± 2 93.3 ± 4.2 77.6 79.5	(298) (298) (406)	MM	[14285-68-8] [83/21] [82/20][74/25] [71/20] [61/10][71/20]
Rh					
$\text{C}_7\text{H}_7\text{O}_4\text{Rh}$	dicarbonyl-2,4-pentanedionatorhodium complex (276–301)	87 ± 2.9	(289)	ME	[14874-82-9] [78/20][87/4]
$\text{C}_9\text{H}_{13}\text{Cl}_2\text{O}_2\text{Rh}$	<i>bis</i> (chloroethylene)-2,4-pentanedionatorhodium complex (275–288)	117.2 ± 7.1	(281)	ME	[12282-04-1] [78/20][87/4]
$\text{C}_9\text{H}_{15}\text{O}_2\text{Rh}$	<i>bis</i> (ethylene)-2,4-pentanedionatorhodium complex (282–301)	97.9 ± 7.1	(292)	ME	[12082-47-2] [78/20][87/4]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
C ₁₀ H ₁₄ O ₄ Rh	<i>bis</i> (2,4-pentanedionato)rhodium(II)	NA			[69047-66-1]
	(383–447)	173.2 ± 7.0	(298)		[94/34]
C ₁₁ H ₁₉ O ₂ Rh	<i>bis</i> (propylene)-2,4-pentanedionatorhodium complex				[12282-38-1]
	(270–296)	86.2 ± 1.7	(283)	ME	[78/20][87/4]
C ₁₃ H ₁₉ O ₆ Rh	<i>bis</i> (vinylacetate)-2,4-pentanedionatorhodium complex				[31724-87-5]
	(309–328)	121.3 ± 3	(319)	ME	[78/20]
C ₁₃ H ₁₉ O ₆ Rh	<i>bis</i> (methyl acrylate)-2,4-pentanedionatorhodium complex				[31724-88-6]
	(311–327)	111.7 ± 4.6	(319)	ME	[78/20]
C ₁₆ O ₁₆ Rh ₆	hexarhodiumhexadecacarbonyl				[28407-51-4]
		117.2 ± 20.0	(298)		[82/20][75/25]
Ru					
C ₁₀ H ₁₀ Ru	<i>bis</i> (cyclopentadienyl)ruthenium				[1287-13-4]
	(383–479)	76.2 ± 1.4			[84/31]
		82.7 ± 1.7	(298)		[84/31]
		77.6 ± 1.6			[67/17]
C ₁₀ H ₁₄ O ₄ Ru	<i>bis</i> (2,4-pentanedionato)ruthenium(II)				[71263-16-6]
	(398–413)	139.7 ± 2.5	(406)	ME	[93/25]
		145.1 ± 2.5	(298)	ME	[93/25]
C ₁₅ H ₃ F ₁₈ O ₆ Ru	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)ruthenium(III)				[16827-63-7]
	(299–313)	114.1 ± 1.0	(306)	ME	[01/13]
		114.5 ± 1.0	(298)	ME	[01/13]
C ₁₅ H ₁₂ F ₉ O ₆ Ru	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)ruthenium(III)				[16702-38-8]
	(350–369)	126.8 ± 1.0	(360)	ME	[01/13]
		129.9 ± 1.0	(298)	ME	[01/13]
	(383–423)	90.0 ± 3.0			[96/30]
C ₁₅ H ₂₁ O ₆ Ru	<i>tris</i> (2,4-pentanedionato)ruthenium(III)				[14284-93-6]
	(423–493)	127.0 ± 0.9			[96/30]
S					
H ₂ S	hydrogen sulfide				[7783-06-4]
	(128–142)	22.5	(135)	MG	[51/16]
	(165–188)	22.5	(176)		[36/5]
SF ₆	sulfur hexafluoride				[2551-62-4]
		23.2 ± 0.01	(186)		[94/23]
	(175–207)	23.3	(191)		[32/2]
Sb					
C ₁₅ H ₃₀ N ₃ S ₆ Sb	<i>tris</i> (N,N-diethyldithiocarbamate)antimony(III)				[22914-48-3]
		160 ± 2	(298)		[94/31]
C ₁₈ H ₁₅ Sb	triphenylantimony				[603-36-1]
		106.3 ± 8.4	(298)		[82/20][60/11]
C ₂₁ H ₄₂ N ₃ S ₆ Sb	<i>tris</i> (dipropyldithiocarbamate)antimony(III)				[226980-30-9]
		169.5 ± 6.1		DSC,E	[99/34]
C ₂₇ H ₅₄ N ₃ S ₆ Sb	<i>tris</i> (N,N-dibutyldithiocarbamate)antimony(III)				[14907-93-8]
		179 ± 3	(298)		[94/31]
C ₂₇ H ₅₄ N ₃ S ₆ Sb	<i>tris</i> (N,N-diisobutyldithiocarbamate)antimony(III)				[41594-79-0]
		157 ± 3	(298)	DSC,E	[97/31]
Sc					
C ₁₅ H ₃ F ₁₈ O ₆ Sc	<i>tris</i> (1,1,1,5,5,5-hexafluoro-2,4-pentadionato)scandium(III)				[18990-42-6]
	(333–363)	55		TGA	[00/35]
	(313–348)	60.2 ± 1.2		I	[78/26]
C ₁₅ H ₁₂ F ₉ O ₆ Sc	<i>tris</i> (1,1,1-trifluoro-2,4-pentadionato)scandium(III)				[14634-68-5]
	(373–403)	78		TGA	[00/35]
	(363–433)	117.6 ± 1.7			[85/16]
	(366–413)	53.2 ± 1.0		I	[78/26]
C ₁₅ H ₁₅ Sc	<i>tris</i> (cyclopentadienyl)scandium				[1298-54-0]
		97.1 ± 3.5	(298)		[82/20][74/23]
C ₁₅ H ₂₁ O ₆ Sc	<i>tris</i> (2,4-pentanedionato)scandium(III)				[14284-94-7]
	(413–443)	95		TGA	[00/35]
	(393–453)	58.2 ± 0.8		I	[78/26]
		99.6 ± 0.8	(298)	HSA	[70/9][70/17]
C ₃₃ H ₅₇ O ₆ Sc	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)scandium(III)				[15492-49-6]
	(413–443)	90		TGA	[00/35]
		79.6 ± 2.4			[97/3]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
Se					
CSe ₂	carbon diselenide (218–229)	46.3	(224)		[506-80-9] [87/4][66/12]
C ₄ H ₄ N ₂ O ₂ Se	selenobarbituric acid (449–486)	141 ± 4.0	(466)	TE	[92754-59-1] [99/6]
C ₄ H ₄ Se	selenophene (208–243)	47.1	(225)		[288-05-1] [51/17]
C ₈ H ₆ N ₂ Se	4-phenyl-1,2,3-selenadiazole (275–343)	91.2 ± 1.7 94.1 ± 0.8	(309) (298)	ME GS	[25660-64-4] [74/5] [73/25]
	(327–345)	90.7	(336)		[87/4]
C ₁₂ H ₁₂ Se ₂	diphenyl diselenide (302–324)	116.7 ± 2.5	(313)	ME	[1666-13-3] [80/33]
C ₁₄ H ₁₄ Se ₂	dibenzyl diselenide (291–330)	130.5		ME	[1482-82-2] [74/5][73/25]
SeF ₆	selenium hexafluoride (194–226)	24.96 ± 0.04 23.5	(205) (210)	C	[7783-79-1] [96/22] [32/2]
Si					
(CH ₃ Cl ₃ Si)–2(C ₆ H ₁₅ N ₃)	<i>bis</i> -1,3,5-trimethyl-1,3,5-triazacyclohexane-methyltrichlorosilane (298–354)	74.0 ± 2.8			[84/5]
CH ₃ NSi	isocyano silane (253–304)	48.8	(279)		[18081-38-4] [87/4][56/16]
C ₂ H ₃ F ₃ O ₂ Si	silyl trifluoroacetate (273–293)	30.7 (liq)	(283)		[6876-44-4] [87/4][67/20]
C ₂ H ₉ NSi	dimethylaminosilane (228–264)	58.8	(246)		[2875-98-1] [87/4][54/11]
C ₆ H ₁₈ O ₃ Si ₃	hexamethylcyclotrisiloxane (297–335)	55.2 ± 0.4	(316)		[541-05-9] [53/11]
C ₇ H ₁₅ NO ₃ Si	1-methyl-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 82 ± 0.8				[2288-13-3] [89/12]
C ₈ H ₁₅ NO ₃ Si	1-ethenyl-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 85 ± 0.8				[2097-18-9] [89/12]
C ₈ H ₁₇ NO ₃ Si	1-ethyl-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 81 ± 0.9				[2097-16-7] [89/12]
C ₈ H ₁₇ NO ₃ Si	1,7-dimethyl-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 92 ± 0.8				[18225-19-9] [89/12]
C ₈ H ₁₇ NO ₄ Si	1-ethoxy-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 81 ± 0.8				[3463-21-6] [89/12]
C ₈ H ₂₄ O ₄ Si ₄	octamethylcyclotetrasiloxane	64 ± 2		B	[556-67-2] [53/11][60/1] [70/1]
C ₈ H ₂₄ O ₁₂ Si ₈	octamethyldecaaoxooctasilicon (463–563)	110.5	(513)		[57348-79-5] [87/4][75/29]
C ₉ H ₁₉ NO ₃ Si	1-propyl-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 84 ± 0.8				[26053-77-0] [89/12]
C ₉ H ₁₉ NO ₃ Si	1-(1-methylethyl)-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 92 ± 0.9				[2097-17-8] [89/12]
C ₉ H ₁₉ NO ₃ Si	1,3,7-trimethyl-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 101 ± 0.8				[56492-01-4] [89/12]
C ₁₀ H ₂₁ NO ₃ Si	1,3,7,10-tetramethyl-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 115 ± 0.9				[71229-51-1] [89/12]
2(C ₆ H ₁₅ N ₃) · Cl ₄ Si	<i>bis</i> -1,3,5-trimethyl-1,3,5-triazacyclohexane · tetrachlorosilane (298–354)	76.1 ± 4.6			[84/5]
C ₁₂ H ₃₆ Si ₅	<i>tetrakis</i> (trimethylsilyl)silane	83.7 ± 20.9	(298)		[4098-98-0] [82/20][72/18]
C ₁₃ H ₁₉ NO ₄ Si	1-(2-phenoxy)methyl)-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 108 ± 0.8				[63071-93-2] [89/12]
C ₁₄ H ₁₉ NO ₅ Si	2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane-1-methanol benzoate ester 109 ± 0.9				[79791-55-2] [89/12]
C ₁₄ H ₂₁ NO ₃ Si	1-(2-phenylethyl)-2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undecane 108 ± 0.9				[63330-92-7] [89/12]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
C ₁₅ H ₂₁ NO ₅ Si	4-methylbenzoic acid 2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undec-1-ylmethyl ester	123 ± 0.9			[100446-65-9]
C ₁₅ H ₂₁ NO ₆ Si	4-methoxybenzoic acid 2,8,9-trioxa-5-aza-1-silabicyclo[3.3.3]undec-1-ylmethyl ester	143 ± 0.9			[89/12] [94697-86-6]
C ₂₀ H ₂₀ OSi	ethoxytriphenylsilane	142.7 ± 1.0			[1516-80-9] [88/27]
C ₂₄ H ₂₀ Si	tetraphenylsilane (428–484)	51.2 156.9 ± 1.7 149.4 ± 1.7	(456) (298) (298)	ME, TE MG	[1048-08-4] [87/4] [78/24] [74/12] [73/14] [72/22][86/17]
C ₂₄ H ₂₀ O ₄ Si	tetraphenoxysilane	51.0 124.7 ± 1.2	(298)		[1174-72-7] [88/27]
C ₃₂ H ₁₆ Cl ₂ N ₈ Si	silicon phthalocyanine dichloride	115.3 ± 11.5			[19333-10-9] [72/31]
C ₃₆ H ₃₀ Si ₂	hexaphenyldisilane	209.2 ± 2.1	(298)	ME, TE	[1450-23-3] [74/12]
SiCl ₄	silicon tetrachloride (175–204)	43.3 ± 0.1		MG	[10026-04-7] [64/12]
SiF ₄	silicon tetrafluoride (148–183)	25.8			[7783-61-1] [30/4]
F ₃ H ₃ Si ₂	1,1,1-trifluorodisilane (195–209)	39.2	(202)		[15195-26-3] [72/29]
Sm					
C ₁₅ H ₁₅ Sm	<i>tris</i> (cyclopentadienyl)samarium(III) (513–633)	109.6 ± 1.7			[1298-55-1] [73/31]
C ₁₅ H ₂₁ O ₆ Sm	<i>tris</i> (2,4-pentanedionato)samarium(III) (293–413)	U 20 ± 2			[14589-42-5] [85/18]
C ₃₀ H ₃₀ F ₂₁ O ₆ Sm	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)samarium(III) (379–394)	158.6 ± 1.7		ME	[17631-69-5] [71/25]
C ₃₃ H ₅₇ O ₆ Sm	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)samarium(III) (378–418) (430–468)	149.7 ± 3.3 180.7 150.6	(298) (398) (447)	DSC ME BG	[15492-50-9] [99/33] [81/21] [69/19]
Sn					
C ₁₆ H ₁₈ Sn	1,1-diphenylstannolane	106.8 ± 5.5	(298)	B	[53561-93-6] [88/11]
C ₁₇ H ₂₀ Sn	hexahydro-1,1-diphenylstannin	75.0 ± 1.5 (liq)	(298)	ME	[19814-46-1] [88/11]
C ₂₀ H ₁₈ Sn	triphenyl vinyl tin	114.1			[2117-48-8] [85/12]
C ₂₄ H ₂₀ Sn	tetraphenyl tin (393–461)	151.7 161.1 ± 4.2 152.5 ± 0.6 151.8 ± 1.1	(427) (298)	TE ME	[595-90-4] [87/4] [82/20][69/18] [69/16] [69/16] [72/22][86/17]
C ₂₇ H ₂₀ Sn	triphenyl phenylethynyl tin (298–316)	U 66.0 ± 21.2	(298)	ME	[62/6][70/13] [1247-08-1] [85/12]
C ₃₂ H ₁₆ Cl ₂ N ₈ Sn	tin(IV) phthalocyanine dichloride	137.6			[18253-54-8] [70/7]
C ₃₂ H ₁₆ N ₈ Sn	tin(II) phthalocyanine	218.4 ± 17.6		ME	[15304-57-1] [70/7]
C ₃₆ H ₃₀ Sn ₂	hexaphenyl ditin	123.4 ± 10.0		ME	[1064-10-4] [69/16]
C ₄₄ H ₂₆ N ₈ Sn	diphenyl tin(IV) phthalocyanine	188.3 ± 4.2	(298)	ME, TE	[219130-47-0] [70/7]
SnBr ₄	tin(IV) bromide (257–299)	174.9 ± 18.8 62.4	(278)	ME	[7789-67-5] [41/7]
SnI ₄	tin(IV) iodide				[7790-47-8]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
Sr	(366–414)	75.6	(390)		[41/7]
SrCl ₂	strontium chloride	328.9 ± 4.8	(298)	LE	[10476-85-4] [65/17]
Ta					
C ₅ H ₁₅ O ₅ Ta	tantalum pentamethoxide	88.3 ± 13.4		ME,E	[865-35-0] [72/23][77/21]
TaBr ₅	tantalum(V) pentabromide	127 ± 18	(298)		[13451-11-1] [96/25]
		121.9	(298)		[96/25][91/22]
TaI ₅	tantalum(V) pentaiodide	120.9			[14693-81-3] [78/30]
Tb					
C ₁₅ H ₁₅ Tb	<i>tris</i> (cyclopentadienyl)terbium(III)	103.8 ± 1.7			[1272-25-9] [73/32]
C ₃₂ H ₄₀ F ₁₂ NaO ₈ Tb	sodium <i>tetrakis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)-terbate	163 ± 3	(445)	T	[12576-88-4] [93/4]
C ₃₃ H ₅₇ O ₆ Tb	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)terbium(III)	138.4 ± 2.6	(298)	DSC	[15492-51-0] [99/33]
	(373–420)	173.6	(396)	ME	[81/21]
	(420–433)	151.0	(426)	ME	[81/21]
	(420–454)	141.5	(437)	BG	[69/19]
Te					
TeCl ₄	tellurium tetrachloride	105 ± 2	(298)		[10026-07-0] [94/32]
TeF ₆	tellurium hexafluoride	25.6	(214)		[7783-80-4] [32/2]
Th					
C ₂₀ H ₁₆ F ₁₂ O ₈ Th	<i>tetrakis</i> (1,1,1-trifluoropentane-2,4-dionato)thorium(IV)	154.6	(298)	GS,HA	[17500-72-0] [86/9]
C ₄₀ H ₄₀ F ₂₈ O ₈ Th	<i>tetrakis</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)thorium (IV)	151.2	(298)	GS,HA	[23841-30-7] [86/9]
α form		130.6	(298)	GS,HA	[86/9]
β form		138.5 ± 3.3	(355)	ME	[70/23]
C ₄₄ H ₇₆ O ₈ Th	<i>tetrakis</i> (2,2,6,6-tetramethylheptane-3,5-dionato)thorium(IV)	152.3 ± 3.3	(400)	ME	[70/23]
Ti					
C ₂ H ₃ Cl ₄ NTi	titanium trichloride–acetonitrile (1:1 complex)	29.4 ± 5.0		T	[13682-81-0] [70/18]
C ₂ H ₃ Cl ₄ NTi	titanium trichloride–acetonitrile (1:2 complex)	41.0 ± 5.0		T	[15227-64-2] [70/18]
C ₄ H ₈ Cl ₄ OTi	titanium trichloride–tetrahydrofuran (1:1 complex)	33.5 ± 5.0		T	[15005-09-1] [70/18]
C ₄ H ₈ Cl ₄ OTi	titanium trichloride–tetrahydrofuran (1:2 complex)	49.1 ± 5.0		T	[31011-57-1] [70/18]
C ₄ H ₈ Cl ₄ STi	titanium trichloride–tetrahydrothiophene (1:1 complex)	29.7 ± 5.0		T	[14281-72-2] [70/18]
C ₄ H ₈ Cl ₄ STi	titanium trichloride–tetrahydrothiophene (1:2 complex)	43.3 ± 5.0		T	[16893-00-8] [70/18]
C ₅ H ₅ Cl ₃ Ti	cyclopentadienyltitanium trichloride	89.8	(379)		[1270-98-0] [87/4]
	(354–404)	104.6 ± 8.4	(298)		[82/20][77/21] [77/23]
C ₅ H ₁₀ Cl ₄ OTi	titanium trichloride–tetrahydropyran (1:1 complex)	89.1 ± 0.8		T	[22538-12-1] [70/18]
C ₅ H ₁₀ Cl ₄ OTi	titanium trichloride–tetrahydropyran (1:2 complex)	73.0 ± 5.0		T	[31011-56-0] [70/18]
C ₈ H ₈ Cl ₄ OTi	titanium trichloride–acetophenone (1:1 complex)	39.1 ± 5.0		T	[31011-60-6] [70/18]
C ₈ H ₈ Cl ₄ OTi	titanium trichloride–acetophenone (1:2 complex)	66.4 ± 5.0		T	[31011-61-7] [70/18]
C ₁₃ H ₁₀ Cl ₄ OTi	titanium trichloride–benzophenone (1:1 complex)				[23368-15-2]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
C ₁₃ H ₁₀ Cl ₄ OTi	titanium trichloride–benzophenone (1:2 complex)	59.6 ± 5.0		T	[70/18]
		68.8 ± 5.0		T	[31011-63-9]
C ₁₀ H ₁₀ Ti	<i>bis</i> (cyclopentadienyl)titanium				[70/18]
		58.5 ± 8.0	(298)		[1271-29-0]
C ₁₀ H ₁₀ Cl ₂ Ti	<i>bis</i> (cyclopentadienyl)titanium dichloride				[82/20][71/22]
		124.4 ± 2.9	(298)	ME	[1271-19-8]
	(418–533)	124.4	(475.5)		[01/3]
		118.8 ± 2.1	(298)		[87/4]
		111.7 ± 1.7			[82/20][77/21]
		96.2			[77/23]
		102 ± 13	(298)		[69/20]
C ₁₂ H ₁₀ O ₂ Ti	<i>bis</i> (cyclopentadienyl)dicarbonyl titanium				[68/19][01/3]
		84.2 ± 3.5	(298)	ME	[12129-51-0]
C ₁₂ H ₁₆ Ti	<i>bis</i> (cyclopentadienyl)dimethyltitanium				[87/18]
		79.5 ± 8.4	(298)		[1271-66-5]
C ₁₄ H ₁₀ F ₆ O ₄ Ti	<i>bis</i> (cyclopentadienyl)titanium <i>bis</i> (trifluoroacetate)				[82/20][77/21]
		108.0 ± 8.0	(298)		[1282-45-7]
C ₂₂ H ₂₀ Ti	<i>bis</i> (cyclopentadienyl)diphenyltitanium				[82/20][81/17]
		88 ± 8			[1273-09-2]
C ₂₄ H ₂₀ O ₄ Ti	<i>bis</i> (benzoato) <i>bis</i> (η^5 -2,4-cyclopentadien-1-yl)titanium				[82/22]
		112 ± 8			[12156-48-8]
C ₂₄ H ₂₄ Ti	<i>bis</i> (cyclopentadienyl)dibenzyltitanium				[81/17]
		83.7 ± 8.4	(298)		[See Note]
	[Note: There is no reference to [77/21] in <i>Chemical Abstracts</i> under the given chemical name. Rather, <i>Chemical Abstracts</i> lists the paper under <i>bis</i> (cyclopentadienyl)diphenyltitanium.]				[82/20][77/21]
C ₃₀ H ₂₈ Fe ₂ Ti	<i>bis</i> (cyclopentadienyl)diferrocenyl titanium				[65274-19-3]
		150 ± 15			[82/22]
TiBr ₄	titanium(IV) tetrabromide				[7789-68-6]
	(283–306)	62.4	294		[41/7]
TiF ₃	titanium(III) fluoride				[13470-08-1]
	(759–865)	237.2 ± 1.7	(810)		[67/13]
Tl					
C ₃ H ₉ Tl	trimethyl thallium				[3003-15-4]
	(258–304)	57.3	(285)	CATH	[65/13][87/4]
TlF	thallium(I) fluoride				[7789-27-7]
		142.7	(298)		[67/14]
Tm					
C ₁₅ H ₁₅ Tm	<i>tris</i> (cyclopentadienyl)thulium				[1272-26-0]
		111.3 ± 3.5	(298)		[82/20][74/23]
		98.7 ± 1.7			[73/32]
	(338–438)	109.2 ± 2.1		ME	[71/32][71/33]
C ₃₃ H ₅₇ O ₆ Tm	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)thulium(III)				[15631-58-0]
		131.3 ± 2.9	(298)	DSC	[99/33]
	(363–418)	156.1	(390)	ME	[81/21]
	(410–446)	131.4	(428)	BG	[69/19]
U					
C ₆ H ₁₈ O ₆ U	uranium hexamethoxide				[69644-82-2]
		102.9 ± 8.4			[91/23]
C ₁₀ H ₂ F ₁₂ O ₆ U	<i>bis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)uranium dioxide complex				[67316-66-9]
	(370–425)	147	(397.5)		[87/4]
	(370–425)	147 ± 4			[78/28]
C ₁₅ H ₁₅ ClU	<i>tris</i> (cyclopentadienyl)uranium chloride				[11087-14-2]
	(338–348)	115.9 ± 2.1		ME	[71/32][71/33]
C ₁₆ H ₁₆ U	<i>bis</i> (cyclooctatetraene)uranium				
	(400–500)	107.9 ± 3.3			[79/31][77/26]
		114.2 ± 4.8	(298)		[79/31][77/26]
C ₂₀ H ₂₀ F ₃₀ O ₁₀ U ₂	<i>bis</i> [<i>pentakis</i> (trifluoroethoxy)]diuranium				[137220-74-7]
		NA			[91/19]
C ₂₀ H ₂₂ Cl ₂ F ₁₂ O ₆ U	<i>bis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)dichlorouranium- <i>bis</i> (tetrahydropyran)				[136211-24-0]
	(316–387)	79.1	(352)	T	[91/16]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{20}\text{H}_{28}\text{O}_8\text{U}$	<i>tetrakis</i> (pentane-2,4-dionato)uranium(IV)	148.1 ± 4.6			[65137-03-3]
$\text{C}_{22}\text{H}_{38}\text{O}_6\text{U}$	<i>bis</i> (2,2,6,6-tetramethylheptane-3,5-dionato)dioxouranium (370–412)	151.6 ± 1.9 156.9 ± 1.9 126 ± 9	(404) (298)	ME	[91/23] [50707-86-9] [93/2] [93/2] [78/28]
$\text{C}_{40}\text{H}_{40}\text{F}_{28}\text{O}_8\text{U}$	<i>tetrakis</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione) uranium(IV)				[23797-50-4]
		68.2		BG	[77/27]
		64.3 ± 3.2		C	[77/27]
$\text{C}_{40}\text{H}_{68}\text{O}_{12}\text{U}$	(343–367) <i>tetrakis</i> (2,6-dimethyl-2-methoxy-3,5-heptanedionato)uranium(IV)	143.5 ± 1.3	(355)	ME	[70/23]
$\text{C}_{44}\text{H}_{76}\text{O}_8\text{U}$	(344–377) <i>tetrakis</i> (2,2,6,6-tetramethyl-3,5-heptanedionato)uranium(IV)	121.7 ± 18	(350)		[133952-93-9] [91/11]
	(372–478)	152.2 ± 3.3	(425)	ME	[77/29]
	(392–409)	149.0 ± 1.3	(400)	ME	[70/23]
$\text{C}_{44}\text{H}_{76}\text{O}_{12}\text{U}$	<i>tetrakis</i> (2,6,6-trimethyl-2-methoxy-3,5-heptanedionato)uranium(IV) (387–428)	160.7 ± 6.3	(408)		[133952-92-8] [91/11]
V					
$\text{C}_{10}\text{H}_8\text{F}_6\text{O}_5\text{V}$	<i>bis</i> (1,1,1-trifluoro-2,4-pentanedionato)oxovanadium(IV) (423–473)	119.7 ± 0.8			[52081-94-4] [85/16]
$\text{C}_{10}\text{H}_{10}\text{Cl}_2\text{V}$	<i>bis</i> (cyclopentadienyl)vanadium dichloride				[12083-48-6]
		140.1 ± 7.4	(298)	ME	[01/3]
$\text{C}_{10}\text{H}_{10}\text{V}$	<i>bis</i> (cyclopentadienyl)vanadium (323–338)	57.4	(330.5)		[1277-47-0] [87/4]
		58.6 ± 4.2	(298)		[82/20][71/22]
$\text{C}_{10}\text{H}_{14}\text{O}_5\text{V}$	<i>bis</i> (2,4-pentanedionato)oxovanadium(IV)				[3153-26-2]
		140.7 ± 4.0	(493)	DSC	[87/14]
	(418–443)	91.5	(430.5)		[87/4]
		140.4 ± 1.1	(298)	C	[86/13]
$\text{C}_{10}\text{H}_{17}\text{NO}_5\text{V}$	amine <i>bis</i> (pentane-2,4-dionato)oxovanadium	29.0	(370)	DSC	[122343-53-7] [89/22]
$\text{C}_{12}\text{H}_{12}\text{V}$	<i>bis</i> (benzene)vanadium				[12129-72-5]
		70 ± 10			[82/20]
$\text{C}_{14}\text{H}_{16}\text{V}$	benzene(ethylbenzene)vanadium (453–483)	69.5	(468)		[36955-47-2] [72/30]
$\text{C}_{15}\text{H}_{12}\text{F}_9\text{O}_6\text{V}$	<i>tris</i> (1,1,1-trifluoro-2,4-pentanedionato)vanadium(III) (383–433)	118.4 ± 2.1			[15695-88-2] [85/16]
$\text{C}_{15}\text{H}_{18}\text{V}$	benzene(isopropylbenzene)vanadium (453–483)	83.7	(468)		[36955-49-4] [72/30]
$\text{C}_{15}\text{H}_{18}\text{BrNO}_5\text{V}$	3-bromopyridine <i>bis</i> (acetylacetonato)oxovanadium				[24263-16-9]
		59.4	(402)	DSC	[89/22]
$\text{C}_{15}\text{H}_{19}\text{NO}_5\text{V}$	pyridine <i>bis</i> (acetylacetonato)oxovanadium	47.8	(404)	DSC	[24263-31-8] [89/22]
$\text{C}_{15}\text{H}_{21}\text{O}_6\text{V}$	<i>tris</i> (2,4-pentanedionato)vanadium(III)				[13476-99-8]
		102.9 ± 0.8	(298)	HSA	[70/9][70/17]
$\text{C}_{16}\text{H}_{18}\text{N}_2\text{O}_5\text{V}$	3-cyanopyridine <i>bis</i> (acetylacetonato)oxovanadium	79.0	(391)	DSC	[24263-13-6] [89/22]
$\text{C}_{16}\text{H}_{18}\text{N}_2\text{O}_5\text{V}$	4-cyanopyridine <i>bis</i> (acetylacetonato)oxovanadium	75.6	(399)	DSC	[24263-14-7] [89/22]
$\text{C}_{16}\text{H}_{21}\text{NO}_5\text{V}$	4-methylpyridine <i>bis</i> (acetylacetonato)oxovanadium	56.9	(421)	DSC	[24263-33-0] [89/22]
$\text{C}_{16}\text{H}_{20}\text{V}$	<i>bis</i> (ethylbenzene)vanadium (453–483)	72.0	(468)		[36955-48-3] [72/30]
$\text{C}_{18}\text{H}_{24}\text{V}$	<i>bis</i> (isopropylbenzene)vanadium (453–483)	86.2	(468)		[36472-53-4] [72/30]
W					
$\text{C}_6\text{O}_6\text{W}$	tungsten hexacarbonyl (265–288)	77.7	(276)	TE	[14040-11-0] [95/36]
	(338–423)	74.9 ± 1.3			[93/28]
	(333–433)	74.4	(348)		[87/4]
	(250–292)	78.9 ± 1.1	(271)	ME	[80/34][79/19]
		73.2	(298)	C	[75/20]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
		76.5 ± 1.3			[75/22]
	(339–410)	69.7			[52/7]
		74.1			[35/2]
C ₇ H ₃ NO ₅ W	acetonitrile tungsten pentacarbonyl (271–303)	98.1 ± 2.0	(298)		[15096-68-1] [80/31]
C ₈ H ₄ N ₂ O ₅ W	pyrazole(pentacarbonyl)tungsten (287–327)	112.5 ± 2.4	(307)	ME	[39017-11-3] [79/19]
C ₈ H ₆ N ₂ O ₄ W	bis(acetonitrile)tetracarbonyltungsten (294–313)	131.0 ± 6.0	(298)		[16800-45-6] [80/31]
C ₈ H ₉ NO ₅ W	trimethylamine(pentacarbonyl)tungsten	89.1 ± 2.1			[15228-32-7] [79/19][80/32] [80/34]
C ₈ H ₉ O ₅ PW	trimethylphosphine(pentacarbonyl)tungsten (283–327)	93.8 ± 1.5	(305)	ME	[26555-11-3] [80/34]
C ₉ H ₄ N ₂ O ₅ W	pyrazine(pentacarbonyl)tungsten (287–321)	108.4 ± 1.3	(304)	ME	[65761-19-5] [79/19]
C ₉ H ₄ N ₂ O ₅ W	pyridazine(pentacarbonyl)tungsten	106.4 ± 2.5			[65761-20-8] [79/19][80/32] [80/34]
C ₉ H ₉ N ₃ O ₃ W	tris(acetonitrile) tungsten tricarbonyl (308–333)	103.4 ± 6.0	(298)		[16800-47-8] [80/31]
C ₁₀ H ₅ NO ₅ W	pyridine(pentacarbonyl)tungsten (285–313)	109.7 ± 2.7	(299)	ME	[14586-49-3] [79/19]
C ₁₀ H ₈ O ₃ W	cycloheptatrienetungstentricarbonyl	92.0	(298)	C	[12128-81-3] [77/22][82/20]
C ₁₀ H ₁₀ Cl ₂ W	dichlorobis(η^5 -2,4-cyclopentadien-1-yl)tungsten	120.7 ± 8.6 104.6 ± 4.2	(298)	ME	[12184-26-8] [01/3] [76/20]
C ₁₀ H ₁₀ I ₂ W	bis(η^5 -2,4-cyclopentadien-1-yl)diiodotungsten	104.6 ± 4.2			[12184-31-5] [76/20]
C ₁₀ H ₁₁ NO ₅ W	piperidine(pentacarbonyl)tungsten (289–327)	106.4 ± 1.0	(308)	ME	[31082-68-5] [79/19]
C ₁₀ H ₁₂ W	dicyclopentadienyltungsten dihydride (313–323)	84.6 ± 1.6 96.2 ± 2.1	(298)	ME	[1271-33-6] [90/30] [82/20][79/24] [76/20]
C ₁₂ H ₁₂ W	dibenzene tungsten	106	(298)	ME	[12089-23-5] [74/37]
C ₁₂ H ₁₆ W	bis(η^5 -2,4-cyclopentadien-1-yl)dimethyltungsten	74.6 ± 4.2			[39333-53-4] [80/37]
C ₁₂ H ₃₆ N ₆ W	hexakis(dimethylamino)tungsten	89.1 ± 7	(298)	C	[68941-84-4] [79/18]
C ₁₂ H ₃₆ N ₆ W ₂	hexakis(dimethylamino)ditungsten	113.3 ± 6	(298)	C	[54935-70-5] [79/18]
C ₂₃ H ₁₅ O ₅ PW	triphenylphosphine(pentacarbonyl)tungsten (340–364)	162.2 ± 8.3	(352)	ME	[26555-11-3] [80/34]
C ₂₃ H ₁₅ O ₈ PW	triphenylphosphite(pentacarbonyl)tungsten (308–348)	120.2 ± 6.6	(328)	ME	[23306-41-4] [80/34]
WCl ₄ O	tungsten(IV) oxychloride (396–447)	63.7 ± 1.7	(421)	56	[13520-78-0] [83/30]
Y					
C ₁₅ H ₃ F ₁₈ O ₆ Y	tris(1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)yttrium(III) (310–365)	91.6 ± 8.5		ME	[18911-76-7] [99/39]
C ₁₅ H ₁₅ Y	tris(cyclopentadienyl)yttrium	111.7 ± 3.5 99.2 ± 3.3	(298)		[1294-07-1] [82/20][74/23] [73/32]
C ₁₅ H ₂₁ O ₆ Y	tris(2,4-pentanedionato)yttrium(III)	98 ± 16			[15554-47-9] [84/34]
C ₃₃ H ₅₇ O ₆ Y	tris(2,2,6,6-tetramethylheptane-3,5-dionato)yttrium(III) (358–387) (357–377) (403–433)	151.0 ± 0.8 153.1 ± 0.4 135.9 117	(372) (366)	TE TE TG,DTA	[15632-39-0] [01/6] [01/6] [97/8] [97/3]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
	(382–412)	126	(397)	T	[96/4]
	(353–433)	117			[93/9]
	(353–433)	115.7			[93/9]
		138.5		GS	[90/15]
		66.5 (liq)		GS	[90/15]
	(363–418)	156.9	(388)	ME	[81/21]
		130.8		ME	[73/18]
$\text{C}_{32}\text{H}_{40}\text{F}_{12}\text{O}_8\text{NaY}$	sodium <i>tetrakis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)-yttrate				[12576-89-5]
	(418–503)	130 ± 3	(460)	T	[93/4]
	(463–503)	142 ± 12	(483)		[93/4]
Yb					
$\text{C}_{15}\text{H}_{15}\text{Yb}$	<i>tris</i> (cyclopentadienyl)ytterbium				[1295-20-1]
		108.8 ± 3.5	(298)		[82/20][74/23]
		96.2 ± 2.9			[73/32]
$\text{C}_{15}\text{H}_{21}\text{O}_6\text{Yb}$	<i>tris</i> (2,4-pentanedionato)ytterbium(III)				[14284-98-1]
	(364–404)	93.3			[81/7]
$\text{C}_{30}\text{H}_{30}\text{F}_{21}\text{O}_6\text{Yb}$	<i>tris</i> (1,1,1,2,2,3,3-heptafluoro-7,7-dimethyloctane-4,6-dione)ytterbium(III)				[18323-96-1]
	(339–356)	154.8 ± 3.3		ME	[71/25]
$\text{C}_{33}\text{H}_{57}\text{O}_6\text{Yb}$	<i>tris</i> (2,2,6,6-tetramethylheptane-3,5-dionato)ytterbium(III)				[15492-52-1]
		131.1 ± 2.7	(298)	DSC	[99/33]
	(363–413)	156.9	(388)	ME	[81/21]
	(410–444)	133.3	(427)	BG	[69/19]
Zn					
$\text{C}_4\text{H}_{16}\text{Cl}_2\text{N}_8\text{S}_4\text{Zn}$	<i>trans</i> -dichloro- <i>tetrakis</i> (thiourea) zinc(II)				[28813-20-9]
	(351–382)	90 ± 20			[70/11]
$\text{C}_{10}\text{H}_{14}\text{O}_4\text{Zn}$	<i>bis</i> (2,4-pentanedionato)zinc(II)				[14024-63-6]
		132.6 ± 8	(298)	C	[85/5]
		117 ± 3			[80/30]
$\text{C}_{10}\text{H}_{20}\text{N}_2\text{S}_4\text{Zn}$	<i>bis</i> (diethyldithiocarbamate)zinc(II)				[14324-55-1]
		115 ± 15	(298)	DSC,E	[00/40]
	(401–444)	143.1	(422.5)		[87/4]
		142.7 ± 2.5		GC	[76/21]
$\text{C}_{14}\text{H}_{28}\text{N}_2\text{S}_4\text{Zn}$	<i>bis</i> (dipropyldithiocarbamate)zinc(II)				[15694-56-1]
		147 ± 2	(298)	DSC,E	[92/19]
$\text{C}_{18}\text{H}_{12}\text{N}_2\text{O}_2\text{Zn}$	<i>bis</i> (8-hydroxyquinolino)zinc(II)				[13978-85-3]
		183.2 ± 6.3	(298)	ME	[94/16]
	(473–513)	167.9 ± 6	(493)	ME	[84/12]
		178 ± 6	(298)		[84/12]
$\text{C}_{18}\text{H}_{36}\text{N}_2\text{S}_4\text{Zn}$	<i>bis</i> (dibutyldithiocarbamate)zinc(II)				[136-23-2]
		107 ± 3	(298)	DSC,E	[91/15]
$\text{C}_{18}\text{H}_{36}\text{N}_2\text{S}_4\text{Zn}$	<i>bis</i> (diisobutyldithiocarbamate)zinc(II)				[36190-62-2]
		283 ± 2	(298)	DSC,E	[94/33]
$\text{C}_{20}\text{H}_{16}\text{N}_2\text{O}_2\text{Zn}$	<i>bis</i> (8-hydroxy-2-methylquinolino)zinc(II)				[14128-73-5]
	(437–556)	172.0 ± 5.0	(541)	ME	[98/8]
		179.4 ± 5.0	(298)		[98/8]
$\text{C}_{22}\text{H}_{38}\text{O}_4\text{Zn}$	<i>bis</i> (2,2,6,6-tetramethylheptane-3,5-dionato)zinc(II)				[14363-14-5]
		136.0		ME	[73/18]
$\text{C}_{22}\text{H}_{44}\text{N}_2\text{S}_4\text{Zn}$	<i>bis</i> (dipentyldithiocarbamate)zinc(II)				[15337-18-5]
		127 ± 3	(298)	DSC,E	[00/40]
$\text{C}_{44}\text{H}_{28}\text{N}_4\text{Zn}$	5,10,15,20-tetraphenylphosphine zinc (II)				[14074-80-7]
	(563–663)	213 ± 3			[94/38][01/2]
		109	(666)	UV/Vis	[71/29][01/2]
Zr					
$\text{C}_{10}\text{H}_{10}\text{Cl}_2\text{Zr}$	<i>bis</i> (cyclopentadienyl)zirconium dichloride				[1291-32-3]
		108.5 ± 4.6	(298)	ME	[01/3]
	(393–457)	100.3	(425)		[87/4]
		105.0 ± 2.1	(298)		[82/20][76/14]
		100.4 ± 1.7			[77/23]
		96.7			[69/20]
		103 ± 13	(298)		[68/19][01/3]
$\text{C}_{12}\text{H}_{16}\text{Zr}$	<i>bis</i> (cyclopentadienyl)dimethylzirconium				[1291-32-3]
		81.2 ± 2.1	(298)		[82/20][76/14]
$\text{C}_{20}\text{H}_4\text{F}_{24}\text{O}_8\text{Zr}$	<i>tetrakis</i> (1,1,1,5,5,5-hexafluoro-2,4-pentanedionato)zirconium(IV)				[19530-02-0]

TABLE 7. Enthalpies of sublimation of organometallic and inorganic compounds, 1910–2001—Continued

Element	Compound	$\Delta_{\text{sub}}H_{\text{m}}/\text{kJ mol}^{-1}$	(T_{m}/K)	Method	CAS registry number
Molecular formula/polymorph	Temperature range (K)				Reference
$\text{C}_{20}\text{H}_{16}\text{F}_{12}\text{O}_8\text{Zr}$	(333–363)	59		TGA	[00/35]
	(366–456)	$48.6 \pm 0.6(\text{liq})$	(411)	T	[96/2]
	<i>tetrakis</i> (1,1,1-trifluoro-2,4-pentanedionato)zirconium(IV)				[17499-68-2]
	(373–403)	94		TGA	[00/35]
	(368–398)	133.6 ± 2.0	(383)	SMZG	[96/2]
$\text{C}_{20}\text{H}_{28}\text{O}_8\text{Zr}$		118.7 ± 3.1	(298)	C	[92/7]
	(383–438)	126.4 ± 1.7		GS	[85/16]
	(383–438)	119.2 ± 1.7		GS	[85/16]
	<i>tetrakis</i> (2,4-pentanedionato)zirconium(IV)				[17501-44-9]
	(413–443)	126		TGA	[00/35]
	(403–433)	138.8 ± 2	(418)	SMZG	[96/2]
		125.8 ± 2.9	(298)	C	[92/7]
		132.0 ± 6.8	(463)		[87/14]
$\text{C}_{22}\text{H}_{20}\text{Zr}$	<i>bis</i> (cyclopentadienyl)diphenylzirconium				[51177-89-0]
$\text{C}_{32}\text{H}_{40}\text{F}_{12}\text{O}_8\text{Zr}$		92.0 ± 4.2			[76/14]
	<i>tetrakis</i> (1,1,1-trimethyl-5,5,5-trifluoro-2,4-pentanedionato)zirconium(IV)				[56044-44-1]
$\text{C}_{44}\text{H}_{28}\text{N}_4\text{Zn}$	(388–423)	134.9 ± 1.6	(406)	SMZG	[96/2]
	5,10,15,20-tetraphenylporphine zinc(II)				[14074-80-7]
$\text{C}_{44}\text{H}_{76}\text{O}_8\text{Zr}$		208 ± 4		GS	[00/36]
	<i>tetrakis</i> (2,2,6,6-tetramethylheptane-3,5-dionato)zirconium(IV)				[18865-74-2]
ZrCl_4	(413–443)	120		TGA	[00/35]
	zirconium tetrachloride				[10026-11-6]
ZrF_4	(405–518)	98.9 ± 0.5	(512)	T	[94/22]
	zirconium tetrafluoride				[7783-64-4]
	(696–856)	240.0 ± 0.1	(298)	TE	[94/36]
	(796)	243.0	(298)	MS	[65/14][94/36]
	(983–1177)	241.1 ± 0.1	(298)		[64/9][94/36]
	(681–913)	242.6 ± 1.7	(298)	MS	[63/9][94/36]
	(713–873)	232.3 ± 1.2	(298)		[63/8][94/36]
	(983–1081)	239.9 ± 0.2	(298)		[58/11][94/36]
	(890–1150)	241.8 ± 0.6	(298)	GS	[54/9][94/36]

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Unpublished Results

U/1 C. G. DeKruif (unpublished results).

U/2 C. H. D. van Ginkel, C. G. DeKruif, and F. E. B. de Waal (unpublished results).

U/3 H. Mackle, W. V. Steele, and D. V. McNally (unpublished results).

U/4 D. Stull (unpublished results).

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